The Wedge Method

A Builder's Guide to Timber Geodesic Domes

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Generated with code @ https://github.com/dkzeanah/DomeSimulator

Disclaimer

This book documents construction methods, geometric principles, experimental designs, and prototype concepts. It is not a substitute for site-specific structural engineering. Building codes, loads, materials, soil conditions, permitting requirements, electrical rules, plumbing rules, and fire-safety requirements vary by location. Structural designs intended for permanent occupancy should be reviewed for the applicable site, loads, materials, connections, foundation, and local requirements by appropriately qualified professionals and authorities.

Introduction

A geodesic dome looks complicated until you stop looking at the whole dome.

From a distance it looks like a curved structure made from dozens of parts pointing in different directions. Up close, it becomes something much simpler. It is a collection of straight pieces arranged into triangles.

That difference is important.

The dome looks curved, but the individual structural members do not have to be curved. A builder can create a large rounded structure using straight pieces of lumber, repeatable cuts, and a relatively small number of different part dimensions.

Most geodesic dome systems solve the problem of connecting those pieces with some type of hub, plate, bracket, flattened strut end, or specially manufactured connector.

The wedge method approaches the problem differently.

Instead of treating the strut as a rectangular member that must somehow be forced into the geometry of the dome, the shape of the structural member itself becomes part of that geometry.

The member is cut as a wedge.

The narrow portion of the wedge faces toward the inside of the dome and the wider portion faces toward the outside. When the members are arranged into triangles and the triangles begin joining one another, the changing thickness of the members helps create the angular relationship between neighboring faces.

This does not eliminate geometry. It uses geometry.

That distinction is the basis of this book.

The purpose of this book is not to describe one dome with one set of measurements. The goal is to explain a method of building geodesic structures so that the builder understands what the parts are doing, how they can be produced, how the system can be scaled, and how different construction approaches can grow from the same basic idea.

Some versions may begin with ordinary dimensional lumber. Others may begin with larger timbers. A builder with access to logs may choose to divide the log longitudinally and recover multiple wedge-shaped structural members from the same piece of wood.

Other versions could use laminated members, manufactured components, metal reinforcement, removable panels, insulated shells, or entirely different enclosure systems.

The frame and the enclosure also do not necessarily have to be treated as the same thing.

A structural dome can support plywood panels, fiberglass panels, transparent panels, insulated composite panels, or a removable shell. Doors and windows can be introduced by reinforcing openings in the triangular structure. Utilities can enter through the floor, through a dedicated service location, or through a central utility system.

Once the geometry and construction method are understood, the dome stops being one particular building and becomes a platform.

That is where this book is headed.

We will begin with the basic geometry of the dome and the wedge-shaped structural member. From there we will move through material selection, fabrication, repetitive cutting, connections, assembly, foundations, panels, openings, utilities, and several variations of the system.

A reference dome will be used throughout the book so that the geometry does not remain abstract. The same principles will then be scaled and modified to show how the method can be applied to structures of different sizes and purposes.

Some information in this book is mathematical. Some comes from fabrication and construction. Some sections explore design possibilities that may require additional testing or engineering before being used in an occupied structure.

Those categories will be identified rather than treated as if they are the same thing.

A dome intended to hold tools is not necessarily designed to the same requirements as a permanent residence. Snow, wind, soil, foundations, fasteners, fire safety, electrical systems, plumbing systems, and local building requirements can all change what is acceptable for a particular project.

This book therefore focuses on understanding and constructing the system rather than replacing project-specific structural engineering or local code requirements.

The basic idea, however, is simple:

Start with straight material.

Understand the triangle.

Shape the member to participate in the geometry.

Repeat the part accurately.

Connect the triangles.

And allow many simple pieces to become one structure.

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Figures

Every picture in this book was made by running one of this project's own tools and photographing the result. A figure is the raw-wedge solver, operated with the settings that make one idea visible — the frame opened far enough to see into a seam, the wireframe switched off, the camera stood off the shell at a stated distance. A plate is a frame of one of the project's films, taken at a named chapter of it, and is how the book shows the things the solver does not model: a doorway, a foundation, the utility column, a 3V dome.

Nothing here was drawn by hand, and every one can be made again from the recipe stored beside it.

Chapter 1: 5 (3 figures, 2 plates)
Chapter 2: 10 (7 figures, 3 plates)
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Part I - Understanding the System

1. Why Build a Dome?

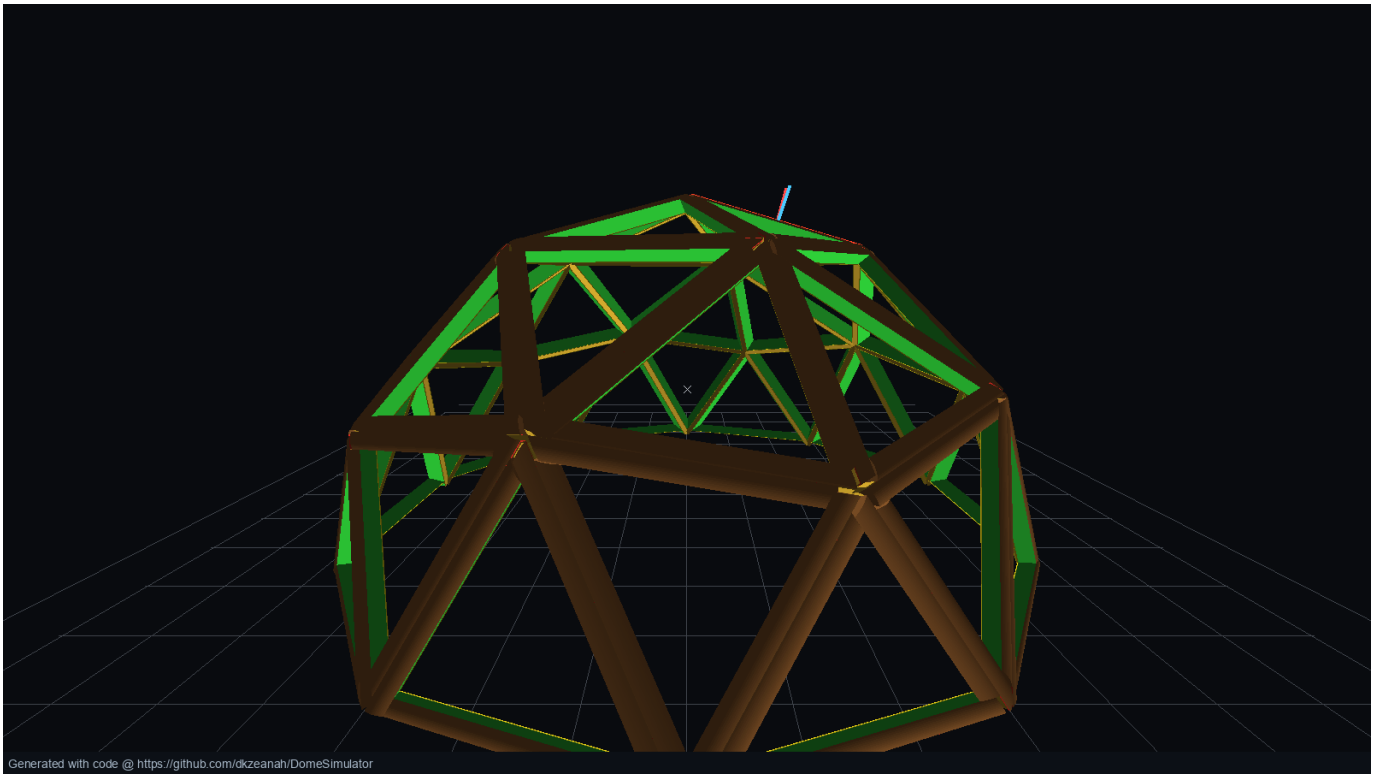

Figure 1. The reference build

A 2V dome on a six-foot longest member, wedge point inward, raw trapezoid seams. This is the dome the rest of the book assumes unless it says otherwise. Generated with code @ https://github.com/dkzeanah/DomeSimulator

What a geodesic dome is

A geodesic dome is a rounded structure built from a network of straight members.

That is the first thing to understand about it.

The finished structure may look curved, but the individual pieces making up the frame do not have to be curved. Instead, straight members are connected into a repeating network of triangles. As more of those triangles are joined together at slightly different angles, the overall frame begins to follow the shape of a sphere.

In practical terms, we are using a collection of flat triangular faces to approximate a curved surface.

A simple way to picture this is to imagine replacing the smooth surface of a ball with a large number of flat panels. A ball made from only a few large flat faces would look obviously angular. Divide those faces into smaller pieces and the shape begins to look progressively rounder.

A geodesic dome works on the same basic idea.

The frame is not one continuous curved beam. It is a collection of straight structural members whose lengths and connection geometry place them at the correct points in space.

This matters to a builder because straight pieces are much easier to produce repeatedly than large curved ones. A saw can cut another member to the same length. A jig can hold another piece in the same position.

A stop can reproduce a cut. A template can reproduce a panel. The finished building may have a complicated three-dimensional shape, while much of the actual fabrication is reduced to making the same small group of parts over and over.

GEOMETRY

A complete sphere can be described using points, straight lines between those points, and the triangular faces created by those lines. A dome uses only part of that spherical structure.

The exact portion of the sphere used depends on the design.

Some domes are approximately hemispherical. Others stop above or below the halfway point. A dome may also be placed on a vertical wall, a raised base, or another supporting structure.

For that reason, saying that two structures are both "geodesic domes" does not mean they automatically use the same strut lengths, number of triangles, base shape, or connection angles.

The underlying geometry matters.

The frequency of the dome also matters. Frequency describes how far the original triangular faces used to develop the sphere have been subdivided. A higher-frequency dome generally contains more members and smaller triangular faces.

The full frequency calculations and the exact strut families used in this book are covered later. For now, the important point is that the rounded shape is produced by a calculated arrangement of straight pieces rather than by bending the frame into a curve.

The reference structure used throughout this book is a 20-foot-diameter 2V dome. That description tells us two useful things. The 20-foot dimension tells us the intended diameter of the reference structure.

The 2V designation tells us the frequency of the geodesic subdivision.

It does not, by itself, tell us every dimension needed to build the dome. The exact sphere geometry, truncation, base configuration, strut lengths, and joint geometry still have to be defined before a reliable cut list can be produced.

That distinction is important throughout this book. A drawing that looks like a 2V dome is not enough information to start cutting lumber.

The geometry has to be established first.

THE WEDGE METHOD

The wedge method does not change the basic definition of a geodesic dome. We still have straight members forming triangular faces and those faces joining together to create a rounded three-dimensional structure. What changes is the shape of the structural member and the way we use that shape at the joints.

A conventional wood dome can begin with rectangular lumber. The builder then has to deal with the fact that neighboring triangular faces do not all lie in the same plane. Their edges meet at angles.

Different dome systems handle that problem in different ways. They may use metal hubs, flattened pipe ends, plates, brackets, compound cuts, flexible connections, or other hardware.

The wedge method approaches the same three-dimensional problem by making the cross-section of the timber itself part of the system. Instead of every member remaining rectangular from its inside face to its outside face, the member is formed as a wedge. The narrow portion is oriented toward the interior of the dome and the wider portion toward the exterior.

When those members are assembled into the frame, the wedge shape becomes part of how neighboring faces transition around the curved surface. That does not mean the wedge automatically determines every connection angle. It also does not mean that any wedge-shaped piece of lumber can be assembled into a dome.

The strut lengths, triangle geometry, vertex arrangement, member orientation, and connection details still have to agree with one another. The useful idea is that the shape of the timber can participate in the three-dimensional geometry instead of being treated only as a straight rectangular stick that needs separate hardware to resolve every change in direction. From a builder's point of view, this gives us two geometries to think about.

The first is the geometry of the dome itself: where every vertex and strut belongs in space. The second is the geometry of the individual wedge member: its width, thickness, taper, orientation, end treatment, and connection to the neighboring pieces. The rest of the method comes from making those two geometries work together.

This is also why the wedge method can have several variations. The starting material could be dimensional lumber. It could be larger sawn timber.

It could be material recovered from a log by dividing the log longitudinally into usable wedge-shaped sections. A wedge could be solid, laminated, reinforced, or incorporated into a larger panel system.

Those are different fabrication strategies built around the same basic idea.

DESIGN VARIATION

The frame should also be thought of separately from the outer shell.

A geodesic frame defines the structural geometry. What is attached to that frame can vary.

A dome might use plywood triangles, fiberglass panels, sheet material, transparent sections, insulation assemblies, removable panels, or another enclosure system. That distinction becomes useful later because changing the enclosure does not necessarily require changing the entire geometric system underneath it.

The same is true for floors, foundations, doors, windows, and utility routing. These systems have to interact with the dome, but they are not what makes the structure geodesic.

At its simplest, the dome can be understood as a three-dimensional framework of calculated straight members. Everything else is built around that framework. For this book, that gives us a useful order of operations.

First we establish the dome geometry. Then we determine what the wedge-shaped member has to do inside that geometry. Then we work out how to manufacture those members repeatedly.

After that, we can deal with the connections, assembly sequence, openings, shell, floor, foundation, utilities, and the different ways the same basic system can be modified. The dome may look complicated when viewed as one complete object. For fabrication, we will break it back down into individual parts.

That is where the system becomes manageable.

Figure 2. Standing in it

The same dome from the middle of the floor, looking up at the apex. Every triangle you can see is one of two shapes. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Why triangles are used

Triangles are used because three connected sides can define a shape without needing another member to keep that shape from folding sideways. That makes them useful anywhere we want a frame to hold a predictable geometry. Take four straight pieces and connect them into a rectangle.

If the corners are allowed to rotate, the rectangle can lean sideways and become a parallelogram without changing the length of any of the four sides. Nothing has to stretch. Nothing has to get shorter.

The joints only have to rotate. Now add a diagonal brace across that rectangle. The diagonal divides it into two triangles.

At that point, changing the shape requires one or more of the member lengths to change, a connection to slip, or a member to bend. That is the basic reason triangular framing appears throughout a geodesic dome.

A triangle is determined by its geometry in a way that a four-sided frame is not. If we know the lengths of all three sides of a valid triangle, its angles are also determined.

For sides a, b, and c, the angle C opposite side c can be found with the law of cosines:

$$c^2 = a^2 + b^2 - 2ab\cos(C)$$

Rearranging it gives:

$$C = \cos^{-1}((a^2+b^2-c^2) / 2ab)$$

The other two angles can be calculated the same way.

This matters in a dome because the strut lengths are not just pieces we cut to fill space. Their lengths establish the triangular faces.

Once those lengths are correct, the triangle has a defined shape. The next problem is then how that triangle meets the triangles around it in three-dimensional space. That second problem is where the dome's curvature appears.

The individual triangle can be treated as a flat geometric face. The neighboring triangle is also a flat face. The angle between those two faces turns the collection of flat triangles into a rounded structure.

A dome is therefore not made by bending each triangle. The curvature comes from changing the orientation of one triangle relative to the next.

For the wedge method, that distinction is especially useful.

We can first think about the geometry inside one triangular face:

- What are its three side lengths?
- Which strut family forms each side?
- What are its three internal angles?
- Is it equilateral, isosceles, or scalene?

Then we deal with the three-dimensional relationship between that face and the faces beside it.

The wedge-shaped member is part of solving that second problem.

TRIANGLES AS REPEATABLE BUILDING UNITS

From a fabrication standpoint, triangles also give us manageable units. Instead of trying to position every member of the dome independently in open space, we can think in terms of repeated triangular relationships. If a dome uses two triangle types, for example, we do not have to invent forty different shapes just because the finished dome contains forty triangular faces.

We identify the repeating geometry and make those parts consistently. This is one of the places where geodesic construction begins to make more sense from a builder's point of view. The finished frame is complicated.

The list of unique parts may not be. A group of accurately repeated strut lengths can produce a much larger three-dimensional form when those struts are assembled in the correct pattern. The triangle becomes the bridge between those two scales.

At the fabrication scale, we are cutting individual pieces. At the assembly scale, those pieces become triangles.

At the building scale, the triangles become the dome.

WHY THIS MATTERS TO THE WEDGE METHOD

In a conventional rectangular timber frame, the member is usually treated as having the same basic cross-section from one end to the other. The connection hardware or the end cuts then have to deal with how one face of the structure changes direction relative to another. The wedge method gives us another variable to work with.

The member itself has a narrow side and a wide side. When the narrow side is oriented toward the interior and the wider side toward the exterior, the cross-section can participate in the transition between neighboring triangular faces. The triangle is still defined by its strut geometry.

The wedge does not replace that geometry. It changes how we can physically realize that geometry in timber. This means there are several different angles in the system that should not be confused with one another.

There are the internal angles inside a triangular face. There are the angles created where struts meet at a vertex. There are the angles between neighboring triangular faces.

There are also the angles created by the wedge-shaped cross-section of the timber itself. They are related, but they are not interchangeable. A common mistake would be to calculate one of those angles and assume it is automatically the saw setting needed for another part of the joint.

We will keep them separate throughout the book.

REFERENCE BUILD

The 20-foot-diameter 2V reference dome will contain repeating triangular faces made from the strut families established by its exact geodesic geometry. Once those strut lengths are formally defined, we can calculate the internal angles of each triangle directly from their side lengths.

For a triangular face with verified side lengths a, b, and c, the calculation will be:

$$C = \cos^{-1}((a^2+b^2-c^2) / 2ab)$$

That gives the flat internal angle of the triangle. It does not yet give us the angle between that triangle and the neighboring face.

That relationship is handled separately when we get into the three-dimensional dome geometry and the wedge itself. This separation is important because it keeps the fabrication logic clean. First determine the correct strut lengths.

Then verify the triangle. Then determine how the triangles meet. Then determine what the wedge member has to do at those connections.

If we mix those steps together, it becomes much easier to cut the right number at the wrong angle.

For now, the main point is simple:

Triangles let us create repeatable, fixed geometric units from straight members.

The dome comes from arranging those units correctly in three-dimensional space.

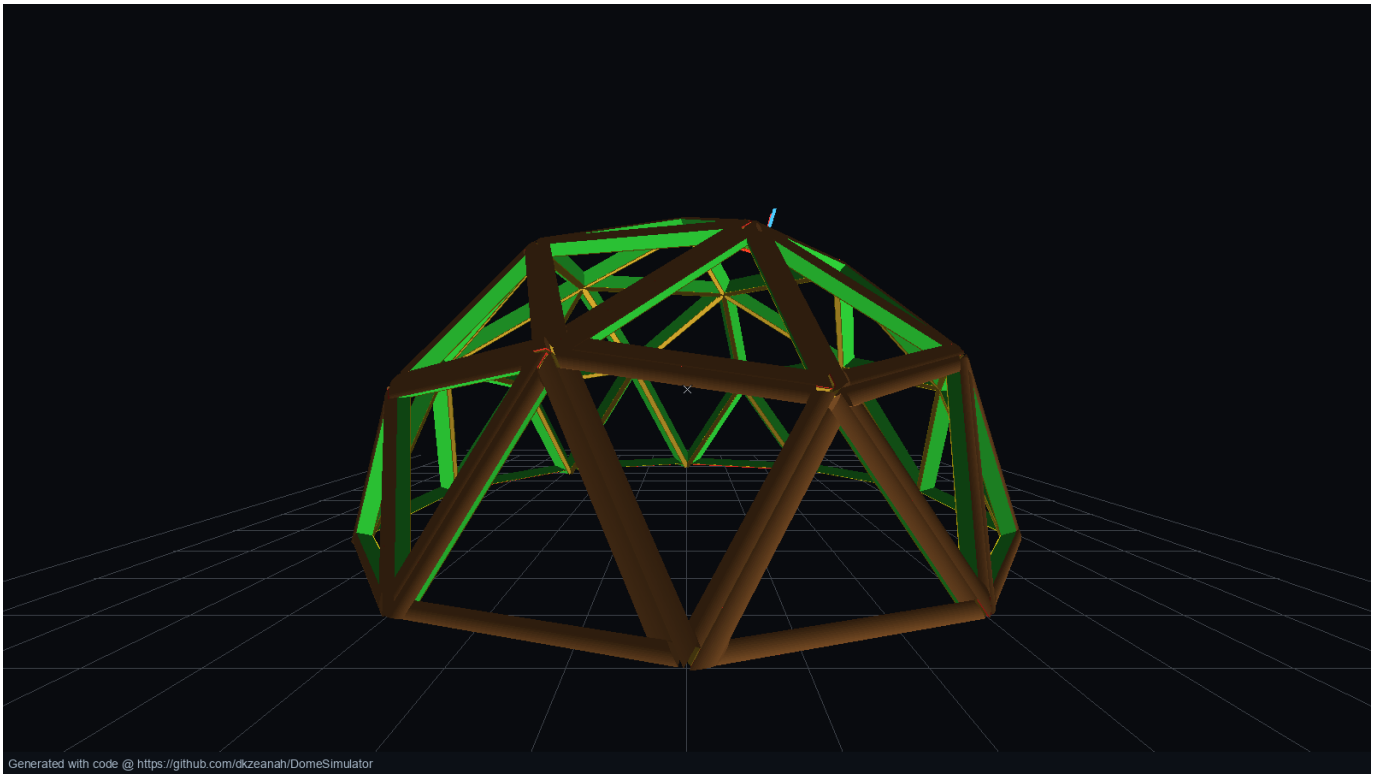

Figure 3. Half a sphere, ten sides on the ground

The dome level-on, which is the view that shows a hemisphere is what this is: the apex is the radius and the base is a decagon. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Frequency

Frequency is how many times the original triangle has been divided.

A geodesic dome starts as an icosahedron: twenty identical triangles wrapped round a ball. That is frequency one. Divide every edge in two and push the new points out to the sphere, and you have frequency two. Divide in three, and you have frequency three.

Each division makes the dome rounder and makes everything about it smaller and more numerous.

WHAT IT COSTS TO GO UP ONE

frequency	struts	lengths	panels	vertices	
1V	25	1	15	11	
2V	65	2	40	26	<- this book
3V	165	5	105	61	
4V	250	6	160	91	

Those are counted by building each one, at the same radius, rather than read off a table.

WHY THIS BOOK IS 2V

Three reasons, and only the first is about the shape.

Two lengths. A 2V dome has 2 strut lengths. A 3V has 5. Every extra length is another saw setting, another batch to keep separate, another chance to reach for the wrong stick on the forty-first panel.

A number of parts one person can handle. 65 geodesic edges becomes 120 members once the pinwheel joint duplicates each edge into the two panels that share it. At 3V that would be 165 edges and 304 members, which is a different kind of project.

The member is the right size. At the reference build's diameter a 2V long member is 72.000 inches — a six-foot stick, which is what comes out of a twelve-foot log section, what one person carries, and what fits in a pickup. Going to 3V at the same diameter makes every member about three feet, which wastes the length a tree gives you for free.

WHEN TO GO UP

Above about twenty feet across. The 2V's flat panels get large enough to start carrying bending they should not, the faceting becomes obvious, and the member length passes comfortable one-person handling. Chapter 19 says where that line is and what this book does and does not know about it.

WHAT FREQUENCY IS NOT

It is not a measure of quality. A 4V dome is not a better building than a 2V one; it is a rounder one with 185 more struts and 4 more lengths to keep track of. Pick the frequency that gives you a member you can make and carry, and let the shape follow.

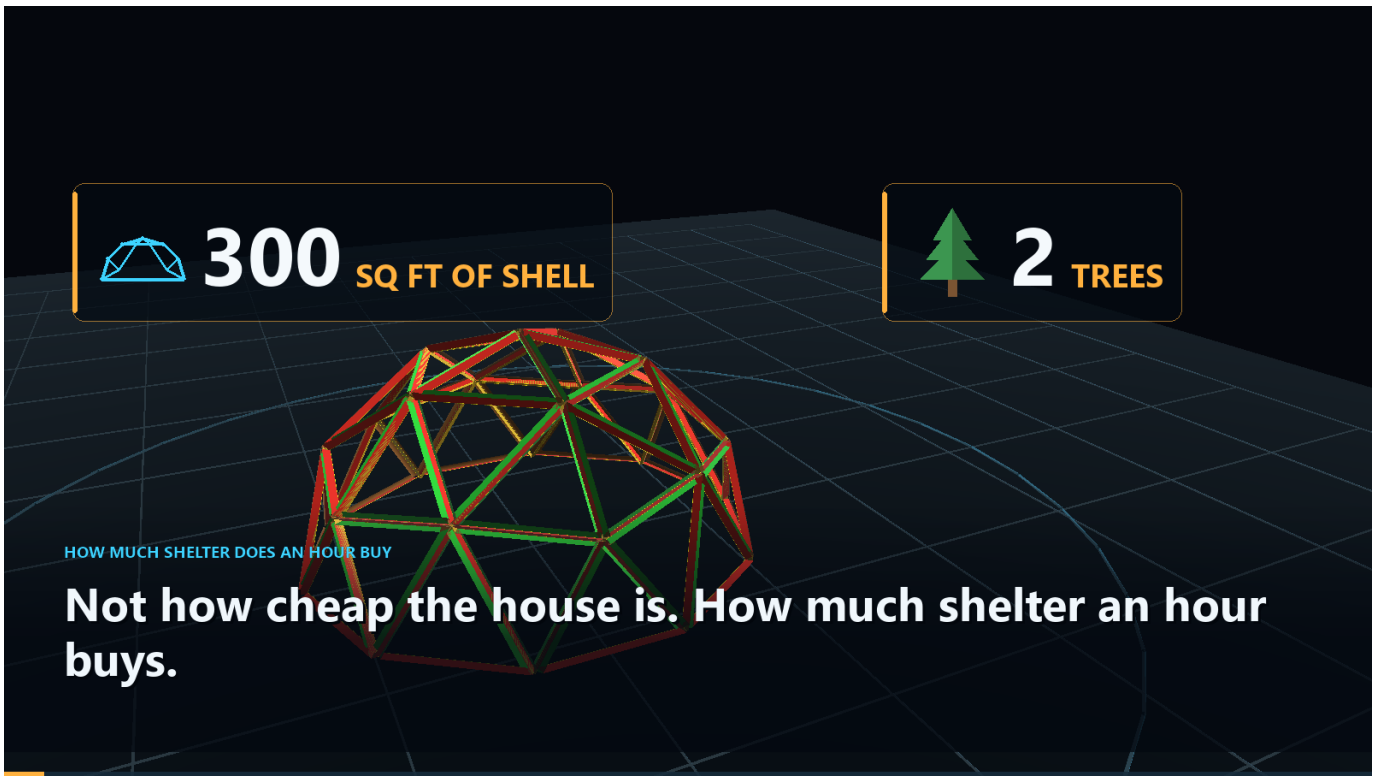

Figure 4. How much shelter does an hour buy

Not how cheap the house is. How much shelter an hour buys. The case for this dome is not that it makes a house cheap. It is that it changes the trade between a person, their time, raw wood and shelter. This is one shell, 300 square feet, from 2 trees. The question worth asking is not how cheap the house is, but how much shelter one hour of your life can buy, and this way of building changes the answer. From Why Build This Way, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Chords versus curved surfaces

A dome may follow a curved surface, but the timber member between two connection points is straight. That difference is where the chord comes in. If we mark two points on a circle or sphere, there are two basic ways to describe the distance between them.

One is to follow the curved surface. The other is to connect the two points directly with a straight line.

That straight line is a chord.

GEOMETRY

For dome construction, the structural member normally follows the chord between two vertices rather than following the curved surface of the sphere.

If two points on a circle are separated by a central angle θ, and the sphere or circle has radius R, the straight chord length between those points is:

$$c = 2R \sin(\theta / 2)$$

where:

- c is the straight chord length,
- R is the radius,
- and θ is the central angle between the two points.

The curved distance along the circle between the same two points is the arc length:

$$s = R\theta$$

when θ is measured in radians. Those two distances are not the same.

The arc follows the curve. The chord cuts straight across between the endpoints. For any normal dome segment, the chord is shorter than the arc connecting the same two points.

That matters because we are cutting straight structural members. If a calculator gives us the spherical surface distance but we cut a straight timber to that number, the member will be too long for the chord it is supposed to occupy. The dome geometry therefore has to distinguish between the mathematical sphere we are approximating and the straight pieces we are actually building from.

STRAIGHT MEMBERS, CURVED RESULT

A geodesic dome does not need its struts bent into the spherical radius. Each individual strut is straight. The curved form appears because the endpoints of those straight struts are positioned on or near the intended spherical geometry.

A single triangle is flat. The triangle beside it is also flat. But the two triangles do not remain in the same plane.

They meet at an angle. Repeat that relationship over many faces and the collection of flat triangles begins to surround a volume that approximates part of a sphere.

This gives us three different things that are easy to accidentally treat as one:

1. the ideal curved sphere, 2. the flat triangular faces approximating that sphere, 3. the straight structural members forming the edges of those faces.

They are related, but they are not geometrically identical. A drawing of a smooth sphere represents the intended overall surface. A geodesic frame replaces that smooth surface with flat triangular facets.

The timber then forms the straight edges of those facets.

THE GAP BETWEEN A CHORD AND THE CURVE

We can also describe how far a chord falls inside the ideal curved surface. For a circular section, the maximum distance between a chord and the corresponding arc occurs at the midpoint of the chord.

That distance is commonly called the sagitta.

If the radius is R and the chord length is c, it can be calculated as:

$$h = R-\sqrt{(R^2-(c / 2)^2)}$$

where:

- h is the sagitta,
- R is the radius,
- and c is the chord length.

This gives us a useful way to think about frequency. Longer chords create larger flat facets across the curved surface. Shorter chords generally reduce the distance between each flat facet and the ideal spherical surface.

That is one reason increasing frequency makes the finished form appear smoother. We are not making the members curved. We are using more, smaller straight sections to approximate the curve more closely.

WHY THIS MATTERS WHEN BUILDING

This distinction affects more than the mathematical drawing. It affects where the shell sits. It affects how neighboring panels meet.

It affects the outer profile of the dome. It can also affect how we define the diameter. Suppose somebody says they are building a 20-foot dome.

That sounds specific, but we still need to know what surface the 20 feet refers to.

It could mean:

- the diameter through the theoretical spherical vertex locations,
- the outside face of the structural frame,
- the inside face of the structural frame,
- the outer surface of the enclosure,
- or another construction reference.

Those numbers do not have to be identical once the frame has real thickness. For mathematical work, the cleanest starting point is usually a defined spherical radius and a defined set of vertex locations. Physical

timber dimensions can then be developed around that geometry.

The alternative is dangerous: allowing the thickness of the material to silently change the geometry while still using calculations based on another radius.

THE WEDGE METHOD AND THE SPHERICAL SURFACE

This distinction becomes particularly important with wedge-shaped members. A rectangular strut has parallel major faces. A wedge member does not.

Its cross-section changes from a narrow interior side to a wider exterior side. Because of that, we need to be explicit about which line through the member corresponds to the calculated geodesic chord. Possible reference lines could include the interior edge, exterior edge, centerline, fastener centerline, or another deliberately chosen construction datum.

They cannot all represent the exact same spherical radius. If the calculated chord is based on vertex centerlines but the physical cut length is measured along the outside edge of a thick wedge, those two measurements may differ. That does not mean the geometry is wrong.

It means the mathematical model and the physical member are being measured from different references. The solution is to establish one reference system and keep using it. For example, the geometry may define every vertex from a common spherical center and establish each theoretical strut as the straight chord between those vertices.

The physical wedge can then be positioned around that theoretical line according to the joint design. Once that relationship is fixed, other dimensions can be derived from it instead of being guessed independently.

This is particularly useful when we start generating cut lists.

A cut list should not simply say:

"A strut = this long."

It should eventually make clear what that dimension represents.

Is it:

- theoretical vertex-to-vertex chord length,
- bolt-center distance,
- full timber length before end cuts,
- finished timber length,
- inside-edge length,
- or outside-edge length?

If we do not define that reference, two builders can use the same number and still manufacture different parts.

REFERENCE BUILD

For the 20-foot-diameter 2V reference dome, the process will be:

First, define exactly what the 20-foot diameter refers to.

Then establish the radius:

$$R = D / 2$$

If the defined mathematical diameter is 20 feet, then:

$$R = 10 \text{ ft}$$

or:

$$R = 120 \text{ in}$$

That radius alone is not enough to calculate every member. We still need the verified angular relationship between each pair of neighboring vertices in the selected 2V geometry.

Once a central angle θ for a particular strut family is known, its theoretical chord can be calculated from:

$$c = 2R \sin(\theta / 2)$$

That gives us the straight geometric distance between those vertices. The physical wedge member can then be developed from that chord according to the connection method and whichever construction reference line we decide to use.

This order matters. We should not begin with the outside surface of a piece of lumber and try to make the spherical geometry conform to it afterward. We establish the geometry first.

Then we decide how the real material fits around it. That keeps the ideal dome, the straight chord, and the finished wedge member as three separate but connected parts of the same system.

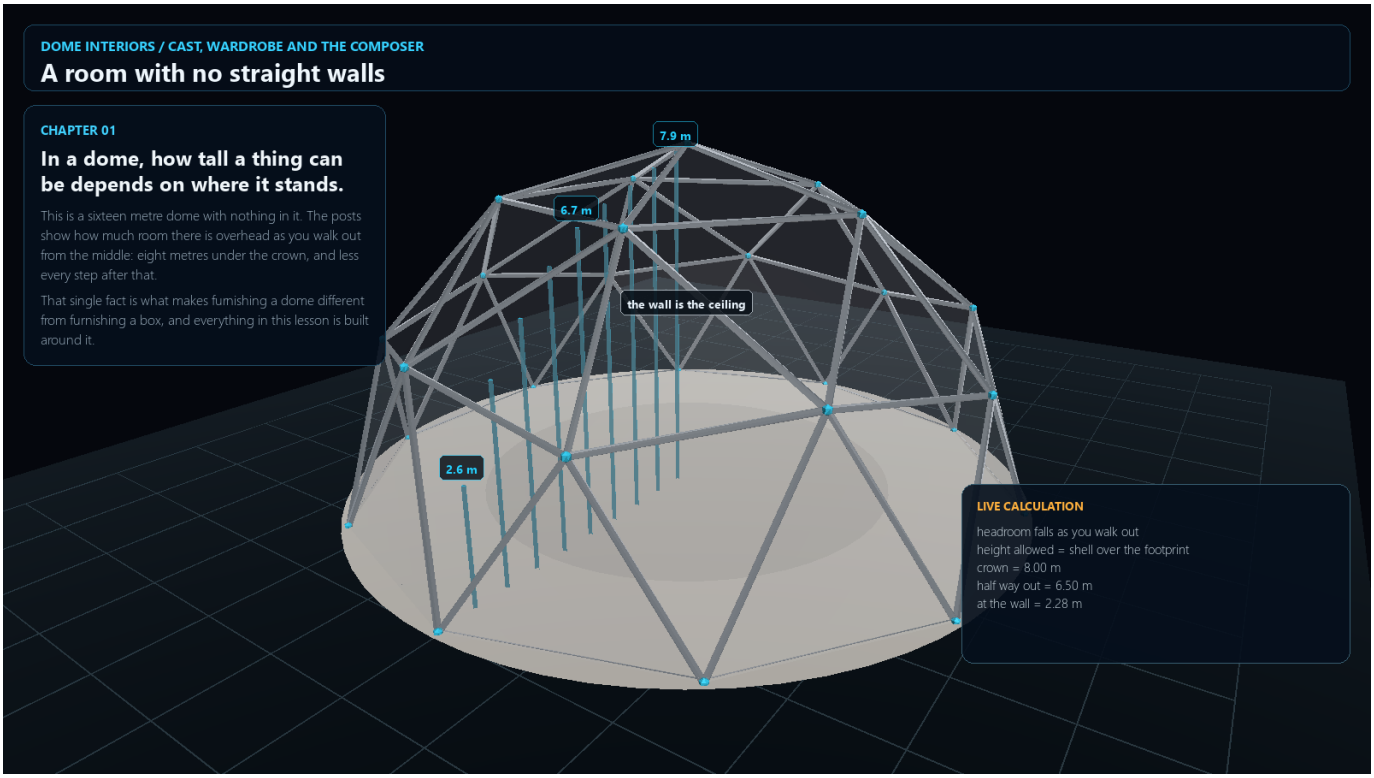

Figure 5. A room with no straight walls

In a dome, how tall a thing can be depends on where it stands. This is a sixteen metre dome with nothing in it. The posts show how much room there is overhead as you walk out from the middle: eight metres under the crown, and less every step after that. From The Dome House Lookbook, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Traditional strut/hub construction

Traditional geodesic dome construction usually treats the struts and the hubs as two separate parts of the system. The struts establish the distances between vertices. The hubs solve the problem of bringing several struts together at one point in three-dimensional space.

That separation is useful because it lets the strut itself remain relatively simple. A strut can be a piece of wood, tube, pipe, angle, or another straight structural member. The complicated directional change happens at the hub.

In practical terms, a hub is the connection point where several members meet. Depending on the dome geometry, a hub may receive five, six, or another number of struts. Those struts do not all approach from one flat plane.

They arrive from different directions because each one belongs to a different triangular face of the dome.

The hub has to accommodate that three-dimensional relationship.

TRADITIONAL HUB LOGIC

A simple flat frame can be bolted together through one plate because the members may all lie in nearly the same plane. A dome is different. At a dome vertex, the neighboring triangles wrap around the surface.

That means the struts meeting at the vertex point outward in different directions.

A connection system has to deal with two basic things:

1. where the end of each strut belongs, 2. what direction each strut leaves the vertex.

The mathematical geometry can define those directions precisely. The physical hub then has to reproduce them closely enough for the structure to assemble. Different dome systems solve this in different ways.

Some use a central metal hub with separate tabs or arms. Some flatten the ends of metal tubing and bolt the flattened ends together. Some use steel plates on one or both sides of wood members.

Some use pipe sections, rings, fabricated sockets, brackets, or custom-machined connectors.

The exact hardware changes, but the general idea is the same:

Keep the long structural members relatively simple and concentrate the connection geometry at the vertices.

STRUT LENGTH AND HUB GEOMETRY ARE RELATED

The geometric chord between two theoretical vertices is not always the same dimension as the physical piece of material that gets cut. That depends on the connection design. Suppose two theoretical vertices are separated by a chord length c.

If a hub occupies some physical distance at each end of that chord, the strut may stop short of the actual vertex location.

A simple conceptual relationship is:

$$L_{physical} = c - e_1 - e_2$$

where:

- $L_{physical}$ is the physical member length between its connection interfaces,
- c is the theoretical vertex-to-vertex chord,
- e_1 is the connection offset at one end,
- e_2 is the connection offset at the other end.

That is not a universal dome-cutting formula. It is a reminder that the connector has physical size. If the connection reference is bolt center to bolt center instead, then the useful shop dimension may be the distance between bolt centers rather than the overall timber length.

This is why dome plans need to state what their strut dimensions actually measure.

A list of numbers without a connection reference can be misleading.

HUBS AS GEOMETRY CONVERTERS

One useful way to think about a hub is as a geometry converter. The strut itself may be a rectangular piece of lumber. The dome around it is not rectangular.

The hub sits between those two conditions. It lets a straight member with a simple cross-section participate in a structure where neighboring members leave the connection at different angles. That can make fabrication easier in one part of the project.

Instead of shaping every long member to resolve the three-dimensional joint, we manufacture or purchase a connector designed to do it. The tradeoff is that the connector becomes a specialized component. Now the build depends on both the struts and the hubs being correct.

If the strut lengths are right but the hub geometry is wrong, the dome will not fit correctly. If the hub is right but the struts are cut to the wrong reference length, the same problem occurs. The connection system and the chord geometry have to be treated as one coordinated design.

WOOD STRUTS WITH METAL HUBS

For timber domes, one common approach is to keep the wood members rectangular and attach their ends to metal hardware.

The hardware can provide:

- a bolt location,
- spacing between members,
- rotational freedom during assembly,
- a defined connection angle,
- and reinforcement around the joint.

This can simplify the wood fabrication. Instead of cutting a complicated three-dimensional joint into every timber, the builder may only need to cut struts to length, drill them accurately, and attach the hardware. That is a strong manufacturing advantage when the hub system is already available and correctly designed.

It also introduces another manufacturing problem. The hubs themselves now have to be bought or made. If the dome uses several different vertex conditions, the connector system may need multiple hub types, different tab angles, or enough flexibility to accommodate those differences.

That means the structure may have only a few strut families but still require a carefully organized connection inventory.

ASSEMBLY WITH HUBS

Hub systems can also change the assembly process. Individual struts can often be connected directly at vertices without first producing complete triangular panels. That allows the dome to grow as a network of members.

One strut connects to a hub. Another strut connects to the next hub. As more pieces are added, the triangular pattern begins locking the structure into its intended form.

During this stage, the partially assembled frame may still be flexible. Fasteners may be left slightly loose so the geometry can settle into position before final tightening. Temporary supports or bracing may also be needed depending on the assembly sequence.

The connection system has to allow enough practical movement for assembly without relying on uncontrolled looseness as part of the finished structural design.

SAFETY / ENGINEERING

A hub is not just a positioning device. In a finished structure, it is part of the load path. Forces moving through one strut have to pass through the connection and into neighboring members.

That means hub plates, bolts, screws, welds, timber around holes, edge distances, washers, bearing surfaces, and connection geometry can all matter structurally. A connection that successfully holds the pieces together during assembly is not automatically adequate for a permanent occupied structure. The required connection capacity depends on the structure, loads, materials, fasteners, geometry, and applicable building requirements.

This book does not assign structural capacity to a hub simply because the parts can be assembled.

WHY THE WEDGE METHOD TAKES A DIFFERENT APPROACH

The wedge method starts from the same geometric problem but changes where some of that geometry is handled. Traditional strut-and-hub construction can keep a rectangular strut simple and place more of the three-dimensional connection problem into the hub. The wedge method moves part of that problem into the timber member itself.

Instead of using only a rectangular cross-section and asking the connector to accommodate the entire change between neighboring faces, the timber is shaped so that its interior and exterior widths are different. The wedge therefore participates in the face-to-face transition. This does not automatically eliminate every plate, bolt, screw, or connector.

It also does not mean that hubs are a bad solution. They are simply a different way of dividing the same geometric problem.

We can think of the two approaches like this:

Traditional strut/hub construction:

simple member + specialized vertex connection

Wedge construction:

geometry-shaped member + connection adapted to that member

That distinction is central to the rest of this book. The question is not whether a dome needs connection geometry.

It does. The question is where we choose to put that geometry. We can concentrate more of it in a hub.

We can shape more of it into the timber. Or we can build a hybrid system that uses both. Understanding traditional hub construction first makes the wedge method easier to understand, because both systems are solving the same physical problem from different directions.

Panelized domes

A geodesic dome does not have to be built as a loose frame first and skinned later. Another approach is to build the dome from panels. A panelized dome treats each triangular face, or sometimes groups of faces, as a prefabricated building unit.

Instead of assembling only struts and hubs, the builder assembles complete sections that already contain some combination of frame, sheathing, insulation, weather barrier, interior finish, or connection hardware. That changes the entire workflow. A strut-built dome starts with lines.

A panelized dome starts with surfaces.

DESIGN VARIATION

The simplest triangular panel might be nothing more than a sheet cut to the shape of one geodesic face.

A more developed panel might contain:

- structural edge members,
- an exterior skin,
- insulation,
- an interior skin,
- gasket material,
- attachment hardware,
- window openings,
- or service penetrations.

At that point the triangle is no longer just geometry. It becomes a prefabricated wall or roof section. The same basic geodesic layout is still underneath everything.

The triangle dimensions still come from the dome geometry. The difference is that more construction work happens before the panel reaches the dome. That can be useful when repeatability matters.

If twenty panels share the same geometry, the builder can make one accurate jig and use it repeatedly. If several panel types exist, each type can have its own fixture, cut list, and assembly process.

The manufacturing problem becomes:

Make Panel A correctly. Make Panel B correctly. Make however many of each type the dome requires.

Then assemble those panels in the correct pattern.

PANEL GEOMETRY STILL COMES FIRST

A panel cannot be designed correctly until the triangular face underneath it is known.

For each panel type we eventually need to know:

- the three edge lengths,
- the three internal angles,
- the orientation of the panel in the dome,
- the angle between that panel and its neighbors,
- the connection method along each edge,

- and how the panel thickness relates to the intended dome surface.

This is especially important when the panel is thick. A flat sheet has an inside face and an outside face that are close together. A thick insulated panel may move the exterior skin noticeably farther from the theoretical geodesic chord.

If neighboring panels are simply made from the same two-dimensional triangle and pushed together, the exterior edges may not meet the way the builder expects. The edge geometry has to account for thickness. That can mean bevels, wedges, gaskets, external covers, internal battens, or another joint system.

This is the same three-dimensional problem seen in strut construction.

We have just moved it into the panel edge.

FRAME-FIRST PANELS

One panelized method is to build a triangular timber frame first. The three edge members define the panel. Sheathing is then attached to that frame.

The frame may be built from rectangular lumber, wedge-shaped lumber, laminated members, or another section. This approach gives the panel its own structural border before it reaches the dome. In a wedge-method system, the wedge can become part of that panel edge.

Instead of using a rectangular perimeter and then cutting a separate bevel into the panel connection, the timber edge itself can help establish the transition between neighboring faces.

That creates a useful manufacturing possibility:

The wedge strut can act as both a structural member and a panel-edge geometry component. The panel skin can then be attached to the outside of the triangular frame. When two complete panels are brought together, their wedge-shaped edge members meet at the dome joint.

The connection system then has to clamp, screw, bolt, plate, or otherwise join those edges in a controlled way.

SKIN-FIRST PANELS

Another method starts with the sheet material. A triangular panel is cut first and framing is attached behind it. This may be practical for thin sheathing or lightweight assemblies where the skin itself provides the easiest reference surface during fabrication.

The advantage is that the visible panel shape can be controlled directly from a template or CNC layout. The disadvantage is that the rear framing still has to agree with the three-dimensional joint geometry. A flat triangle is easy to cut.

Making many thick flat triangles close into a dome without gaps is the harder part. That is why panel edge design matters as much as panel face dimensions.

PANEL JOINTS

When panels meet, several things have to happen at once.

The joint has to:

- locate the panels,
- hold them together,
- deal with the angle between faces,
- control water entry,
- control air leakage where required,
- allow whatever installation tolerance the system needs,
- and transfer structural forces if the panel is part of the load-bearing system.

One connection may not need to do every job by itself. For example, a structural fastener might hold the panels together while a separate gasket handles weather sealing. An exterior cap might cover the seam.

An interior batten might reinforce it. A removable design might use bolts or clamps instead of permanent adhesive. The important part is to stop treating "the panel joint" as one vague detail.

It is several problems occupying the same edge.

PANELIZATION AND REPEATABILITY

Panel construction makes the most sense when we take advantage of repetition. If every panel is built differently in the field, much of the benefit disappears.

The useful approach is to identify the repeating panel families early.

For each family we can eventually create:

- a panel ID,
- a drawing,
- a cut list,
- a jig,
- an edge orientation,
- an assembly sequence,
- and a location map for the finished dome.

That turns the dome into something closer to a manufactured kit. The site work becomes assembly of previously defined parts instead of fabrication of every part in place. This does not automatically make the project faster.

The fabrication time still exists. It has simply moved into a more controlled stage of the build. The advantage is that repeatable work can be organized around fixtures, templates, stops, and inspection.

A bad first panel can be corrected before twenty more are made from the same setup. A good setup can then be repeated without re-solving the geometry each time.

OPENINGS AND SPECIAL PANELS

Doors and windows interrupt the repeating triangle system. A panelized dome gives us one way to handle that interruption. Instead of modifying random pieces during erection, special panels can be designed before assembly.

For example:

- one triangular panel could contain a window,
- several neighboring triangles could be replaced by a reinforced door module,
- one panel could contain a ventilation opening,
- or a utility panel could contain controlled service penetrations.

These special pieces should be treated as separate panel types. They may look similar to the normal panels from outside, but structurally and functionally they are not the same. That means they need their own drawings and connection details.

A hole cut into a completed generic panel is not automatically equivalent to a panel designed around that opening.

REMOVABLE PANELS

DESIGN VARIATION

Panelization also makes removable sections possible. If the joint system is mechanical rather than permanently bonded, an individual panel could potentially be removed for repair, replacement, modification, or access.

That could be useful for:

- replacing damaged exterior sections,
- changing transparent and opaque panels,
- adding future openings,
- accessing utilities,
- or experimenting with different enclosure systems on the same structural frame.

This idea becomes more useful when panel dimensions and connections are standardized. If every panel is custom-fit during construction, replacing one later may require rebuilding that exact piece by hand. If the panel system is repeatable, replacement becomes a manufacturing problem rather than a reconstruction problem.

That does not mean every structural panel should be casually removable. If the panel participates in the load path, removing it may change the behavior of the structure. The connection between structural frame and removable shell therefore needs to be clearly defined.

SEPARATING FRAME FROM SHELL

One useful design decision is whether the panel is part of the structure or only part of the enclosure. Those are two different systems. In one version, the dome has a complete timber frame.

Panels attach to that frame but are not intended to be the primary structural system. In another version, the panel and its perimeter framing act together as a structural module. In another, the skin itself may participate in transferring forces between the edges.

Those systems may look similar after the dome is finished. They should not be assumed to behave the same way. For the wedge method, separating the frame and shell can keep the basic structural concept

easier to understand.

The wedge members establish the geodesic frame. The panel system then attaches to that frame. Later variations can explore whether the panel can take on additional structural work.

SAFETY / ENGINEERING

A panel that carries structural load needs more than correct triangle dimensions.

Its behavior can depend on:

- panel material,
- thickness,
- edge framing,
- fastener spacing,
- connection design,
- openings,
- moisture,
- buckling,
- wind pressure,
- suction,
- and the way forces enter and leave the panel.

For that reason, this book does not treat a panel as structural simply because it fits into the geodesic pattern. A panel can be geometrically correct and still be structurally inadequate. Geometry tells us where the piece belongs.

Engineering determines what that piece can safely carry.

REFERENCE BUILD

For the 20-foot 2V reference dome, panelization will eventually be tied directly to the verified triangle families. Once those triangle types are fixed, each repeated face can be assigned a panel type.

A basic workflow will be:

1. identify the verified triangle families, 2. assign a panel ID to each family, 3. determine the edge geometry, 4. determine how panel thickness relates to the theoretical dome surface, 5. design the joint between neighboring panels, 6. define normal and special panels, 7. create fabrication jigs or templates, 8. label the panels, 9. assemble them according to a dome map.

That gives us a panel schedule instead of a pile of triangles. The important idea is that panelization does not replace the dome geometry.

It packages the geometry into larger, repeatable building units.

Where the wedge method differs

The wedge method changes where the dome's three-dimensional geometry is handled. A conventional timber dome often starts with rectangular struts and then uses hubs, plates, brackets, compound end cuts, or other connection hardware to make those rectangular members meet correctly in space. The wedge method moves part of that problem into the member itself.

Instead of treating every strut as a rectangular stick with parallel inside and outside faces, the timber is shaped so the inside edge is narrower than the outside edge. The member becomes part of the angular transition between neighboring triangular faces. That is the main difference.

The dome still depends on correct geodesic geometry. The struts still have to be the correct lengths. The vertices still have to occur in the correct locations.

The triangles still have to assemble into the intended spherical pattern. The wedge does not replace any of that.

It changes how the physical timber occupies the geometry.

GEOMETRY

A geodesic frame can be thought of as a network of theoretical straight lines joining vertices.

Those lines are the chord geometry.

A conventional rectangular strut is then placed along one of those lines. Its cross-section does not naturally change from the interior side of the dome to the exterior side.

That means the joint has to handle the difference in direction between the triangular face on one side and the triangular face on the other. A wedge-shaped member gives us another way to handle that transition. Viewed in cross-section, the member has an interior width and an exterior width.

The two side faces are therefore not parallel.

That creates an included wedge angle.

If the difference between the outside width and inside width is:

$$\Delta w = w_{outside} - w_{inside}$$

and the distance between the interior and exterior reference faces is t, then a symmetrical wedge can be described by:

$$\alpha = 2 \tan^{-1}((\Delta w) / 2t)$$

where:

- α is the total included wedge angle,
- $w_{outside}$ is the outside width,
- w_{inside} is the inside width,
- and t is the thickness between the inside and outside reference faces.

This equation describes the shape of the wedge. It does not tell us automatically what wedge angle a particular dome needs. That angle depends on the actual dome geometry, the reference line used through the timber, the relationship between neighboring faces, and the connection method.

The important part is that the timber now contains a controlled angular relationship in its own cross-section.

RECTANGULAR MEMBER VERSUS WEDGE MEMBER

Take two neighboring triangular faces. If their common edge is made from a rectangular timber, the member has approximately the same width at the inside and outside of the frame. The two triangular faces still need to turn relative to one another.

That change may be handled with a gap, bevel, plate, hub, bracket, angled fastener arrangement, or some combination of those. Now replace that rectangular member with a wedge. The inside edge is narrower.

The outside edge is wider. The widening cross-section gives the adjoining faces more separation toward the exterior while allowing them to remain closer together toward the interior. Instead of asking the connection to create the entire face-to-face transition, some of that geometry is already built into the timber.

That is the central construction idea behind the wedge method. The member is not only spanning from one vertex to another.

Its cross-section is also doing geometric work.

THE WEDGE IS NOT A HUB

It would be easy to describe the system simply as hubless construction, but that description can be misleading. Removing a traditional central hub does not remove the need for a connection. The members still have to be held together.

The vertex still has to stay in the correct location. Loads still have to pass from one member into the others. The difference is that a large specialized hub does not necessarily have to establish the primary angular relationship between the surrounding faces.

Depending on the version of the system, the remaining connection could use:

- structural screws,
- through-bolts,
- washers,
- internal plates,
- external plates,
- straps,
- brackets,
- direct timber contact,
- or combinations of these.

The wedge method therefore redistributes the geometry rather than eliminating connection design. Some of the work normally done by a hub is moved into the timber.

THE MEMBER BECOMES A REPEATED GEOMETRIC PART

This changes the fabrication problem. In a traditional hub system, the strut can often remain a relatively simple component while the specialized hub handles more of the spatial geometry. In the wedge method, the strut becomes a more specialized component.

That means its manufacturing process matters. If one wedge profile is correct, we want to reproduce that profile instead of laying out every part independently.

A practical production process should eventually define:

- the starting stock,
- the reference face,
- the fence or jig position,
- the workholding method,
- the order of cuts,
- the final length reference,
- the drilling reference,
- the labeling method,
- and the inspection method.

The goal is to stop treating each wedge as a custom carving. It should become a manufactured part. If twenty members are supposed to be identical, the twentieth member should come from the same references and setup as the first.

That is where jigs, fences, stops, templates, and gauges become more useful than repeated measuring.

FABRICATION ERRORS

The wedge creates another set of dimensions that have to remain controlled. A rectangular strut can be the correct length and still have minor width variation without completely changing its basic geometry. With a wedge, the width relationship is part of the intended shape.

If the outside width changes while the inside width stays the same, the wedge angle changes. If the timber is flipped during assembly, the taper works in the opposite direction. If the wrong face becomes the reference during a cutting operation, the same mistake can be reproduced across an entire batch.

The common errors therefore need to be kept separate:

- incorrect strut length,
- incorrect inside width,
- incorrect outside width,
- incorrect wedge angle,
- reversed wedge orientation,
- incorrect end geometry,
- incorrect fastener-hole position,
- incorrect member family,
- and incorrect labeling.

These errors do not all produce the same result. A wrong strut length changes the vertex position. A wrong wedge profile changes how the surrounding faces meet.

A reversed wedge puts the wide side toward the interior instead of the exterior. A misplaced bolt hole can change the effective connection point even when the overall piece of timber is the correct length. This means quality control eventually has to check more than one measurement.

BUILD NOTE

The actual reference-build fabrication procedure should be recorded from the shop rather than invented from the geometry.

For each wedge family, that record should include:

- nominal starting stock,
- actual measured stock dimensions,
- tool used,
- fence or guide setup,
- workholding,
- first cut,
- second cut if required,
- reference face after each cut,
- final length operation,
- drilling operation,
- labeling,
- target dimensions,
- measured finished dimensions,
- and variation across multiple completed parts.

Until those measurements exist, the book can explain how the operation should be organized without claiming a production tolerance that has not been measured. The same rule applies to saw settings. A drawing can show the intended wedge.

The builder-facing procedure needs the actual setup used to reproduce it.

DIMENSIONAL LUMBER

One version of the wedge method starts with ordinary rectangular lumber. The wedge profile is ripped from that rectangular blank. The cutting layout matters because the offcut may represent a large portion of the original board.

That does not automatically mean the offcut is waste. Depending on the final wedge geometry, another useful member may be recovered from it.

The same material might produce:

- another wedge,
- a smaller structural member,

- blocking,
- panel cleats,
- spacers,
- trim,
- or another repeated component.

This makes material yield part of the system. The fastest way to produce one wedge is not necessarily the most efficient way to process all the lumber required for a dome. The better question is how many useful finished parts can be recovered from each starting board while maintaining repeatable reference surfaces.

LOGS AND LARGER TIMBER

The wedge method also creates another possibility when the starting material is a log or large timber. A log is already radial in cross-section. Instead of converting the entire log into rectangular boards first, the log can potentially be divided longitudinally into rough wedge-like sectors.

Those sectors can then be processed into controlled structural members. This does not mean that a raw radial slice from a tree is automatically a finished dome strut.

The original log may have:

- taper,
- sweep,
- knots,
- checks,
- changing diameter,
- moisture variation,
- irregular grain,
- and irregular exterior surfaces.

The rough wedge still needs controlled reference surfaces and verified dimensions.

But the material layout can change substantially.

Instead of:

log arrow rectangular lumber arrow wedge

the process can potentially become:

log arrow rough wedge sectors arrow finished wedges

That is more than a different connection detail. It changes how the structural material can be recovered from the original timber.

CONNECTION GEOMETRY STILL MATTERS

Shaping the member can reduce how much specialized three-dimensional geometry has to be concentrated at the vertex. It does not make the connection trivial. The connection still has to answer

practical questions.

Where is the theoretical vertex? Where is the fastener center? Which timber surfaces actually touch?

Are the wedges clamped directly together? Is there a controlled gap? Does a plate bridge several members?

Can the connection be tightened after erection? Can a damaged member be replaced? Does an exterior panel attach across the same joint?

How does the connection affect the physical cut length compared with the theoretical chord?

Those questions connect the wedge profile to the actual building system.

The two problems therefore have to be developed together:

member geometry

and

connection geometry

A good wedge profile attached through a poorly defined connection is not a complete system.

Neither is a good connection attached to incorrectly manufactured wedges.

REPEATABILITY IS THE LARGER IDEA

The wedge is the obvious visual difference, but repeatability is the more important fabrication idea behind the method. A dome may contain a large number of pieces while containing only a relatively small number of unique part families.

Once those families are defined, each one can eventually have:

- a profile,
- a length,
- a drilling pattern,
- a label,
- a manufacturing fixture,
- an inspection gauge,
- and a known location in the finished dome.

That changes the question during assembly.

Instead of asking:

"How do I make this piece fit here?"

we should be asking:

"Which defined part belongs here, and was that part made correctly?" That difference matters. Custom fitting allows one error to be hidden by another adjustment.

Repeatable fabrication makes the error easier to find. If a correctly labeled part does not fit where the geometry says it belongs, something is wrong with the part, the assembly position, the connection, or the

underlying model. That gives us a system that can be checked instead of continuously improvised.

DESIGN VARIATION

The wedge concept can also extend into panelized construction. A wedge member can form the perimeter of a triangular panel. The outside face of the wedge can support exterior sheathing while the inside edge defines the interior side of the frame.

Neighboring wedge-framed panels can then meet along their prepared edges. That creates the possibility of prefabricating larger triangular modules instead of assembling every individual strut on site. Other versions could keep the wedge frame permanent while making the exterior panel removable.

Still others could combine the wedge with metal reinforcement or laminated timber. These are variations of the same underlying idea: make the member participate in the dome geometry instead of requiring every spatial relationship to be created entirely by separate hardware.

SAFETY / ENGINEERING

Changing the member shape does not prove structural capacity.

The wedge affects the physical cross-section of the timber and may also affect:

- fastener placement,
- available edge distance,
- bearing surfaces,
- connection geometry,
- bending orientation,
- and the amount of wood remaining at different parts of the section.

Those effects have to be evaluated for the actual material, joint, and loading condition. A wedge that fits the geometry correctly is not automatically a structurally adequate wedge. The same applies to log-derived members.

Material efficiency does not override timber quality, grading, defects, moisture, connection requirements, or structural analysis. The geometry tells us what shape we are trying to build. Engineering determines whether the resulting parts and connections can carry the required loads.

WHERE THE METHOD DIFFERS

The wedge method differs from conventional timber geodesic construction in several connected ways. It makes the shape of the timber part of the three-dimensional dome geometry. It can reduce the amount of specialized geometry concentrated in a traditional hub.

It turns cross-sectional accuracy into part of the fabrication process. It can be produced from rectangular lumber, larger sawn timber, or potentially timber recovered directly from logs. It can form the edge geometry of prefabricated triangular panels.

Most importantly, it gives us a system that can be broken into defined and repeatable manufacturing operations. That is the point where the wedge stops being only a shape.

It becomes a building method.

2. The Wedge Method

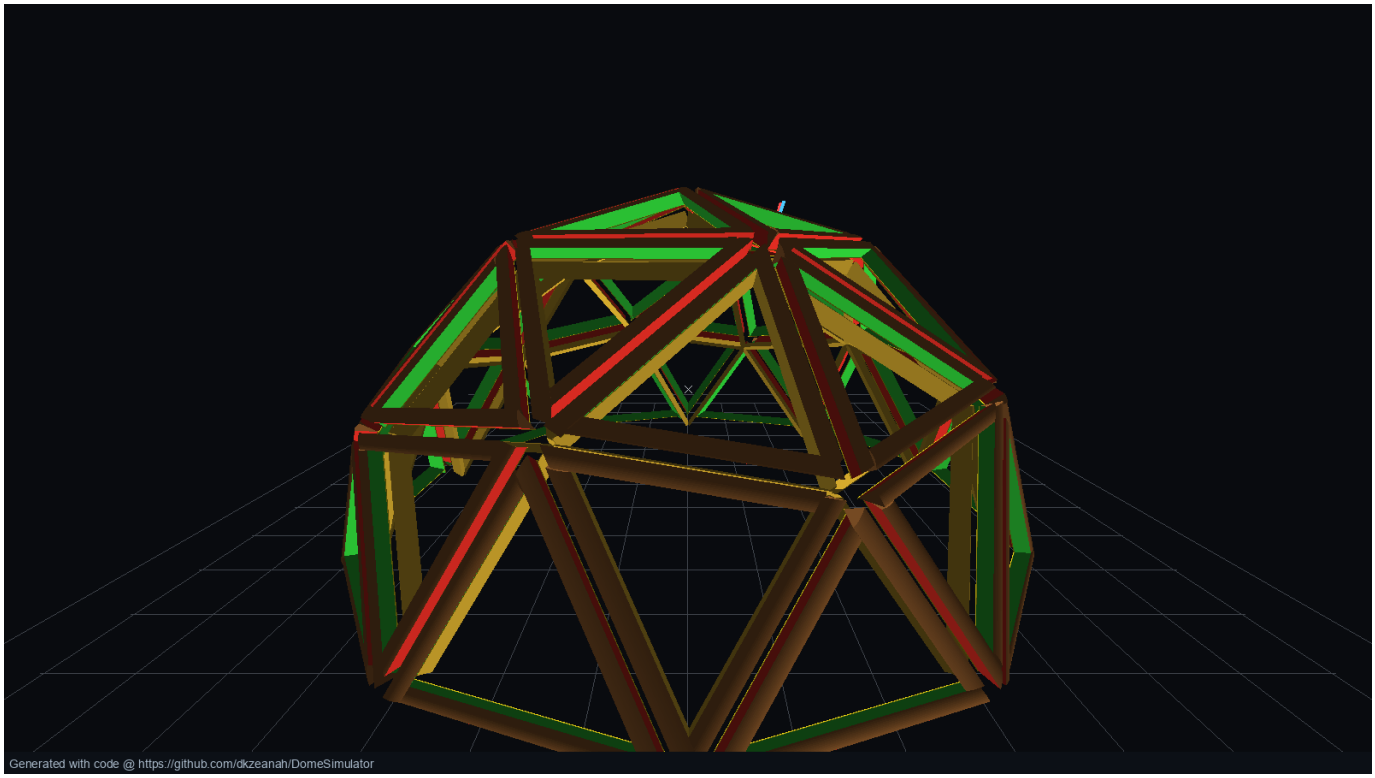

Figure 6. The same dome, opened up

The reference build with the panels pushed eight inches apart. Nothing has moved in the geometry -- this is the same solve, drawn with a gap -- and it is the only way to see what happens between two panels. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Basic concept

The wedge method starts with one simple change:

Instead of building the dome from rectangular struts and solving most of the three-dimensional geometry at the joints, we shape the struts so the timber itself participates in the dome geometry. That is the core idea. A normal rectangular board has approximately the same width on its inside and outside faces.

A wedge-shaped member does not. Its narrow side faces toward the interior of the dome. Its wider side faces toward the exterior.

When those members are arranged into triangles, the changing width of the wedge helps neighboring triangular faces turn relative to one another. The member is therefore doing two jobs at once. It spans between vertices.

It also helps establish the angular relationship between the faces on either side of it.

GEOMETRY

The theoretical geodesic frame can still be represented as straight chord lines connecting vertices. The wedge does not replace those chord lines. It is built around them.

That distinction is useful because it lets us separate the geometry into two layers.

The first layer is the mathematical dome.

That layer tells us:

- where the vertices are,
- which vertices connect,
- the chord lengths,
- the triangle families,
- and the face-to-face relationships.

The second layer is the physical timber.

That layer tells us:

- how wide the member is on the inside,
- how wide it is on the outside,
- how thick it is,
- how it is oriented,
- where the fasteners pass through it,
- and how neighboring members actually meet.

The wedge method works when those two layers agree. If the theoretical dome is correct but the wedge profile is wrong, the physical frame will not follow the intended geometry correctly. If the wedge profile is correct but the strut lengths are wrong, the vertices will be misplaced.

Both parts matter.

THE WEDGE ORIENTATION

The basic orientation is:

narrow side inward, wide side outward. This should remain visually obvious during fabrication and assembly.

The reason is geometric. The interior side of the dome is closer to the center of the sphere. The exterior side is farther from the center.

As neighboring triangular faces wrap around the dome, they spread apart toward the exterior. The wedge shape gives us a member whose cross-section follows that general relationship. A reversed wedge works against the intended orientation.

If a member is installed wide-side inward and narrow-side outward, it is not merely cosmetically reversed. Its cross-section is now pointing in the wrong direction relative to the dome. That is why every finished member should eventually have an obvious inside/outside marking.

A simple label, arrow, burn mark, paint mark, stamped letter, or other repeatable identifier can prevent an orientation error during assembly.

THE TRIANGLE STILL CONTROLS THE FRAME

The wedge does not make the dome by itself. The triangles do. The strut lengths still define the triangle edges.

The triangle geometry still determines the shape of each flat face. The wedge then helps those faces meet in three dimensions. This distinction should stay clear throughout the book.

A wedge angle is not a substitute for the correct triangle. A correct triangle does not automatically tell us the correct wedge profile. And the correct wedge profile does not automatically tell us the correct connection.

We can treat those as three linked design problems:

triangle geometry

wedge geometry

connection geometry

The complete system comes from making all three agree.

ONE MEMBER, TWO NEIGHBORING FACES

Each wedge member can be thought of as lying along the shared edge between two neighboring triangular faces. One face exists on one side of the strut. Another face exists on the other side.

Those faces are not normally coplanar. They meet at an angle. That means the edge between them has to work in three dimensions.

A rectangular member gives both sides approximately parallel reference faces. A wedge member gives the two sides a controlled divergence between the interior and exterior. That is the physical feature we are trying to use.

The wedge is therefore best understood from a cross-section perpendicular to the strut length. Looking only at the long side of the timber can hide the entire concept. From the side, the wedge may look like an ordinary straight board.

From the end, the difference becomes obvious.

That end view should be used repeatedly throughout this book.

VERTICES

A dome vertex is where several struts meet. The number of members depends on the location and the geometry. Some vertices may have five members meeting.

Others may have six. Boundary conditions can create other arrangements. The wedge method does not eliminate those vertex patterns.

It changes the shapes of the members entering them. At a vertex, several wedge-shaped struts approach from different directions. The connection must still locate those members correctly and hold them together.

The wedge profile helps the surrounding faces occupy the correct spatial relationship, but the vertex still needs a practical connection system. That connection might be very simple. It might not be.

The final detail depends on how the member ends are cut, how much timber remains around the fasteners, whether the members contact one another directly, whether plates are used, and where the theoretical vertex lies relative to the physical wood. For that reason, the vertex should be thought of as an assembly of several controlled parts rather than a point where we simply force the lumber together.

THREE-MEMBER RELATIONSHIPS

Before thinking about a full five-way or six-way vertex, it is useful to reduce the problem. Take three members and build one triangle.

Now the basic relationships are visible.

Each member has:

- a length,
- an inside face,
- an outside face,
- a left side,
- a right side,
- and two ends.

That orientation needs to remain consistent. If the triangle is assembled correctly, the narrow faces point toward the interior side of the future dome. The wider faces point toward the exterior.

That triangle becomes one flat face of the larger structure. Add another triangle along one of its edges and the three-dimensional relationship becomes easier to see. The shared wedge member sits between the two faces.

Repeat this around the structure and the dome emerges. This is a useful way to think about assembly because the complete vertex can look complicated while the individual relationships remain simple. Start with one member.

Then one triangle. Then two neighboring triangles. Then the complete vertex.

Then the dome.

That sequence keeps the geometry understandable.

THE WEDGE PROFILE

A finished wedge member can be described with a small group of dimensions.

At minimum, we need:

- inside width,
- outside width,
- thickness,
- length,

- and end geometry.

If holes or other connection features are added, we also need:

- hole diameter,
- hole location,
- edge distance,
- and reference surface.

Those dimensions define the physical part.

A basic symmetric wedge cross-section can be described by:

$$\Delta w = w_{outside} - w_{inside}$$

where:

- $w_{outside}$ is the exterior width,
- w_{inside} is the interior width.

If the taper is symmetric about the member centerline, each side accounts for half of that difference:

$$(\Delta w) / 2$$

If t is the distance from the inside face to the outside face, one side taper angle is:

$$\beta = \tan^{-1}((\Delta w/2) / t)$$

and the total included wedge angle is:

$$\alpha = 2\beta$$

or:

$$\alpha = 2 \tan^{-1}((\Delta w) / 2t)$$

This gives us a way to describe a finished wedge once the required widths and thickness are known. It does not yet tell us what those dimensions should be for the reference dome.

That has to come from the verified dome and joint geometry.

MANUFACTURING LOGIC

Once a wedge profile is established, it should become a repeatable shop operation.

The goal is not:

measure a strange shape, cut it, check whether it fits,

then adjust the next one.

The goal is:

define the part, define the process, repeat the process.

That may mean a fence setting. It may mean a dedicated sled. It may mean a taper jig.

It may mean splitting or ripping larger stock along predetermined lines. It may mean using one finished surface as the datum for every later operation. The actual production method can vary.

The principle should not. Every repeatable dimension should come from a repeatable reference. That becomes especially important when the dome requires many parts.

A small error repeated once may be manageable.

A small error repeated forty times becomes a pattern.

BUILD NOTE

The builder-facing version of this method should eventually record the actual production setup.

For the reference build, that means documenting:

- the stock used,
- actual measured stock dimensions,
- cutting tool,
- jig or fence arrangement,
- feed direction,
- workholding,
- cut order,
- datum surface,
- final length operation,
- drilling operation,
- marking method,
- and measured variation between finished parts.

Those details should come from the actual fabrication process. Until they are measured, they remain development work rather than finished production specifications.

MATERIAL DOES NOT HAVE TO START AS A BOARD

The wedge shape can be produced from different starting materials. One method starts with dimensional lumber and removes material until the required cross-section remains. Another can begin with larger sawn timber.

Another can begin with a log and recover wedge-like sectors directly from the original round material.

The finished requirement is the same:

the structural member must end up with controlled geometry. How we reach that geometry can change. This gives the wedge method an important manufacturing flexibility.

The geometric part is defined first. The raw material strategy can then be chosen based on what is available.

DESIGN VARIATION

A wedge can also become more than a bare strut. It can become the edge of a triangular panel. It can be laminated from several pieces.

It can be reinforced with metal. It can include a groove, rabbet, panel seat, gasket channel, or other feature. It can be produced as part of a larger prefabricated module.

Those variations should all be treated as later branches of the basic system.

The simplest version is still the best place to start:

a straight structural member, with a controlled wedge-shaped cross-section, oriented narrow-side inward and wide-side outward,

placed along a verified geodesic edge,

and connected to the neighboring wedge members at the correct vertices.

REFERENCE BUILD

For the 20-foot-diameter 2V reference dome, we will eventually reduce the structure to a defined set of member families.

Each family will need:

- a verified chord relationship,
- a physical cut length,
- a wedge profile,
- an orientation,
- a connection pattern,
- a quantity,
- and a label.

Those values should not be mixed together prematurely. First the exact 2V geometry must be frozen. Then the connection reference has to be defined.

Then the wedge profile can be derived around that system. Once those decisions are fixed, the book can move from concept to a real shop table.

That table should eventually allow a builder to look at a member labeled, for example, "A," and know exactly what that means:

which stock to start with, which wedge profile to cut, how long to make it,

where to drill it, which direction faces inward, and where it belongs in the dome.

That is the basic concept of the wedge method in practical terms. Turn the geometry into defined parts. Turn the defined parts into repeatable operations.

Then assemble those operations back into the dome.

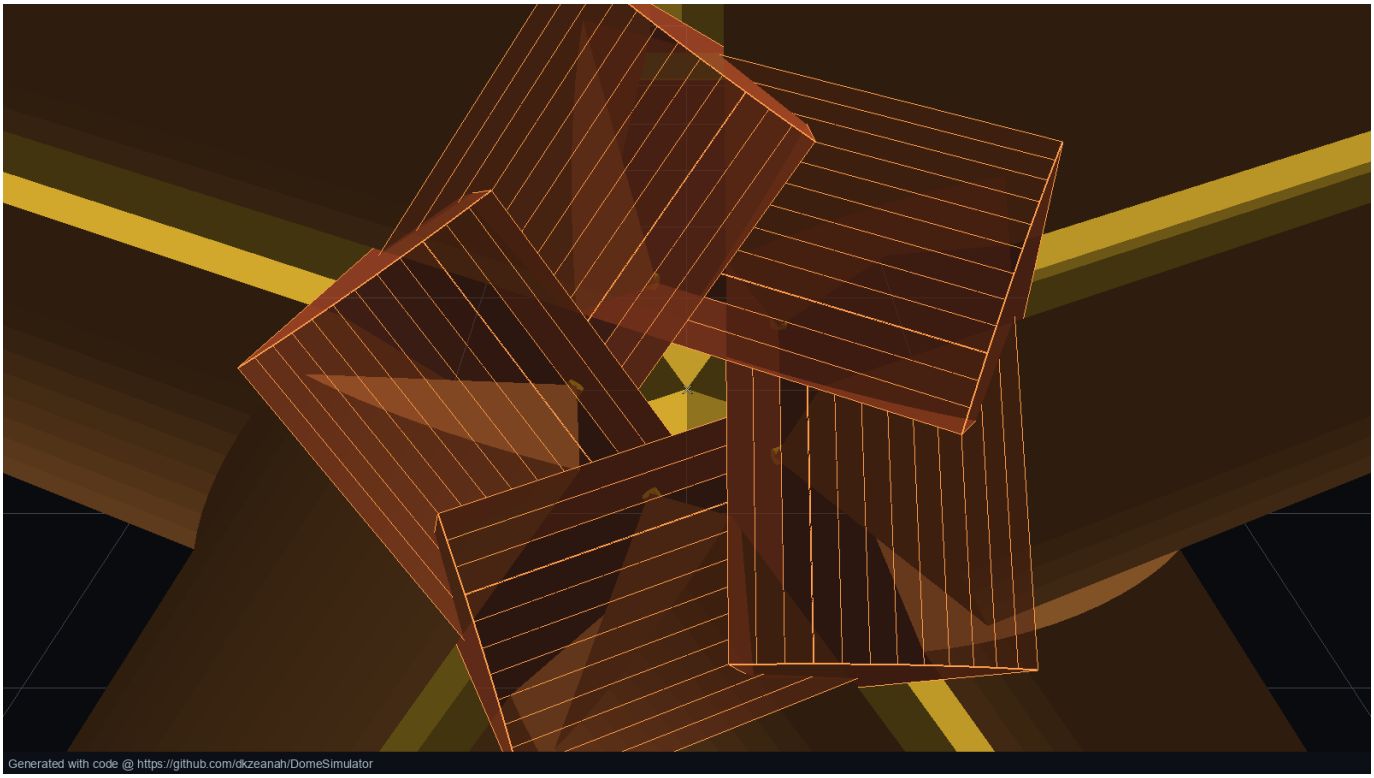

Figure 7. One wedge, end on

The end of a single member, close enough to read the section: a 45-degree slice of a 12-inch log, two sawn radial faces and a bark face. Nothing has been machined -- this is what the froe leaves. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Wedge-shaped structural members

A wedge-shaped structural member is still a strut. It still has to connect two vertices, hold its place in the triangle, and fit the other members around it. The difference is in the cross-section.

Instead of carrying the same width from the inside of the dome to the outside, the member becomes wider toward the exterior. That shape gives us another geometric variable to work with. We are no longer defining a strut only by length.

We are defining it by length and profile. For the wedge method, that profile is part of the building system.

GEOMETRY

A basic wedge member can be described with four primary dimensions:

- member length,
- inside width,
- outside width,
- thickness.

The difference between the outside and inside widths is:

$$\Delta w = w_o - w_i$$

where:

- w_o is the outside width,
- w_i is the inside width.

For a symmetric wedge, half of that width change occurs on each side of the member:

$$x = (w_o - w_i) / 2$$

If the distance between the interior and exterior faces is t, the taper angle on each side is:

$$\beta = \tan^{-1}(x / t)$$

or:

$$\beta = \tan^{-1}((w_o - w_i) / 2t)$$

The full included wedge angle is:

$$\alpha = 2\beta$$

which gives:

$$\alpha = 2 \tan^{-1}((w_o - w_i) / 2t)$$

Those equations describe the physical cross-section once its dimensions are known. They do not determine what the correct wedge should be for the dome.

That comes from the dome geometry and the way the neighboring faces are intended to meet. A wedge can be mathematically consistent as a piece of timber and still be the wrong wedge for the structure.

THE MEMBER HAS TWO DIFFERENT GEOMETRIES

It helps to separate the strut into longitudinal geometry and cross-sectional geometry. Longitudinal geometry runs from one end of the member to the other.

That includes:

- theoretical chord length,
- physical cut length,
- end cuts,
- fastener locations,
- and the relationship between the finished timber and the two vertices it connects.

Cross-sectional geometry is what we see when looking directly at the end of the member.

That includes:

- inside width,
- outside width,
- thickness,

- side tapers,
- centerline,
- and any grooves, rabbets, bevels, or panel seats added later.

Those two geometries solve different problems. The length helps place the vertices. The wedge profile helps the member occupy the space between neighboring faces.

Changing one does not automatically correct the other. A member can have the correct wedge profile and still be too long. It can have the correct length and still have the wrong taper.

For production purposes, both have to be inspected.

STARTING STOCK

A finished wedge may begin as rectangular dimensional lumber, a larger sawn timber, a laminated blank, or a rough section recovered from a log. Whatever the source, the starting material needs a known reference. With dressed lumber, that means recording the actual measured dimensions rather than relying only on the nominal lumber name.

The nominal size tells us what class of material we started with. The actual measured size tells us what material is physically available for cutting the wedge. Those are not interchangeable in the fabrication record.

If the wedge calculation requires a specific outside width, inside width, and thickness, the stock has to contain enough material to produce those dimensions after cutting and cleanup.

The first question is therefore not:

"What size board is this called?"

It is:

"What dimensions do I actually have?"

REFERENCE SURFACES

A repeatable wedge needs repeatable reference surfaces. If every cut is referenced from a different irregular face, errors begin stacking on top of one another.

A controlled process should establish which surface is the datum.

That might be:

- the interior face,
- the exterior face,
- one side face,
- a centerline,
- or a fixture that locates the part independently of an unfinished edge.

Once that datum is chosen, later operations should reference it intentionally. This becomes even more important when the starting material comes from rough-sawn timber or a log-derived sector. A rough wedge may already have approximately the right shape, but "approximately wedge-shaped" is not enough

for a repeated structural part.

At least some surfaces have to be brought under dimensional control.

SYMMETRIC AND ASYMMETRIC WEDGES

The easiest wedge to describe is symmetric.

Its left and right sides taper equally away from the centerline.

If the total width difference is:

$$\Delta w = w_o - w_i$$

then each side accounts for:

$$(\Delta w) / 2$$

That gives the member a centered profile.

A wedge does not have to be symmetric, however.

DESIGN VARIATION

An asymmetric wedge could place more of the taper on one side than the other. That might become useful if the connection system, panel geometry, fabrication method, or dome-face relationship requires it.

In that case, the left and right offsets should be recorded separately:

$$x_L + x_R = w_o - w_i$$

where:

- x_L is the width increase on the left,
- x_R is the width increase on the right.

If:

$$x_L = x_R$$

the wedge is symmetric. If they differ, the member is asymmetric.

This distinction matters because an asymmetric member also gains a left-hand and right-hand orientation. That increases the possibility of installing a geometrically correct part in the wrong direction. Unless a later design requires asymmetry, a symmetric member is easier to manufacture, label, inspect, and reverse end-for-end.

THE CENTERLINE

The wedge needs a clearly defined centerline. That centerline gives us a stable way to describe the part even though its side faces are angled.

It can be used to locate:

- fastener holes,

- end cuts,
- labels,
- panel features,
- and the member relative to the theoretical chord.

The physical timber does not automatically have to be centered on the mathematical chord. That is a design decision. But once that relationship is chosen, it should remain fixed.

For example, if the theoretical chord passes through the cross-sectional center of the wedge, that becomes one kind of member definition. If the fastener-center line is offset from the geometric center of the wedge, that becomes another. Both can be modeled.

What creates problems is changing references between calculations and shop drawings without stating it.

END GEOMETRY

The wedge profile describes the long body of the member.

The ends are another problem.

A member might terminate with:

- a square end,
- an angled end,
- a compound cut,
- a relieved end,
- a drilled connection zone,
- a plate interface,
- or another prepared joint.

The end treatment can change the physical overall length even when the theoretical chord remains unchanged.

For that reason, the final member drawing should eventually distinguish between at least:

$$L_c = \text{theoretical chord length}$$

and:

$$L_f = \text{finished physical member length}$$

If fastener centers define the structural reference, we may also need:

$$L_b = \text{bolt-center to bolt-center distance}$$

Those three numbers should not be assumed to be equal. A production drawing has to tell the builder which one is being measured.

HOLES AND CONNECTION FEATURES

Drilling a wedge is also different from merely drilling a rectangular board if the angled side surfaces are being used as references.

A hole location should be defined from controlled dimensions.

That can include:

- distance from the end,
- distance from the member centerline,
- distance from the inside or outside face,
- hole diameter,
- drilling direction,
- and which surface the drill or jig references.

A hole that is shifted along the member changes the effective connection location. A hole that wanders sideways may change edge distance or interfere with neighboring hardware. A hole drilled at the wrong angle may prevent plates, washers, or neighboring members from seating correctly.

For repeated production, the drilling operation should eventually use a jig or fixed reference instead of being independently laid out on every member.

THE PROFILE HAS TO SURVIVE THE FULL LENGTH

A wedge is not useful if the correct dimensions exist only at one end. The profile has to remain controlled along the member. If the inside width changes unexpectedly from one end to the other, then the member contains a second taper running along its length.

That may happen intentionally with some future design. For the basic wedge member, it should not happen accidentally.

This gives us another inspection problem.

Checking only one cross-section may miss:

- saw drift,
- stock taper,
- bow,
- twist,
- uneven feed,
- or a jig that moved during the cut.

A finished member may therefore need width checks at more than one location.

A practical inspection record might include:

- end A,
- midpoint,
- end B.

If the target profile is supposed to be constant, those measurements should show whether the production process is actually maintaining it.

TWIST

Twist deserves separate attention. A member can have the correct width at both ends and still have one end rotated relative to the other. That matters because the wedge has a defined inward and outward orientation.

If the cross-section twists along the member, the side surfaces no longer remain in the intended planes. Some twist may originate in the stock. Some may appear as timber dries.

Some may come from poor workholding during machining. Whatever the cause, a member that cannot sit consistently in the intended geometry should not be treated as correct merely because its tape-measure dimensions match.

BUILD NOTE

The final inspection process for the reference build should include an actual method for checking twist. That could be a flat inspection table, winding sticks, a dedicated fixture, a straightedge arrangement, or another repeatable method. The method and acceptable measured variation should come from the actual build record rather than being invented here.

MEMBER FAMILIES

Different strut lengths do not automatically require different wedge profiles. That depends on the geometry. It is possible for several length families to share the same cross-section.

It is also possible that different edge relationships require different profiles. Until the reference geometry is fully modeled, we should not assume either result.

This creates two separate classification systems.

A member can belong to a length family:

A, B, C, and so forth.

It can also belong to a profile family:

W1, W2, W3, and so forth.

If every strut uses the same profile, the system becomes simple.

For example:

A-W1

B-W1

If different profiles are required, the labeling system can still keep them distinct:

A-W1

A-W2

B-W1

The point is not the exact naming scheme. The point is to avoid hiding multiple geometric properties inside one ambiguous letter. A member family should tell us enough to manufacture the correct part.

FABRICATION FAILURE MODES

Several mistakes can produce a part that looks close but is not actually the correct member.

Wrong inside width:

The wedge may be too narrow or too wide at the interior.

Wrong outside width:

The face-to-face relationship changes even if the inside dimension is correct.

Wrong thickness:

The entire cross-sectional relationship changes and panel or connection dimensions may no longer align.

Unequal side taper:

A member intended to be symmetric becomes offset.

Longitudinal taper:

The profile changes from one end to the other.

Twist:

The wedge orientation rotates along the member.

Wrong finished length:

The triangular geometry changes.

Wrong end treatment:

The theoretical geometry may be correct while the physical members interfere or leave unintended gaps.

Wrong drilling:

The connection reference moves.

Wrong label:

A perfectly manufactured part can still be installed in the wrong position.

Reversed orientation:

The wide side and narrow side trade places relative to the dome. Those failures should eventually be incorporated into a shop inspection sheet. A finished member should be accepted because it matches its definition, not because it looks close enough to the previous piece.

REPEATABLE FABRICATION

The production goal is to convert all of those dimensions into controlled operations.

One possible sequence is:

1. inspect the starting stock, 2. establish the primary datum, 3. cut the wedge profile, 4. verify the profile, 5. cut the member to its family length, 6. prepare the ends, 7. drill or machine connection features, 8. mark inward and outward orientation, 9. apply the member-family label, 10. perform final inspection.

The exact order may change once the actual tooling is developed. For example, it may be easier to cut long continuous wedge stock first and crosscut individual struts afterward. Another process may cut blanks to length before tapering.

A log-derived process may require an entirely different sequence. That is why the actual shop setup has to be documented instead of assumed. The final method should be selected based on repeatability, workholding, material yield, tool access, and the ability to inspect the result.

DESIGN VARIATION

Once the basic wedge is controlled, other features can be built into it. The exterior portion could receive a panel seat. The interior face could receive a finish strip.

A groove could locate a gasket. A machined recess could locate a plate. A laminated wedge could place stronger or more durable material where the connection occurs.

A replaceable exterior attachment could be added without changing the underlying chord geometry. Those features create new member types, but they should all grow from the same controlled base profile. The more functions a single timber performs, the more important its drawing becomes.

At that point, it is no longer enough to call it a "wedge strut."

It is a manufactured component with defined interfaces.

SAFETY / ENGINEERING

Removing material to create a wedge changes the physical cross-section of the timber. Adding bolt holes, notches, grooves, rabbets, or other features changes it again. Those changes can affect structural behavior.

The fact that a profile fits the dome geometry does not establish its capacity. Structural evaluation has to use the actual finished section, actual material, actual connections, and actual loading conditions. The same applies to defects in the starting timber.

A knot, split, check, or other defect does not disappear because the outside dimensions of the wedge are correct. Dimensional inspection and structural material inspection are separate requirements. For this book, the geometry defines the intended part.

Structural engineering determines whether that part is adequate for a particular building and load condition.

THE USEFUL WAY TO DEFINE A WEDGE MEMBER

By the time the system is fully developed, a wedge member should be describable without needing to hold the physical piece in our hands.

A complete member definition should eventually tell us:

- what stock it starts from,

- its actual finished length,
- its theoretical chord relationship,
- its inside width,
- its outside width,
- its thickness,
- its taper,
- its end geometry,
- its connection features,
- its inward/outward orientation,
- its member-family ID,
- its quantity,
- its inspection dimensions,
- and where it belongs in the dome.

Once all of those values are known, the wedge becomes something we can reproduce. That is the point. We are not trying to make a collection of similar-looking pieces of timber.

We are defining structural parts.

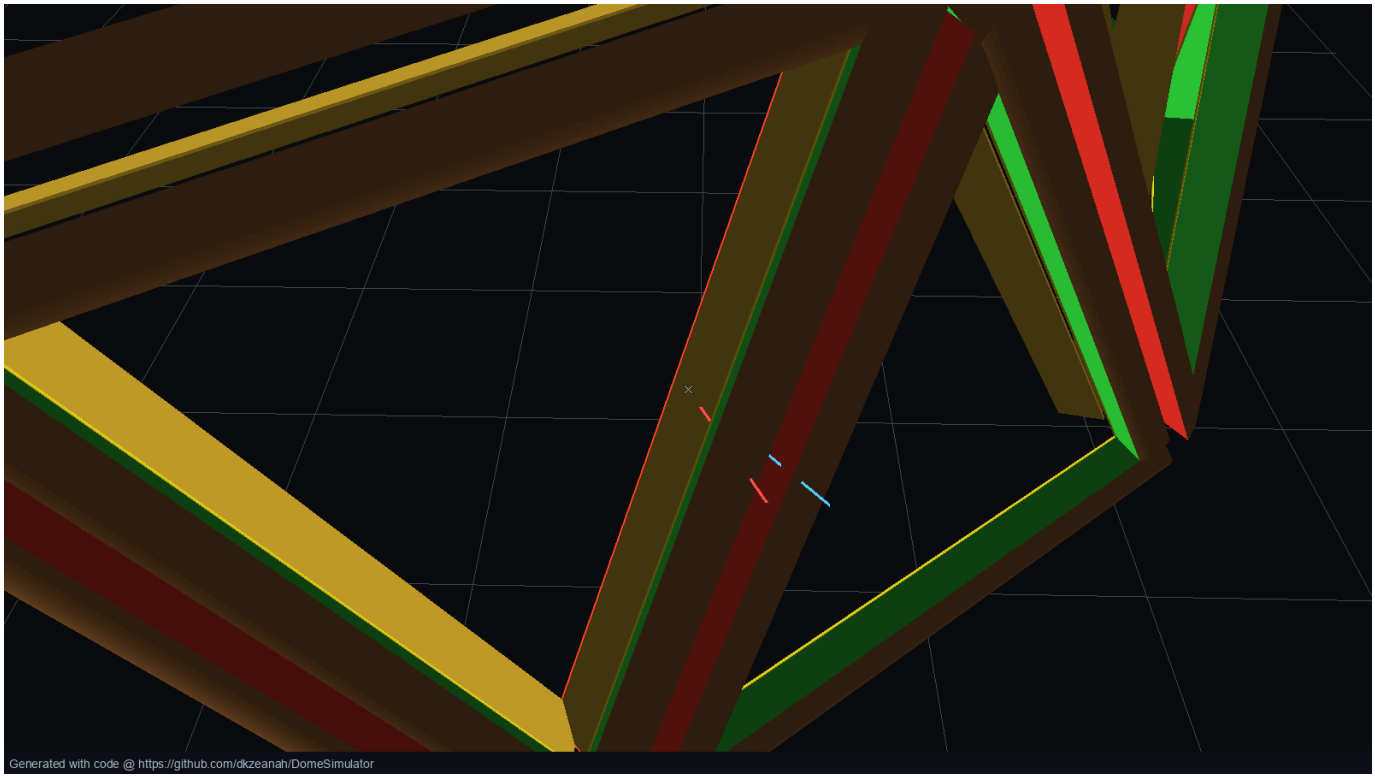

Figure 8. The point faces in

The reference build's orientation, read off a seam: both wedges' points aim at the centre of the sphere, so the wide bark faces are outside and the seam opens 27.0 degrees toward the weather. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Wide side / narrow side orientation

The basic orientation rule for the wedge method is simple:

The narrow side points inward. The wide side points outward. That rule should stay obvious from the moment the member is made until the moment it is installed.

The reason is geometric. The dome wraps around a center point. The inside of the frame is closer to that center.

The outside of the frame is farther away. As neighboring triangular faces move outward from the interior toward the exterior, they need more separation between them. The widening wedge gives us that extra width.

That is why the member is shaped the way it is.

GEOMETRY

A wedge member has two primary widths:

$$w_i$$

for the inside width,

and:

$$w_o$$

for the outside width.

For the standard inward-pointing orientation:

$$w_o > w_i$$

The difference is:

$$\Delta w = w_o - w_i$$

That difference is what creates the taper.

If the wedge is symmetrical, half of that change appears on each side:

$$x = (w_o - w_i) / 2$$

The wider face therefore spreads the neighboring surfaces farther apart toward the exterior. The narrow face allows them to remain closer together toward the interior.

That is the part that matters physically. The wedge is not being installed wide-side outward because it looks right. It is installed that way because its widening cross-section is intended to follow the direction in which the dome faces diverge.

THINK FROM THE CENTER OF THE DOME

The easiest way to understand orientation is to ignore the individual strut for a moment and think about the center of the sphere. Every point on the dome surface exists some distance from that center. The

inside of the timber is closer to the center.

The outside of the timber is farther from it. If we take a cross-section through a wedge member, the narrow face points generally toward that center.

The wider face points away from it.

That gives us a reliable visual rule:

narrow toward the center, wide away from the center. For an assembled dome, "toward the center" is normally the interior. That same rule can be used during panel fabrication, preassembly, and field erection.

If a triangular module is lying flat on a work table, the installer may not be able to think in terms of "up" and "down."

The better reference is still:

Which face will point toward the center of the completed dome?

That is the narrow side.

REVERSING A WEDGE

A wedge installed backward does not become another valid version of the same part. Its geometry is reversed. Instead of becoming wider toward the exterior, it becomes narrower.

That changes how the neighboring faces are allowed to sit around it.

The result may appear as:

- an unexpected gap,
- interference between adjacent members,
- a triangle that will not settle into position,
- a vertex that has to be forced,
- an exterior panel that no longer fits,
- or accumulated distortion farther away from the reversed member.

The exact symptom depends on the connection system. The important point is that the error may not show up directly beside the incorrectly installed piece. Geodesic structures distribute geometry across connected triangles.

One reversed member can push an error into the next face, then the next connection, and eventually appear somewhere else in the dome.

That makes orientation marking worth doing before assembly begins.

ORIENTATION SHOULD NOT DEPEND ON MEMORY

A production system should assume that a member will eventually be handled by somebody who did not make it. Even if one person cuts and assembles the entire dome, the same member may sit in a stack for days or weeks between operations.

The orientation should be physically marked.

Possible marking methods include:

- an arrow pointing inward,
- an arrow pointing outward,
- the word "IN,"
- the word "OUT,"
- a paint mark,
- a burned mark,
- a stamped symbol,
- a routed identification mark,
- or a consistent label location.

Whatever system is used, it should be difficult to interpret backward. A single line on one face may be enough if the convention is fixed. A more complete production label can identify both orientation and part family.

For example:

A-W1-IN

could indicate:

- A = length family,
- W1 = wedge-profile family,
- IN = marked interior face.

The exact naming convention can change.

The useful part is making the information survive handling.

LEFT AND RIGHT SIDES

For a symmetric wedge, the left and right side tapers are the same. That makes orientation simpler. The important distinction is only inside versus outside.

For an asymmetric wedge, the problem becomes more complicated.

Now the member may have:

- inside,
- outside,
- left,
- right,
- end A,
- end B.

That creates more ways to install the correct member incorrectly. If asymmetric wedges are ever introduced, the labeling system has to preserve all of those relationships. A simple arrow may no longer be

enough.

Unless the geometry gives us a reason to do otherwise, symmetry reduces the number of orientation mistakes that can occur.

THREE-MEMBER TRIANGLE

The orientation becomes easier to see when three wedge members are assembled into one triangular face. Each member should point its narrow side toward the same side of the triangle. That side becomes the interior-facing side of the module.

The wider faces all point toward the opposite side. When the triangle is viewed from the interior, we should see the narrow side of each edge member. When viewed from the exterior, we should see the wider side.

This gives us a simple inspection before the triangle ever reaches the dome. Lay out the three members. Identify the intended interior face.

Check that all three narrow sides face that direction. Then connect them. If one member shows its wide side while the other two show their narrow sides, something is wrong before any fastener is tightened.

TWO TRIANGLES SHARING AN EDGE

Now add a second triangle beside the first. The shared wedge member sits between the two triangular faces. Both faces extend away from the narrow side toward the wider side.

This is the relationship the wedge is designed to support. The interior edges remain closer together. The exterior faces spread farther apart.

This is where the orientation stops being merely a label and becomes part of the geometry. If that shared member is reversed, the two faces are being asked to do the opposite. The member becomes narrower in the direction where the structure expects it to widen.

That is why orientation has to remain consistent across the whole frame.

VERTEX ORIENTATION

At a five-member or six-member vertex, multiple wedges meet around one point. Every one of them should follow the same inward/outward logic. Their narrow sides converge toward the dome interior.

Their wider sides expand toward the exterior shell. From the interior, the vertex should therefore read as a group of narrow wedge faces converging around the joint. From the exterior, those same members present their wider portions.

This is also useful for checking a partially assembled dome. An installer should be able to inspect the frame visually and identify a member that appears opposite to the others. The connection hardware may hide some of the cross-section later.

That is another reason the orientation marks should remain visible until the frame has been verified.

BUILD NOTE

The final assembly process should include an orientation check before each joint is permanently tightened.

A practical sequence can be:

1. identify the part family, 2. identify the marked inside face, 3. position the narrow side toward the dome center, 4. confirm the neighboring members use the same orientation, 5. loosely connect the joint, 6. verify the triangle or vertex, 7. then complete the fastening operation.

This check is simple enough that it can become part of the normal assembly rhythm. It is cheaper to catch a reversed member before the next several triangles depend on it.

PANEL ORIENTATION

If wedge members are used as the edges of prefabricated panels, the orientation rule should be built into the panel jig.

The jig can establish:

- interior face downward or upward,
- exterior face downward or upward,
- member family,
- wedge orientation,
- and panel skin location.

That removes another decision from the operator. Instead of asking which way each wedge belongs every time, the fixture itself can make the correct orientation obvious. For example, a panel jig might always be loaded with the narrow face toward the table center or toward a marked "INTERIOR" side.

The exact method depends on the manufacturing setup. The useful principle is that orientation should be controlled by the process, not remembered separately for every part.

DESIGN VARIATION

There may be later systems where the wedge is intentionally reversed or reoriented for another purpose. For example, a special edge condition, opening, base transition, or nonstandard panel might require geometry different from the main dome field. If that happens, the special member should receive a separate part definition.

It should not be treated as a normal wedge installed backward. That distinction matters. A reversed standard part is an assembly error.

A deliberately opposite or modified part is a different component and should have its own label, drawing, and installation location.

SAFETY / ENGINEERING

Orientation can also affect more than fit.

If the wedge contains:

- drilled holes,

- plates,
- notches,
- panel seats,
- rabbets,
- fastener recesses,
- or other asymmetric connection features,

reversing it may move those features into different structural positions. That can change edge distances, bearing surfaces, connection alignment, or the way loads pass through the joint. For that reason, a completed structural member should be treated as directional whenever its geometry is directional.

A part that fits physically when reversed is not automatically structurally interchangeable.

REFERENCE BUILD

For the 20-foot 2V reference dome, every production member should eventually carry a clear orientation convention.

The simplest standard should be:

NARROW = IN WIDE = OUT That convention can be printed on drawings, shown in every assembly diagram, and physically marked on the timber.

If a member is shown in cross-section, the interior side should always appear consistently in the same direction where practical. If a panel diagram is shown, the reader should always be able to tell which face points toward the dome center. If a vertex is shown, the narrow sides should visibly converge toward the interior.

This may seem repetitive. That is intentional. Orientation is simple enough that it should never become a source of confusion.

The wedge only performs its intended geometric job when it points the right way.

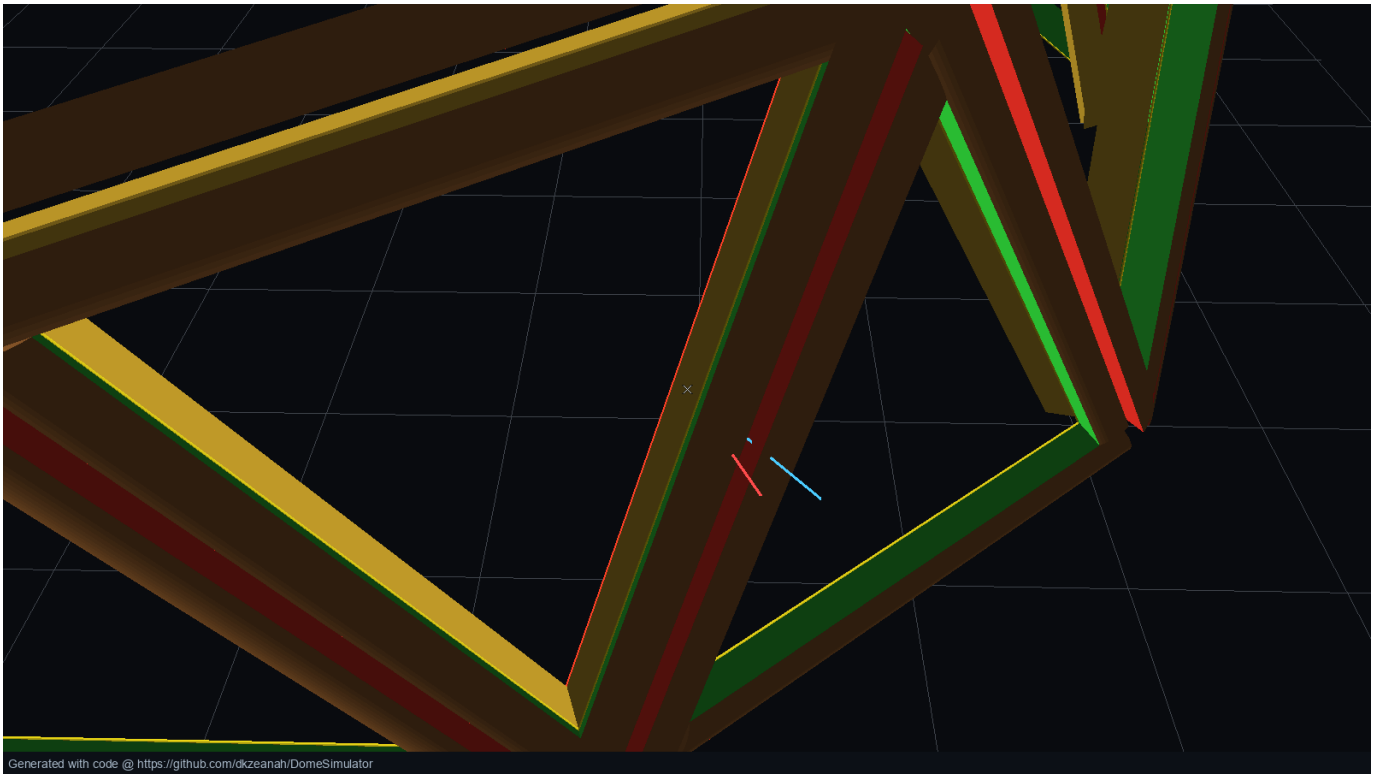

Figure 9. Both points inward, toward the centre

Both points inward, toward the centre. The A seam then wants a key 2.80 in across its base. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Inward-pointing wedge geometry

The wedge does not actually have to come to a sharp point for its geometry to have a point.

That distinction is useful.

The physical member has a narrow inside face and a wider outside face. If we extend the two tapered sides inward beyond the actual timber, those sides eventually intersect.

That imaginary intersection is the virtual point of the wedge. For the basic wedge method, that point is directed toward the interior of the dome. The timber itself is therefore a truncated wedge.

We are using only the useful section between the narrow interior face and the wider exterior face instead of continuing the material all the way to a sharp edge.

GEOMETRY

Look at a wedge member directly from the end. For a symmetric wedge, the two side faces spread away from a centerline by equal amounts.

The narrow width is:

$$w_i$$

The outside width is:

$$w_o$$

The distance between those two faces is:

$$t$$

The total widening is:

$$\Delta w = w_o - w_i$$

Because the wedge is symmetric, each side moves outward by:

$$(\Delta w) / 2$$

over the thickness t.

The half-angle of the wedge is therefore:

$$\beta = \tan^{-1}((w_o - w_i) / 2t)$$

and the total included wedge angle is:

$$\alpha = 2\beta$$

or:

$$\alpha = 2 \tan^{-1}((w_o - w_i) / 2t)$$

That is the same basic wedge relationship used earlier, but now we can give it a physical meaning. If we extend the tapered sides inward, they meet at the virtual point.

The narrow face of the timber is simply stopping before the wedge reaches that point.

THE VIRTUAL APEX

For a symmetric wedge, we can calculate how far the narrow face sits from the virtual apex.

Let that inward distance be:

$$r_i$$

At that distance, the width of the wedge is:

$$w_i = 2r_i \tan(\alpha / 2)$$

so:

$$r_i = (w_i) / (2 \tan(\alpha / 2))$$

The outside face is another thickness t farther from the virtual apex:

$$r_o = r_i + t$$

and its width becomes:

$$w_o = 2r_o \tan(\alpha / 2)$$

Because both widths belong to the same pair of taper lines, another useful form is:

$$r_i = (t,w_i) / (w_o-w_i)$$

and:

$$r_o = (t,w_o) / (w_o-w_i)$$

These relationships are useful because they show what "pointing inward" actually means. The wedge sides can be imagined as radiating from a common line or point located inward from the physical timber.

The member gets wider as we move away from that virtual apex.

A SHARP WEDGE IS NOT REQUIRED

If the member actually continued all the way to its theoretical point, the inside width would become zero.

That would give us:

$$w_i = 0$$

For structural timber, that is generally not the shape we are trying to produce.

We need real wood remaining on the interior side.

We may need room for:

- fasteners,
- bearing surfaces,
- plates,
- panel attachment,
- edge distance,
- and practical handling.

So the useful member is a truncated wedge. We preserve the angular relationship without cutting the timber down to a knife edge.

This also gives us another way to design the member.

Instead of thinking only:

"What saw angle makes this wedge?"

we can think:

"What angular relationship are we trying to represent, and where do we want to truncate that wedge into a usable timber section?"

Those are different questions.

THE VIRTUAL POINT IS NOT AUTOMATICALLY THE CENTER OF THE DOME

This is an important distinction. The wedge points inward, but that does not automatically mean that extending every wedge side will make it intersect at the exact center of the geodesic sphere.

There are several different geometric references in the structure:

- the sphere center,
- the theoretical dome vertices,
- the theoretical chord lines,
- the triangular face planes,
- the physical timber,
- the panel surfaces,
- and the wedge's virtual apex.

Those relationships have to be modeled. They should not be assumed. A useful local wedge can point generally toward the dome interior without its extrapolated taper lines literally intersecting at the global center of the sphere.

The exact relationship depends on how the wedge is defined around each geodesic edge. This matters because a dome is not made from radial spokes running from the center to the surface. The struts run between neighboring vertices.

They are chords.

The inward-pointing wedge geometry exists around those chord directions.

LOCAL GEOMETRY

It is useful to think of each wedge as having its own local coordinate system. Along the member is the longitudinal direction. Across the member is the side-to-side direction.

From the narrow face toward the wide face is the inward-to-outward direction.

That gives us three useful local axes:

L = along the strut

W = across the strut

R = inward/outward through the wedge

The geodesic model can tell us where the strut belongs in global three-dimensional space. The local member definition tells us what the timber looks like around that line. This is similar to defining a part in CAD.

First define the coordinate system. Then define the section. Then place the finished part into the assembly.

That prevents the shape of the timber from becoming confused with the position of the timber.

THE WEDGE AND THE FACE-TO-FACE ANGLE

Two neighboring dome triangles are not normally coplanar. They meet with a change in direction along their shared edge. That face-to-face relationship is one of the reasons the wedge exists.

The tempting shortcut would be to take the dihedral angle between those faces and call that the required wedge angle.

That should not be done automatically.

The physical wedge angle depends on how the timber is positioned relative to:

- the theoretical shared edge,
- the neighboring face planes,
- the panel surfaces,
- the timber thickness,
- and the connection datum.

If the tapered side faces of the wedge are deliberately designed to align with particular panel or face planes, then there will be a direct geometric relationship between the wedge and those planes. If the panels sit on another surface of the timber, or if the timber is offset from the theoretical chord, the relationship changes.

So the correct process is:

calculate the face geometry, define the physical reference system, then derive the wedge from that model.

Do not simply copy a dome angle onto the table saw and assume the same number applies.

THREE MEMBERS

With three wedge members forming one triangle, each member follows the same general rule. The narrow portion is directed toward the interior. The wider portion extends outward.

But the important geometry is easier to see when we stop looking at the whole triangle and inspect one shared edge. That edge has one triangular face on one side and another face somewhere else in the completed dome on the other side. The wedge occupies physical space around that edge.

Its profile gives us a controlled way to transition outward from the narrower interior frame toward the larger exterior envelope. When several wedges are connected into one triangle, the interior perimeter and exterior perimeter are therefore not exactly the same shape or size. The exterior side of the triangular assembly occupies a larger envelope because every edge member widens outward.

That becomes important later when we design skins and panels. A panel fitted to the exterior cannot automatically use the same dimensions as a panel fitted to an interior reference surface.

FIVE- AND SIX-MEMBER VERTICES

The inward-pointing idea becomes easier to see at a complete vertex. Imagine several wedge members meeting around one point. From the interior, their narrow sides gather around the connection.

Moving outward, each member becomes wider. The group therefore expands as it approaches the exterior side of the structure. At a five-member vertex, five wedges participate in that pattern.

At a six-member vertex, six do. The exact angles around those vertices depend on the selected geodesic geometry.

The general wedge orientation stays the same:

narrow toward the inside, wide toward the outside. This gives us a useful visual check.

If one member at a vertex appears to flare inward while every neighboring member flares outward, that member deserves inspection before the assembly continues.

THE WEDGE CREATES TWO DIFFERENT ENVELOPES

Once the member has thickness and taper, the structure has more than one geometric envelope. There is an interior structural envelope. There is an exterior structural envelope.

There may also be a theoretical chord or centerline geometry somewhere between them.

If panels or skins are added, more surfaces appear:

- interior finish surface,
- insulation boundary,
- exterior sheathing surface,
- weather surface.

These surfaces are offset from one another. They do not have identical triangle dimensions. This is one of the reasons the wedge should be modeled as a three-dimensional member instead of only as a line on a dome calculator.

A line model is enough to determine the underlying geodesic topology. It is not enough to determine every physical panel dimension. The more functions we add to the wedge, the more those offsets matter.

DESIGN VARIATION

The virtual-apex idea also gives us a way to compare different wedge profiles. Two wedges can have the same included angle but different inside and outside widths. They simply represent different truncated portions of the same theoretical wedge.

For example, one member could begin closer to the virtual apex and therefore have a narrower inside face. Another could use the same taper angle but be shifted farther outward, producing both a larger inside width and a larger outside width. The angular relationship would remain the same while the amount of timber changes.

This can become useful when evaluating:

- different structural member sizes,
- larger connection zones,
- panel attachment requirements,
- different timber thicknesses,
- or laminated sections.

The wedge angle and the wedge size are related, but they are not the same thing.

A wedge should therefore not be specified only by saying:

"Cut this angle." We also need to know where that angle is truncated into the finished member.

MATERIAL YIELD

This also matters when wedges are cut from rectangular lumber. Suppose the required taper angle is already known. There may still be several possible ways to position that wedge inside a rectangular blank.

One layout might produce the widest possible interior face. Another might allow two mirrored wedges to be recovered from one board. Another might leave an offcut useful for another part.

The geometric requirement sets limits. The material layout determines how efficiently we produce that geometry. The same principle applies to a log.

Several rough wedge sectors can radiate around the log center. Those rough sectors already resemble the general inward-pointing geometry, but their final dimensions still have to be brought under control. The fact that a log naturally divides into radial sectors is useful.

It does not eliminate the need to define the finished wedge. FABRICATION ERROR: MOVING THE VIRTUAL APEX A useful way to understand wedge-profile error is to think about what happens to the virtual apex.

If the outside width is cut too large while the inside width and thickness stay fixed, the wedge angle increases. The extrapolated sides meet closer to the timber. If the outside width is too small, the wedge angle decreases.

The virtual apex moves farther away. If one side is cut differently from the other, the virtual apex shifts sideways. So a fabrication error is not only changing the dimensions of a piece of lumber.

It is changing the geometry that piece represents. This is why a wedge jig should control both sides from a known reference. Repeatedly producing a slightly wrong profile means repeatedly producing the same wrong angular relationship.

BUILD NOTE

When the first controlled wedge batches are produced, the inspection record should include enough information to reconstruct the actual cross-section.

At minimum:

- inside width,
- outside width,
- thickness,
- left-side taper,
- right-side taper,
- measurements at more than one point along the member.

From those measurements, the actual wedge angle can be calculated. If a symmetric target was intended, left and right measurements can also show whether the finished profile remained centered. This gives us a better quality-control method than simply recording the table-saw setting.

The saw setting is an input. The finished timber is the output.

The output is what belongs in the dome.

TOOL SETUP

There are several possible ways to create an inward-pointing wedge. A rectangular blank can be run through a taper jig. A fence can be positioned to remove a controlled amount from each side.

A long blank can be processed into continuous wedge stock and then crosscut into individual struts. Larger timber can be resawn. A rough radial sector can be recovered from a log and then refined.

The actual reference-build procedure should eventually document:

- what surface is established first,
- how the stock is restrained,
- which side is cut first,
- how the second cut references the first,
- how the centerline is maintained,
- how long material is supported,
- and how the final profile is checked.

A wedge can be geometrically simple on paper and awkward to manufacture safely if the stock is poorly supported.

The jig and workholding are therefore part of the method.

SAFETY / ENGINEERING

The inward-pointing shape should not be confused with proof of structural behavior. A wider exterior section and narrower interior section change the timber cross-section.

The narrow side may become important around:

- bolt holes,
- screw locations,
- notches,
- bearing areas,
- plates,
- and panel connections.

The usable narrow width cannot be reduced indefinitely simply because the taper geometry continues mathematically toward a point. At some point there may not be enough material for the intended connection or structural demand. That limit has to come from the actual member and connection design.

The mathematical wedge can reach zero width.

The structural timber does not need to.

REFERENCE BUILD

For the 20-foot 2V reference dome, the wedge should eventually be defined from a full three-dimensional model.

For every relevant edge condition, the model should establish:

1. the theoretical chord, 2. the neighboring triangular faces, 3. the chosen physical datum, 4. the inward/outward direction, 5. the member thickness, 6. the required wedge taper, 7. the inside width, 8. the outside width, 9. the connection geometry, 10. and the resulting panel or shell surfaces.

Once those relationships are fixed, the inward-pointing wedge stops being a concept and becomes a shop dimension. We will be able to say exactly what part to cut and why it has that shape.

Until then, the important geometric rule is this:

The physical member is a truncated wedge. Its virtual point lies inward. Its section becomes wider as it moves outward.

And its exact dimensions have to be derived from the physical dome system, not guessed from the appearance of the finished structure.

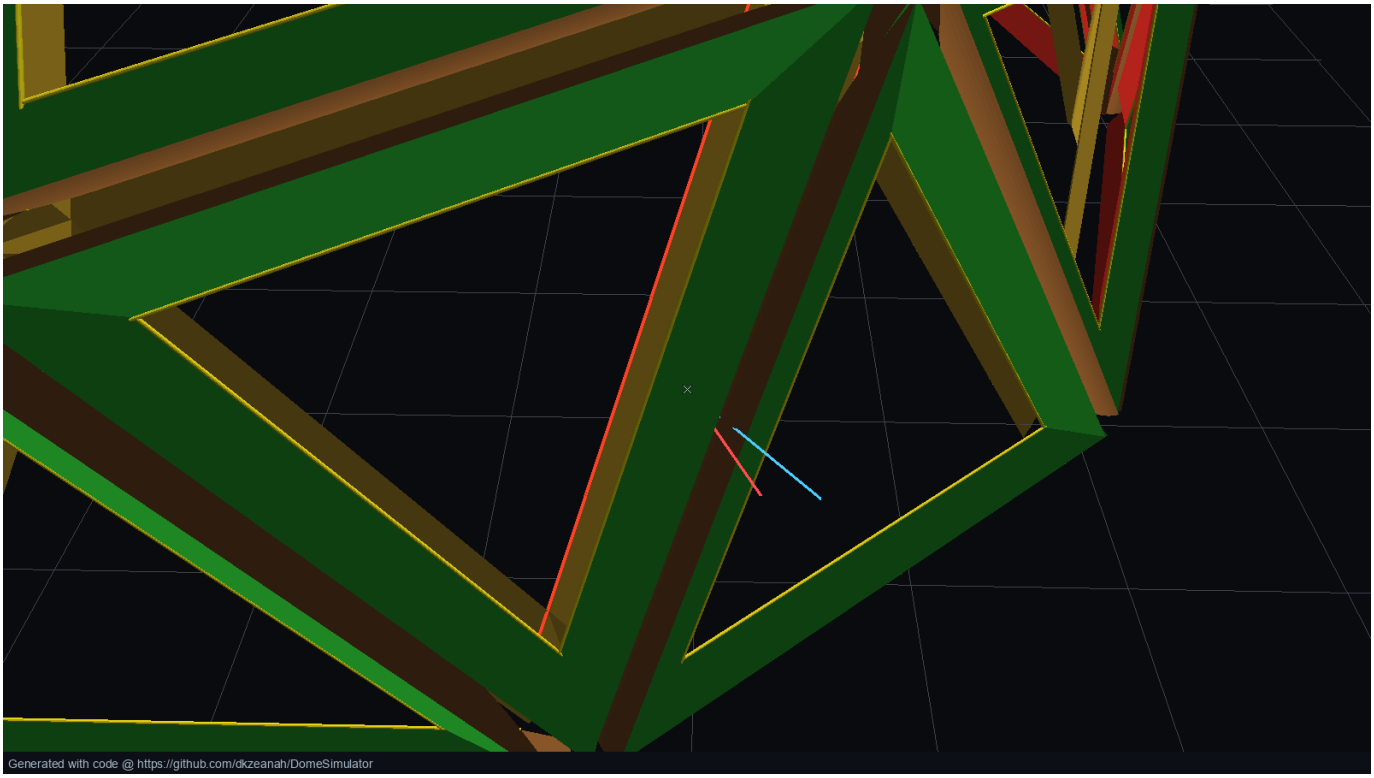

Figure 10. Points apart, each into its own triangle

Points apart, each into its own triangle. The A seam then wants a key 11.67 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Hubless or reduced-hardware construction

The wedge method can reduce how much specialized hardware is needed at a dome vertex. That does not mean the structure has no connection system. It means the timber itself is doing more of the geometric work, so the connection does not necessarily have to do all of it.

That is the difference between saying "hubless" and saying "no hardware." A hubless dome can still use bolts, screws, plates, washers, straps, or other fasteners. The useful distinction is that we are no longer depending on one large central hub to establish the entire three-dimensional relationship between the members.

THE TRADITIONAL HUB PROBLEM

In a conventional strut-and-hub dome, several straight members meet at one vertex. Those members arrive from different directions. A hub sits between them and provides the physical geometry that allows those directions to coexist.

That hub may have:

- arms,
- tabs,
- holes,
- plates,
- sockets,
- welded features,
- or other geometry.

The more work the hub has to do, the more specialized the hub becomes. That is not automatically a problem. A well-designed hub can make assembly much easier.

But it also means the dome depends on a manufactured component that may be harder to make than the struts themselves. The wedge method changes that balance. By shaping the timber to participate in the face-to-face geometry, the connector may be allowed to become simpler.

MOVING GEOMETRY INTO THE MEMBER

The wedge gives each strut a controlled inward-to-outward taper. That taper helps define how neighboring triangular faces spread apart. If the member itself is already carrying part of that relationship, then the connector does not have to create the whole angle from scratch.

Conceptually, the system changes from:

rectangular strut+geometry-heavy hub

to:

geometry-shaped strut+simpler connection

The exact amount of simplification depends on the final joint. It may be substantial.

It may be modest. The important point is that the geometry can be distributed between the timber and the connection instead of concentrated almost entirely in the hub.

DIRECT MEMBER-TO-MEMBER CONNECTION

One possible form of reduced-hardware construction is direct timber-to-timber connection. In that approach, the wedge members meet physically at the vertex and are held together with fasteners.

The connection could use some combination of:

- structural screws,

- through-bolts,
- washers,
- dowels,
- pins,
- or local reinforcement.

The main advantage is obvious. There is no large fabricated hub occupying the center of the joint. But direct contact also creates new design questions.

Which timber surfaces touch? Do the ends meet fully? Do they stop short of the theoretical vertex?

Is there a controlled gap? Do several members bear against one another? Does the fastener clamp the members together or only locate them?

Can the connection be assembled without forcing the surrounding geometry? A direct timber joint can look simple while still containing several geometric decisions.

PLATE-ASSISTED CONNECTIONS

Another approach uses plates without using a traditional central hub. A flat or formed plate can connect several wedge members while allowing the wedge profiles to establish much of the face geometry.

The plate might exist:

- on the interior,
- on the exterior,
- on both sides,
- between members,
- or locally around a vertex.

That creates a hybrid system. The timber carries part of the geometry. The plate provides restraint, load transfer, alignment, or reinforcement.

This is different from a hub whose arms or sockets establish the full strut direction. The plate may simply connect members that are already shaped to occupy the required geometry. That can make plate fabrication simpler.

It may also make replacement easier because the plate does not necessarily have to be a highly specialized three-dimensional part.

BOLTS

Through-bolts are attractive because the connection can be tightened and disassembled. A bolt also gives us a clear geometric reference. If the design uses bolt centers as the structural datum, then the distance between bolt centers can become the practical reference length between connections.

That can simplify dimensional control.

But the bolt does not solve the whole joint by itself.

We still need to define:

- bolt diameter,
- hole diameter,
- hole location,
- washer size,
- plate geometry if used,
- timber thickness around the hole,
- edge distance,
- member overlap,
- and tightening method.

If several members share one bolt, the stack-up also matters. The physical thickness of every member and washer changes the joint. If each member uses its own fastener, then those individual fastener locations still have to agree with the theoretical vertex geometry.

STRUCTURAL SCREWS

Structural screws offer a different assembly method. They can sometimes avoid the need to drill a complete through-hole across several members. They may also be installed from one side.

That can make field assembly easier in certain joint layouts. But a screw-driven joint still has to be deliberately designed. The screw path matters.

The angle matters. The member thickness around the screw matters. The available edge distance matters.

The receiving timber matters. A screw placed wherever there appears to be enough wood is not a repeatable connection detail. If structural screws become part of the reference build, their position should eventually be jigged or otherwise referenced just like a bolt hole.

HARDWARE REDUCTION SHOULD BE COUNTED

If the goal is reduced hardware, that reduction should eventually be measured instead of described vaguely.

For a reference dome, we can count:

- number of hubs avoided,
- number of plates,
- number of bolts,
- number of screws,
- number of washers,
- number of unique hardware types,
- number of drilled operations,
- and number of fabricated metal parts.

That gives us a real comparison. A system that eliminates forty hubs but replaces them with hundreds of unique steel pieces has not necessarily simplified manufacturing. A system that replaces a complex hub

with one repeated plate and one repeated fastener pattern may have.

The useful metric is not:

"Does this system contain hubs?"

The better questions are:

How many unique parts are required? How difficult are those parts to make? How many operations are required?

How repeatable is the connection? How easy is it to assemble and inspect?

That gives us a more realistic definition of simplification.

ASSEMBLY WITHOUT A CENTRAL HUB

Removing a central hub changes the erection process. A hub can act as a convenient collection point. Several struts can be loosely attached to one component and then positioned as the frame grows.

Without that hub, the wedge members may need to be held directly against one another while the connection is made.

That can require:

- temporary clamps,
- assembly jigs,
- temporary screws,
- straps,
- supports,
- or preassembled triangular modules.

A connection that uses less permanent hardware may therefore require more temporary workholding during erection. That is not necessarily a disadvantage. It just needs to be included in the method.

The permanent joint and the assembly fixture solve different problems.

THREE-MEMBER PREASSEMBLY

One way to reduce difficulty at the full dome vertex is to assemble triangles first. Three wedge members can be placed in a flat or controlled fixture and connected into a triangular module. That gives the builder a larger, already-defined unit.

Those triangles can then be brought together at the dome. This moves some of the assembly work away from the final vertex. Instead of positioning every loose strut independently in space, the builder handles repeated modules.

Whether that is better depends on:

- triangle size,
- weight,
- connection access,
- lifting method,

- and the design of the shared edges.

The wedge method can support either individual-strut erection or panel/module assembly. The joint design should be developed with the intended erection method in mind.

FIVE- AND SIX-MEMBER VERTICES

The challenge becomes more obvious at a five-member or six-member vertex.

Several wedge members approach the same region from different directions.

If there is no central hub, the members may:

- meet directly,
- overlap,
- terminate around a shared fastener,
- connect through plates,
- connect in pairs,
- or use another repeated arrangement.

The final geometry has to keep the intended vertex from becoming a random pile of timber ends.

A clean connection should define where every member belongs.

That can be done with:

- drilling jigs,
- vertex plates,
- temporary assembly fixtures,
- indexed holes,
- machined end features,
- or another repeatable locating system.

The less geometry the permanent hardware provides, the more important the fabrication accuracy of the timber becomes.

That is one of the central tradeoffs of the wedge method.

SIMPLER HARDWARE REQUIRES BETTER PARTS

A large hub can absorb some variability. A member may rotate slightly within a connection. A tab may flex.

A bolt hole may provide some installation clearance. When the timber itself is expected to establish more of the geometry, the timber has to be more consistent. That means reduced hardware and increased fabrication precision can go together.

If the wedge profile is wrong, the connection has less ability to hide that mistake. If the length is wrong, the vertex moves. If the end geometry is wrong, several members may no longer meet cleanly.

If a hole location is wrong, a simple plate may no longer fit.

The manufacturing process therefore becomes part of the connection system.

BUILD NOTE

For the reference build, the connection-development record should include more than photographs of the finished joint.

It should document:

- member IDs entering the vertex,
- member orientation,
- contact surfaces,
- connection hardware,
- hole pattern,
- plate dimensions if used,
- assembly order,
- temporary workholding,
- tool access,
- tightening sequence,
- disassembly method,
- and any interference observed during assembly.

That information is what turns a concept into a repeatable joint. A connection that works once after several pieces are forced into place is not yet a production method.

REPLACEABILITY

Reduced-hardware construction can also be designed around replacement. If the structure uses mechanical fasteners instead of permanent adhesive, an individual strut may potentially be removed later.

That could help with:

- damaged-member replacement,
- modification,
- panel access,
- transportation,
- or experimental configurations.

But replaceability has to be designed into the assembly sequence. A member can be bolted in place and still be impossible to remove after neighboring members trap it. A plate can be removable and still be inaccessible after the shell is installed.

So the question is not simply:

"Can the fastener be removed?"

It is:

"Can the entire part be removed from the assembled structure without damaging the surrounding system?" That becomes a useful design criterion if modularity is one of the goals.

DESIGN VARIATION

Several connection levels can exist within the same wedge system. A minimal connection might use direct timber contact and mechanical fasteners. A reinforced version might add small plates.

A heavy-duty version might add larger plates or brackets while still keeping the wedge geometry. A removable prototype might use bolts where a permanent version uses another connection. A panelized version might connect complete triangular modules instead of loose struts.

These are all variations of the same system. The presence of hardware does not make the method stop being a wedge method. The defining feature is that the wedge member still participates in establishing the geometry.

SAFETY / ENGINEERING

The connection is one of the most critical structural parts of the dome. Forces carried by the struts have to pass through the vertices. A reduced-hardware joint therefore cannot be evaluated only by counting how few pieces it uses.

The actual connection may depend on:

- timber bearing,
- bolt shear,
- bolt bending,
- screw withdrawal,
- screw shear,
- plate bending,
- washer bearing,
- hole spacing,
- edge distance,
- group effects,
- splitting,
- moisture movement,
- and the direction of the force relative to the grain.

Those requirements depend on the actual material, fastener, geometry, and loading. This book should not claim that fewer connectors means a stronger or weaker joint without the engineering to support that conclusion. A simple connection is useful only if it is also adequate for the job.

HUBLESS DOES NOT MEAN CONNECTIONLESS

This point is worth keeping explicit. A geodesic dome cannot be made from a group of correctly shaped pieces that merely touch one another. The members still have to act together as a structure.

The wedge can reduce the amount of connection geometry that must be manufactured separately.

It cannot remove the need to transfer forces between members.

So when this book uses the term "hubless," it should mean:

no conventional centralized geometry-defining hub

rather than:

no connection hardware of any kind. That definition leaves room for bolts, screws, plates, and other practical connection methods while keeping the distinction from traditional strut-and-hub construction clear.

REFERENCE BUILD

For the 20-foot 2V reference dome, the primary connection should eventually be frozen as one repeatable vertex system.

The process should define:

1. where the theoretical vertex lies, 2. where each wedge member terminates, 3. which surfaces make contact, 4. where the fasteners are located, 5. what hardware is required, 6. how the joint is temporarily held during assembly, 7. what order the pieces are installed, 8. how the connection is tightened, 9. how it is inspected, 10. and whether the joint can later be disassembled.

Once that joint exists, the hardware can be counted and compared against a traditional hub system. Until then, reduced-hardware construction should be treated as a design objective rather than a claim that the dome has already eliminated a particular percentage of connectors. The wedge gives us the opportunity to simplify the joint.

The finished connection determines whether that opportunity actually turns into a better construction system.

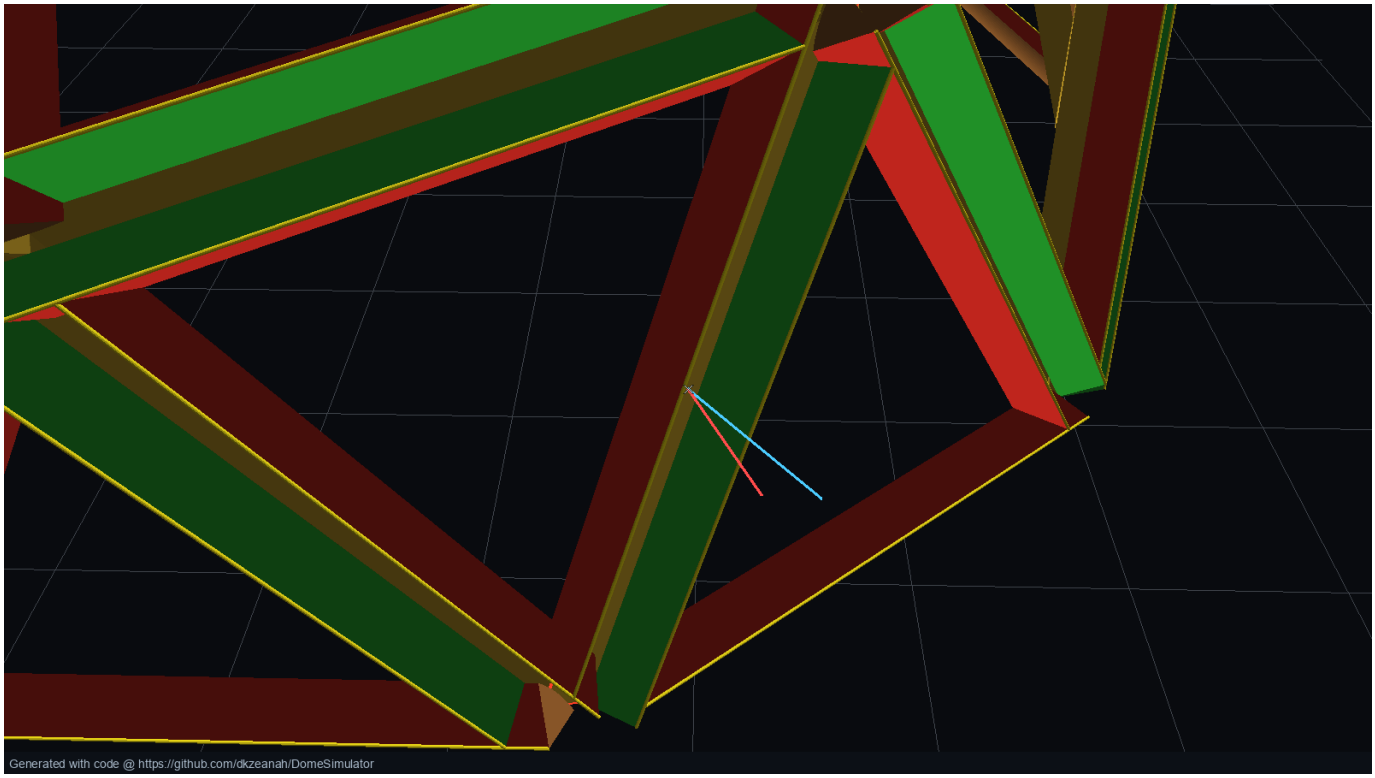

Figure 11. Both points outward, at the sky

Both points outward, at the sky. The A seam then wants a key 6.27 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Using lumber versus logs

The wedge method can start from two very different kinds of material. One approach starts with lumber that has already been sawn into rectangular boards or timbers. The other starts much closer to the tree.

A log can be divided lengthwise into rough wedge-shaped sectors and those sectors can then be refined into structural members. Both approaches can produce a wedge. They just solve the manufacturing problem from opposite directions.

With dimensional lumber, we start with a rectangle and remove material until a wedge remains. With a log, we can start with radial material that already resembles a wedge and then create controlled reference surfaces from it. That difference affects material yield, tooling, drying, workholding, inspection, and how repeatable the finished members are.

DIMENSIONAL LUMBER

Dimensional lumber gives us a convenient starting point because the stock already has flat reference surfaces. That is useful in a fabrication system. A flat face can ride against a fence.

A square edge can sit against a stop. A jig can locate the same board repeatedly. The starting stock may therefore require less preparation before the actual wedge cut begins.

The basic process is:

rectangular stock arrow wedge profile arrow finished strut

That looks simple, but the layout matters. Suppose a rectangular timber has a starting width W and thickness T.

The final wedge requires an inside width w_i and an outside width w_o.

The starting stock has to contain the complete finished profile:

$$W \geq w_o$$

before allowing for saw kerf, cleanup, irregular stock, or another machining allowance. The actual board should be measured.

A nominal lumber designation is not enough to develop the final wedge dimensions. If a board is sold under a nominal size, the finished dressed dimensions may be smaller.

The shop drawing should therefore distinguish between:

nominal stock size and actual measured stock size.

The wedge is cut from the actual wood.

CUTTING A WEDGE FROM A RECTANGLE

A simple wedge can be laid out inside the rectangular section. If the finished wedge is symmetric, the narrow face remains centered beneath the wide face.

Material is removed from both sides.

The amount removed on each side at the narrow face is:

$$x = (w_o - w_i) / 2$$

The actual saw setup depends on how the blank is oriented during the cut. One process might tilt the workpiece.

Another might use a taper sled. Another may guide the stock along an angled fence while keeping the blade vertical. The geometry of the finished piece matters more than which specific machine configuration is used.

The production setup should eventually be chosen based on:

- safe workholding,
- adequate stock support,
- repeatability,
- available tooling,
- material length,
- cut quality,
- and the ability to verify the result.

The saw-angle setting by itself is not the specification.

The finished wedge dimensions are the specification.

MATERIAL REMOVED FROM LUMBER

Starting with rectangular lumber means some material has to be removed to create the taper. For one rectangular blank, the offcuts may appear wasteful. Across an entire dome, those offcuts represent enough material that they should be considered during the design of the cutting pattern.

There are several possibilities.

An offcut may become:

- another wedge,
- a smaller strut,
- panel blocking,
- battens,
- cleats,
- spacers,
- trim,
- jig material,
- or another repeated building component.

A cutting pattern should therefore be evaluated as a complete material layout.

The question is not only:

"How much wood does one wedge remove?"

The better question is:

"How much useful material can we recover from the starting timber?"

MIRRORED WEDGES

Some rectangular stock may allow two complementary or mirrored wedge-like parts to be recovered from the same blank. Whether that is practical depends on the required profile and the saw kerf. Conceptually, if one diagonal or tapered cut divides the stock into two useful pieces, material efficiency improves because the removed section becomes another part rather than scrap.

But the two resulting pieces have to be inspected independently. Saw kerf removes material between them. Any bow or taper in the starting stock may affect each piece differently.

And if the required finished dimensions include cleanup passes, the theoretical nesting pattern has to include enough material for those operations. A perfect CAD layout with zero-width saw cuts is not a real cutting plan.

Kerf has to be included.

LOG-DERIVED WEDGES

A log offers a different starting geometry. Viewed from the end, the log is approximately circular. Lines running outward from the center divide that circle into radial sectors.

Those sectors already have the basic relationship we want:

narrower toward the center, wider toward the bark. That makes the wedge method naturally compatible with radial breakdown.

Instead of first producing rectangular boards, we can potentially divide the log lengthwise into wedge-like sections.

The process becomes:

log arrow radial sectors arrow controlled wedge blanks arrow finished struts

This can remove one intermediate manufacturing step. It also creates several new ones.

A log is not a manufactured blank.

THE LOG IS NOT A CYLINDER

A drawing usually shows a log as a straight cylinder. A real log may not behave that way. Its diameter can change from one end to the other.

It may curve. Its center may not follow a perfectly straight line. The cross-section may not be circular.

The pith may not sit at the geometric center. The surface may contain butt swell, knots, branch transitions, damage, or other irregularities. This means a radial division that looks equal at one end may not produce the same dimensions at the other.

The first step in log processing is therefore understanding the material that actually exists.

For a controlled wedge system, useful measurements can include:

- diameter at several positions,
- log length,
- centerline or pith location,
- sweep,
- major knots,
- checks,
- visible decay or damage,
- and the intended usable section of the log.

Only then does it make sense to decide how many sectors can be recovered.

RADIAL BREAKDOWN

Suppose the usable log section is divided into n equal angular sectors.

The nominal sector angle is:

$$\theta_s = (360°) / n$$

For example, dividing the cross-section into eight equal sectors gives:

$$\theta_s = (360°) / 8 = 45°$$

That does not mean an eight-sector log automatically produces the correct wedge profile for a dome. It only describes the rough radial breakdown.

The structural wedge may require a different included angle, different truncation, different width, or further resawing. The radial sector is raw material. The finished wedge is the controlled part.

This distinction matters because it allows us to choose the log breakdown for material recovery without forcing the final dome geometry to equal the rough sector angle. One rough sector may become one structural wedge. It may also be split again.

Or it may be squared and machined to another profile. The number of sectors recovered from the tree and the number of finished dome members recovered from those sectors are separate decisions.

REMOVING THE PITH

The center of a log deserves special attention. If several wedges are divided exactly through the center, their narrow tips may all contain the pith or material immediately surrounding it. That region may not be where we want the finished narrow structural face to remain.

The rough wedge can instead be truncated. Material near the center can be removed to create a flat interior reference surface. That operation turns a sharp radial sector into the kind of truncated wedge used throughout this system.

Conceptually:

sharp radial sector arrow remove inner tip arrow flat narrow face

The amount removed should come from the final wedge dimensions and material condition rather than from an arbitrary distance from the log center. The result gives us a surface that can be measured, jigged, drilled, and connected more easily than a sharp point.

ESTABLISHING REFERENCES ON A ROUGH WEDGE

A wedge cut from dressed lumber starts with useful flat surfaces. A wedge cut from a log may not. The bark-side surface is curved.

The radial cuts may vary. The log itself may taper along its length.

So a log-derived wedge needs a process for creating datums.

One possible order is:

1. divide the log into rough sectors, 2. remove the bark or irregular outer material as required, 3. establish one controlled reference surface, 4. establish the narrow interior face, 5. reference the remaining cuts from those surfaces, 6. bring the outside width under control, 7. verify the profile along the full length, 8. cut the member to final length, 9. machine the connection features, 10. label and inspect the part.

The final shop process may use a different order. The important part is that rough surfaces do not remain the dimensional reference indefinitely. At some point the work has to transition from tree geometry to manufactured-part geometry.

TAPER ALONG THE LOG

A log gets smaller as it approaches the top of the tree. That natural taper creates a problem if the goal is a constant wedge cross-section. A rough sector from a tapered log will also tend to become smaller along its length.

If we simply cut radial sectors and leave them untouched, the wide face at one end may not match the wide face at the other. For a member intended to have a constant profile, that taper has to be removed or deliberately accounted for. This may mean cutting the finished member to dimensions that can be maintained across the smallest usable portion of the blank.

In other words, the limiting end of the log may determine the recoverable finished wedge size. That can affect yield. A large butt diameter does not automatically mean the entire length can produce members sized from the butt.

The top end has to contain the required profile too.

LENGTH RECOVERY

Log length and member length also interact. If one log is long enough to produce several struts end-to-end, the log can potentially be divided longitudinally first and then each rough wedge can be crosscut into several shorter members. Another process could crosscut the log into manageable lengths before radial breakdown.

Each approach changes material handling. Longitudinal-first processing may preserve more flexibility for choosing final member positions around defects. Crosscut-first processing may make the material easier to handle on smaller equipment.

The best method depends on the tools, log size, final strut lengths, and defects. This is another place where material yield should be treated as an optimization problem rather than one fixed recipe.

GRAIN DIRECTION

One major difference between lumber and log-derived members is how clearly we can choose the wedge relative to the original tree. A radial sector preserves a direct relationship to the log center. The grain pattern visible in the cross-section may therefore be different from a wedge ripped out of commercially sawn rectangular stock.

That can affect:

- drying behavior,
- checking,
- movement,
- machining,
- and structural characteristics.

Those effects depend on species, moisture, sawing pattern, grade, dimensions, and the individual timber. They should not be reduced to a simple claim that one orientation is always better. The book should document the orientation being used and rely on appropriate timber references when structural consequences are stated.

MOISTURE AND DRYING

A freshly processed log and purchased dry lumber may begin at very different moisture conditions. That matters because the wedge is a geometric part. If the member changes dimensions after fabrication, its profile may also change.

Drying can affect:

- width,
- thickness,
- straightness,
- twist,
- checking,
- and connection fit.

That does not mean all log-derived material has to be handled in one particular way. It means moisture condition belongs in the fabrication record. If wedges are cut green and later assembled after drying, their final dimensions should be verified after drying.

If they are assembled green, the design has to account for whatever dimensional movement occurs afterward. The same principle applies to dimensional lumber, even if the amount of change is different. The actual moisture condition matters more than the label attached to the material.

BUILD NOTE

For the reference build, material records should eventually include:

- material source,
- species where known,
- nominal lumber size if applicable,
- actual starting dimensions,
- log diameter measurements if applicable,
- moisture measurement,
- visible defects,
- breakdown method,
- recovered blank dimensions,
- finished wedge dimensions,
- and rejected pieces.

That gives us real yield information. Without those records, statements about how many wedges come from a board or tree are only estimates.

TOOLING FOR LUMBER

Rectangular stock works naturally with common shop machines because the material already has flat surfaces.

Possible tools include:

- table saw,
- band saw,
- circular saw with guide,
- track saw,
- planer,
- jointer,
- router,
- drill press,
- and dedicated jigs.

The exact tool is less important than the workholding and reference system. Long wedge stock can become difficult to control if most of its weight hangs beyond the machine. Support tables, rollers, extensions, or another fixture may be needed so the cut is not affected by the operator trying to hold the stock level.

A repeatable cut requires the jig to control the wood. The operator should not have to manually steer the taper by eye.

TOOLING FOR LOGS

Logs introduce another scale of workholding. Before a rough sector can be treated like lumber, the log itself has to be restrained.

Possible processing methods could involve:

- sawmill equipment,
- band saw systems,
- chainsaw milling,
- circular saw systems,
- guide rails,
- cradles,
- dogs,
- clamps,
- or purpose-built fixtures.

The important part is that the log cannot be allowed to roll or shift during a longitudinal cut. A radial layout line is only useful if the saw follows it while the material remains fixed. Once the log has been divided into manageable rough wedges, those pieces can transition to smaller fixtures and more conventional shop operations.

The log-processing system and the wedge-finishing system may therefore be two different stages with different tools.

MATERIAL YIELD

The log approach creates the possibility of recovering many wedge blanks from one original tree. That is one of the main reasons to investigate it. But yield should be calculated from usable finished parts, not

from a theoretical circle.

A theoretical cross-section may suggest that eight, ten, twelve, or another number of sectors fit around the center.

Real yield may be lower because of:

- saw kerf,
- bark and sapwood removal if required,
- defects,
- pith removal,
- log taper,
- sweep,
- minimum finished width,
- minimum finished thickness,
- and rejected pieces.

The useful yield equation is therefore not simply:

number of radial sectors = number of finished struts

Instead:

finished yield = (usable finished members) / (starting material)

The starting material can be measured by board feet, cubic volume, log count, or another consistent quantity depending on what we are comparing. That gives us a way to compare lumber and logs with actual production data later.

LABOR IS PART OF YIELD

Material efficiency is only one part of the comparison. A log may contain a large amount of useful wood while requiring more handling and preparation. Commercial lumber may cost more per unit of wood but arrive straight, surfaced, graded, and ready to fixture.

A useful comparison should eventually include:

- starting material cost,
- transport,
- drying,
- equipment,
- setup time,
- number of cuts,
- material handling,
- rejected material,
- finishing operations,
- and finished usable part count.

This is important because "free tree" does not mean "free wedge." The processing work still exists. On the other hand, access to local timber and the ability to process it directly may make the log route practical in situations where buying large quantities of finished lumber is not.

The book should show both paths rather than pretending one is always the correct choice.

REPEATABILITY

Dimensional lumber has an obvious advantage when repeatability is the priority. The starting material is already closer to a standardized blank. That reduces the number of variables the jig has to remove.

Log-derived material can still become repeatable.

It simply requires more processing before the member reaches that state.

A good way to think about the transition is:

irregular natural material arrow controlled blank arrow controlled part

Once a log sector has been converted into a controlled blank, the later operations should look much more like the lumber process. At that stage we should no longer be compensating individually for the shape of the tree.

The jig should be manufacturing the defined wedge.

FAILURE MODES WITH LUMBER

Dimensional lumber can still create bad wedge members.

Common fabrication problems can include:

- using nominal dimensions in place of actual measurements,
- insufficient stock width for the required profile,
- bow,
- twist,
- cup,
- taper,
- incorrect fence setup,
- wrong jig orientation,
- saw drift,
- inconsistent feed,
- and labeling errors.

The flat starting surfaces reduce some problems.

They do not remove the need for inspection.

FAILURE MODES WITH LOG-DERIVED MEMBERS

Log processing adds another group of possible errors:

- incorrect radial layout,
- shifting log during cutting,
- following the bark instead of a controlled reference,
- sector angle changing along the log,
- pith off-center,
- excessive taper,
- sweep,
- hidden defects,
- irregular drying,
- insufficient finished dimensions at the small end,
- and assuming the rough radial surface is already accurate enough.

The rough wedge should therefore be treated as stock. Not as the finished strut. That one distinction can prevent a large amount of cumulative geometric error.

DESIGN VARIATION

There is also no requirement that a build choose only one material path.

A hybrid system could use:

- log-derived wedges for the main frame,
- dimensional lumber for special openings,
- laminated blanks for highly controlled connection areas,
- and smaller offcuts for panel framing.

Another build might use commercial lumber for the reference dome and treat log-derived construction as a later variation. The geometry can remain the same while the manufacturing path changes. That is useful because it separates the definition of the part from the source of the material.

The dome should not need a new mathematical model simply because the timber came from a different saw.

The finished part still has to meet the same dimensional definition.

SAFETY / ENGINEERING

Material source does not establish structural capacity. A wedge cut from a large tree is not automatically stronger because it came directly from a log. A commercially sawn board is not automatically appropriate simply because it has straight edges.

Structural use can depend on:

- species,
- grade,
- grain,
- knots,

- slope of grain,
- checks,
- splits,
- decay,
- moisture,
- member size,
- connection details,
- and the applicable design requirements.

Log-derived members may also fall outside ordinary assumptions used for standard dimensional lumber unless they are evaluated under an appropriate design method. For structural applications, the finished member has to be evaluated as the material and section it actually is.

REFERENCE BUILD

For the 20-foot 2V reference dome, the cleanest development path is to define the wedge member independently from the raw stock. The finished-part drawing should specify what the dome needs. Then we can develop at least two manufacturing routes.

Route A — Dimensional lumber

commercial blank arrow wedge rip arrow family length arrow connection machining arrow finished member

Route B — Log-derived timber

log arrow rough radial sector arrow controlled blank arrow finished wedge profile arrow family length arrow connection machining arrow finished member

Both routes should end at the same inspection drawing if they are producing the same member family. That is the important part.

The tree and the board are starting materials.

The wedge is the part.

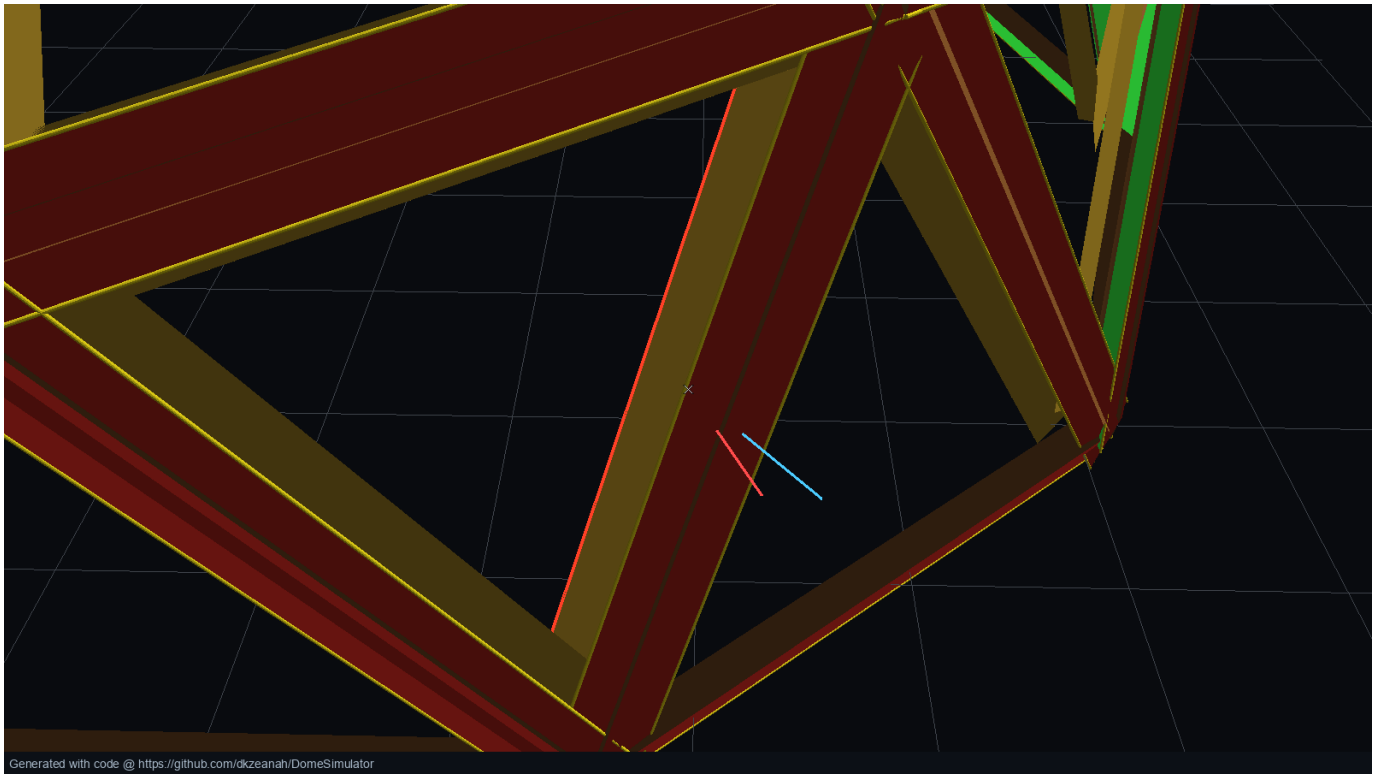

Figure 12. Points toward each other, across the seam

Points toward each other, across the seam. The A seam then wants a key 11.67 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

How the wedge influences the final shell geometry

The wedge does more than affect how two pieces of timber meet. It changes where the inside and outside surfaces of the finished dome end up. That matters once we stop thinking about the frame as a collection of centerlines and start thinking about actual wood, actual panels, actual insulation, and an actual weather surface.

A geodesic calculator can give us a network of ideal points and straight chord lengths. That is enough to define the mathematical frame. It is not enough to define the finished shell.

The wedge adds thickness and taper around those lines. Once that happens, the dome has several different surfaces instead of one.

GEOMETRY

At minimum, we need to distinguish between:

- the theoretical geodesic geometry,
- the interior structural surface,
- the exterior structural surface,
- the panel or sheathing surface,
- and the final weather surface.

Those surfaces are offset from one another. They do not necessarily have the same radius. They also do not necessarily use identical triangle dimensions.

That distinction becomes important as soon as the wedge has enough thickness to matter.

THEORETICAL GEOMETRY

The cleanest mathematical model starts with vertices located on a defined sphere. Straight lines connect those vertices. Those straight lines are the geodesic chords.

If we stopped there, every member would have zero thickness.

The model would tell us:

- where the vertices are,
- how long the chords are,
- which triangles exist,
- and how the triangular faces meet.

That is the skeleton of the dome. The wedge is physical material wrapped around that skeleton. Once we add the real timber, one side of the member lies closer to the dome center and another lies farther away.

That immediately creates more than one envelope.

THE INSIDE ENVELOPE

The narrow side of the wedge faces inward. If we connect those narrow faces around the dome, they create an interior structural envelope. That envelope is not automatically the same surface as the theoretical geodesic sphere.

Its exact location depends on where the wedge is positioned relative to the theoretical chord. For example, if the mathematical chord passes through the center of the timber section, then part of the wedge lies inside that line and part lies outside it. If the chord is aligned with another datum, such as a bolt-center line or interior reference face, then the offsets change.

That is why the physical datum has to be fixed before final dimensions are generated.

THE OUTSIDE ENVELOPE

The wide side of the wedge creates the exterior structural envelope. Because the member widens toward the outside, the exterior edge occupies a larger local envelope than the interior edge. This is part of what makes the wedge useful.

It gives the structure a controlled transition from a smaller interior perimeter to a larger exterior perimeter. But it also means that an exterior panel cannot automatically be cut to the same dimensions as an interior panel. The outside triangle is physically farther from the dome center.

Its edges may therefore have different finished dimensions depending on how the panel lands on the wedge.

ONE TRIANGLE, MULTIPLE TRIANGLES

A single geodesic face can produce several different physical triangle outlines.

We can think of them as:

1. the theoretical triangle, 2. the interior wedge-face triangle, 3. the exterior wedge-face triangle, 4. the exterior panel triangle, 5. the finished weather-surface triangle.

Those outlines may be close to one another. They are not automatically identical. This becomes especially important in larger members or thick insulated assemblies.

If the difference is ignored, gaps can appear where panels meet. Edges can overhang. Fastener locations can move away from the intended framing.

A panel that fits perfectly on a computer-generated theoretical triangle may not fit the actual outside of the timber frame.

THE WEDGE AS AN OFFSET GENERATOR

A useful way to think about the wedge is that it generates an offset between the interior and exterior shell geometry.

Suppose the distance from the inside face of the wedge to the outside face is:

$$t$$

That thickness moves the exterior surface outward from the interior surface.

At the same time, the wedge widens from:

$$w_i$$

to:

$$w_o$$

The width difference is:

$$\Delta w = w_o - w_i$$

The wedge therefore changes the shell in two directions at once. It moves the exterior surface outward.

It also spreads the neighboring face edges apart. That means the shell is not created by simply taking the interior triangle and moving it outward by one constant dimension. The face plane moves outward, and the edge geometry changes with it.

This is why a full three-dimensional model becomes useful.

FACE PLANES

Each triangular face can be represented as a plane. The neighboring face has another plane. Those two planes intersect along their shared edge.

The wedge occupies space around that edge. If the wide surfaces of the wedge are intended to support exterior paneling, then the outside panel planes have to agree with the wedge geometry. There are several possible ways to handle this.

One approach is to shape the wedge so its side faces directly establish the neighboring panel planes. Another approach is to let the panels sit on another prepared surface. Another is to use battens, spacers, clips, or a secondary frame over the wedge structure.

Each approach creates a different shell geometry. The important part is that the relationship is deliberate. A panel should not be forced into whatever space remains after the frame is built.

THE PANEL CONTACT SURFACE

If a sheet panel sits directly on the wide side of the wedge, we need to know exactly what surface supports it.

The panel may bear on:

- the entire exterior face,
- a narrow flat land,
- a rabbet,
- a machined seat,
- a secondary strip,
- or a separate panel frame.

That contact surface becomes part of the shell datum. Once it is defined, the panel dimensions can be derived from it. Without that definition, the phrase "exterior panel size" is incomplete.

The geometry depends on where the panel actually sits.

PANEL THICKNESS

The panel itself also has thickness. A plywood skin, fiberglass panel, composite panel, or insulated assembly adds another offset outside the structural frame.

If the panel thickness is:

$$p$$

then the finished exterior surface lies farther outward than the panel-support surface. If another weather layer is added, that creates another offset.

This means the finished weather radius can be larger than the structural radius used for the original geodesic calculation. That is not a problem. It simply has to be understood.

The reference diameter of the dome should always state which surface it describes.

A "20-foot dome" could refer to:

- theoretical vertex diameter,
- structural centerline diameter,
- inside structural diameter,
- outside structural diameter,
- panel-support diameter,
- or finished exterior diameter.

Those are different measurements once the structure has real thickness.

PANEL SEAMS

The wedge also affects what happens where panels meet. Two flat panels on neighboring faces approach each other at an angle.

Their edges can be handled several ways.

They may:

- butt together,
- leave a controlled gap,
- overlap,
- receive an exterior cap,
- receive a gasket,
- sit inside a channel,
- or meet against a cover strip.

The wedge can help create the space required for that joint. For example, a wider exterior portion may allow a panel edge to land fully on timber while leaving room for a seam cover or gasket. The amount of usable exterior bearing width therefore becomes part of the panel-joint design.

A narrow wedge may fit the geometry but provide too little practical panel attachment area. A wider wedge may improve the interface but use more material. That is another example of geometry and construction requirements meeting in the same part.

THREE-MEMBER TRIANGLE

Take one triangular wedge frame. On the interior side, the three narrow edges define one perimeter. On the exterior side, the three wider edges define another.

If the wedge profiles are symmetrical and consistent, the exterior envelope expands around the triangle. A panel attached outside the frame therefore follows the exterior triangle, not the interior one. That sounds obvious once the physical member is visible.

It becomes easy to miss when working from a line drawing where every strut is represented by one thin line. This is why the book should show the same triangle in several layers.

The reader should be able to see:

the chord triangle, the timber triangle,

and the panel triangle.

THE SHELL DOES NOT HAVE TO FOLLOW THE WEDGE DIRECTLY

DESIGN VARIATION

The exterior shell does not have to sit directly on the wedge.

A secondary shell system can be added outside the frame.

For example, the wedge frame could support:

- furring strips,
- panel clips,
- an offset rail,
- a secondary triangular frame,
- insulation blocks,
- or another attachment system.

That can create a more continuous exterior surface. It can also decouple the structural wedge geometry from the final weather skin.

This may be useful if:

- the wedge is optimized for structure,
- the panel needs a larger bearing surface,
- exterior insulation is added,
- removable panels are desired,
- or the shell needs a drainage cavity.

The tradeoff is additional material and another layer of geometry to manage. The cleanest system is not automatically the one with the fewest layers. It is the one where every layer has a defined job and a defined reference.

INTERIOR FINISH

The same problem exists on the inside. If interior finish is attached directly to the narrow faces, those narrow faces define the interior panel geometry. If a secondary interior frame is added, the visible interior surface moves inward.

That can change:

- interior diameter,
- usable floor area near the wall,
- window and door trim,
- utility routing,
- and the relationship between the interior finish and the structural frame.

The shell should therefore be thought of as a layered system. The wedge is one layer.

It influences the others.

INSULATION

An insulated dome can place material:

- between wedge members,
- outside the frame,

- inside the frame,
- inside panel assemblies,
- or in combinations of these locations.

Each approach changes the final shell thickness. If insulation fills the space between neighboring wedge frames, the face geometry of that cavity matters. If insulation is continuous outside the frame, then panel clips or secondary members may move the weather shell farther outward.

If insulation is inside the structure, the interior envelope moves inward. The geometry of the building should therefore be tracked through the complete wall section rather than only through the strut centerlines.

DOORS AND WINDOWS

Openings make these offsets more obvious. A door frame or window frame has to pass through some portion of the shell thickness.

If the structure has:

- wedge frame,
- exterior panel,
- insulation,
- weather layer,
- interior finish,

then the opening detail has to connect all of those surfaces. The frame depth cannot be designed from the theoretical geodesic line alone. The real shell has thickness.

That shell thickness may also vary depending on whether a secondary panel system is used.

Special opening modules should therefore reference the actual finished wall section.

BASE GEOMETRY

The wedge can also influence the base of the dome. At the bottom course, the exterior structural envelope and interior structural envelope may intersect the floor or foundation at different locations.

That can affect:

- base plates,
- floor edge details,
- flashing,
- exterior drainage,
- insulation transitions,
- and anchor locations.

If the mathematical base ring is defined only by chord centerlines, the physical timber can extend inward or outward from that ring. That extension needs to be accounted for when the floor and foundation are designed. Otherwise the frame may be mathematically correct while the physical base does not land where

the foundation detail expects it.

SHELL SMOOTHNESS

The wedge does not turn the geodesic dome into a continuously curved surface. The shell is still made from flat or nearly flat triangular faces unless another curved skin system is added. The wedge influences how those flat faces meet.

It can make the transition between faces cleaner and more controlled. Increasing dome frequency still reduces the size of the facets. The wedge does not replace frequency.

It works within the chosen frequency. A low-frequency dome with wedge members still has large visible triangular faces. A higher-frequency dome still uses smaller facets.

The wedge helps define the physical edge between them.

FABRICATION ERROR AND SHELL FIT

Errors in wedge geometry can show up dramatically at the shell. Suppose the wedge is slightly too narrow on the outside. One isolated member may not look badly wrong.

But when panels are installed, the joint may close more tightly than intended. If the wedge is too wide, a seam may open. If the profile varies along the member, the panel gap may change from one end to the other.

If the wedge twists, the supporting surface may no longer remain in the intended panel plane. This gives us another inspection method. The wedge can be checked as an individual part.

But a completed triangular frame or sample panel assembly can reveal whether the combined geometry actually produces the intended shell surfaces.

BUILD NOTE

Before committing to a complete dome shell, the reference build should include a full-scale mockup of at least two neighboring triangular faces.

That mockup should include:

- actual wedge members,
- the actual connection,
- the intended exterior panel material,
- the intended seam or gasket,
- and enough surrounding structure to reproduce the real face-to-face angle.

The mockup can then be checked for:

- panel bearing,
- seam width,
- fastener access,
- edge alignment,
- exterior continuity,

- interior clearance,
- and whether the panels sit naturally without being forced.

A three-dimensional mockup can expose problems that do not appear in a flat panel drawing. That is especially valuable before producing an entire dome worth of panels.

TOLERANCE STACK-UP

Shell geometry also gives us a reason to track cumulative error. Imagine a panel system where every wedge is slightly too wide on the exterior. One joint may absorb the error.

Several repeated joints can move the final position of a distant vertex or cause the last panel in a course to fit poorly. The same can happen with panel dimensions. A small seam-width error repeated around a ring can become a large closure error.

For that reason, tolerances should eventually be assigned to the complete assembly instead of only to individual pieces.

Useful checks can include:

- individual wedge profile,
- completed triangle dimensions,
- distance between known vertices,
- panel seam width,
- course circumference or polygon dimensions,
- and closure error at a known reference point.

The part tolerance and the assembly tolerance solve different problems.

DESIGN VARIATION

The wedge can also be deliberately modified to create shell features.

Possible variations include:

- a wider outside face for more panel bearing,
- a rabbet for panel edges,
- a groove for a gasket,
- a drainage channel,
- a removable cover strip,
- a clip attachment,
- a secondary insulation seat,
- or an interior finish seat.

Each added feature changes the physical member. That feature should therefore be included in the member drawing rather than added casually during installation. A structural wedge with a panel rabbet is a different manufactured part from a plain wedge of the same overall size.

EXPERIMENTAL

A more developed version of the system could treat the wedge as an interface between structure and replaceable shell panels. The frame could remain permanent while exterior triangular panels are mechanically attached to standardized wedge edges. A damaged panel could then potentially be replaced without replacing the frame.

Opaque, transparent, insulated, or specialized utility panels could share a common edge interface. This depends heavily on the final joint design. Weather sealing, structural contribution, drainage, tolerances, panel restraint, and long-term movement would all have to be developed and tested.

The geometry makes the concept possible.

It does not prove the finished enclosure system.

SAFETY / ENGINEERING

The shell can also participate structurally depending on how it is built. A thin nonstructural cover attached to a complete frame is different from a structural panel intended to transfer load between members.

The same is true for:

- plywood skins,
- composite panels,
- stressed skins,
- diaphragms,
- insulated structural panels,
- and mechanically connected shells.

If the enclosure contributes to structural performance, its fasteners, edge conditions, material properties, openings, and load path require engineering appropriate to that system.

The wedge shape alone does not establish shell capacity.

REFERENCE BUILD

For the 20-foot 2V reference dome, the shell should eventually be modeled as a set of separate but related surfaces.

At minimum:

$$S_0 = \text{theoretical geodesic/chord geometry}$$

$$S_1 = \text{interior structural surface}$$

$$S_2 = \text{exterior structural/panel-support surface}$$

$$S_3 = \text{exterior panel surface}$$

$$S_4 = \text{finished weather surface}$$

An interior finish surface can be added separately if required.

The exact offsets between those surfaces will depend on the final wedge size and enclosure design. Once those layers are defined, we can generate different dimensions for different jobs. The strut cut list comes from the structural geometry.

The exterior panel cut list comes from the panel-support geometry. The floor interface comes from the physical base geometry. The window and door depth comes from the complete shell section.

The finished outside diameter comes from the weather surface. Those are all parts of the same dome. They are not the same measurement.

That is the main way the wedge influences the shell. It takes the dome from a network of mathematical lines and gives that network thickness. Once the frame has thickness, every layer attached to it needs its own defined geometry.

Figure 13. Stop squaring. Start splitting.

The same log, opened a different way. Now the other approach. A round tree divides naturally. From Eight Cuts to a House: A Dome Straight From the Tree, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

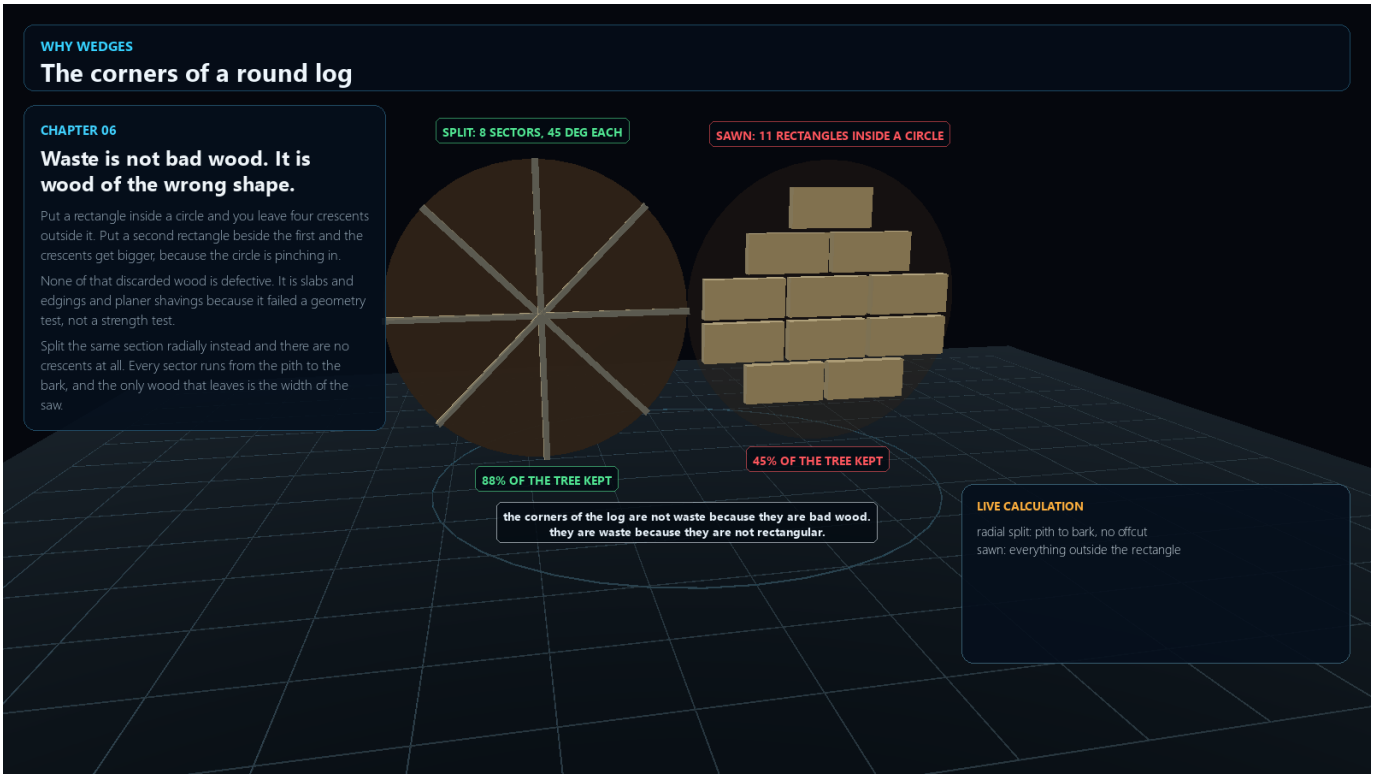

Figure 14. The corners of a round log

Waste is not bad wood. It is wood of the wrong shape. Put a rectangle inside a circle and you leave four crescents outside it. Put a second rectangle beside the first and the crescents get bigger, because the circle is pinching in. From Why Wedges: A Dome That Does Not Need a Sawmill, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

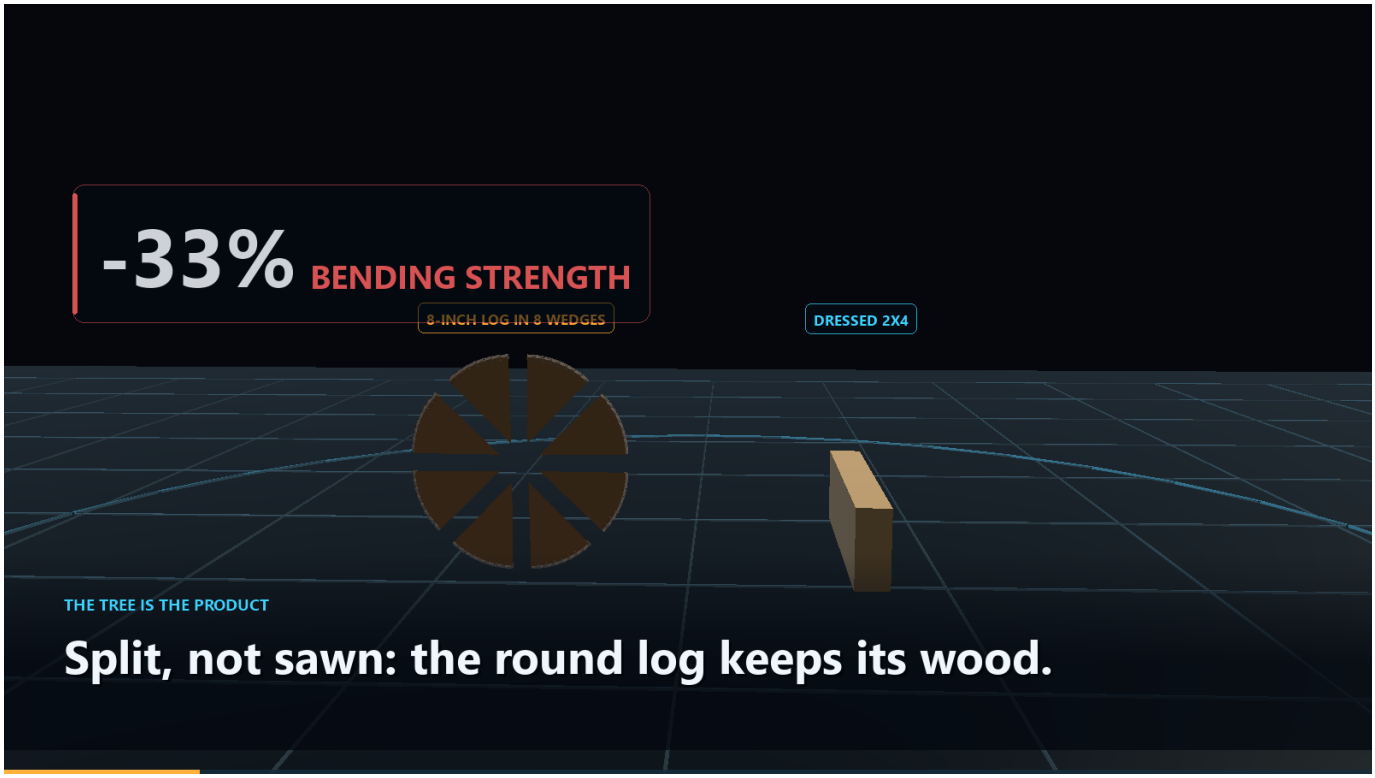

Figure 15. The tree is the product

Split, not sawn: the round log keeps its wood. A sawmill forces a round tree into rectangles. The wedge does not. An 8 inch log splits into 8 sectors, and each one keeps 6.28 square inches of wood, against 5.25 in a dressed two by four. That does not make every wedge stronger than every board; from a log that small, a single wedge is the weaker stick in bending. What it means is that very little happens between the tree and the frame: 8 operations, where a graded board takes 14. From Why Build This Way, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

3. Understanding Dome Geometry

The wedge method is a way of building a dome, but a dome is a shape first and a method second. This chapter is the shape.

None of it is specific to wedges. Everything here is true of any 2V dome built by anybody out of anything, and if you have built one before you can skim it. What you cannot skip is the last section but one, on dihedral angles, because the whole method rests on a number that appears there and does not appear anywhere else.

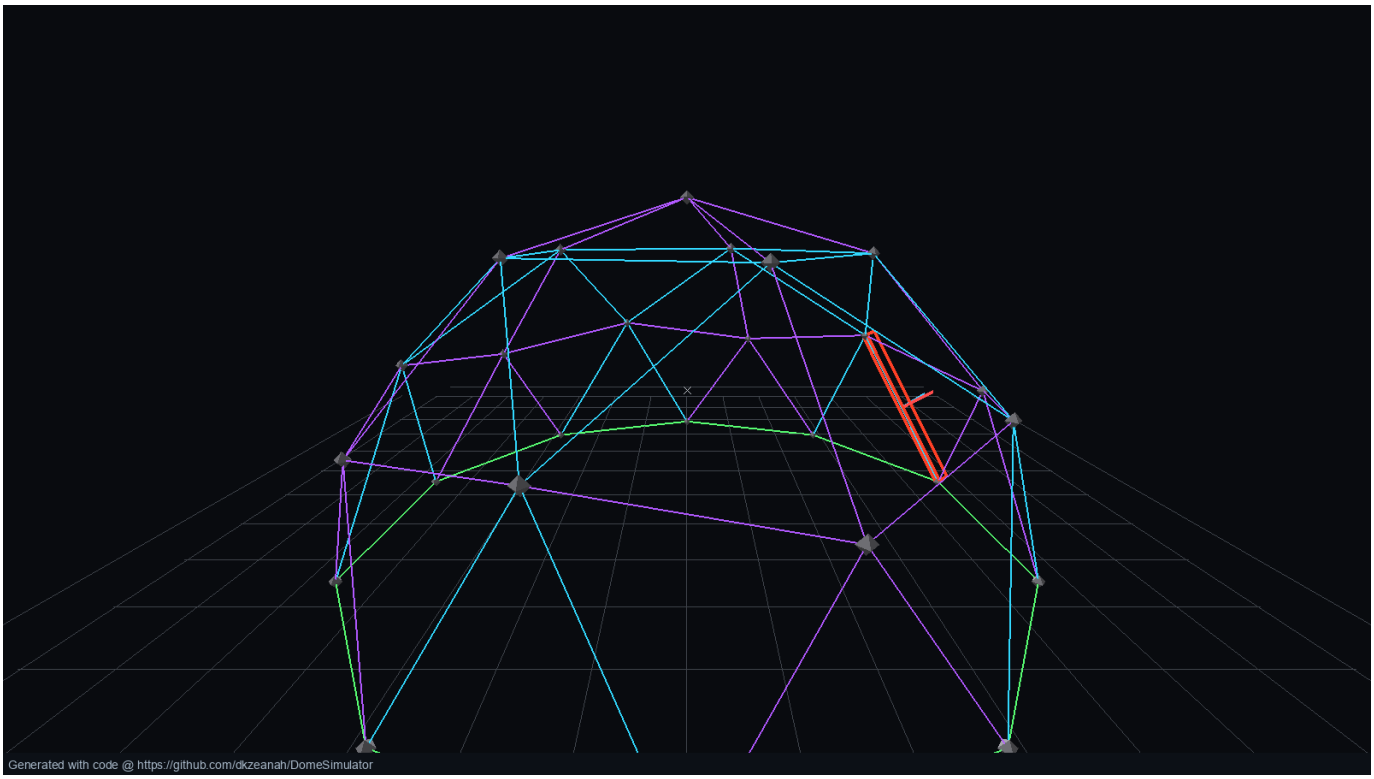

Figure 16. The 2V solution, alone

The mathematical dome with no wood in it: 26 vertices, 65 edges, 40 triangles. The long A edges and the short B edges are coloured apart, and the ten base edges again. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Radius and diameter

Everything in a geodesic dome is measured from one point that is not there.

The centre of the sphere. On the reference build it sits on the floor, in the middle of the room, and nothing is bolted to it. But every vertex in the frame is exactly the same distance from it, and that distance is the only number the whole design really has.

On the reference build that number is 116.4984 inches. Call it a radius of nine feet eight and a half.

The dome is a hemisphere, so the diameter across the floor is twice that: 19.416 feet. Nineteen feet five. And because it is a hemisphere and not a taller or shorter slice, the height is the same as the radius: 9.708 feet.

Those three numbers are not three decisions. They are one decision, written three ways.

WHERE THE RADIUS ACTUALLY COMES FROM

Here is the part that surprises people, and it is worth getting straight before anything else.

You do not choose the radius.

You choose the longest stick you are willing to cut, and the radius falls out of it.

On the reference build the longest member is six feet. Seventy-two inches. That is not a geometric choice at all — it is a decision about what fits on a truck, what one person can carry, and what comes out of the log you have. From that, the geometry gives you a radius of 116.4984 inches, and from the radius you get the diameter, the height, the floor area, and every other stick in the building.

Six feet in, nineteen feet five out.

If you want a bigger dome you do not scale the radius. You pick a longer stick and let the radius follow. That is the direction the arithmetic runs, and people who try to run it the other way end up with a beautiful radius and a member length nobody can cut.

WHAT THE RADIUS DOES NOT TELL YOU

The radius is the distance to the vertices. It is not the distance to the skin, and it is not the distance to the inside face of the wall.

A member on this dome is six inches deep. It sits with its outer face near the sphere and its inner face six inches closer to the middle. So the room inside is smaller than the sphere by roughly the depth of the wall, everywhere.

That sounds obvious written down. It stops being obvious at two in the afternoon when you are trying to work out whether a piece of furniture fits, and you measure from a drawing of the sphere rather than from the inside of the wood.

Measure the room. Not the maths.

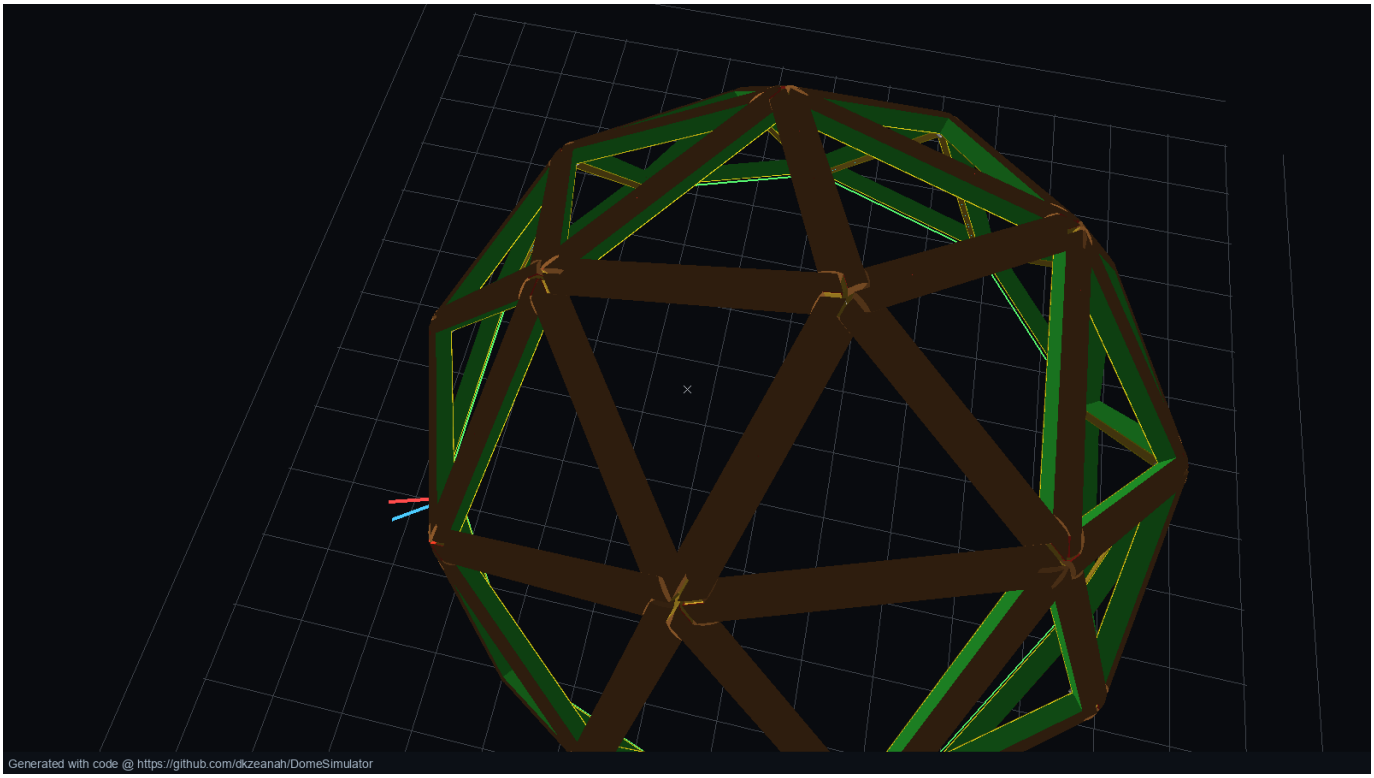

Figure 17. The wireframe over the frame

The same mathematical edges drawn over the members that approximate them. No member lies on its edge: each one is offset by the sector's point, and neither of its ends reaches a vertex. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Sphere versus dome

A geodesic sphere is a closed ball of triangles. A geodesic dome is the part of one you kept.

That is the whole distinction, and it matters more than it sounds.

The reference build is a hemisphere: exactly half a sphere, cut on the equator. It sits on the ground on ten flat feet and the cut is level all the way round.

WHY A HEMISPHERE AND NOT SOMETHING ELSE

You can cut a 2V sphere at the equator and you get a clean edge. Ten base vertices, all at the same height, all level. Nothing has to be shimmed and nothing is left hanging.

Cut it anywhere else and you do not get that. A three-eighths dome or a five-eighths dome means cutting through triangles rather than between them, and every one of those cuts is a new angle, a new length, and a new opportunity to be wrong.

The hemisphere is the cut the geometry offers you for free. The reference build takes it.

WHAT YOU GIVE UP

Honesty first: a hemisphere is not the most useful shape for a house.

The wall at the bottom is already leaning in. Stand at the edge of a nineteen-foot dome and the ceiling is at your shoulder. The floor area says 277 square feet, but the floor area you can stand upright on is

considerably less than that, and a wardrobe against the wall does not fit against the wall.

A taller cut — five-eighths of a sphere rather than a half — gives you a straighter wall and more usable room. It also gives you a base ring that is no longer level, a set of triangles that are no longer the two families you know, and a lot of new arithmetic.

The reference build chooses the easy cut and pays for it in headroom near the edge. If you want the other trade, the geometry will do it. You will just be solving a different dome.

THE FLOOR IS NOT A CIRCLE

One more thing that catches people.

The base of a 2V hemisphere is not round. It is a ten-sided polygon, because there are ten base vertices and straight members between them.

The circle that would pass through those ten corners encloses 296 square feet. The decagon actually enclosed by the members encloses 277.

Nineteen square feet of difference. Not enough to matter for most things, and exactly enough to matter if you are buying flooring or pouring a slab and you measured the circle.

Frequency

Frequency is how many times you cut each edge of the original shape before you push the points out onto the sphere.

That is it. That is the whole idea.

Start with an icosahedron — twenty triangles, the most round thing you can make out of identical flat faces. Leave it alone and you have a 1V dome: twenty big triangles, very angular, very cheap, and not very round.

Cut every edge in half and you get a 2V. Each original triangle becomes four smaller ones. Push all the new points out until they sit on the sphere, and the shape gets rounder.

Cut every edge in three and you get a 3V. Each triangle becomes nine. Rounder again.

Higher frequency means more triangles, smaller triangles, more sticks, more joints, and a shape closer to a true sphere.

WHAT FREQUENCY COSTS

Every step up in frequency is a step up in parts count, and the parts count goes up faster than the roundness does.

A 2V hemisphere has 65 edges. A 3V has substantially more, and every one of them is a stick to cut, a joint to make, and a place to be out by a sixteenth.

The reference build is a 2V because 2V is where the parts count stops being absurd and the shape is still round enough to behave like a dome. Forty triangles. Sixty-five edges. Twenty-six vertices. Those are numbers a person can hold.

If you are building something forty feet across, 2V will give you triangles too big to panel sensibly and you will want 3V. If you are building a greenhouse over a vegetable bed, 1V is probably enough and you should not let anyone talk you into more.

THE PART NOBODY MENTIONS

Frequency does not change how hard the joints are. It changes how many of them there are.

A 3V dome is not a harder dome to build than a 2V. It is the same difficulty, more times. If your 2V joints are coming out clean, your 3V joints will too. If they are not, going up a frequency will not fix them — it will just give you more of them to be unhappy about.

Get the 2V right first.

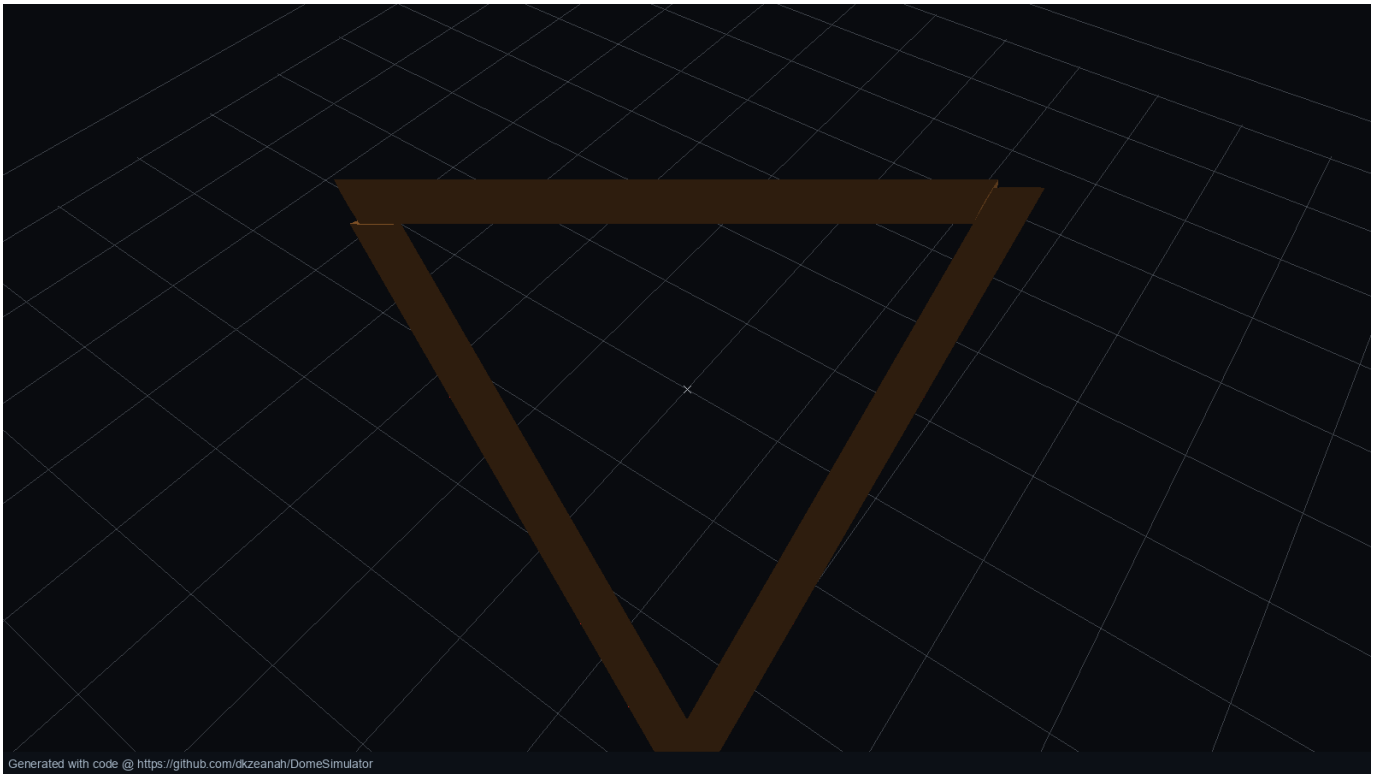

Figure 18. The AAA triangle

One of the ten equilateral panels, alone. Three A members, 72.000 inches to a side as a chord, and an altitude of 62.354 inches. Generated with code @ https://github.com/dkzeanah/DomeSimulator

1V, 2V, 3V, and higher frequencies

Here is what each one actually gives you, in the terms that matter on a build site rather than in a geometry textbook.

1V

Twenty triangles in a full sphere, ten in a hemisphere. All of them identical.

This is the only frequency where every single strut is the same length, and that is a genuinely large advantage. One cut list. One jig setting. No A and B members to keep separated.

It is also barely a dome. The faces are large and flat, the shape is noticeably faceted, and the span you can cover before the members get absurdly long is small. Fine for a play structure, a chicken coop, a garden cover.

2V

Forty triangles in the hemisphere the reference build uses. Two strut lengths, called A and B.

This is the first frequency that looks like a dome and behaves like one, and it is the last frequency where the parts count is small enough to keep in your head. Sixty-five edges, twenty-six vertices, 120 physical members once every triangle owns its own three sticks.

The two lengths are the price of admission. You will cut sixty of one and sixty of the other and you will mix them up at least once.

3V AND UP

Three strut lengths at 3V, more above that. Rounder, stronger for its weight, better at spanning distance.

Also: substantially more sticks, more joints, more sorting, and a much higher cost to any systematic error. If your jig is out by a degree at 2V you will notice at the last panel. At 4V you will notice at the last twelve panels, and you will have already cut them.

WHAT THIS BOOK ASSUMES

Everything from here on assumes 2V unless it says otherwise.

Not because 2V is the best frequency. Because it is the frequency where the wedge method can be explained completely, built by one person, and checked without a computer — and because everything that is true about a wedge at 2V is still true at 3V, with more of it.

Learn the method at 2V. Go up a frequency when the span demands it, not because higher numbers sound better.

Chord lengths

A chord is the straight line between two vertices.

It is not the curve of the sphere. The sphere curves; your timber does not. Every member in the frame is a straight chord cutting across the inside of a curved surface, and the dome is round only in the sense that enough straight lines arranged carefully look round from a distance.

THE TWO CHORDS ON THIS DOME

A 2V dome has two of them. The book calls them A and B, and so does everyone else.

On the reference build:

- A is 72.000 inches — six feet, the longest member
- B is 63.670 inches — five feet three and eleven-sixteenths

There are sixty of each in the frame.

CHORD FACTORS

The lengths above are for one dome at one size. The numbers that travel are the chord factors: the chord divided by the sphere radius.

- A factor: 0.618034
- B factor: 0.546533

Multiply either by your radius and you have your member length. That is the whole calculation, and it is the same calculation for every 2V dome that has ever been built.

The A factor is worth a second look. 0.618034 is the golden ratio minus one, and it is not a coincidence — it falls out of the icosahedron the dome is subdivided from. You do not need to care why. You do need to type it correctly.

GOING THE OTHER WAY

Most people do not start with a radius. They start with the longest stick they can get.

Divide it by 0.618034 and you have your radius. From there, multiply by 0.546533 for the B chord and you have your second length.

Seventy-two inches divided by 0.618034 is 116.4984 inches. Times 0.546533 is 63.670 inches. That is the reference build, derived from one decision about timber.

WHAT THE CHORD IS NOT

The chord is the distance between vertex points — the mathematical corners, floating in space.

It is not the length of wood you cut.

Your actual sticks come out shorter than the chord, because the ends get trimmed back where they meet other members, and how much shorter depends on how the joint is made. On the reference build the A members come off the saw between 68.977 and 70.182 inches of usable stock against a 72.000 inch chord.

That gap between the chord and the stick is where most first-dome errors live. It has its own chapter.

Figure 19. The BAB triangle

One of the thirty isosceles panels: one A edge and two B, with an altitude of 52.516 inches above the long edge -- which is the number that decides whether a sheet of plywood fits. Generated with code @ https://github.com/dkzeanah/DomeSimulator

A/B/C strut families

A strut family is a set of members that are all the same length and all do the same job. Domes are described by how many families they have.

A 2V dome has two: A and B. A 3V has three, usually called A, B and C. A 1V has one.

WHAT A AND B ACTUALLY ARE ON THIS DOME

A members are the longer ones. Sixty of them, 72.000 inches of chord.

B members are the shorter ones. Sixty of them, 63.670 inches of chord.

They are not scattered at random. They sit in a pattern, and the pattern is worth learning because it is how you check your work without counting everything.

THE PATTERN

Look at the top of the dome. There is a ring of five triangles around the apex, and every edge of every one of them is an A. Those are the AAA triangles. There are ten of them in the hemisphere.

Everything else is a BAB triangle: one A and two Bs. There are thirty of those.

Ten plus thirty is forty triangles, which is the whole dome.

So: if you are looking at a triangle and all three sticks are the same length, you are near the top. If two are short and one is long, you are not. That is a check you can do from the ground with your eyes.

WHY THE FAMILIES MATTER MORE IN THIS METHOD

In a normal strut-and-hub dome the families are just two lengths. You cut sixty of each, you keep them in two piles, and the difference between them is eight inches.

In the wedge method they are more than that, because the two families sit at different angles to each other and their seams fold by different amounts.

On the reference build:

- an A seam folds 18.029 degrees
- a B seam folds 22.459 degrees

Same dome, same wedge, four and a half degrees of difference depending on which family you are looking at. That difference is not an error and it does not average out. It is the geometry telling you that an A seam and a B seam are different joints.

Mix up an A and a B and you will not just have a stick of the wrong length. You will have a stick that wants to sit at the wrong angle.

Keep the piles apart. Mark them before they leave the saw.

Triangle types

There are two kinds of triangle in a 2V hemisphere and you can tell them apart across the room.

AAA TRIANGLES

Three A members. All three sides 72.000 inches. Equilateral — genuinely equilateral, not approximately.

There are ten of them, and they live at the top of the dome, in and around the ring at the apex.

These are the easy ones. Every side is the same, every corner is sixty degrees, and if one of them comes out wrong you will see it immediately because it will not be symmetrical.

BAB TRIANGLES

One A and two Bs. Two sides of 63.670 inches and one of 72.000.

There are thirty of them, and they make up everything below the top: the middle band, the base ring, and every triangle that touches the ground.

These are isosceles, not equilateral. They have a long edge and two short ones, and the long edge always points the same way relative to the dome — which is the thing to check when you are assembling, because a BAB fitted upside down looks nearly right and is not.

WHAT THIS MEANS FOR PANELS

If you are building panel by panel — and in the wedge method you are — you are building exactly two kinds of panel.

Ten of one. Thirty of the other.

That is a much better position than it sounds. It means one jig setting for the AAA panels and one for the BAB panels, and it means the fortieth panel is the same job as the first.

It also means your total perimeter of panel edge is two numbers rather than forty. On the reference build an AAA triangle has 216.00 inches of perimeter and a BAB has 199.34 inches. Ten of the first and thirty of the second is every edge in the building, and you will want that number when you buy gasket, tape, or anything else sold by the foot.

THE COUNT, IN ONE PLACE

- 40 triangles
- 10 AAA, 30 BAB
- 26 vertices
- 65 geodesic edges
- 55 of those edges are interior seams, where two triangles meet
- 10 are the base ring, where nothing meets them

Those numbers do not change with size. A nineteen-foot 2V hemisphere and a forty-foot one have exactly the same counts. Only the lengths move.

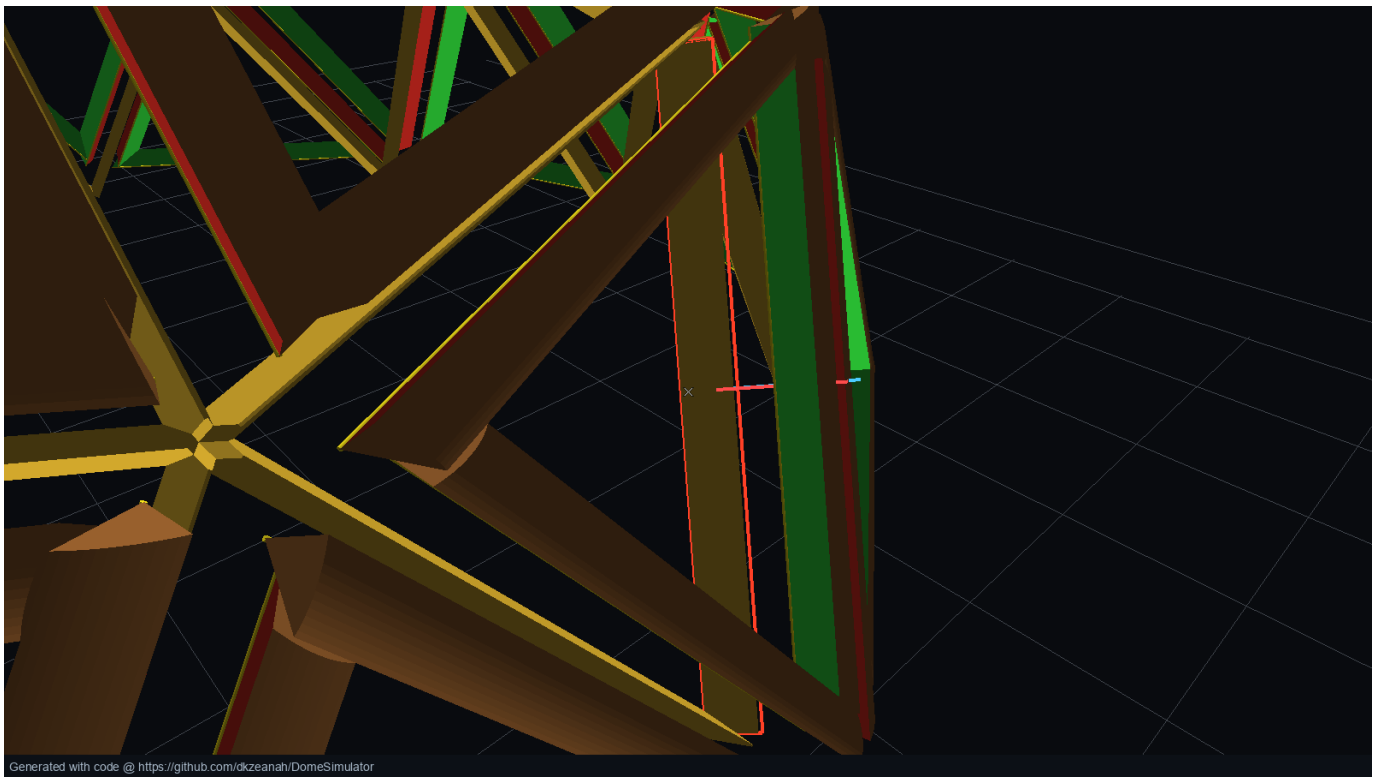

Figure 20. The fold between two panels

Two triangles meeting at a seam, opened so the fold is readable. An A seam folds 18.029 degrees and a B seam 22.459, and the two of them sum to the 45 degrees the log was split at. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Triangle heights

The height of a triangle — the perpendicular distance from one side to the opposite corner — is the number that decides whether your panel fits on a sheet of anything.

It is also the number nobody calculates until they are standing in the plywood aisle.

THE NUMBERS

On the reference build:

an AAA triangle is 72.000 inches on every side, and its height is 62.354 inches a BAB triangle has a 72.000 inch long edge and two 63.670 inch sides, and its height above the long edge is 52.516 inches

Both of those are the minimum width the triangle can be squeezed into. You cannot rotate a triangle to make it narrower than its shortest altitude.

WHY THAT NUMBER RUINED A PLAN

A sheet of plywood is 48 inches wide.

62.354 is not 48. 52.516 is not 48 either.

No panel of this dome fits on a four-foot sheet. Not in any rotation, not with any clever nesting, not if you turn it diagonally. The narrowest panel in the building is still four and a half inches wider than the widest sheet you can buy at a normal yard.

That is not a small inconvenience. It decides how the whole skin gets built. You either buy oversized sheet goods, or you sheet across the frame and cut on site, or you build the panels up from narrower boards.

The reference build sheets across the frame and tapes the joints, because that is the option that does not require a special order.

CHECK THIS BEFORE YOU DESIGN ANYTHING

Take your longest member. Multiply by 0.866 for an equilateral panel — that is your AAA height. Compare it to whatever you were planning to skin the dome with.

If the answer is bigger than your sheet, you have just found out something important, and you have found it out for free rather than in a car park with a sheet of ply that does not fit in the car.

Dihedral relationships

This is the section the rest of the book leans on, so it is worth reading slowly.

A dihedral angle is the angle between two flat faces that share an edge. Open a book halfway and the angle between the two covers is a dihedral angle.

On a dome, every interior seam is a dihedral. Two triangles meet along a line and they are not coplanar — they fold, because that folding is what makes the flat triangles approximate a curved sphere.

HOW MUCH THEY FOLD

On the reference build:

- an A seam folds 18.029 degrees
- a B seam folds 22.459 degrees

That is the angle the two panels turn through, relative to each other, at their shared edge.

Fifty-five interior seams in the hemisphere: twenty-five A and thirty B. The other ten edges are the base ring and they do not fold, because there is nothing on the other side of them.

WHY A WEDGE METHOD CARES MORE THAN A STRUT METHOD

In a strut-and-hub dome the dihedral is handled at the hub. The hub is a casting or a plate with the angles built into it, and the struts just arrive.

In the wedge method there is no hub. The dihedral has to be handled by the timber, and it is handled by the fact that a wedge is wider on one face than the other.

That is the entire mechanism. A wedge, seen end-on, is a sector of a log — a triangle in section. Put two of them side by side along a seam and the difference in width between their inner and outer faces makes the two panels turn relative to each other. The wood does the folding.

THE NUMBER THAT MAKES THIS WORK

Split a log into eight, and each sector is 45 degrees.

Now look at what the dome asks for:

- A seam: folds 18.029 degrees, leaves 26.971 degrees
- B seam: folds 22.459 degrees, leaves 22.541 degrees

Add each pair. Both come to exactly 45.

That is not a coincidence and it is not tuned. The raw 45-degree sector is being split between the fold the dome needs and the gap left over, and the two always add to the sector angle because that is what a sector is.

WHAT THE LEFTOVER IS

The leftover — 26.971 degrees on an A seam, 22.541 on a B — is a gap. Two sawn faces meeting at a dihedral angle do not close flush. There is a wedge-shaped void along every one of those fifty-five seams.

Every other way of building a dome treats that as a defect and machines it out.

This book does not. That gap is the key, it has a chapter of its own, and it turns out to be the most useful thing in the building.

Figure 21. Frequency, and where the dome gets cut

Even frequencies stop at the equator. Odd ones do not. Raising the frequency buys a smoother shell and charges you in. From Every Dome In The World, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Scaling a design

Scaling a 2V dome is one multiplication, and then a list of things that do not scale with it.

WHAT SCALES

Pick your new longest member. Divide by 0.618034 to get the radius. Multiply by 0.546533 to get the B member. Everything geometric follows:

- diameter is twice the radius
- height is the radius, for a hemisphere
- every chord is its factor times the radius

Double the member length and you double the radius, the diameter and the height.

WHAT SCALES FASTER THAN YOU EXPECT

Area goes as the square. Double the dome and the floor goes up four times, and so does the skin you have to buy.

The reference build is 19.416 feet across with 277 square feet of decagonal floor and 592 square feet of spherical envelope. Double it to a member of twelve feet and the floor is over 1,100 square feet. That is not a bigger dome; that is a different building with a different budget.

WHAT DOES NOT SCALE AT ALL

The counts. Forty triangles, sixty-five edges, twenty-six vertices, fifty-five interior seams, 120 members. A nine-foot 2V and a forty-foot 2V have exactly the same parts list, with different numbers written on it.

The angles. Every dihedral is the same at every size — 18.029 degrees on an A seam whether the dome is a playhouse or a hall. Angles are a property of the shape, not the size, and that is why one jig design works across the whole range.

WHAT SCALES THE WRONG WAY

Your timber.

The chords scale linearly, but the stock you need to cut them from does not behave as politely. A six-foot member comes out of an eight-foot log with room. A twelve-foot member does not come out of a sixteen-foot log the same way, because the taper of a real tree means the wedge cross-section changes along its length, and the far end of a long member is coming from a thinner part of the trunk.

There is a size above which you stop being able to get your members out of one tree, and it arrives sooner than the arithmetic suggests.

THE ADVICE

Scale up by going up in frequency, not by stretching the members.

If you want more span, a 3V dome with the same six-foot sticks covers more ground than a 2V with nine-foot ones, and every stick still comes off the same saw, out of the same size log, with the same jig. You get a bigger building without changing anything about how you make the parts.

Stretching the member is the obvious way to get a bigger dome. Going up a frequency is the one that keeps the method intact.

Part II - Making the Structure

4. Materials

The method works with bought lumber, with sawn timber from a local mill, and with logs off your own land. It was designed for the third.

The reason is one number. Split a trunk into wedges and you keep about 88 percent of it; mill the same trunk into 2x4s and you keep about 45. Squaring a round log means throwing the round part away, and this method does not square anything.

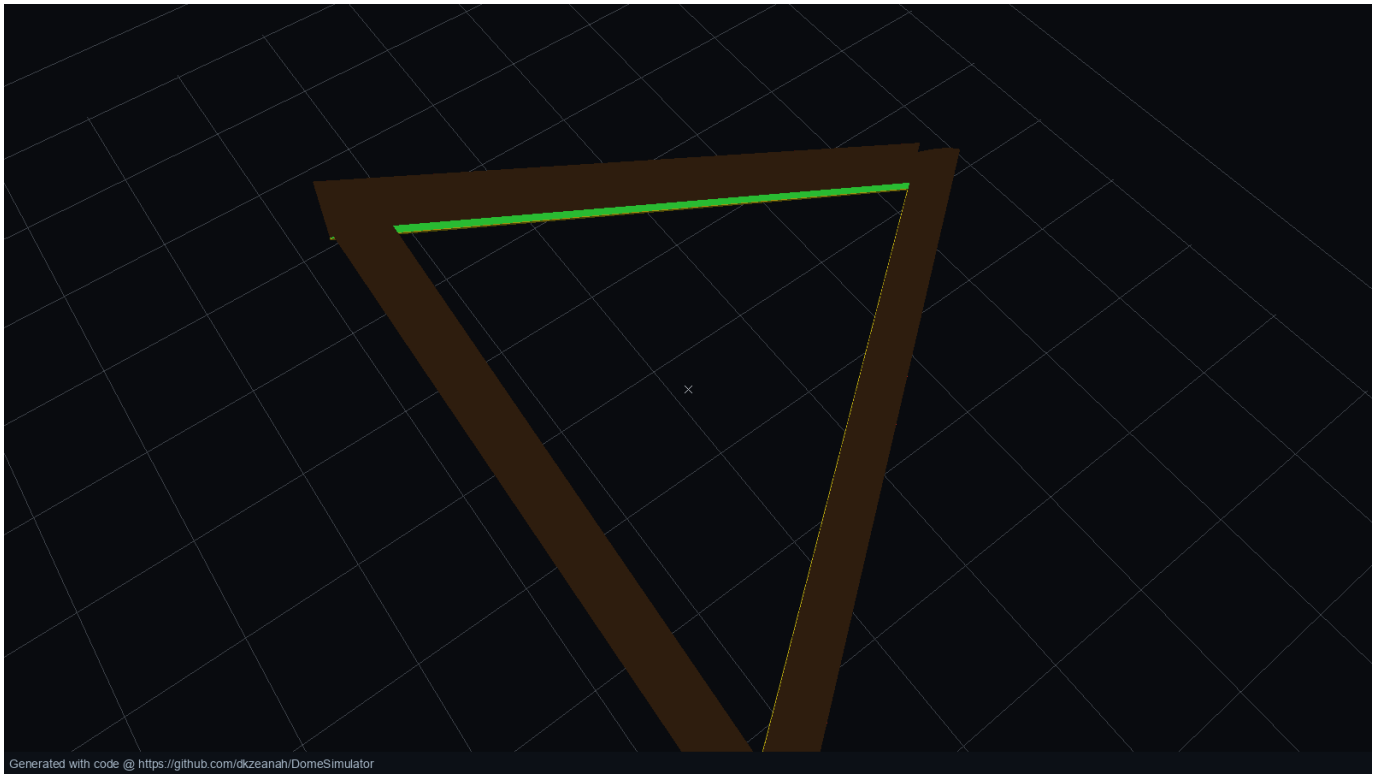

Figure 22. One member, out of the frame

A single stick with the rest of the dome hidden: a log sector with a compound butt at one end and a flush-cut head at the other. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Dimensional lumber

You can build this dome out of bought lumber. It works, it is faster, and it costs more.

WHAT IT MEANS FOR THE METHOD

A 2x6 is not a wedge. It is the same width on both faces, so it does not do the thing the method depends on — the changing width that lets neighbouring triangles turn relative to each other.

So if you build from dimensional lumber you have two choices.

Rip the wedge yourself. Take a 2x6 and rip it at an angle so one face is narrower than the other. You now have a wedge with the profile the geometry wants, and you have thrown away the offcut.

Or give up the wedge and handle the dihedral at the joint instead, with shims, bevels or plates. That is no longer the wedge method; it is a conventional dome built out of conventional lumber, and there are better books about that.

THE COST

At the time of writing, a 2x6x12 at a big-box store is about fourteen dollars, and it rips and crosscuts into four 2x3x6 struts. That is three dollars fifty a stick.

A shared-strut dome needs 65 of them. This method needs 120.

So the frame is about four hundred and twenty dollars in wood if you buy it, against roughly two hundred and thirty for the same dome built the conventional way.

That is the honest number and it is not enormous either way. Wood is not what makes a dome expensive.

WHEN BOUGHT LUMBER IS THE RIGHT ANSWER

When you do not have trees. When you are in a hurry. When the dome is small and the difference is forty dollars. When you want to build one to learn the method before you commit a woodlot to it.

There is no purity requirement here. The method works with a bought wedge; it just does not get the thing that makes it interesting.

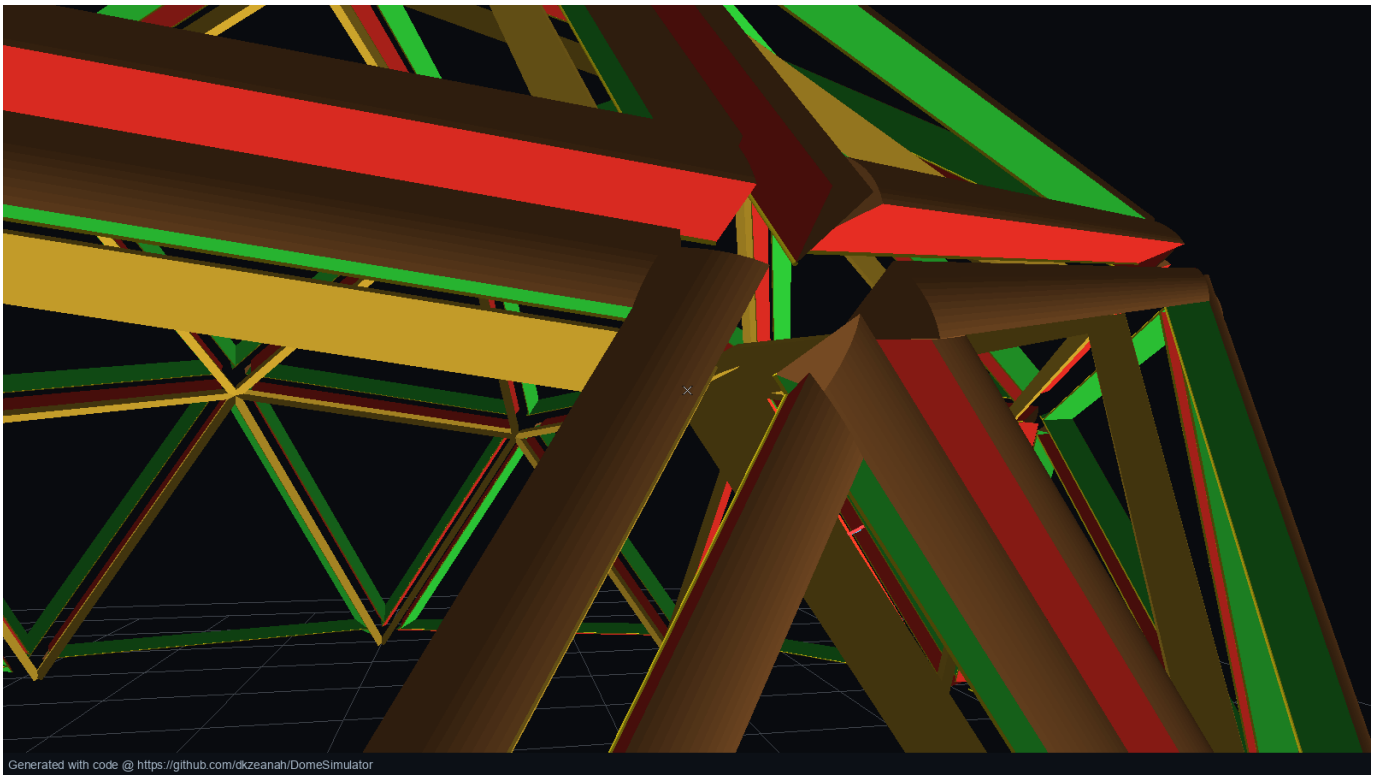

Figure 23. An eight-inch trunk

An eight-inch trunk: a slender frame. The A seam then wants a key 1.87 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Sawn timber

Sawn timber — squared stock from a local mill rather than a big-box store — sits between dimensional lumber and logs, and for a lot of people it is the practical middle.

WHAT YOU GET

Bigger sections than a lumber yard carries. Rough sawn rather than planed, which means the stated size is the actual size, not the dressed size.

Usually green, or air-dried rather than kiln-dried.

Usually local species, which may or may not be what the span tables assume.

And usually cheaper per board foot than dressed lumber, sometimes substantially.

WHY IT SUITS THIS METHOD

A rough-sawn 6x6 or 8x8 rips into wedges very conveniently, and a mill will often sell you the slabwood and offcuts that a dressed-lumber supplier does not have.

The dimensional stability is worse than kiln-dried and that matters less here than it would in a stud wall. The frame is triangulated and the joints are compression joints; a member that moves a sixteenth is not a problem.

WHAT TO ASK THE MILL

Moisture content, or at least when it was cut.

Whether they will cut to a given section rather than a standard one. Most small mills will, and having your wedge profile cut at the mill removes the hardest operation from your shop.

Whether they have anything that has been sitting. Timber that has been in the stack for a year is drier and cheaper than fresh, and a mill usually has some.

THE OPTION MOST PEOPLE MISS

Ask them to rip the wedges.

A mill has a big enough saw to split a log into eight accurately in a fraction of the time it takes you, and they charge by the hour. For the price of a morning of mill time you can skip the entire splitting chapter.

That is not cheating. The method is about what the wedge does in the building, not about who cut it.

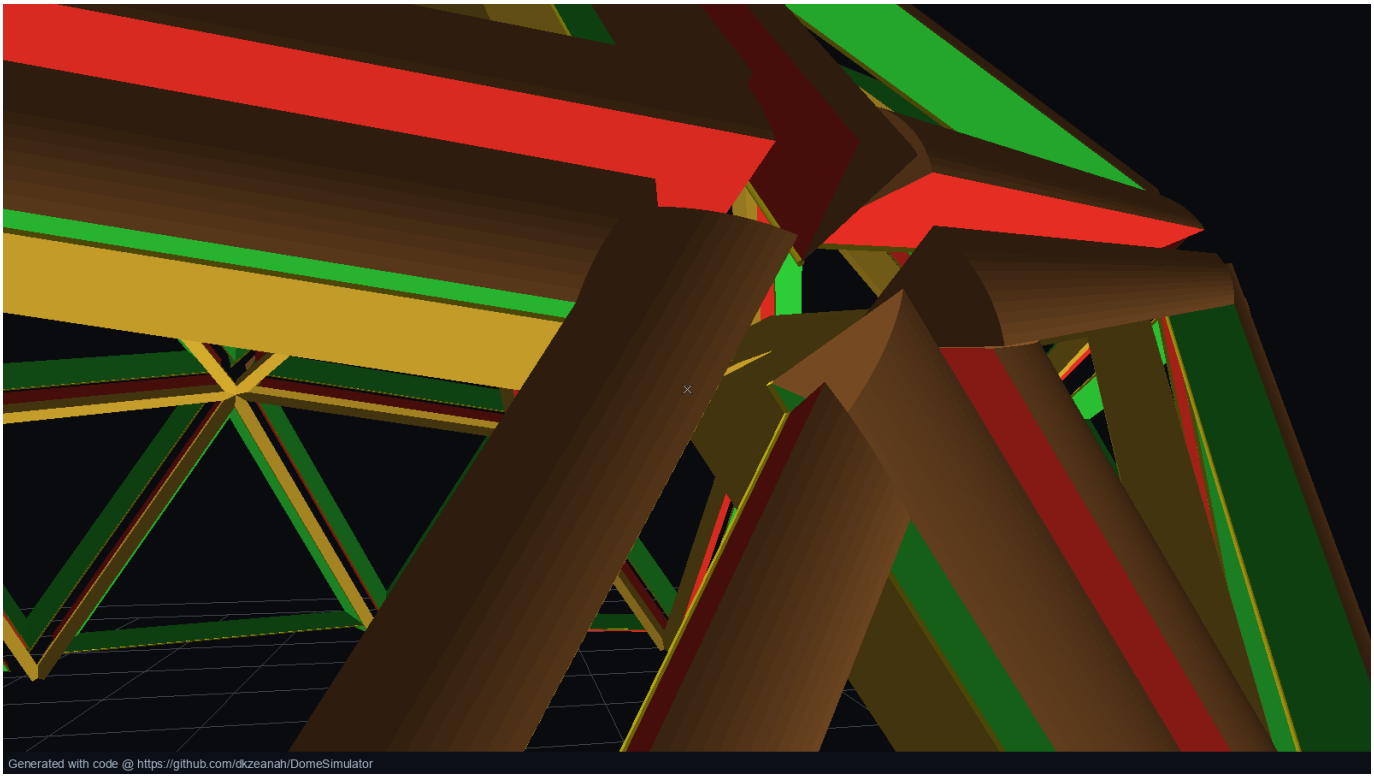

Figure 24. A twelve-inch trunk

A twelve-inch trunk: the reference build. The A seam then wants a key 2.80 in across its base. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Logs

This is what the method is really for.

A log is not a worse version of lumber. It is a different material with a different economics, and the wedge method exists because squaring it is wasteful.

THE NUMBER THAT MATTERS

Split a trunk into wedges and you keep about 88 percent of it — 391 board feet out of a tree that has 442.

Mill the same trunk into 2x4s and you keep about 45 percent. Two hundred board feet.

Nearly twice the building, out of the same tree.

That is the whole argument, and it is not subtle. Squaring a round log means throwing the round part away. The wedge method does not square anything.

WHY IT WORKS OUT THAT WAY

A rectangle inscribed in a circle leaves four corners of waste. Do it once per board and the waste adds up to more than half the log.

A sector of a circle wastes only the saw kerf. Eight sectors out of a round log is eight sectors, minus eight thin cuts.

WHAT A LOG COSTS

If it is standing on your land, it costs your time.

If you are buying, a log is sold by the ton or by the board foot and is substantially cheaper than sawn timber, because you are buying it before anybody has done anything to it.

THE CATCH, AND IT IS A REAL ONE

Green logs are heavy, they are awkward, and they will do things sawn timber does not.

They move as they dry. They can have tension in them that makes a rip cut close on the blade. They have bark, which holds water and insects. They are never quite straight and never quite round.

The chapter on turning timber into wedges is longer than it would be for lumber for exactly these reasons.

WHAT THE REFERENCE BUILD ASSUMES

A trunk about twelve inches across, split into eight, yielding sectors that give a member roughly four and a half inches wide and six inches deep.

Roughly one and a half trees for the whole frame.

Twelve inches is not arbitrary: it is what gives a sector of the right size for a six-foot member on this dome. A smaller trunk gives a thinner wall and a bigger one gives you wood to trim off.

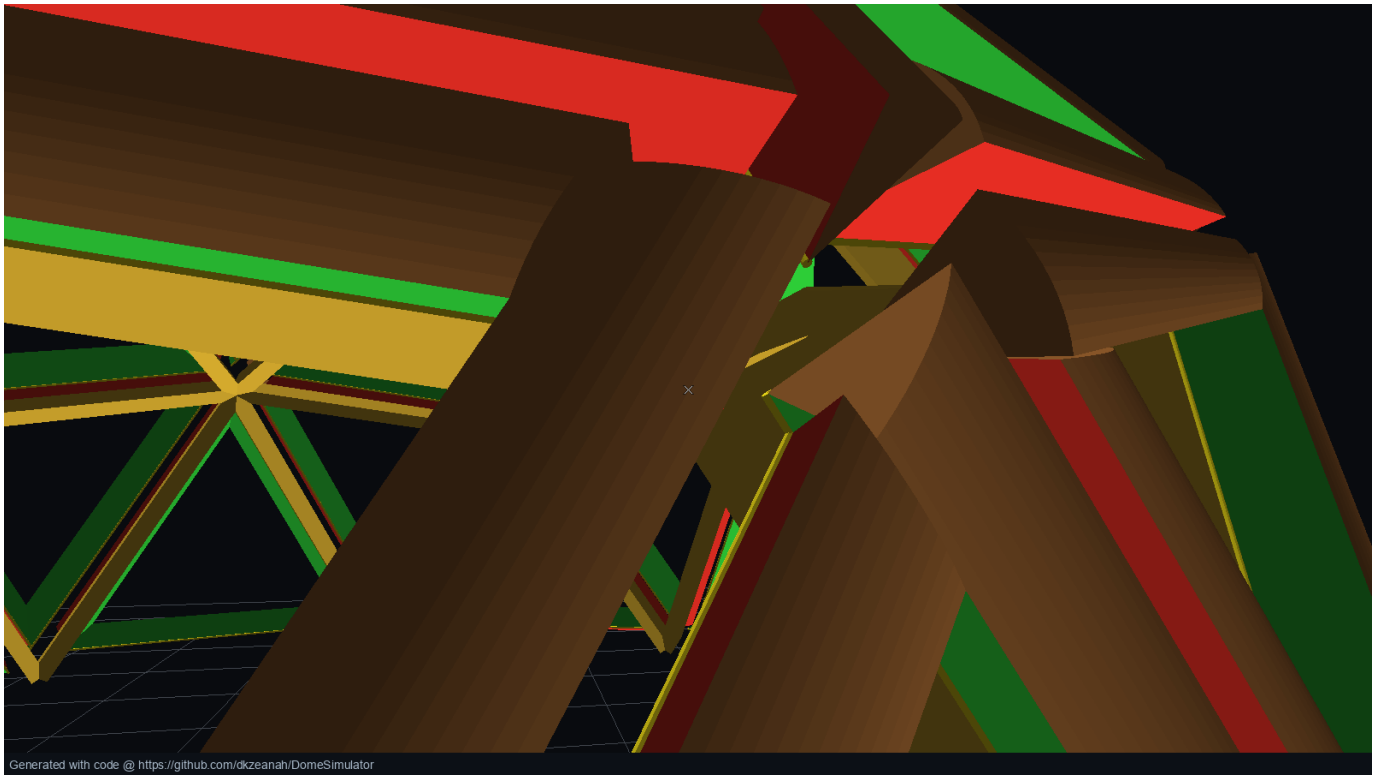

Figure 25. A sixteen-inch trunk

A sixteen-inch trunk: heavy section, fewer trees. The A seam then wants a key 3.73 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Species

The method does not care much which tree. The building does.

WHAT MATTERS, IN ORDER

Splitting behaviour. Some species split cleanly and predictably along the grain; some tear, wander, or spiral. This is the property that decides whether the method is pleasant or miserable, and it is not on any span table.

Rot resistance, if any part of the frame is near the ground or exposed.

Strength, which matters less than people expect, because a triangulated dome does not load its members heavily.

Weight, which matters when you are carrying 120 of them.

WHAT SPLITS WELL

Straight-grained softwoods. Pine, spruce, fir, cedar. These are the species the method was developed around and they split predictably with a froe or on a saw.

WHAT DOES NOT

Anything with interlocked or spiral grain. Elm is famously difficult. Some eucalyptus. Any tree that grew on a windy slope and twisted as it went.

You can tell a lot by looking at the bark: if the bark runs in a spiral up the trunk, the grain does too, and your sectors will come out twisted.

THE ROT QUESTION

Cedar, larch, black locust and osage orange are durable. Pine and spruce are not.

For a frame that is off the ground, dry, and inside an envelope, durability barely matters — the reference build uses pine and expects it to last as long as the roof over it.

For anything in contact with the ground or outside the envelope, use a durable species or do not use wood.

THE ADVICE

Use what is growing where you are.

A local species you can get for free and can split reliably beats an ideal species you have to buy and transport. The method is tolerant, and the frame is not the part of this building that will fail.

Figure 26. Two trees

The whole woodpile is standing up. It will take me 2 whole pine trees to harvest all of the wood a 300 square foot house needs. No lumber yard and no delivery truck, just the trees already standing on the property, and one small chainsaw. From Two Trees, All at Once, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Moisture

Green wood is wet wood, and it will get drier after you build with it.

That is not a reason to avoid it. It is a reason to know what it will do.

WHAT HAPPENS AS IT DRIES

Wood shrinks across the grain and barely at all along it.

For a member, that means it gets narrower and shallower and does not get shorter. Chord lengths stay put. Cross-sections do not.

So a joint that was tight when built gets slightly loose after a season. That is the single most predictable thing about building with green timber and it has one answer: come back and tighten everything after the first summer.

HOW MUCH

Softwood shrinks roughly four to eight percent across the grain going from green to dry, depending on species and direction.

On a member six inches deep that is a quarter to a half inch. Not nothing.

BUILDING GREEN ANYWAY

People have built with green timber for a very long time and the technique is well understood: use joints that can be tightened, expect to tighten them, and do not build anything that depends on a dimension staying exactly put.

The pinwheel joint is well suited to this. It is a compression joint held closed by a fastener, and tightening the fastener takes up the shrinkage.

A joint relying on a precise fit — a mortice, a dovetail, a machined key — is not well suited, and that is worth remembering when you are tempted to get clever at a seam.

WHAT NOT TO DO

Do not seal green timber. Paint, varnish or a tight plastic wrap on wet wood traps the moisture and you get rot instead of drying.

Do not build it into an unventilated cavity while wet.

Do not assume it is dry because the outside feels dry. A six-inch member is wet in the middle for a long time.

THE SCHEDULE THAT WORKS

Split green, because green splits far more easily than dry.

Then sticker the sectors — stacked with spacers, under cover, with air all round — for as long as you can stand. Three months makes a real difference. Six is better.

Then cut to final size, because a member cut dry stays the size you cut it.

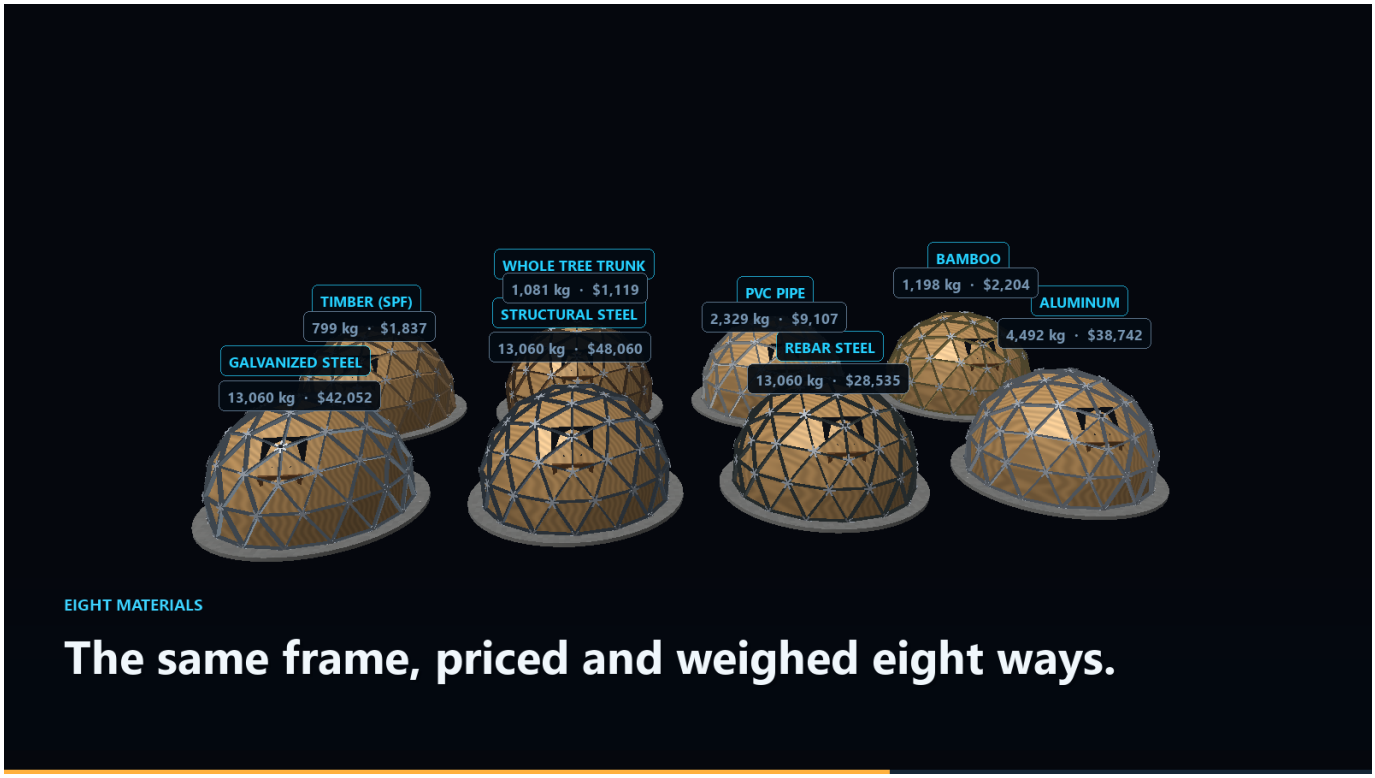

Figure 27. Eight materials

The same frame, priced and weighed eight ways. The material menu carries a density and a price per kilogram, so. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Knots and defects

Every real tree has defects. The question is which ones matter and where.

KNOTS

A knot is where a branch was. The grain flows round it, which makes the wood locally stronger in some directions and much weaker in others.

For this method the important thing about a knot is not strength. It is that knots ruin splits.

A split runs along the grain, and grain goes round a knot. Hit a knot while splitting and the split wanders, forks or stops.

So: look at the log before you split it, mark the knot clusters, and buck the log so the clear lengths become members and the knotty lengths become something else.

WHERE A KNOT IS ACCEPTABLE

In the middle of a member's length, away from the ends, where the member is in compression and nothing is fastened.

WHERE IT IS NOT

Near an end. Every member gets fastened near both ends, and a fastener into or beside a knot splits the wood or gets no grip.

On a sawn face that becomes a seam. The seam faces have to be flat and predictable and a knot there is a lump.

OTHER DEFECTS

Shake — a separation along the growth rings. A member with shake through it can come apart along that plane. Reject it.

Check — surface cracks from drying. Usually cosmetic on a member this size. Fine.

Sweep — a gentle bend along the log. Members are short; a six-foot length out of a long gentle sweep is nearly straight. Usable.

Spiral grain — the whole log twisted. Splits come out twisted and members wind. This is the worst defect for this method and it is worth rejecting the log.

THE RULE

Defects are not a reason to reject wood. They are a reason to decide what each piece becomes.

The clear butt log becomes A members. The knotty top becomes B members and blocking. The bad piece becomes jig stock. Almost nothing is actually waste, which is another part of why the method suits trees you cut yourself.

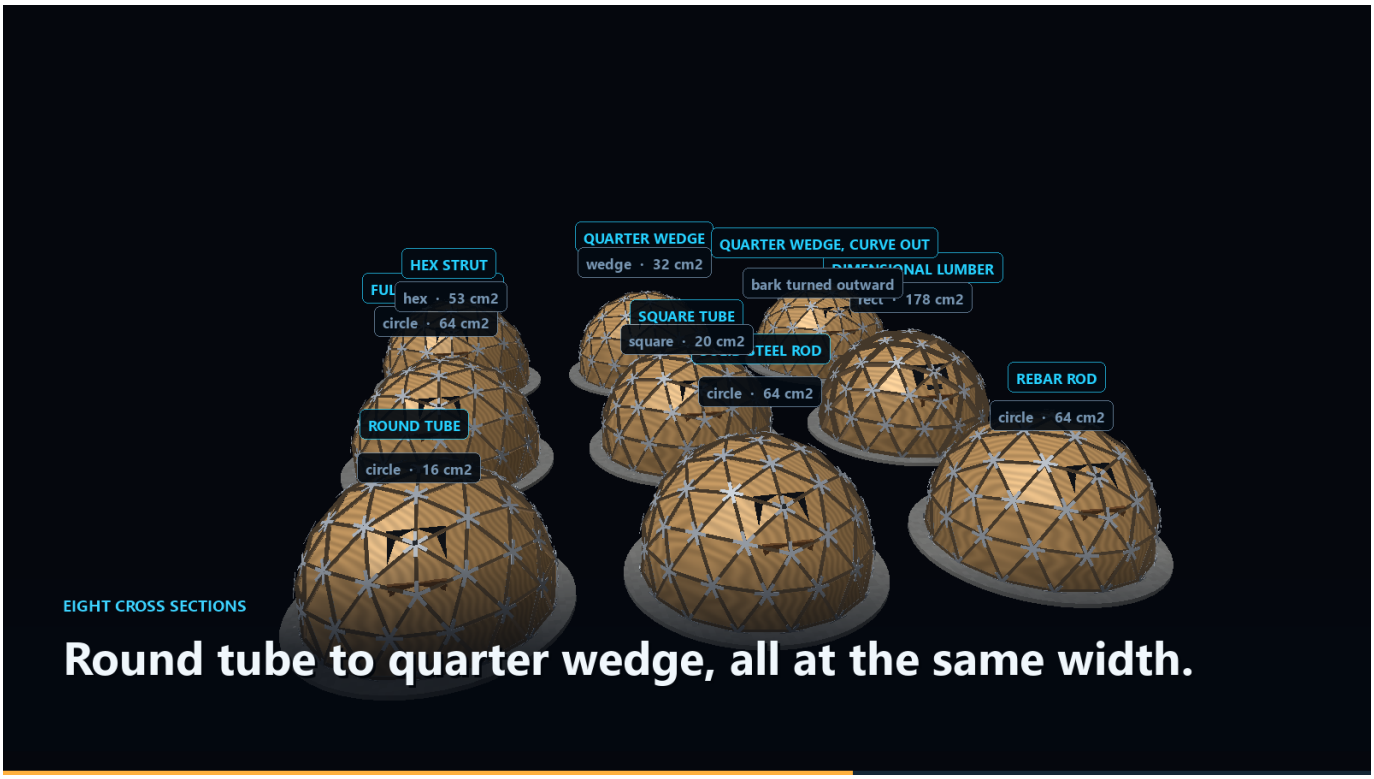

Figure 28. Eight cross sections

Round tube to quarter wedge, all at the same width. The strut shape menu decides what every member is cut from. Round. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Grain direction

Wood is much stronger along the grain than across it, and every decision in this chapter follows from that.

IN A MEMBER

The grain runs along the length. That is automatic when you split, because the split follows the grain — which is one of the quiet advantages of splitting over sawing.

A sawn board can have grain running out the side of it, where the saw cut across a curving grain line. That board is weak and it will break where the grain runs out. A split member cannot have that, because the split went where the grain went.

That is worth saying plainly: a split member is stronger for its size than a sawn one, because splitting cannot cut across the fibres and sawing can.

AT A JOINT

The butting end presents end grain. The receiving face presents side grain.

Compression on end grain is fine — wood is very strong that way. Compression on side grain is weaker but adequate at these loads, and the contact area is large.

A fastener into end grain holds poorly. A fastener into side grain holds well.

So: wherever possible, drive fasteners into the side of a member rather than into its end. The pinwheel joint arranges this naturally, which is another of its quiet advantages.

SPLITTING NEAR AN END

The end of a member is where the grain is free to separate, and it is where every fastener goes.

Pilot every hole. In green timber, near an end, without a pilot, you will split the piece — reliably, not occasionally.

THE ORIENTATION QUESTION

A wedge has a curved bark face, two sawn faces, and a point. Which way it goes in the wall is a real decision and it has its own section in the geometry chapter.

For grain purposes it does not matter much: the fibres run along the member whichever way you rotate it. What changes is which face takes the fastener and which face meets the seam, and those are covered where the geometry is.

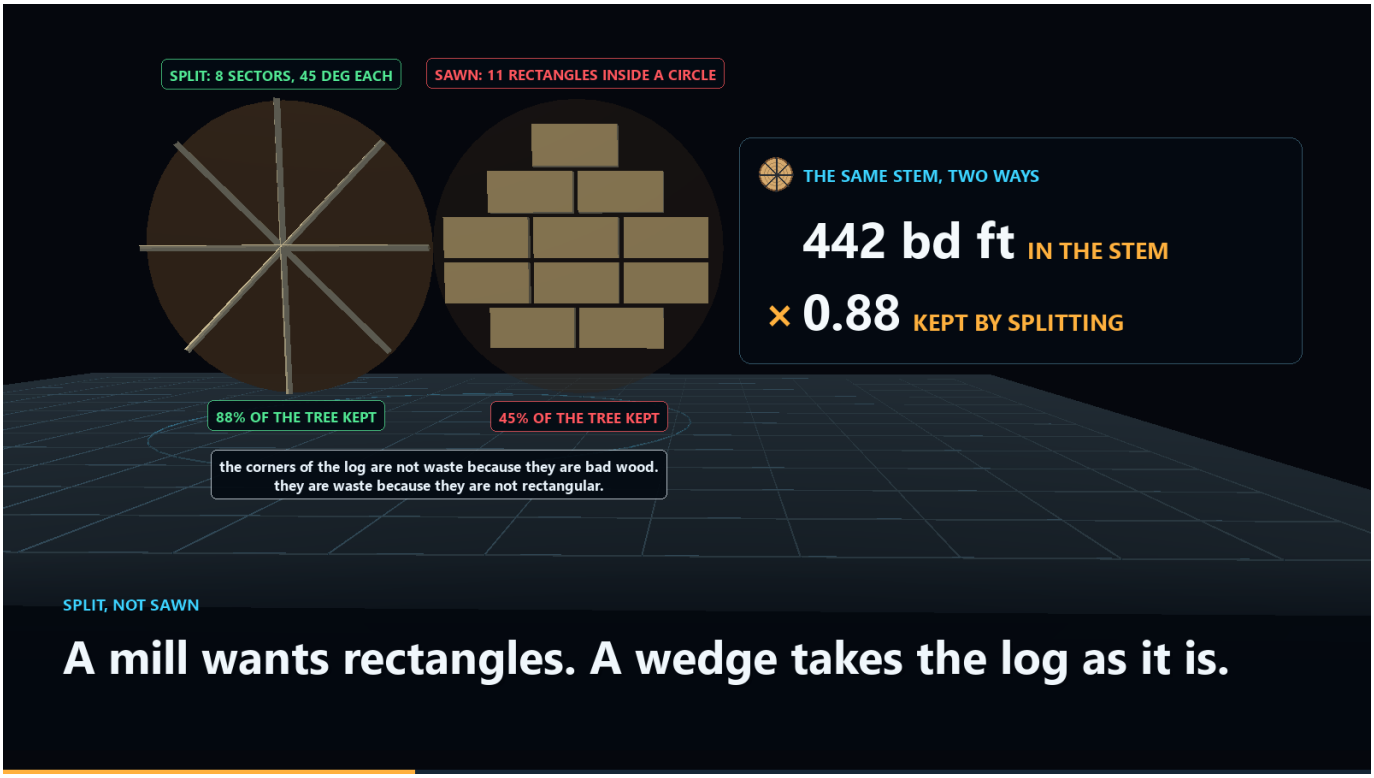

Figure 29. Split, not sawn

A mill wants rectangles. A wedge takes the log as it is. Now shape. A sawmill has to turn a round log into rectangles, and everything outside them leaves as slabs, edgings and sawdust. The picture is the simulator's, packing this log with two by fours, and it keeps only 45 percent. A good mill cutting mixed sizes does better, so I give it 60 percent. Splitting follows the round log instead of fighting it. From the same stem it keeps 88 percent, 389 board feet, where the mill keeps 265. That is 124 more board feet of usable wood from one tree, because the corners of a log are not bad wood. They are only not rectangular. From The $20 Pine, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Fasteners

The frame has 120 joints and every one of them needs fasteners that will still be doing their job in twenty years.

WHAT THE FASTENERS ACTUALLY DO

They do not hold the dome up. The wood does that, in compression, through the butt faces.

The fasteners hold the joint closed so the wood can work. That is a smaller job than it sounds and it changes what you need.

CORROSION

This is the one that decides longevity.

Green timber is wet and many species are acidic. Plain steel in wet acidic wood corrodes, and corroding steel stains the wood black around it and loses section.

Use stainless, or a properly coated exterior screw. Not zinc-plated indoor screws, which will be rust within a couple of damp seasons.

Stainless is expensive at 120 joints and it is the right expense. Work out what it costs to replace every fastener in a standing dome and the premium stops looking large.

QUANTITY

Two or three per joint, 120 joints, so somewhere between 240 and 360 structural fasteners for the frame.

Plus pilot holes for every one of them.

THE INSERT OPTION

A threaded insert set into the receiving member, with a bolt threading into it, gives a joint you can undo and remake indefinitely because the thread lives in metal rather than wood.

Two per joint is 240 inserts. It is more money and more work at build time and it is the difference between a dome you can take apart once and a dome you can take apart repeatedly.

BUY FOR THE DOME AFTER NEXT

If there is any chance you will build a bigger one, buy the bolts long now.

A bolt cut about a third longer than this dome needs still reaches through the thicker members of the next two sizes up. The extra length costs pennies and saves re-buying every fastener in the building.

Plates and reinforcement

The short version: you do not need them, and here is when you might want them anyway.

WHY THE FRAME DOES NOT NEED THEM

The pinwheel joint is a compression joint. Wood pressed against wood, with a large contact area, at loads well within what softwood takes across the grain.

A steel plate does not improve that. The wood is already carrying it.

Plates and brackets exist in other dome systems because those systems put their joints in tension and shear, where wood is weak and steel is not. This one does not.

WHERE REINFORCEMENT IS ACTUALLY WORTH IT

Three places.

The base connection. Uplift is the load case that matters on a dome — it is a very good aerofoil and it is light. Every base vertex wants a strap or an anchor holding it down, and that is a tension connection where steel belongs.

Around openings. Cutting a door or window out of a triangulated shell interrupts the triangulation. The members either side of the hole are carrying what the missing ones used to, and they may want help.

Point loads. A hoist, a heavy fixture, a partition carrying something. The frame takes those locally if you give it something to take them with.

WHAT TO AVOID

Plating every joint "to be safe".

That is not safer. It adds cost, adds thermal bridges through the envelope, adds visible steel to an interior that is mostly visible, and — the part that matters — it hides whether the joints are working.

If the frame needs plates everywhere, something is wrong with the joints and the plates are covering it up rather than fixing it.

THE HONEST LIMIT OF THIS CHAPTER

Nothing here is an engineering specification.

Species, moisture, snow load, wind exposure and local code all change the answer, and no book can know yours. What this chapter can tell you is what the joint is — a butt in compression, fasteners holding it closed — and that is the description an engineer needs to size it for your site.

If the dome is a workshop, build it as described and sleep fine. If people are going to live in it, take this description to somebody who can stamp it.

Every other way: what the frame is made of

The same 120 members, in each material this project can price. The volume is the reference build's own: a 45-degree sector of a 12-inch log, 664 feet of it.

EVERY OTHER WAY OF DOING IT

	cost	weight	
Galvanized Steel	$ 40,593	31,961 lb	7,850 kg/m3 at $2.80/kg
Structural Steel	$ 46,391	31,961 lb	7,850 kg/m3 at $3.20/kg
Rebar Steel	$ 27,545	31,961 lb	7,850 kg/m3 at $1.90/kg
Aluminum	$ 37,398	10,993 lb	2,700 kg/m3 at $7.50/kg
Timber (SPF)	$ 1,773	1,954 lb	480 kg/m3 at $2.00/kg
Whole Tree Trunk	$ 1,080	2,646 lb	650 kg/m3 at $0.90/kg <- the reference build
PVC Pipe	$ 8,791	5,700 lb	1,400 kg/m3 at $3.40/kg
Bamboo	$ 2,128	2,931 lb	720 kg/m3 at $1.60/kg

The dearest of these is 42.9 times the cheapest. That is the size of the decision, and it is why the book states which one it assumes rather than leaving it to a reader to infer.

These change the frame itself. Every other choice in this book can be made after the dome is standing; these cannot be made twice.

WHERE THESE NUMBERS COME FROM

materials.FRAME_MATERIALS, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

Every other way: the shape of the stick

Eight cross sections at the same nominal width, and how much material each one is. In timber at 650 kg per cubic metre, so the column compares shapes rather than materials. Read the wedge row carefully: the Creator's Quarter Wedge is 8.28 sq in because it is a quarter of a log sized by the strut's width, and this book's member is 14.14 sq in because it is an eighth of a 12-inch one. Same family, different stick.

EVERY OTHER WAY OF DOING IT

	weight	
Round Tube	605 lb	20.9 cm2 of section; section 3.23 sq in
Solid Steel Rod	3,101 lb	106.9 cm2 of section; section 16.56 sq in
Rebar Rod	3,101 lb	106.9 cm2 of section; section 16.56 sq in
Full Tree Trunk	3,101 lb	106.9 cm2 of section; section 16.56 sq in
Square Tube	770 lb	26.6 cm2 of section; section 4.12 sq in
Dimensional Lumber	8,685 lb	299.3 cm2 of section; section 46.39 sq in
Hex Strut	2,564 lb	88.4 cm2 of section; section 13.70 sq in
Quarter Wedge	1,550 lb	53.4 cm2 of section; section 8.28 sq in <- the reference build

These change the frame itself. Every other choice in this book can be made after the dome is standing; these cannot be made twice.

WHERE THESE NUMBERS COME FROM

materials.STRUT_SHAPES, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

5. Turning Timber Into Wedges

This is the chapter where a tree becomes a building material, and it is the one that has no equivalent in a normal dome book.

A strut-and-hub builder buys lumber. Here you are making the lumber, and the shape you make it into is what carries the dome's geometry. Get this chapter right and the assembly is straightforward. Get it wrong and no amount of care later will recover it.

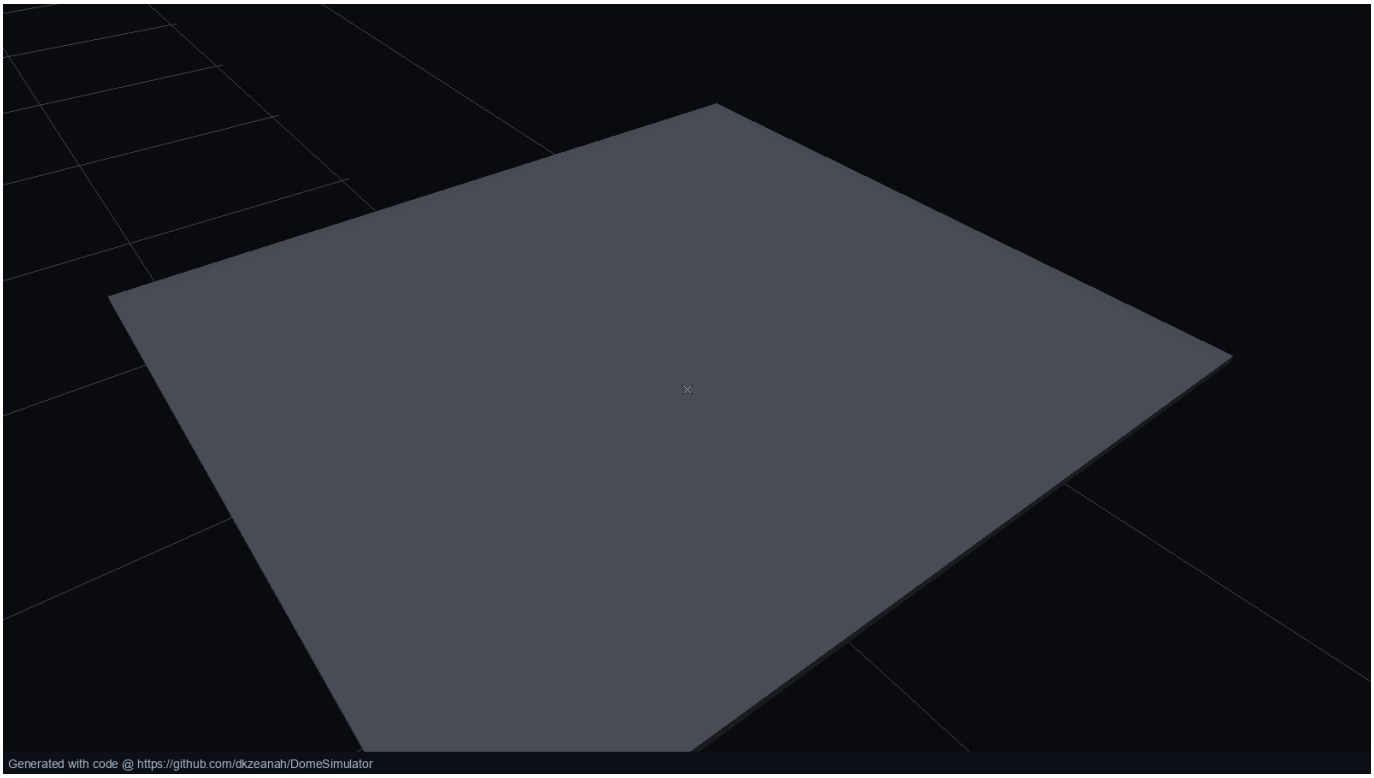

Figure 30. The Bare Bench

One flat sheet on sawhorses, and nothing else. This board is the only thing in the whole build that has to be true. Flatness. Every angle below is measured from this one surface, so the panel cannot wind or twist as long as the board does not. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Layout

Layout is deciding, before any saw runs, which part of the tree becomes which part of the building.

Do it badly and you will run out of good wood halfway through and start using the parts you set aside for a reason.

START AT THE BUTT

The bottom of the trunk is the best wood in the tree. Straightest grain, fewest knots, largest diameter.

Your longest members should come from there. On the reference build that is the sixty A members at six feet each.

The top of the trunk is smaller, knottier, and tapers faster. That is where the short members and the offcuts come from.

This sounds obvious. It is routinely got wrong, because people buck the log into equal lengths first and then decide what each piece is for — by which point the decision has already been made for them.

Decide first. Then cut.

HOW MUCH TREE

The reference build needs about one and a half trees of usable stock for 120 members: 664 feet of member at a six-inch depth and four and a half inch average width.

"One and a half trees" is doing a lot of work in that sentence. It assumes a trunk you can get six-foot clear lengths out of, straight enough to split predictably, and about twelve inches across.

A twelve-inch trunk split into eight gives sectors that are the right size for this dome. A smaller trunk gives you smaller sectors and a thinner wall. A bigger one gives you sectors you will have to trim, which is wasted work and wasted wood.

MARK BEFORE YOU BUCK

Chalk the whole log while it is still one piece.

Mark where each member will come from, mark the direction of the taper, and mark anything you are avoiding — a knot cluster, a bend, a shake.

Once the log is in pieces you have lost the information about which end was up, and which end was up matters for the taper.

THE ONE THAT CATCHES EVERYONE

A tree is not a cylinder. It is a cone.

A six-foot length taken from a twelve-inch butt is twelve inches at one end and perhaps eleven at the other. Split that into eight and every sector is wider at one end than the other.

You cannot avoid that. You can only decide which end of the member the fat end goes, and be consistent about it across all 120 pieces. Pick a rule now — fat end toward the vertex, say — and write it on the wall.

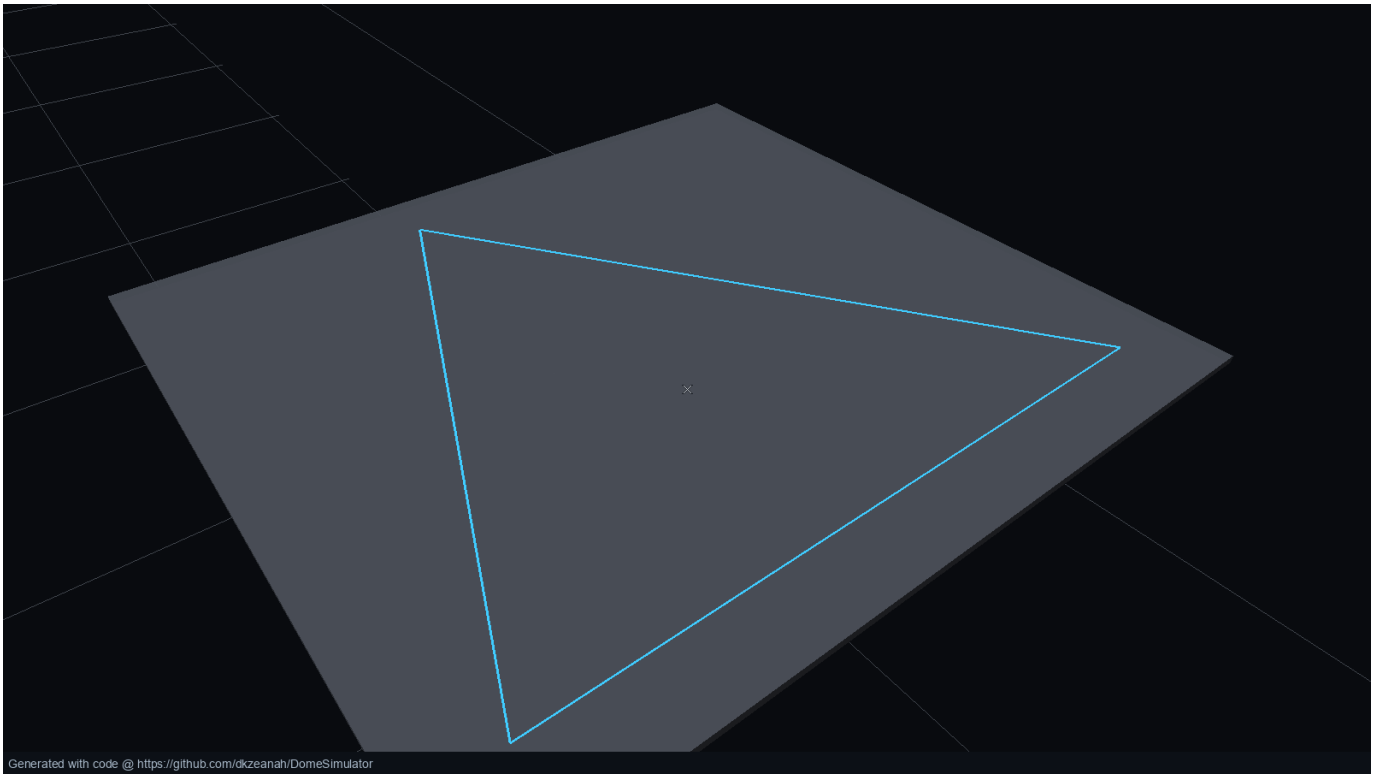

Figure 31. The Triangle, Drawn Once

The panel's mathematical triangle, snapped onto the board. It is drawn one time, from the geodesic solution, and never measured again. Repeatability. Forty panels are laid out from three lines that were struck once, instead of from forty tape measurements. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Ripping

Ripping is cutting along the grain. Everything that turns a log into a wedge is a rip.

THE DIFFERENCE THAT MATTERS

A crosscut goes across the fibres and is over quickly. A rip goes along them, and the fibres want to steer the blade.

That means rips wander. A rip that starts true can finish an eighth of an inch off, and in green timber with tension in it, more.

Everything in this chapter is a way of fighting that.

BLADE

Use a rip blade — fewer teeth, bigger gullets, aggressive hook. A crosscut blade will do it and will burn, bind and wander.

Sharp matters more than anything else. A dull blade in a rip cut is how kickback happens, and green softwood dulls a blade faster than dry hardwood because of the resin.

FEED RATE

Slower than feels right at first, then steady.

The mistake is varying the feed. A rip cut that speeds up and slows down produces a wandering line, because the blade deflects differently under different load. Pick a rate you can hold for the whole cut and hold it.

TENSION

Green logs have internal stress. As you cut into one it can close on the blade, or spring open, and both are dangerous.

Use a riving knife or a wedge in the kerf behind the blade. Always. Not for the first few cuts until you see how the log behaves — from the first cut, because the one that closes is the one you did not expect.

If a cut starts to pinch, stop, back out, and wedge it open before continuing.

THE ORDER

Cut the log in half first, then quarters, then eighths.

Halving a round log is the hardest cut because there is no flat face to reference. Once you have a flat, every subsequent cut has something to sit on, and the whole job gets easier as it goes.

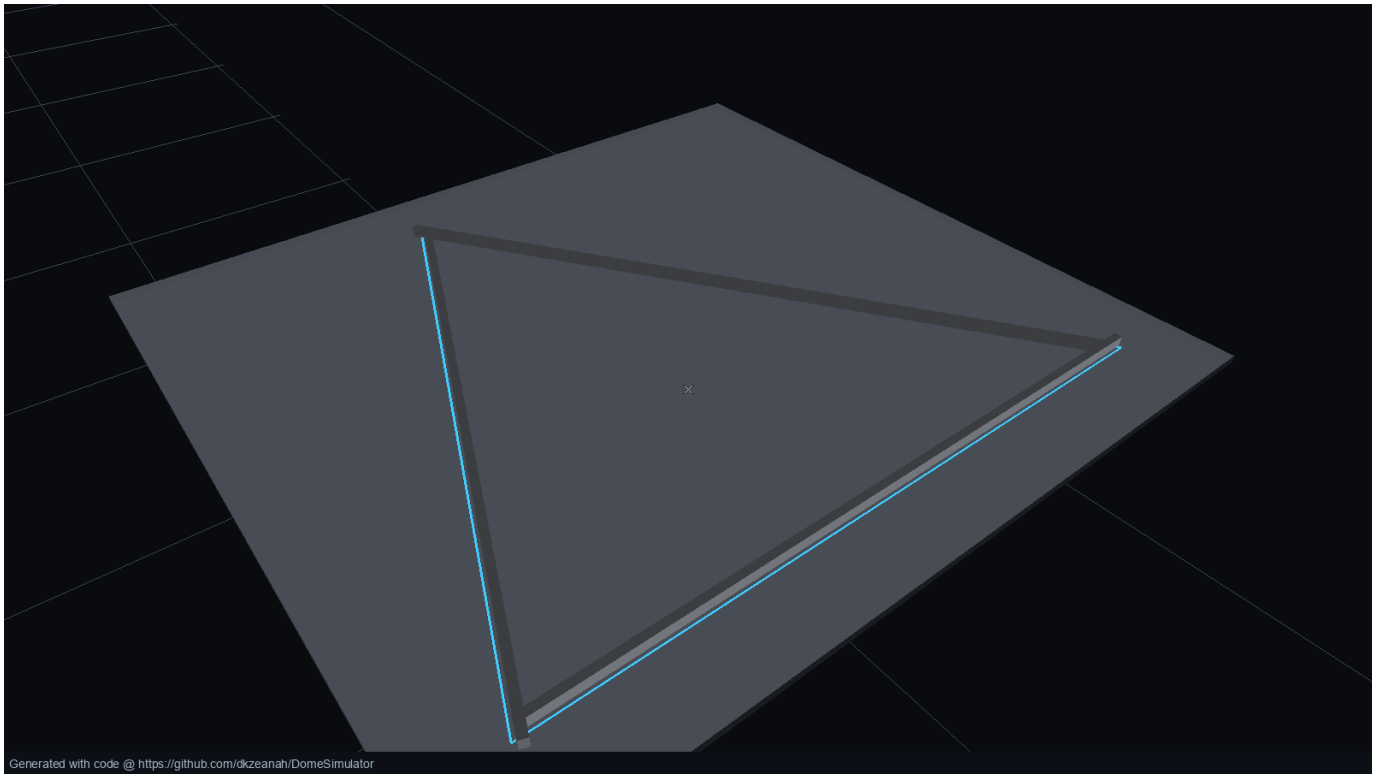

Figure 32. Axis Rails

Three low cleats, one per member, each set parallel to that member's point-axis and just to the seam side of it. Position across the panel. Push the stick against its rail and the sector's point line sits exactly where the solution says. The rail references the sawn point, never the bark, so a lumpy log does not move the joint. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Log division

Splitting a log into eight is the operation the whole method rests on.

Eight sectors of 45 degrees each. Those 45 degrees are what the dome's dihedral angles come out of, which is covered in the geometry chapter and is worth reading before you cut.

WHY EIGHT

Not because the geometry demands it.

Eight is what a person can do accurately. Halving is easy — you find the centre and cut through it. Halving again is easy. Halving a third time gives you eight, and each cut references a flat face that already exists.

Six or twelve means cutting angles that are not repeated halvings, and that means measuring rather than referencing.

The reference build uses eight because of that, not because 45 degrees is special. If you can hold twelve accurately, twelve works and gives you thinner sticks.

FINDING THE CENTRE

Not the centre of the bark. The centre of the wood.

A tree is not round and the pith is not in the middle. Find the pith at both ends, mark it, and run your first cut through both marks. That is the cut that makes the other seven work.

Get this one wrong and every sector is a different size, and you will not find out until you are assembling.

THE SEQUENCE

- find the pith at both ends, mark it
- cut through both marks: two halves
- each half, cut through the pith again: four quarters
- each quarter, cut through the pith again: eight sectors

Every cut after the first has a flat face to run against a fence, which is why this gets easier rather than harder.

WHAT YOU GET

Eight sectors, each a 45-degree wedge in cross-section, each with two sawn faces meeting at the pith and one curved bark face on the outside.

The sawn faces are the ones that matter. They are what meets the neighbouring member at a seam, and the angle between them is the 45 degrees the geometry uses.

The bark face is the outside of the dome. Leave it or take it off — the method does not care structurally, but bark holds moisture and insects, so most people take it off.

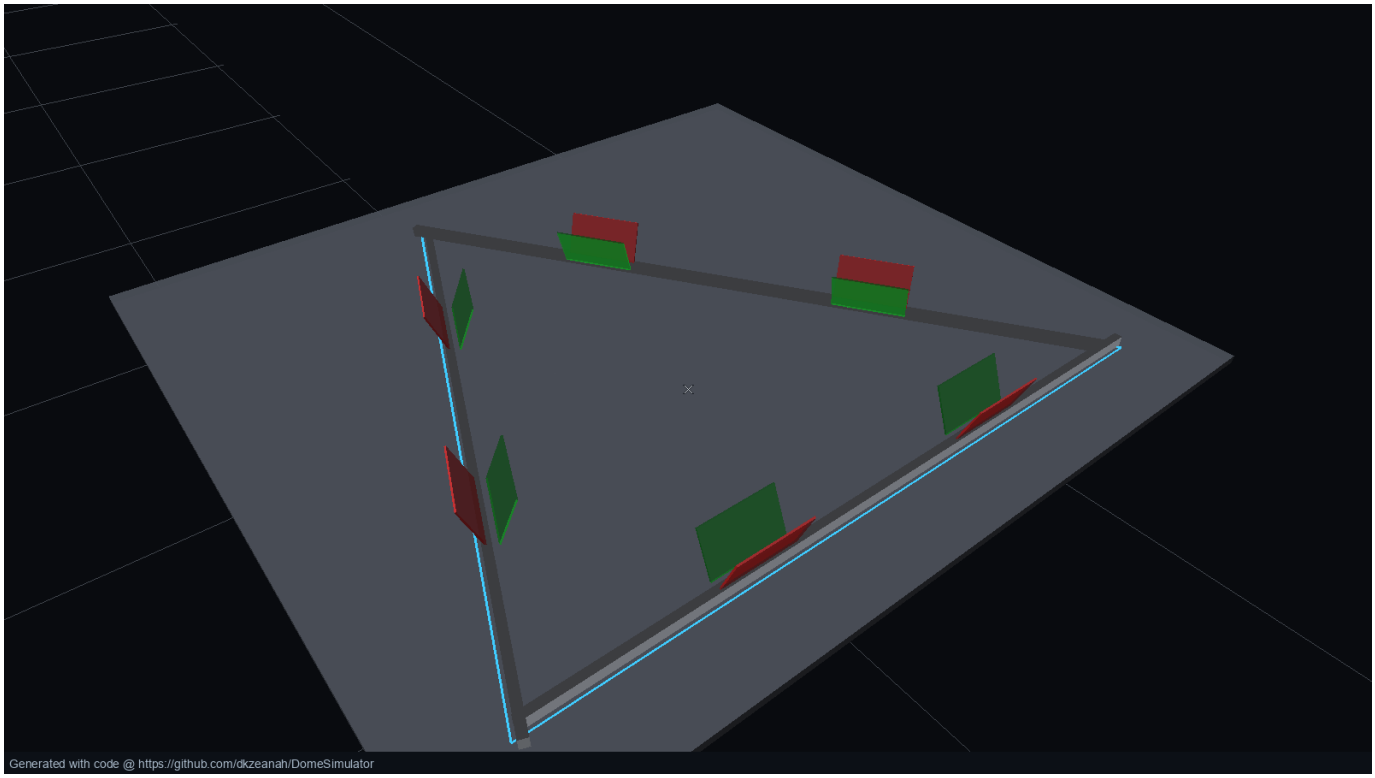

Figure 33. Red / Green Orientation Fences

Two short cradle plates per member, standing against the two sawn radial faces of the wedge. Rotation about the stick's own axis. A 45-degree sector dropped in rotated 90 or 180 degrees still LOOKS fine; these plates make it not fit. Red side to red plate, green side to green plate, or the stick does not seat. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Wedge extraction

Extraction is getting a usable member out of a sector.

The sector is 45 degrees of a tapering cone. The member needs to be a consistent cross-section over its length. Those are not the same thing, and the difference is the work.

THE TAPER PROBLEM

A sector from a log that is twelve inches at the butt and eleven at the top is wider at one end.

For a six-foot member that difference is about half an inch across the width. Small enough to live with in a frame; too big to ignore when you are trying to make forty identical panels.

Two ways to handle it.

TRIM TO THE SMALL END

Take every sector down to the cross-section of its narrow end.

Consistent members, predictable panels, and you throw away the difference. Simplest and most wasteful.

USE IT ORIENTED

Keep the taper and decide, once, which way it faces.

Every member has its fat end toward the vertex, say, and its thin end toward the middle of the edge. Every panel is then identically tapered and the panels still match each other, because they are all wrong in the same direction.

More wood kept, more discipline required, and completely dependent on doing it the same way 120 times.

The reference build trims to the small end, because the first dome is not the place to be clever. Once you have built one, the oriented method is worth the saving.

WHAT COMES OFF

The head end. Where the member meets a vertex, material has to come off to make room for the joint.

That offcut is not scrap — it is the right size for blocking, for the jig itself, and for the keys that fill the seams. Keep a box of it.

THE RULE

Extract every member the same way, in the same order, referencing the same faces.

A sector is an irregular object and there are four different ways to set it on a bench. Three of them are wrong for your cut and all four look reasonable. Mark the reference face on every sector as it comes off the splitting station and always put that face down.

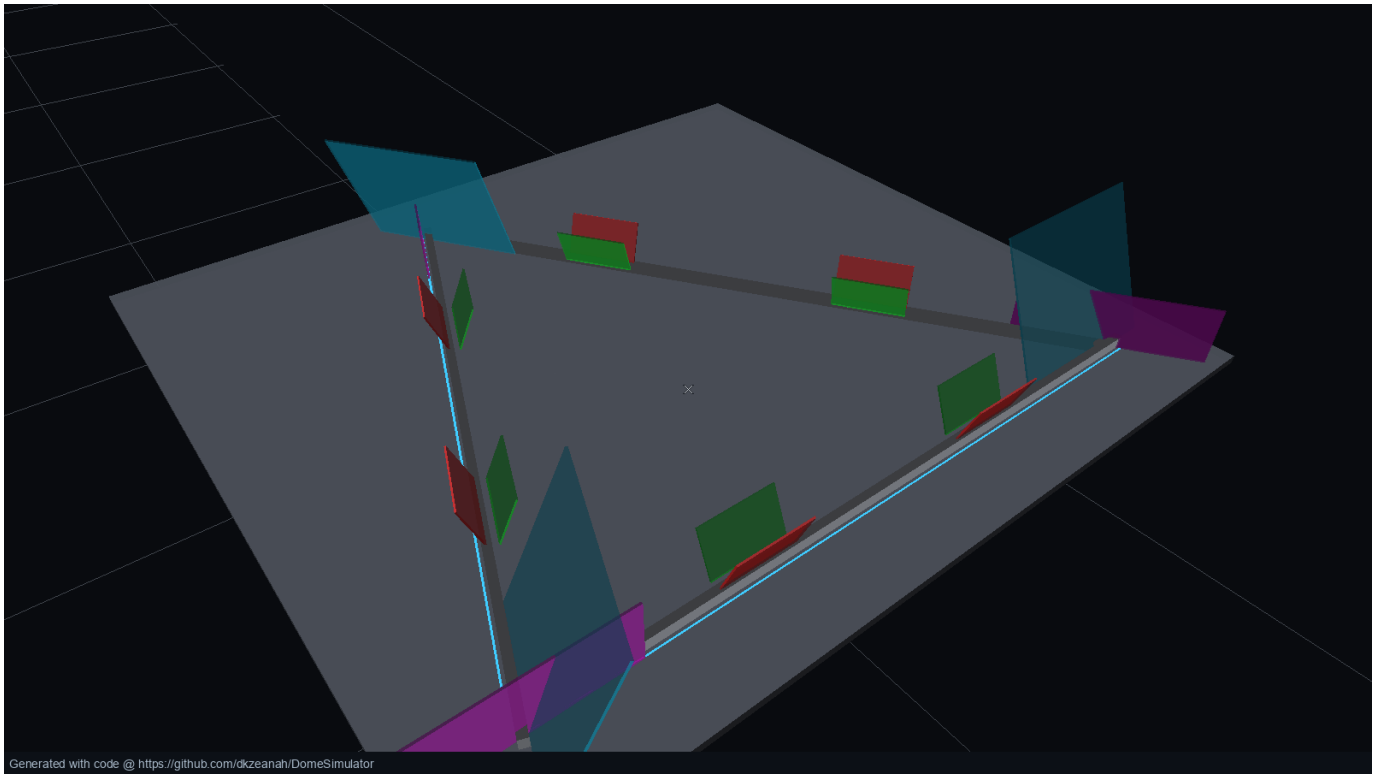

Figure 34. The Two Cut Planes

The magenta plate at each head end is the flush-cut fence. The cyan plate is the receiving face the next stick's butt will land on. The saw. A circular saw or a handsaw run flat against the magenta face cuts the head to the exact plane of the triangle edge, without anyone setting a bevel or a mitre angle. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Saw setups

A setup is a saw, a fence, a stop and a blade height, held constant while you run a batch through it.

The number of setups in this build is small. The number of cuts is large. That ratio is the thing to protect.

THE SETUPS YOU NEED

Splitting: a rip setup with a tall fence and a riving knife.

Length: a crosscut setup with a stop block, one for A members and one for B.

Butting ends: an angled crosscut, referenced off a fence at the butt angle.

Receiving faces: a shallow cut along the side of the member, referenced off the same fence.

That is four setups for 120 members and about 300 cuts.

CHANGE THE SETUP AS SELDOM AS POSSIBLE

Every setup change is an opportunity for the new setting to be slightly different from the old one.

If you do all sixty A lengths, then change to B, then change back because you miscounted, the second batch of A members is not guaranteed to match the first.

Do all of an operation. Count before you change. Then change once.

THE STOP BLOCK

A stop block sets length without measuring, which is the single biggest accuracy win available.

Clamp it, cut one, check it, then cut the rest without measuring anything.

Check it again every tenth piece. Stop blocks creep — the clamp relaxes, the block gets knocked, sawdust builds up behind it. Ten pieces is often enough to catch creep while it is still small.

AND THE THING THAT WILL BITE YOU

Sawdust behind the stop.

A sixteenth of packed dust between the workpiece and the stop makes every subsequent piece a sixteenth long, and it accumulates invisibly over a batch.

Blow it out between pieces. Every piece. It takes a second and it removes an entire category of error.

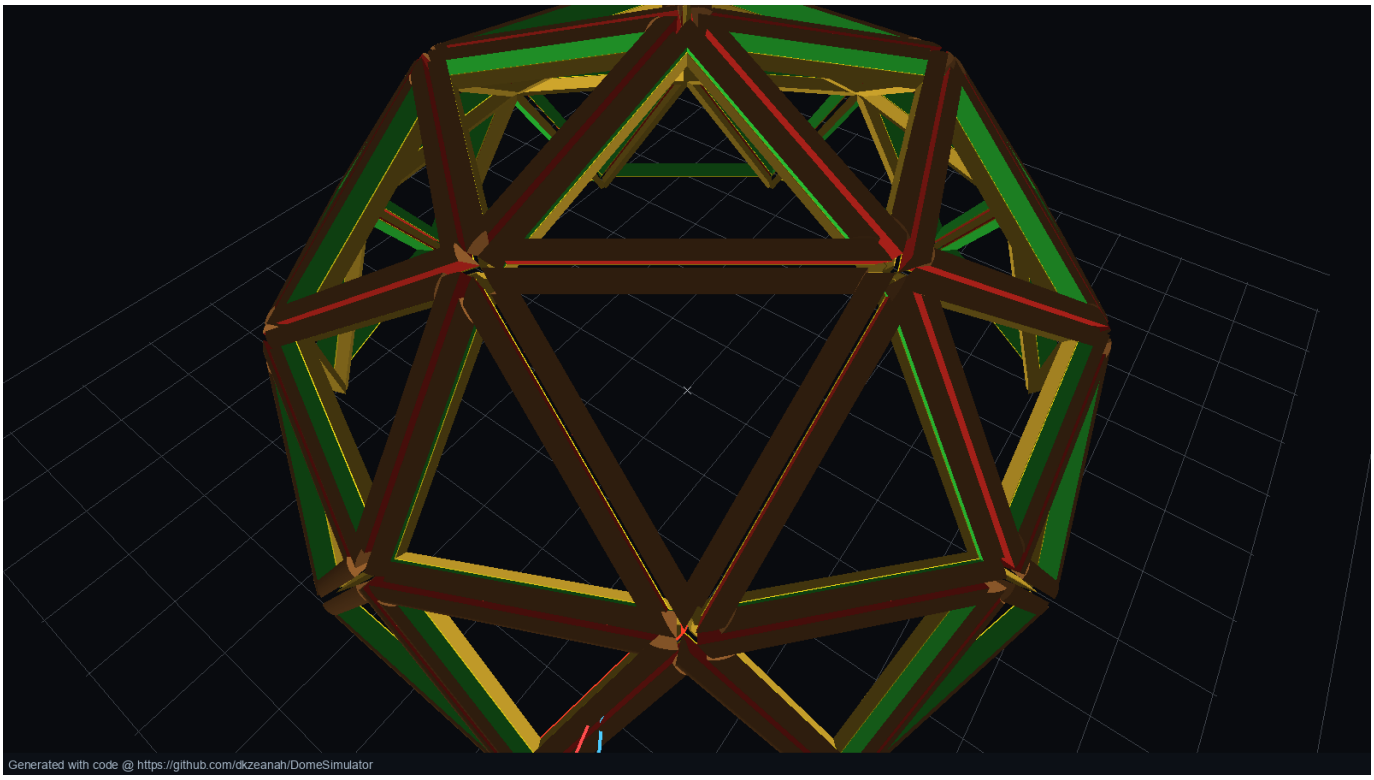

Figure 35. Clockwise, seen from outside

Clockwise, seen from outside. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Jigs

A jig is a fixture that holds a workpiece in a known position so a cut lands in the right place without being measured.

In this method, the jig is the method. One flat jig turns 120 irregular split-log sectors into forty identical panels, and without it the whole approach falls apart.

WHAT IT DOES

It holds three members in the relationship they will have in a finished triangle, and presents each joint to the saw at the right angle.

You do not measure the joint. You put the stick against the fences, and the joint comes out right because the fences are in the right places.

WHY THAT MATTERS SO MUCH HERE

The members are not rectangular. They are wedges, with a curved face and two sawn faces at 45 degrees, and no two are exactly alike because they came out of a real tree.

You cannot reference an irregular object off itself. You have to reference it off something known, and the jig is that something.

THE FENCES

The jig in the tool has coloured fences and they do different jobs:

- the rail sets the stick's point-axis across the panel
- the red fence is the sawn face that must land on the seam side
- the green fence is the sawn face the next butt lands on
- the magenta plate is the head-end flush-cut fence
- the cyan plate is the receiving face the next member's butt meets
- the clamp holds the stick down on rail and fences

Six references, and every one of them is a physical stop rather than a number.

BUILD THE JIG FIRST

Before any member is cut to final size.

The jig is made out of offcut from the splitting, so it costs nothing but time, and it is the thing every other operation depends on. A jig that is a degree out makes 120 members a degree out.

Build it, check it, and check it against the tool's own drawings rather than against your expectations.

THE TEST

Make one panel. If all three corners close and the triangle sits flat on the floor, the jig is right.

If they do not, the jig is wrong, and it is wrong now — before it has made 119 more.

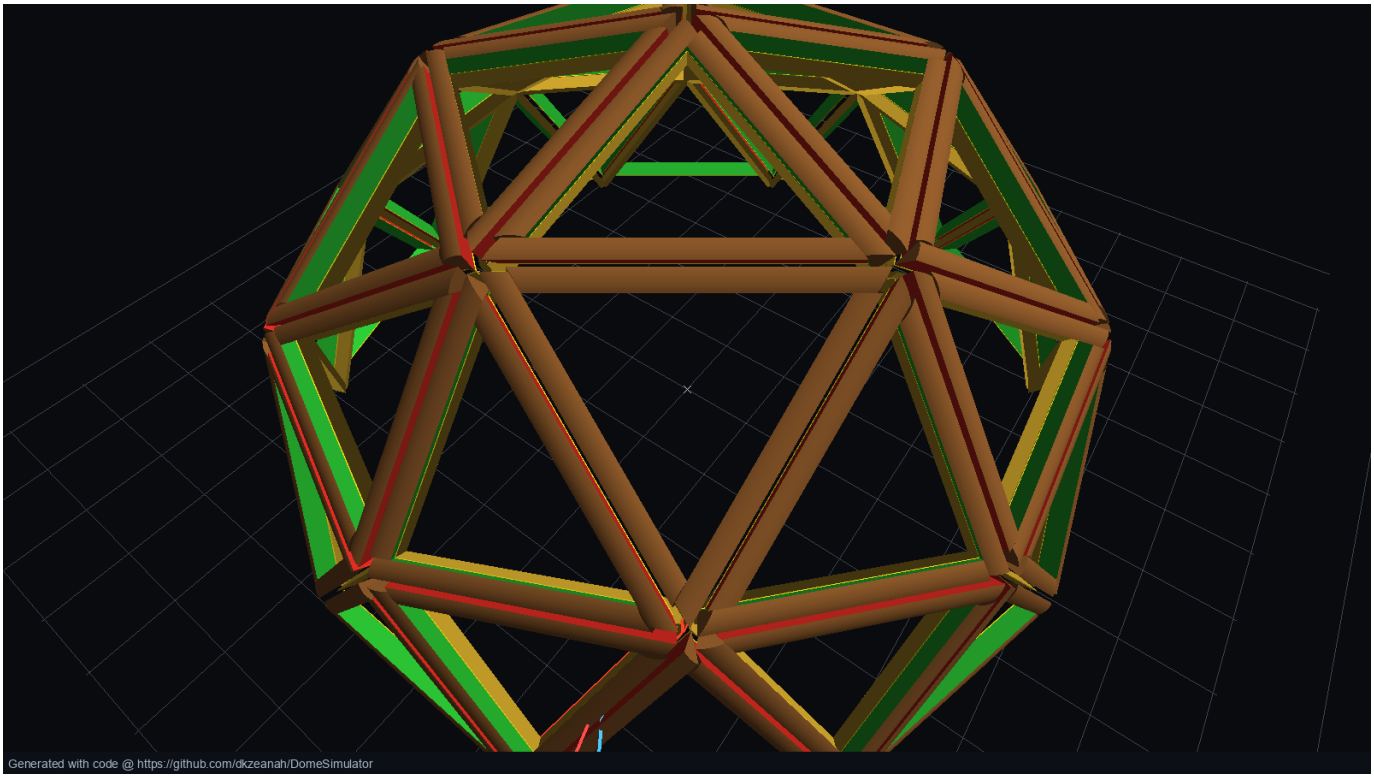

Figure 36. Counterclockwise, seen from outside

Counterclockwise, seen from outside. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Repetitive cutting

The forty-first identical cut is where accuracy goes to die.

Not because the saw changed. Because you did.

WHAT ACTUALLY DEGRADES

Attention first. By the thirtieth piece you are no longer checking that the stick is against the fence; you are assuming it is.

Then technique. You get faster, and faster means less pressure on the fence, and less pressure means the piece can lift.

Then the setup, slowly, from vibration and sawdust and the clamp relaxing.

None of those announce themselves.

THE COUNTERMEASURES

Work in batches of ten, not sixty. After each ten, check one against the master, blow out the stop, and take thirty seconds.

That is not a break for you. It is a checkpoint for the work, and it caps how many pieces a drifting setup can ruin at ten.

Count out loud or make a mark. You will lose count otherwise, and a batch of nine or eleven is how the A and B counts end up uneven.

Stop when you are tired. The last hour of a long session produces most of the errors, and a piece cut wrong at eight in the evening costs more than the hour you would have saved.

THE THING THAT HELPS MOST

Do not think of it as 120 cuts.

Think of it as twelve batches of ten, each with a check at the end. Twelve is a number you can hold. 120 is a number you stop counting.

That is a psychological trick and it works, which is the only test that matters.

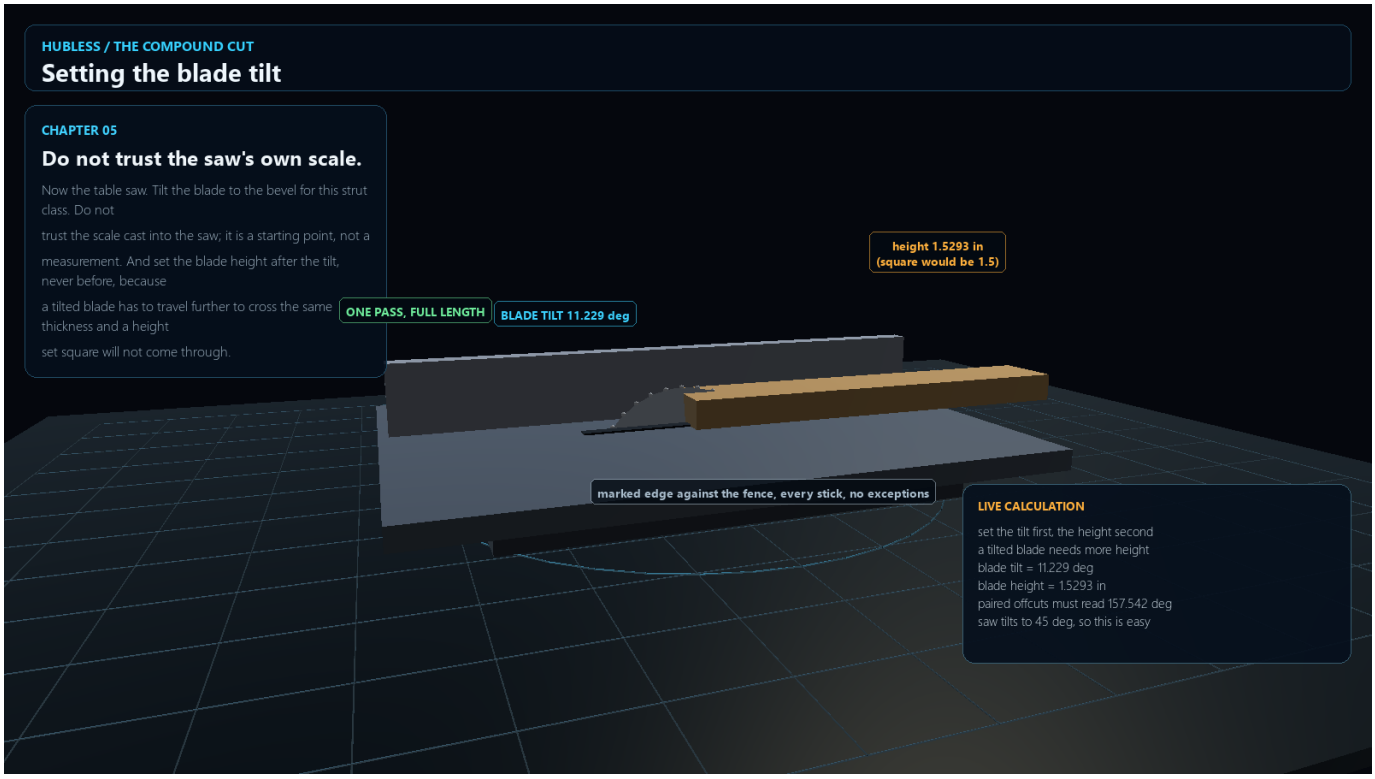

Figure 37. Setting the blade tilt

Do not trust the saw's own scale. Now the table saw. Tilt the blade to the bevel for this strut class. Do not. From The Compound Cut, Both Machines, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Tolerances

A tolerance is how wrong a part is allowed to be before it stops working.

Stating them is worth doing because without a stated tolerance every part is either perfect or scrap, and neither is true.

WHAT THIS BUILD CAN LIVE WITH

Member length: plus or minus a sixteenth of an inch. The joints take up under the fastener and a sixteenth vanishes.

End angles: within half a degree. This is the tight one. An angle error does not take up — it holds the joint open and it accumulates round a ring.

Cross-section: a quarter inch of variation is fine. The members are split from real trees and they are not going to be identical. The frame does not care.

Panel flatness: a triangle that rocks by more than a sixteenth on a flat floor is telling you an end angle is out.

WHY ANGLE IS TIGHTER THAN LENGTH

Because length errors are random and angle errors are systematic.

If your stop block is right and your technique is variable, lengths scatter either side of the target and they cancel around the dome.

If your fence is set half a degree off, every single angle is off by the same half degree in the same direction, and ten of them round a base ring is five degrees of accumulated error with nowhere to go.

Length errors average out. Angle errors add up. That is the whole reason for the different tolerance.

MEASURE THE RIGHT THING

Do not measure a member's length with a tape against the curved bark face.

Measure between the reference faces — the sawn faces the jig uses. Those are the surfaces the geometry actually uses, and the bark face is decoration.

WHEN TO SCRAP A PIECE

If it is long, it is not scrap. Recut it.

If it is short, it may still be a B member, or blocking, or jig stock.

If the angle is wrong, recut the end. If both ends are wrong, it is blocking.

Very little of this actually becomes waste, which is the compensation for using eighty percent more wood than a shared-strut dome.

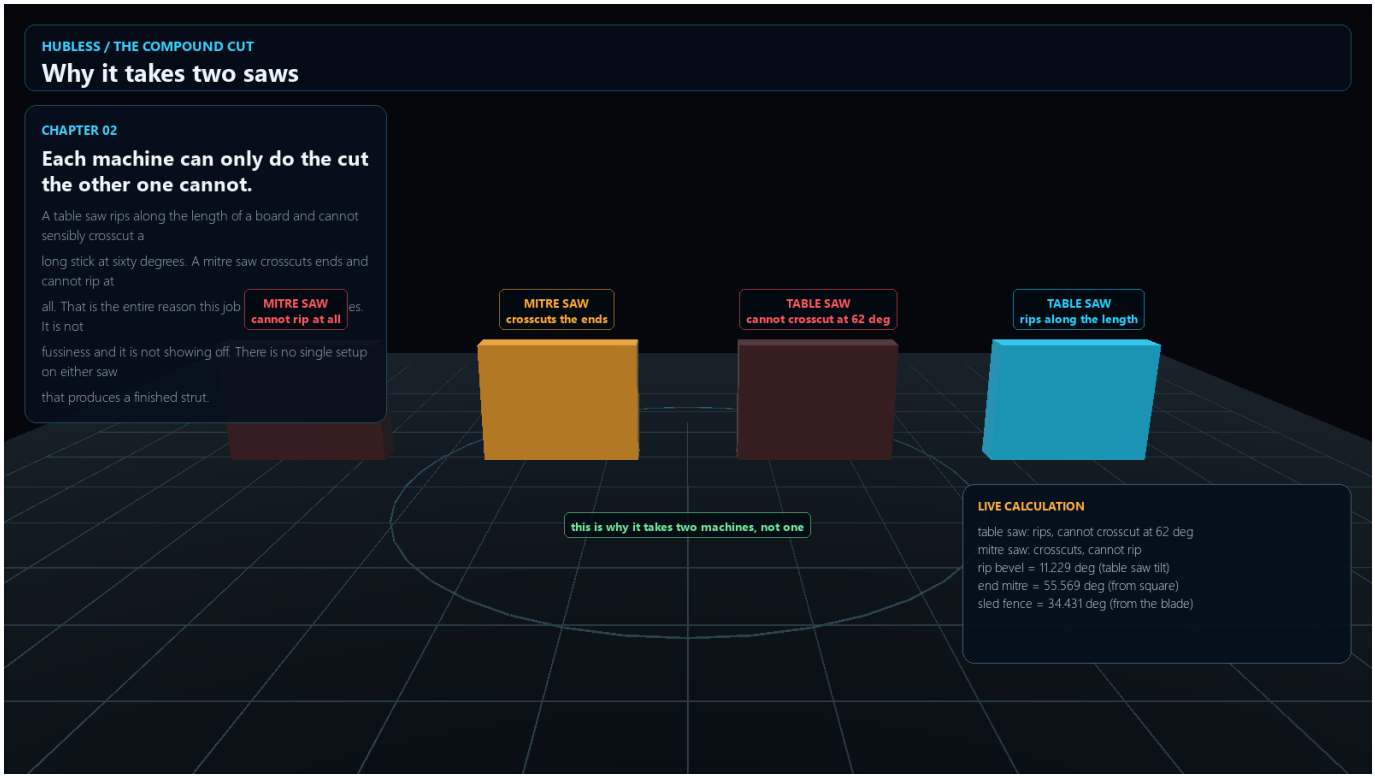

Figure 38. Why it takes two saws

Each machine can only do the cut the other one cannot. A table saw rips along the length of a board and cannot sensibly crosscut a. From The Compound Cut, Both Machines, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Labeling

Label everything, as it is made, with something that survives.

This is the least interesting section in the book and skipping it will cost you a day.

WHAT NEEDS A LABEL

Every member: which family it is, A or B.

Every member: which face is the receiving face. This is not visible once the stick is out of the jig, and fitting a member with its receiving face on the wrong side mirrors the panel.

Every panel: which triangle it is, AAA or BAB, and which way is up. A BAB panel fitted upside down looks nearly right.

Every sector, before it becomes a member: which face is the reference face, and which end was the butt of the tree.

HOW TO MARK

One line for A, two lines for B. Not the letters — they look alike upside down in bad light at the bottom of a pile, and you will be reading them from across the shop.

Use a lumber crayon or a keel, not pencil. Pencil disappears off rough sawn timber the first time it rains or gets handled.

Mark on the end grain where possible. It is the part you see when the stick is in a stack.

WHEN TO MARK

As the piece comes off the saw. Not at the end of the batch.

"I will know which pile is which" is true for about four minutes.

THE PANEL LABEL

Once a panel is assembled, write its number on it — and write it on the inside face, because the outside is going to be covered.

The tool numbers the panels and the fabrication package prints the list. Use those numbers rather than inventing your own, so that when something does not fit you can describe it to somebody in terms they can look up.

THE COST OF NOT DOING THIS

A pile of 120 unlabelled split-log members all look broadly similar, and telling a 72.000 inch chord from a 63.670 inch one by eye is harder than it sounds when both sticks are tapered, irregular, and lying at an angle.

You will end up measuring all of them. That is the day it costs.

6. Making the Struts

Cutting a hundred and twenty sticks is not a hundred and twenty problems. It is about six problems, repeated, and this chapter is the six.

The one worth reading twice is the first. The chord is not the cut, and every dome that has ever been built a hundred and twenty sticks too short was built by somebody who assumed it was.

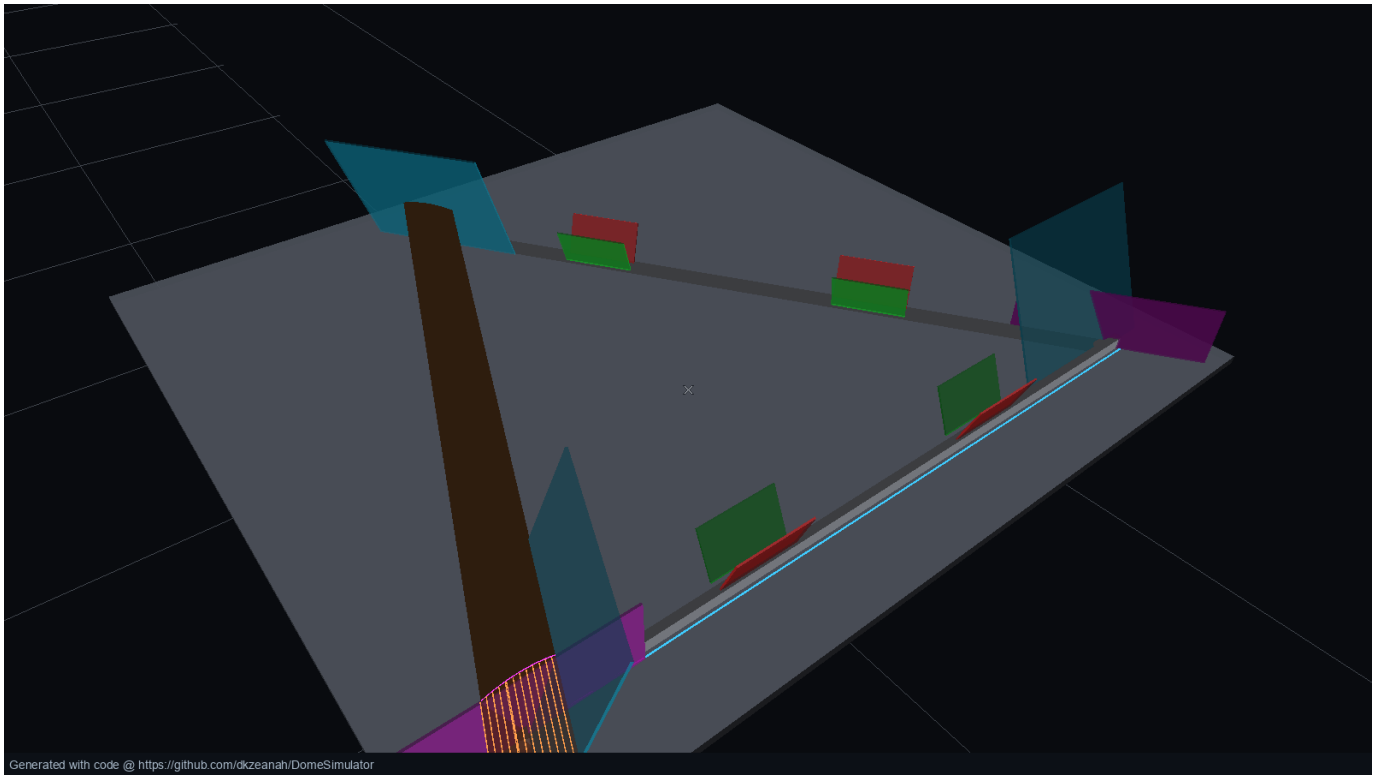

Figure 39. The Butt Cut, Made Off The Jig

One stick, still full length at both ends, laid up to the cyan plate. The amber tail past that plate is the stock the butt cut removes. Batching. This is the ONLY cut made before assembly, and every stick of the same family takes the same one, so it can be run through a saw bench in a batch rather than fitted one at a time. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Cutting length

The chord is not the cut.

This is the single most expensive misunderstanding in dome building and it catches people who have done everything else right.

The chord is the distance between two vertex points. Those points are mathematical — they float in space at the corners of the triangles, and on the reference build they are 72.000 inches apart for an A member.

The stick you cut is shorter than that, because both of its ends get taken back to make room for the members it meets.

THE NUMBERS ON THE REFERENCE BUILD

A members: 72.000 inch chord, and the usable stock comes off the saw between 68.977 and 70.182 inches.

B members: 63.670 inch chord, stock between 60.492 and 62.681 inches.

Look at the range. It is not one number. A members are not all the same length, even though they are all the same chord.

WHY THE RANGE EXISTS

Because the amount taken off an end depends on what that end runs into.

In the pinwheel joint every member butts into the side of its neighbour rather than meeting it point to point. How far it has to be set back depends on how thick the neighbour is at that spot and what angle it arrives at. Those are not the same at every corner of the dome.

So an A member near the apex is trimmed slightly differently from an A member in the middle band, and the difference is up to an inch and a fifth.

WHAT THIS MEANS FOR YOUR CUT LIST

You cannot cut sixty identical A members and sixty identical B members and expect them to go together.

You need the actual per-member list, and the tool generates it. Run the fabrication package and you get every member with its own length, its own identifier, and the panel it belongs to.

THE ONE THING TO DO ANYWAY

Cut long.

Every member is easier to shorten than to lengthen, and the first panel you build is the one that tells you whether your numbers are right. Cut the first three members half an inch over, fit them, measure what you actually needed, and then commit the other 117.

Half an inch of sacrificial length on three sticks is the cheapest insurance in the whole build.

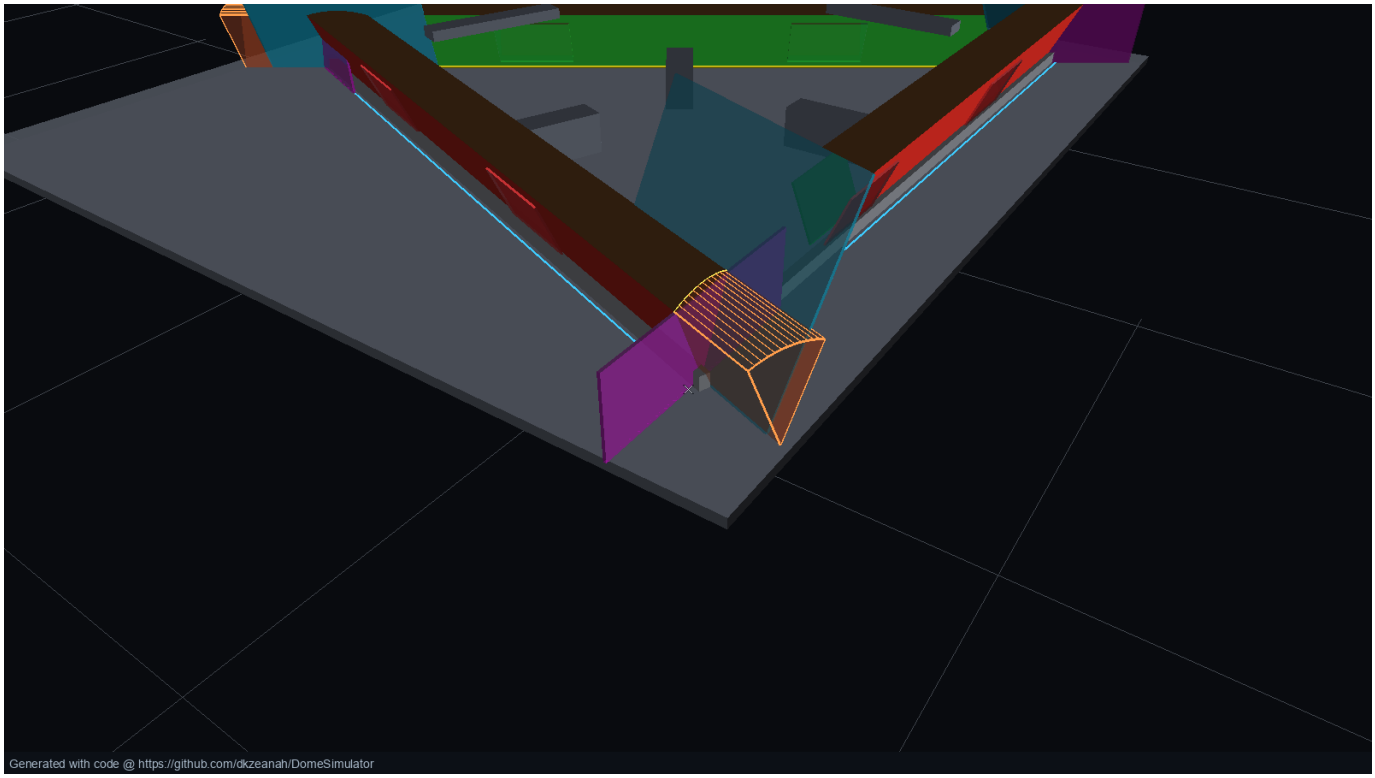

Figure 40. Flush-Cut The Three Heads

Each head is sawn back to its magenta fence. The orange stock hanging past the board is what falls off. This is the whole trick. The head is never measured. It is cut in place, to the plane the neighbouring member has already established, so error in stock length or in the butt angle leaves as offcut instead of accumulating around the triangle. Generated with code @ https://github.com/dkzeanah/DomeSimulator

End geometry

What happens at the end of a member is more important than how long the member is.

A length that is out by a sixteenth makes a slightly loose joint. An end angle that is out by a degree makes a joint that does not sit down, and a panel that will not lie flat, and forty panels that will not agree with each other.

WHAT THE END ACTUALLY HAS TO DO

In the pinwheel joint, each member does two different things at its two ends.

At one end it butts into the side of its neighbour. That end needs a flat face cut square across the stick, at the angle the neighbour presents.

At the other end it receives a butt from a different member. That end needs a flat face on its side for the neighbour to land on.

So every stick has a butting end and a receiving face, and they are not the same operation.

THE CORRECTION

An earlier telling of this — in the films and in an earlier draft of this chapter — said every one of those cuts was a plain angle. No compound mitre, no bevel and mitre at once.

That is wrong, and it is worth saying plainly rather than quietly fixing, because a lot of people heard it.

Here is what the solve actually gives. The butt end is a COMPOUND cut: a bevel and a mitre together, which is the trickiest operation in the whole build.

WHAT IS TRUE INSTEAD, AND WHY IT IS STILL GOOD NEWS

Two things come out of it that make the cut jiggable rather than hand-fitted, and they are better news than the false version was.

The bevel is the same on every member in the dome: 22.5 degrees, which is exactly half the 45-degree sector the log was split into.

Read that again, because it is the whole thing. The bevel is set by HOW THE LOG WAS SPLIT, not by where the member sits in the shell. Set it once, on the first stick, and never touch it again for the other 120.

And the mitre takes only 3 values across all 120 members: 21.138, 30.000, 34.431 degrees.

So the real number to hold on to is 3 saw settings for a whole dome. Not one, as the earlier version claimed — but three, all of them under 50 degrees and so within what an ordinary mitre saw will swing.

THE PART THAT IS GENUINELY AWKWARD

The cut runs a long way along the stick, because it lands on a face that is itself tilted. The sector's point reaches past its bark corner by 2.23 to 2.91 inches.

That is why the blank has to be long on that side, and it is the part that gets judged by eye when there is no fixture. Make the fixture.

WHAT THE PINWHEEL DID REMOVE

Not the mitre. The shared vertex.

No stick is coped, shaved or brought to a point, and no two ends meet each other — every end lands on somebody's flat side. That is a real and large simplification and it is what the method actually buys.

It uses more wood — eighty percent more than a dome that shares its struts — and in return there are three saw settings and no fitted joints.

THE FENCES

The jig holds the stick with the cut faces referenced off fences rather than off measurements.

A fence is a physical stop. You put the stick against it and the cut is right, or the stick does not fit against it and you know immediately. That is a much better failure mode than a measurement, which can be wrong quietly.

Every cut in this method should be referenced off a fence. If you find yourself measuring and marking a cut angle with a bevel gauge, you are doing it the way the method is trying to save you from.

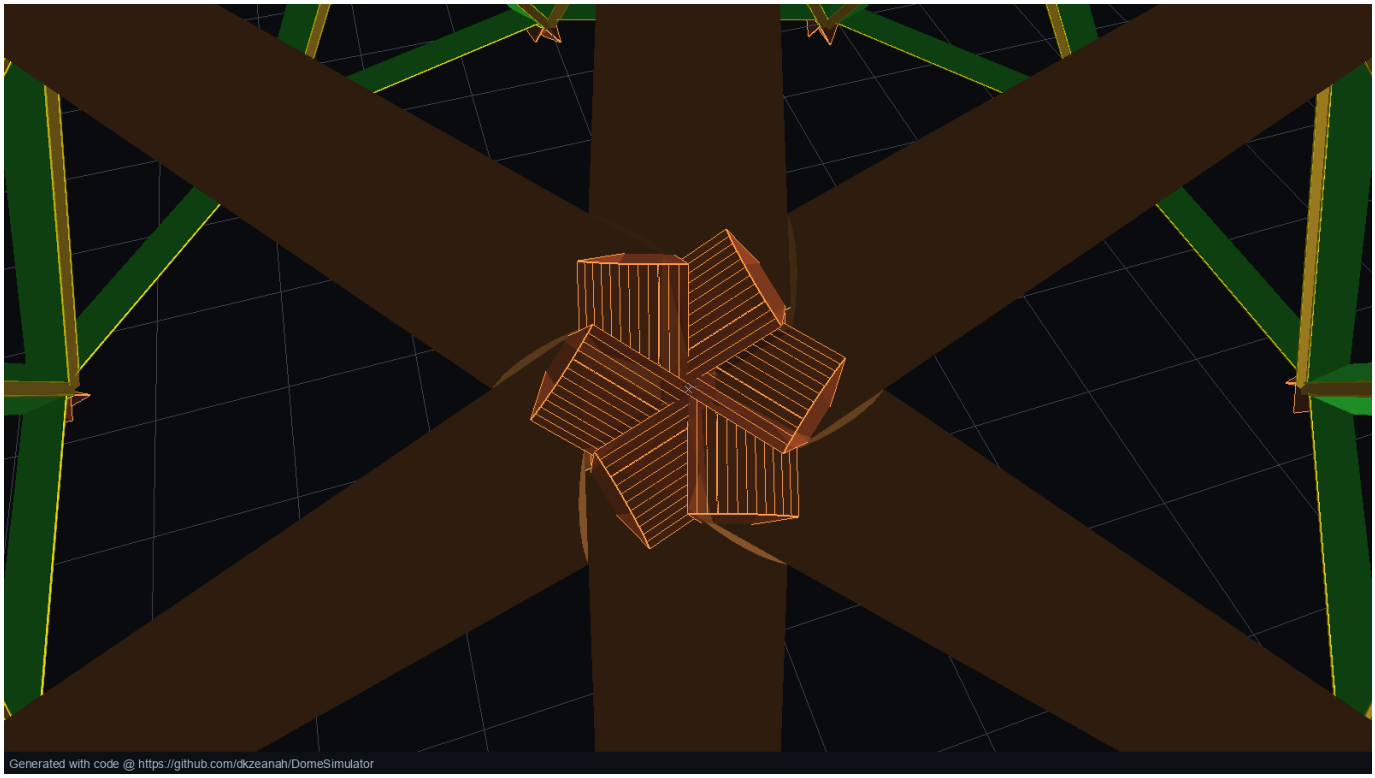

Figure 41. Before the flush cut

The assembled frame with every head end still carrying its six inches of stock. Uncut, the heads trespass across the geodesic vertex into the neighbouring panel -- which is the argument for cutting them in place. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Repeatability

You are making 120 members. The question is not whether you can make one correctly. It is whether the hundred and twentieth is the same as the first.

WHERE DRIFT COMES FROM

Three places, in order of how much damage they do.

A fence that moves. If the stop that sets your length shifts by a thirty-second over forty cuts, the last stick is an eighth short and nothing tells you.

A blade that wanders. A saw with play in the pivot cuts a slightly different angle depending on how hard you push it.

You. Cut fifty of anything and your technique changes — you get faster, you push differently, you stop checking.

WHAT TO DO ABOUT IT

Do not measure from the last piece. Ever.

If you use the stick you just cut to set the length of the next one, every error is inherited and added to. Forty sticks later you have accumulated forty small mistakes into one large one. Always measure from the

fence, or from a master.

Keep a master. Cut one member, check it properly, and hang it on the wall. It is not for building with. It is the thing you compare against every tenth piece, and it is how you catch a fence that has crept without noticing when it started.

Check at intervals, not at the end. Every tenth member, put it against the master. If it matches, you have lost nothing. If it does not, you have nine bad sticks to redo rather than fifty-nine.

THE HONEST BIT

You will still get some wrong.

The point of all of this is not to make a hundred and twenty perfect members. It is to make sure that when one is wrong you find out while it is still one.

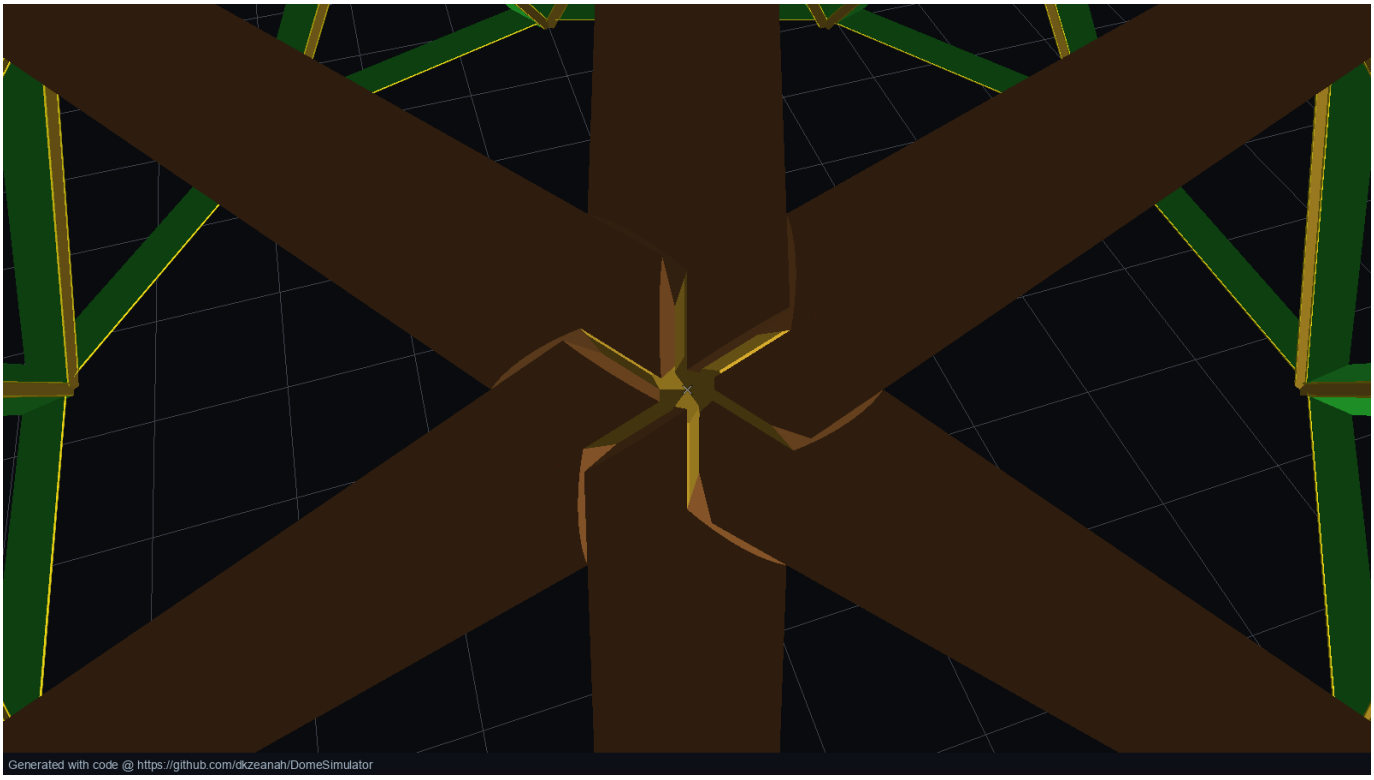

Figure 42. After it

The same corner with the stock gone. Every member now ends inside its own triangle and nothing crosses the vertex. Generated with code @ https://github.com/dkzeanah/DomeSimulator

A and B members

Sixty A members and sixty B members. Eight and a third inches of chord between them.

That difference is big enough to see and small enough to get away from you in a pile.

KEEP THEM APART FROM THE FIRST CUT

Not after. From the first cut.

Two stacks, two locations, and mark every stick as it comes off the saw. Not later, not when you have a few. As it comes off.

A pencil mark on the end grain takes one second and survives the whole build. An unmarked stick in the wrong pile costs you a panel.

WHY THIS IS WORSE THAN IT SOUNDS

If these were just two lengths, mixing them up would be an annoyance. You would offer up a stick, find it too short, and reach for another.

But in the wedge method the two families sit at different angles. An A seam folds 18.029 degrees; a B seam folds 22.459. The member is not just a length, it is a length at an angle, and a B member fitted where an A belongs is wrong in a way that partially fits.

That is the dangerous kind of wrong. It goes in. It looks close. The panel comes out slightly off and you do not know why, and by the time you find it you have built six more.

THE MARKING SCHEME

Use something you cannot misread at a glance:

- A members get one mark
- B members get two

Not "A" and "B" in pencil, which look similar upside down in poor light at the bottom of a pile. One line or two lines. You can count that from across the shop.

Mark the receiving face too, while you are at it, and mark it differently. Which face receives a butt is not obvious once the stick is out of the jig, and getting it backwards mirrors the panel.

THE COUNT

Sixty of each. If you have cut sixty-one of one and fifty-nine of the other, something went in the wrong pile and you want to know now.

Count them before you start assembling. Counting 120 sticks takes two minutes. Finding the miscount at panel thirty-eight takes an afternoon.

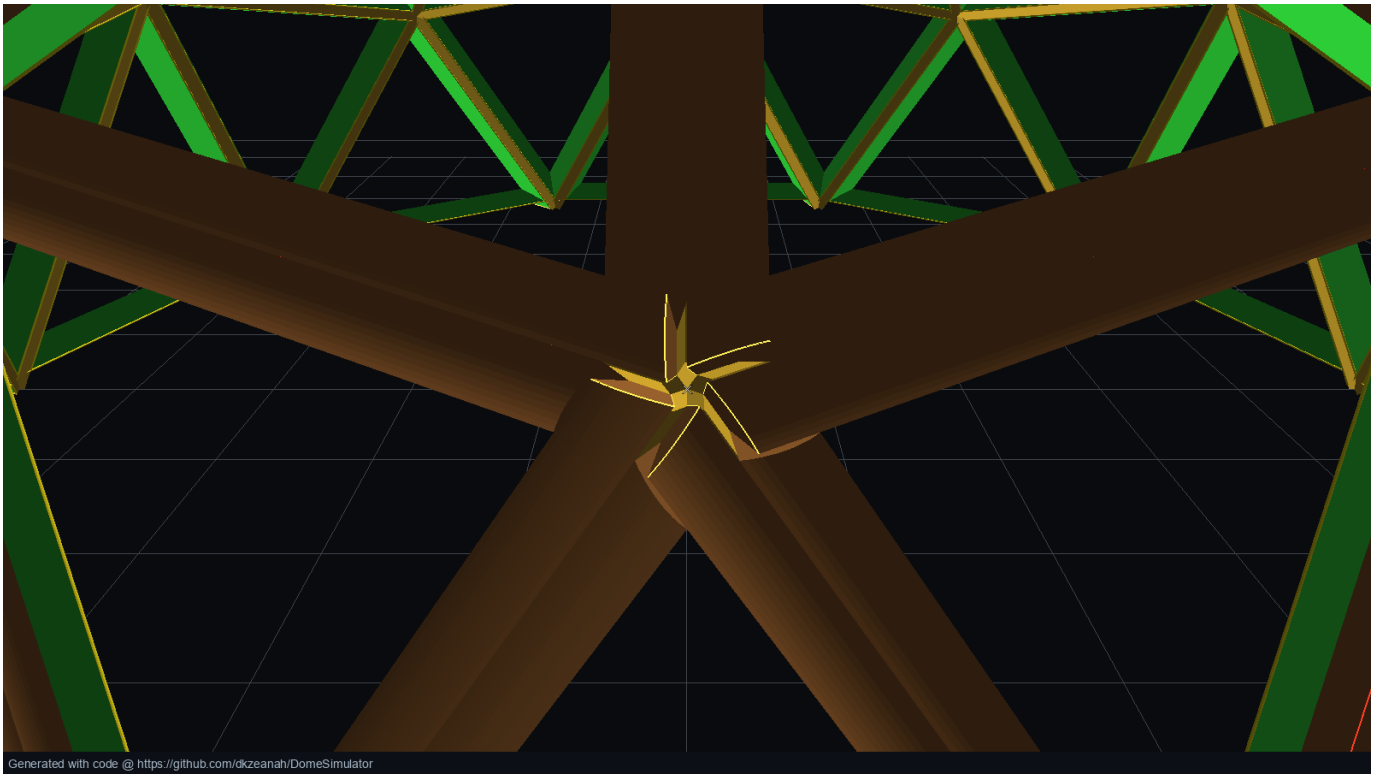

Figure 43. Nothing

Nothing: both ends cut to size, and the error has nowhere to go. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Test fitting

Build one panel before you cut a hundred and twenty members.

That is the whole section. Everything below is why.

WHAT THE FIRST PANEL TELLS YOU

Three members, three joints, one triangle. It takes an hour and it checks every assumption you have made:

- whether your cut lengths are right
- whether your end angles are right
- whether the jig holds what you thought it held
- whether the wedge orientation you chose does what you expected
- whether the joint pulls up tight or leaves a gap

Any one of those being wrong is a thing you want to find out on three sticks rather than on a hundred and twenty.

HOW TO CHECK IT

Lay the finished panel on a flat floor.

It should sit down on all three corners with no rock. If it rocks, one of your end cuts is off and the triangle is not planar.

Measure the three sides between the vertex points. They should match the chords for that triangle type — all 72.000 for an AAA, or 72.000 and two of 63.670 for a BAB.

Measure both diagonals from each corner to the middle of the opposite side. For an AAA they should be identical. If they are not, your triangle is not symmetrical and one member is long.

THEN BUILD THE SECOND ONE

The first panel tells you whether you can build a panel. The second tells you whether you can build the same panel twice.

Put them on top of each other. They should agree everywhere. Any difference between panel one and panel two is a difference that will be in all forty, and it is much easier to find now.

WHAT TO DO WITH THE TEST PANELS

Use them. They are real panels and there is nothing wrong with them.

Unless there is, in which case take them apart and find out why, and be grateful it cost you two panels.

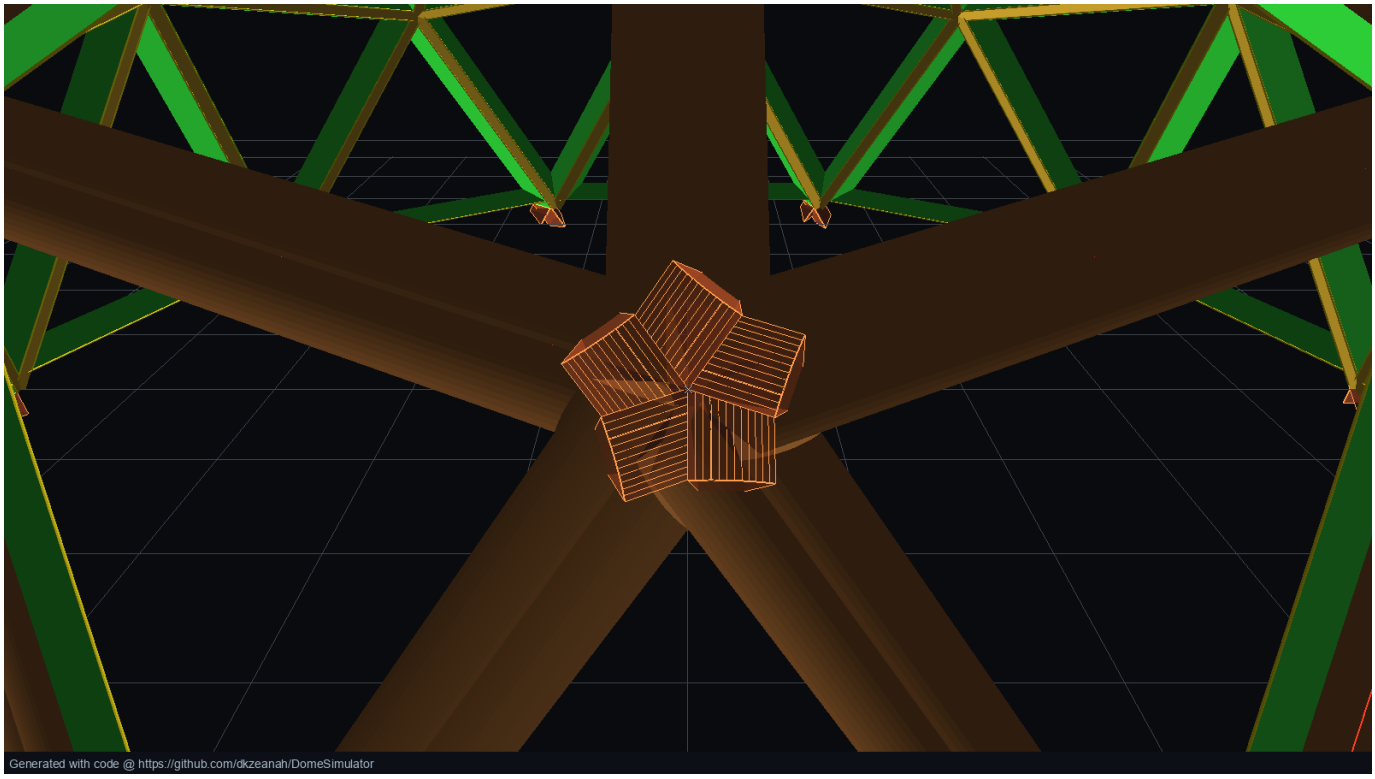

Figure 44. The reference build's six inches

The reference build's six inches. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Error checking

Errors in a dome do not stay where you put them.

A dome is a closed shape. Every panel has to agree with its neighbours all the way round, and when you get back to where you started, the last panel either fits or it does not. There is no slack to absorb a mistake and nowhere to hide it.

That makes the last panel an extremely good, extremely late test.

FINDING IT EARLIER

Check the sticks. Every tenth member against your master, as described earlier.

Check the panels. Every panel against the first good one. They should stack.

Check the rings. As you assemble, the base ring should close on itself and sit level. If the tenth base vertex does not land where the geometry says, stop. Do not force it. Something upstream is wrong and forcing it will push the error into the next ring.

THE ERRORS THAT ADD UP AND THE ONES THAT DO NOT

Random errors mostly cancel. If half your sticks are a sixteenth long and half are a sixteenth short, the dome closes and nobody notices.

Systematic errors do not cancel. If every stick is a sixteenth long because your fence is a sixteenth out, then by the time you have gone round the base ring you are ten sixteenths out — more than half an inch — and the last panel will tell you about it.

So the question to ask when something does not fit is not "how big is this error". It is "is this error in all of them".

Measure three sticks from different points in the run. If all three are long by the same amount, you have a systematic error and a fence to adjust. If they disagree randomly, you have a technique problem and a different fix.

WHEN TO STOP AND WHEN TO CARRY ON

Stop if a ring will not close, if a panel rocks, or if the same measurement comes out differently twice.

Carry on if a joint has a small gap that closes when the fastener pulls up, or if a member is a sixteenth under. Wood moves, joints take up, and a dome that is built to a sixteenth is built better than most houses.

The skill is telling those two apart, and it comes from having checked enough things that you know what normal looks like.

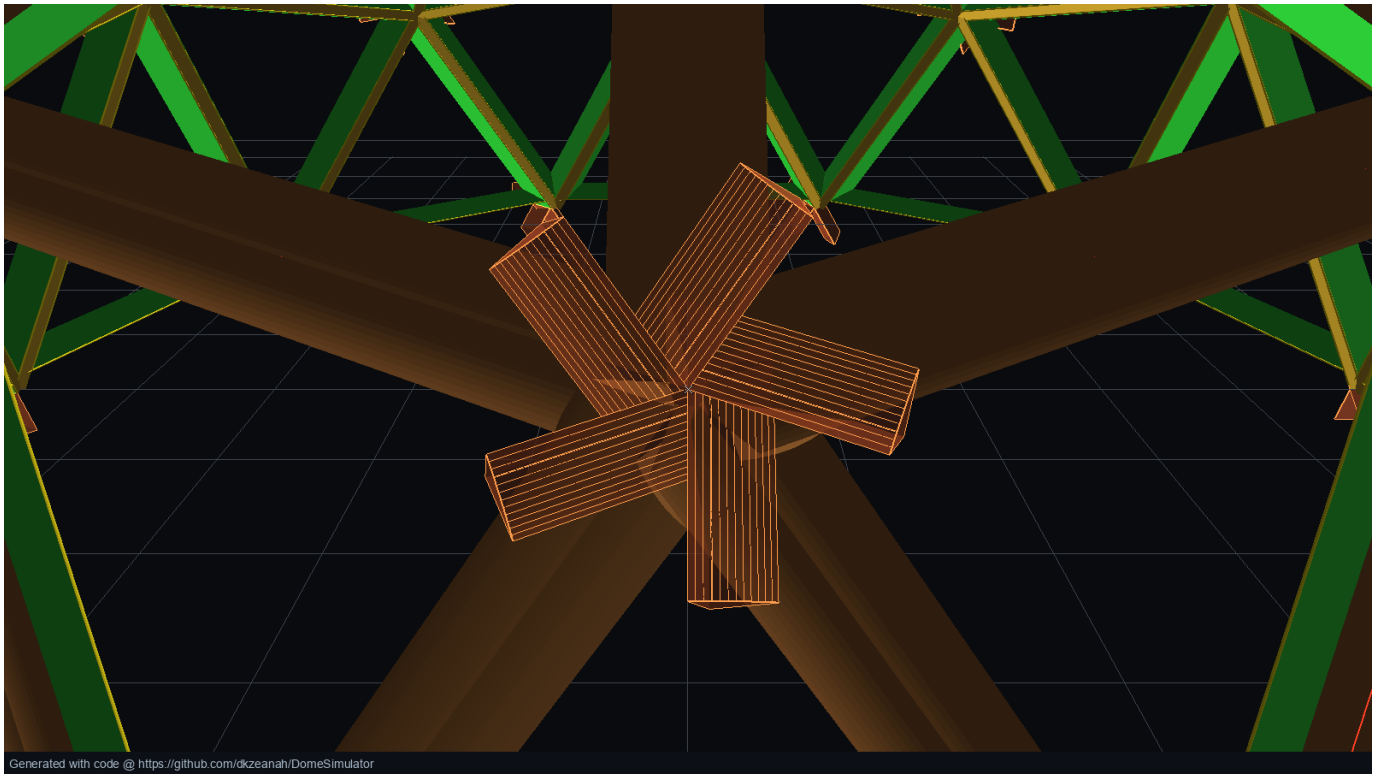

Figure 45. A foot of it

A foot of it: generous, and a foot of offcut a stick. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Mass-production workflow

Forty panels is a production run, not a project. Treat it like one and it goes quickly. Treat it like forty separate builds and it does not.

THE ORDER THAT WORKS

Do every operation on every piece before moving to the next operation.

Split all the wedges. Then cut all the A members to rough length. Then all the Bs. Then do every butting end. Then every receiving face. Then assemble.

This feels slower than building a panel at a time. It is substantially faster, for three reasons.

You set the jig once per operation instead of forty times. Setup is the expensive part of any cut; doing it once for sixty sticks rather than once per panel removes almost all of it.

Your hands learn. The tenth identical operation is faster than the first and the fiftieth is faster again. That only happens if you do them in a run.

Errors show up in batches. If a fence is wrong you find it on the third stick of an operation, not on the third panel — and you have three to redo instead of a panel to take apart.

THE STATIONS

Set up the shop as stations rather than as a bench.

- the splitting station, where logs become sectors
- the length station, where sectors become members
- the joint station, where members get their ends
- the assembly station, where members become panels
- the stack, where finished panels live

Material moves one way through those. Nothing goes backwards. If you find yourself carrying a stick back to a previous station, something is wrong with the sequence and it is worth stopping to fix.

WHERE THE TIME ACTUALLY GOES

Not where you expect.

The cuts are quick. Splitting is quick once you have done a few. What takes the time is handling — picking things up, putting them down, finding the one you want, deciding which pile it goes in.

So the highest-value thing in the whole shop is not a better saw. It is a place to put things down that keeps them sorted.

Two racks, labelled, within reach of the saw, will save you more hours than any tool you could buy.

THE REALISTIC SCHEDULE

The reference build assumes roughly 135 hours of labour for the frame, including splitting the timber.

That is not a weekend. It is three or four solid weeks of evenings, or two people for a week and a half. Anybody who tells you a dome frame goes up in a weekend is talking about a kit with the parts already cut.

7. Connections

The pinwheel joint carries the building through wood pressed against wood. The fasteners are not holding the dome up. They are holding the joint closed so the wood can.

That distinction runs through this whole chapter, and it is why a method with no hubs and no brackets is not missing anything. It is also why the fastener choices here are mostly about getting the joint apart again rather than about strength.

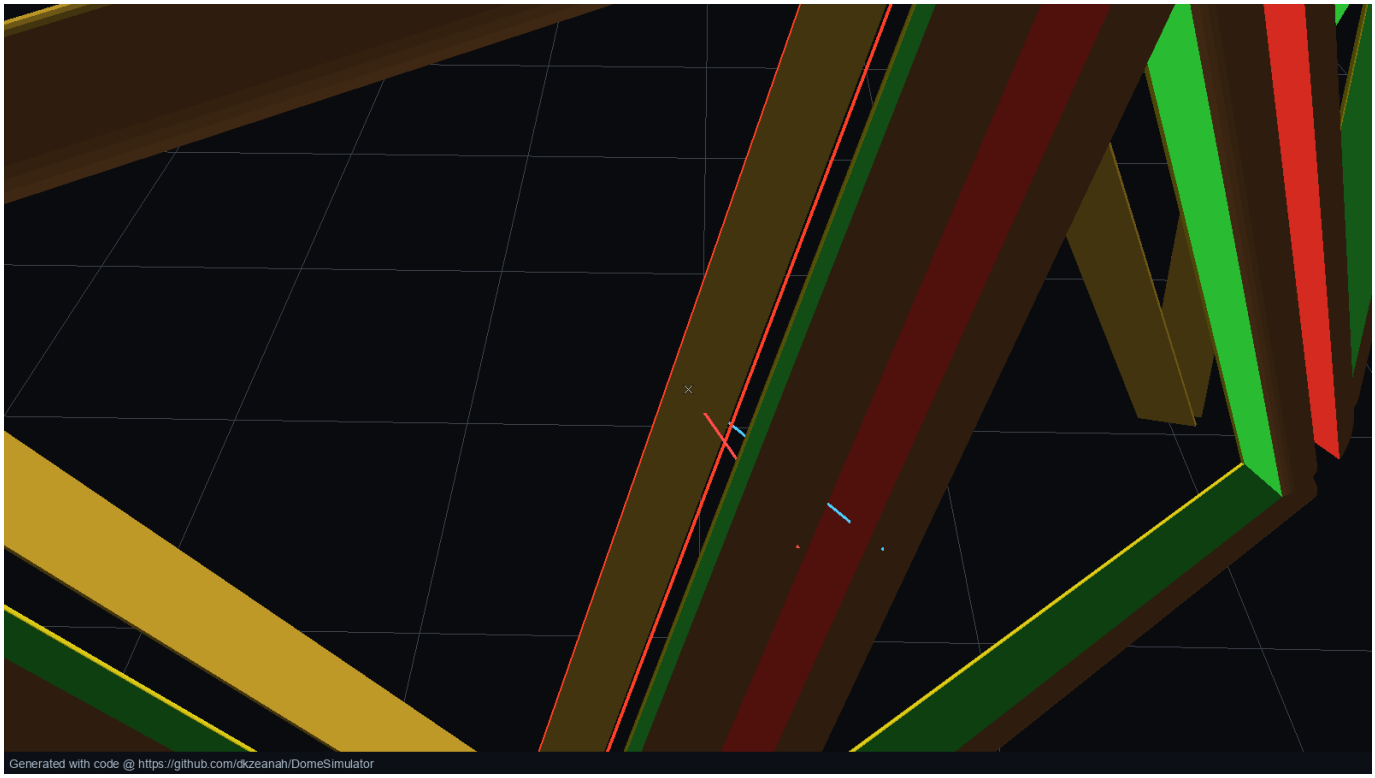

Figure 46. The key, in the space the wedge leaves

Two sawn faces meeting at a dihedral angle do not close flush. The frame is exploded here so you can see into the seam and see what fills it: a key 2.80 inches across its base. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Direct wood joints

The pinwheel joint is wood against wood. No plate, no hub, no bracket.

Three members meet at each corner of a triangle. Instead of all three running to a single point, each one butts into the side of the next, and they chase each other round the corner. That is the pinwheel.

WHAT IT BUYS YOU

Not what an earlier version of this section claimed. It said no end of any member is mitred and every cut is a plain angle, and that is not true. See the correction in the chapter on making the struts: the butt is a compound cut of a 22.5-degree bevel and one of 3 mitres.

What the pinwheel genuinely buys is this.

No shared vertex. Three members reach each corner and none of them meets another end-to-end. Every end lands on the flat side of a neighbour, so nothing is coped, shaved, brought to a point or fitted.

No hub. In a hub dome the angles live in the hub, and the hub is either an expensive casting or a piece of steel somebody has to fabricate accurately.

And a very short list of settings. 3 saw settings for 120 members, with the bevel constant because it comes from the split rather than from the shell.

The pinwheel moves the angle out of the fit and into the arrangement. The joints are dumb; the pattern is clever. That is a smaller claim than the one this section used to make and it is the one that survives contact with a real log.

WHAT IT COSTS

Eighty percent more timber.

A dome that shares one strut between two triangles uses 65 members and 369 feet of stock. The pinwheel gives every triangle its own three sticks: 120 members, 664 feet.

That is not a small tax and this book is not going to pretend it is. You are buying simple cuts with wood.

Whether that is a good trade depends entirely on where your wood comes from. If you are buying dimensional lumber by the foot, it is an expensive trade. If you are splitting your own trees, it is close to free, and the chapter on harvest explains why.

WHERE THE LOAD GOES

Each member is in compression along its length, and it delivers that compression into the side of its neighbour through a flat face.

Wood is strong in compression across the grain at these loads, and the contact area is large — the full width of the member. This is a much better loading condition than a fastener in shear, which is what most bracket systems rely on.

The fasteners in a pinwheel joint are not really carrying the building. They are holding the members against each other so that the wood can do the work. That distinction matters when you choose them.

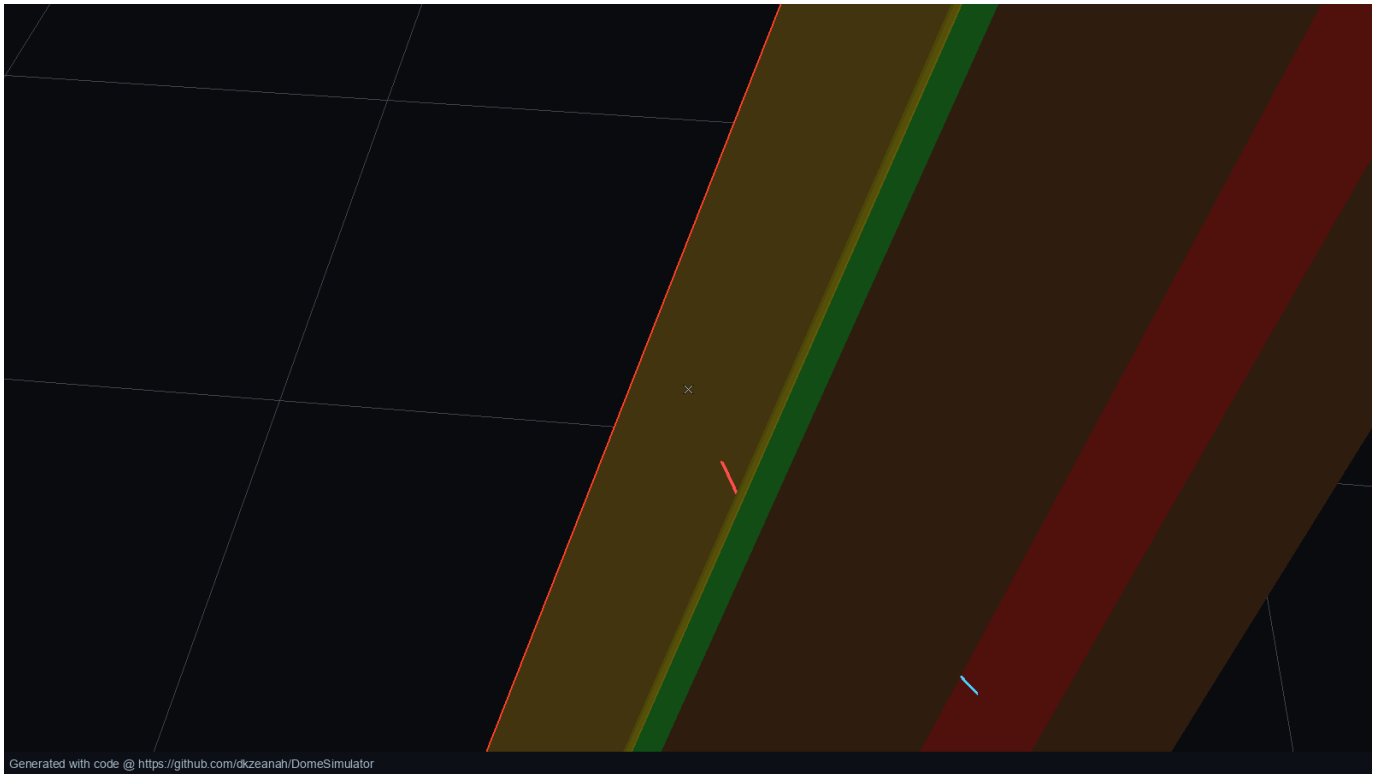

Figure 47. One seam, close

The same seam with the camera in the gap. The key is the truncated point of a raw sector, not a machined part -- it is cut from the same log as the members and at the same angle. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Screws

Screws are the default fastener for this method and they are the right one for most of the building.

WHY SCREWS AND NOT NAILS

You can take a screw out.

That is nearly the whole argument. A dome frame is a thing you will adjust, and the first one you build you will adjust a lot. A nailed joint is a decision; a screwed joint is a proposal.

Screws also pull. A nail holds two pieces where they are; a screw draws them together. In a pinwheel joint that matters, because the joint wants the butt face pressed hard against the receiving face, and a screw does that on its own.

WHAT NOT TO USE

Drywall screws. They are brittle, they snap under torque, and the head shears off leaving a stub you cannot get out. They cost almost nothing and they will cost you an afternoon.

Anything not rated for exterior use, anywhere that will ever be damp.

DRILL THE PILOT

In green timber, near an end, without a pilot hole, you will split the wood.

Not sometimes. Reliably.

Every fastener in a pinwheel joint is going in near the end of a member, which is the worst place for splitting, and split timber at a joint is a joint that has already failed. Pilot every one.

That is 120 joints with two or three fasteners each. It is a lot of holes and it is not optional.

LENGTH

The screw has to pass through the butting member and get well into the receiving one. On a member six inches deep, that means a long screw — and it means the screw is working across most of the joint, which is what you want.

Do not use a fastener that only just makes it. A screw that reaches two threads into the receiving member is a screw that will pull out when the wood dries and shrinks.

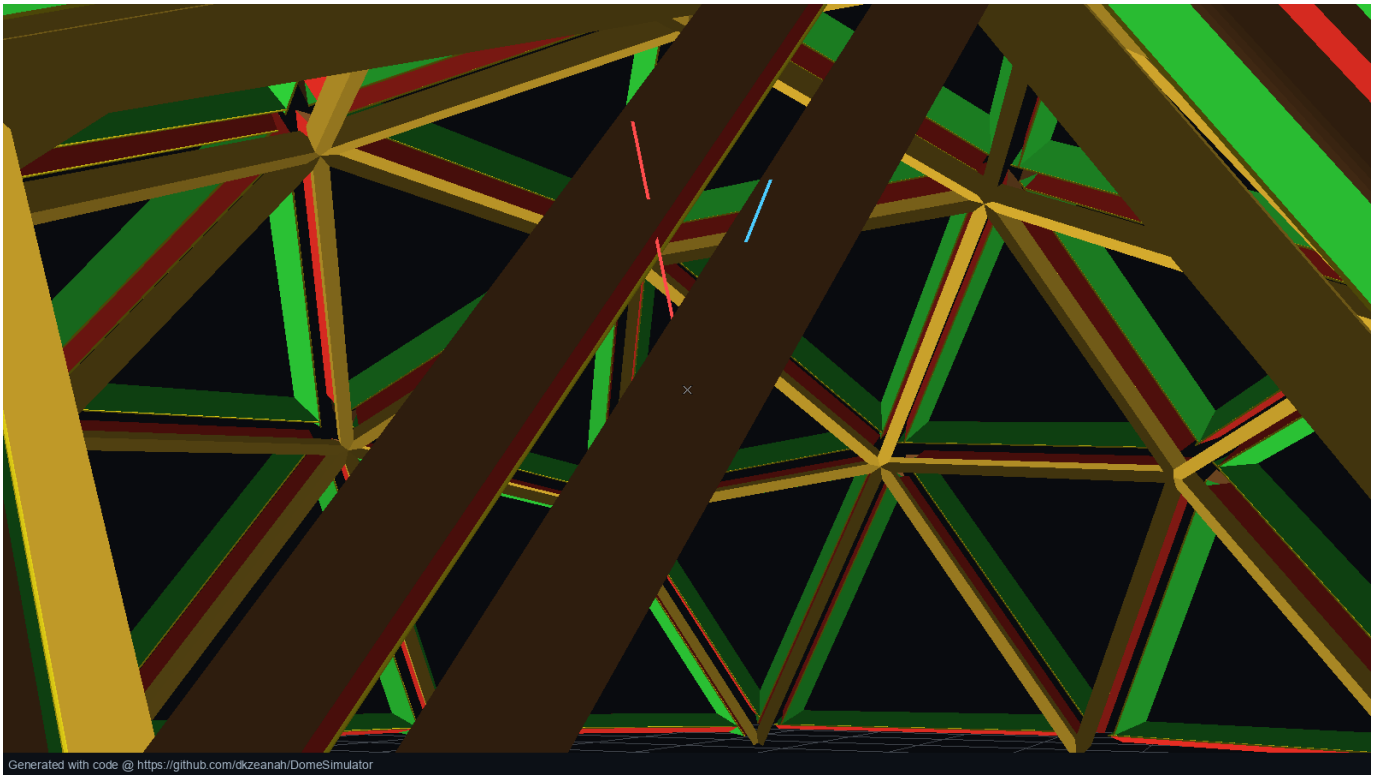

Figure 48. Every seam has one

The frame from underneath, opened eight inches. There are 55 interior seams in this dome and every one of them carries a key. They are not fasteners and there are no fasteners in this picture: the key is what holds the angle. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Structural screws

A structural screw is not a longer wood screw. It is a different product with a published capacity.

WHAT MAKES ONE STRUCTURAL

A stated, tested load rating. That is the difference that matters. A structural screw comes with a number you can design against; a hardware-store screw comes with a box.

They also have a hardened shank, a thread designed for withdrawal rather than just driving, and usually a head that will not cam out under real torque.

Most of them do not need a pilot hole in dimensional lumber. In green split timber near an end, drill one anyway.

WHERE TO USE THEM

At the pinwheel joints. All 120 of them.

This is the load path of the entire building and it is not the place to save four pounds. The joints are the only thing holding the dome's shape, and if one fails the load goes to its neighbours, which are already carrying their own.

WHERE NOT TO BOTHER

Everywhere else.

Battens, blocking, trim, temporary bracing, panel fixings — ordinary exterior-rated screws are fine and considerably cheaper. Using structural screws for everything is a way of spending money without making the building better.

THE HONEST LIMIT

This book cannot tell you what size or how many.

That depends on your timber species, your moisture content, your snow load, your wind exposure and your local code, and anyone who gives you a number without knowing those is guessing. What this book can tell you is that the joint is a butt in compression with fasteners holding it closed, and that is the information an engineer needs to size the fastener for you.

If the dome is a shed, use good structural screws at every joint and sleep fine. If people are going to live in it, get the fastener schedule stamped.

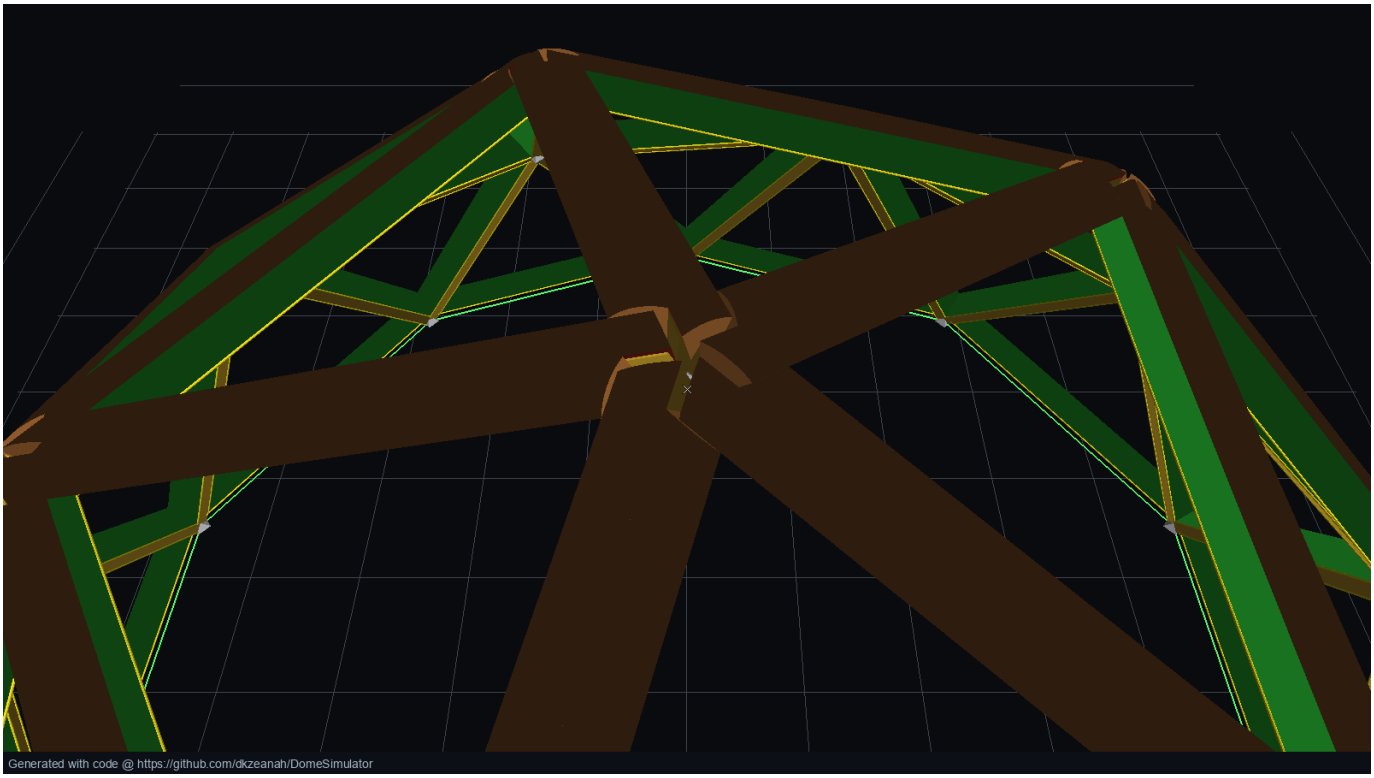

Figure 49. A five-member vertex

Where five triangles meet. Nothing arrives at the point: each member stops inside its own triangle and the corner is empty, which is what the pinwheel buys. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Bolts

A bolt goes all the way through and has a nut on the other side. That is the difference from a screw, and it changes what the joint can do.

WHEN A BOLT IS BETTER

When you want to take it apart and put it back together more than once.

A screw into wood loses holding power every time you remove and refit it — the threads cut a slightly bigger hole each time. A bolt does not care. The bolt bears on the hole and the nut does the clamping, and you can undo it a hundred times.

That matters for a dome you intend to move, and it matters enormously for the first dome you build, which you will take apart at least once.

WHEN A BOLT IS WORSE

When you cannot reach the back of it.

Every bolt needs access to both sides. In a finished dome with panels in, a lot of joints have one accessible side and one buried. A bolt there is a bolt you cannot ever tighten.

Plan the bolted joints before the panels go on, not after.

THE THREADED INSERT

There is a middle option and it is the one the reference build is heading toward.

Set a threaded insert into the receiving member — a metal sleeve with an internal thread, driven into a drilled hole. Then a bolt from the butting member threads into that.

The insert takes the thread so the wood never does. You get the repeatability of a bolt without needing access to the far side, and you can undo and remake the joint indefinitely without chewing out the hole.

Two per joint, 120 joints, 240 inserts.

BUY THE BOLTS LONG

If you think you might ever build a bigger dome, buy the bolts for the bigger one now.

A bolt cut thirty-five percent longer than this dome needs will still reach through the thicker members of the next two sizes up. The extra length costs pennies per bolt and saves re-buying 240 fasteners later.

That is the cheapest forward decision in the whole build and almost nobody makes it.

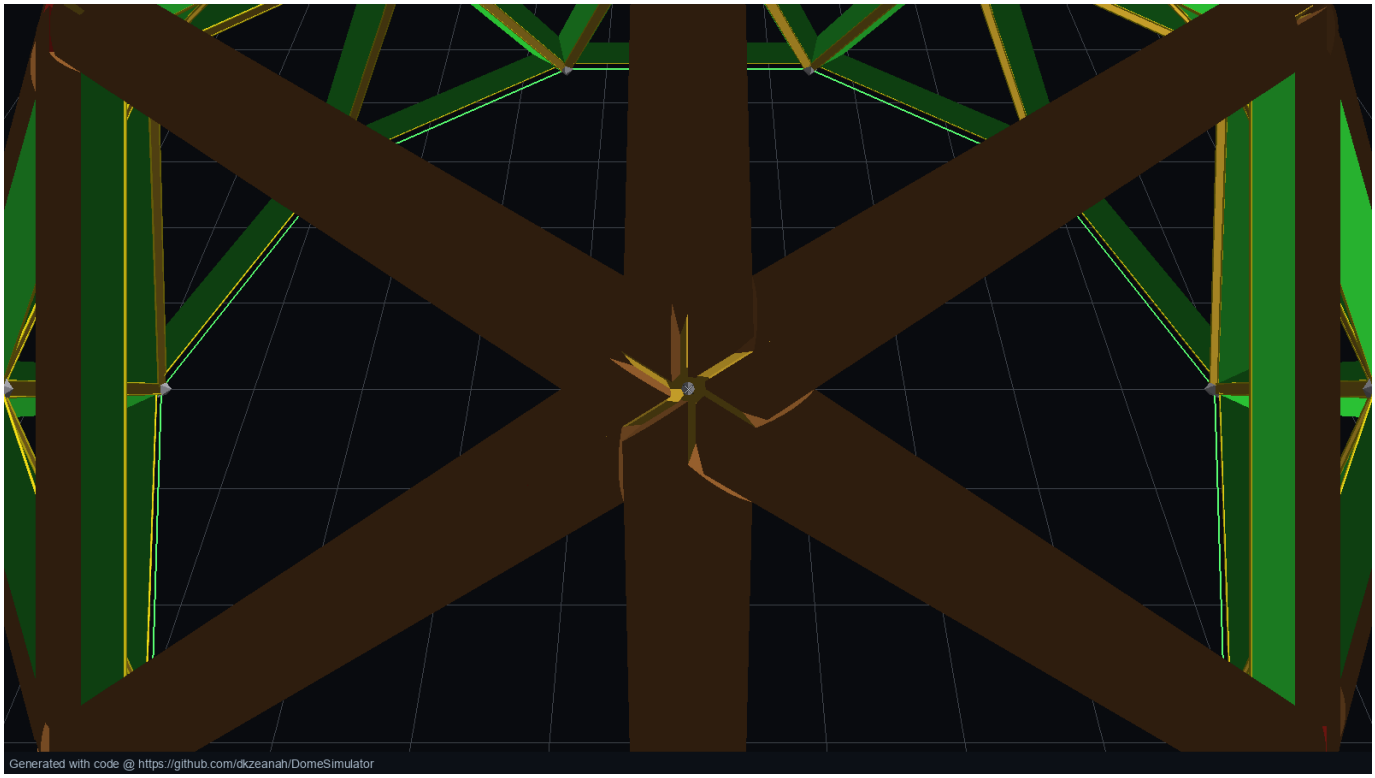

Figure 50. A six-member vertex

The other kind. Six triangles, same empty corner, same three cuts. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Washers

A washer spreads a point load into an area load. In softwood, that is not a refinement — it is the difference between a joint that holds and a fastener that slowly buries itself.

WHY IT MATTERS MORE HERE

Split timber has exposed grain on its sawn faces, and a bolt head bearing directly on end grain or near-end grain will crush into it under load and under time.

Once it has crushed in, the joint is loose. Retightening crushes it further.

A wide washer under every bolt head and every nut puts that same load over several square inches of wood instead of the area of a bolt head.

USE THEM WIDER THAN YOU THINK

The standard washer that comes taped to the bolt is sized for steel.

For timber, go wider. A washer twice the diameter of the standard one costs a few cents more and quadruples the bearing area.

WHERE THEY GO

Under every nut and under every bolt head. Both ends. Not just the nut side, which is the side people remember.

BEVELLED WASHERS

If the surface the washer sits on is not square to the bolt — and on a wedge it often is not, because the faces are sawn at the sector angle — a flat washer bears on one edge only.

A bevelled washer, or a small squared-off pad let into the face, fixes that. On the reference build most joints are close enough to square that a flat washer with a large enough diameter is fine. Check yours rather than assuming; put a washer on and look at whether it sits flat.

THE ONE-SENTENCE VERSION

If you find yourself retightening a bolt more than once, the wood under the washer has crushed, and a bigger washer would have prevented it.

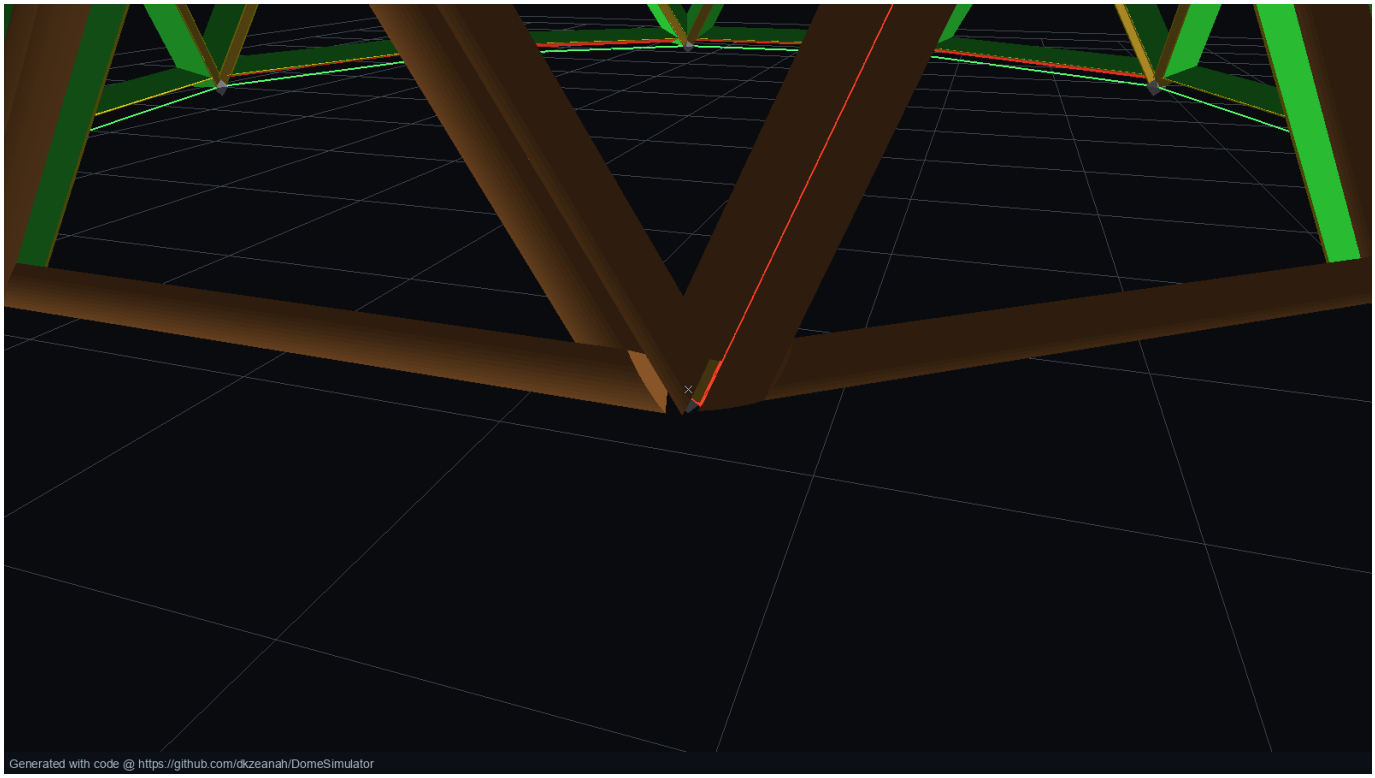

Figure 51. A base vertex

At the rim only three members meet, because there is no panel below. It is the one vertex a builder sets out on the ground. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Plates

A plate is a flat piece of metal spanning a joint, fastened to both members.

The wedge method does not need them. This section is about when you might want them anyway.

WHY THE METHOD DOES NOT NEED THEM

The pinwheel joint carries its load through wood in compression. A plate does not improve a compression joint — the wood is already doing the job, and the plate is along for the ride.

That is the whole point of the method. Hub domes need plates or castings because their joints are in shear and tension. Pinwheel joints are not.

WHEN TO ADD ONE ANYWAY

Three situations.

Repair. If a member has split at a joint, a plate across it is a reasonable way to bring it back into service without taking the dome apart.

Uplift. A dome is light and wind gets under it. The base ring joints — where the frame meets the floor — are the ones that see uplift rather than compression, and those are worth plating or strapping.

Somewhere a load arrives that the geometry did not plan for. A hoist point. A heavy door frame. A partition carrying a floor above. The frame will take those loads locally if you give it something to take

them with.

WHAT PLATES COST YOU

Visibility. A plate is visible from inside, forever, and a dome interior is mostly visible.

Thermal bridging, if the plate passes through the insulated envelope.

And a small honesty problem: a dome with plates everywhere is a dome that has stopped being a wedge dome and started being a bracket dome with wedge-shaped struts. If you find yourself plating every joint, the method is not working and the reason is worth finding rather than covering.

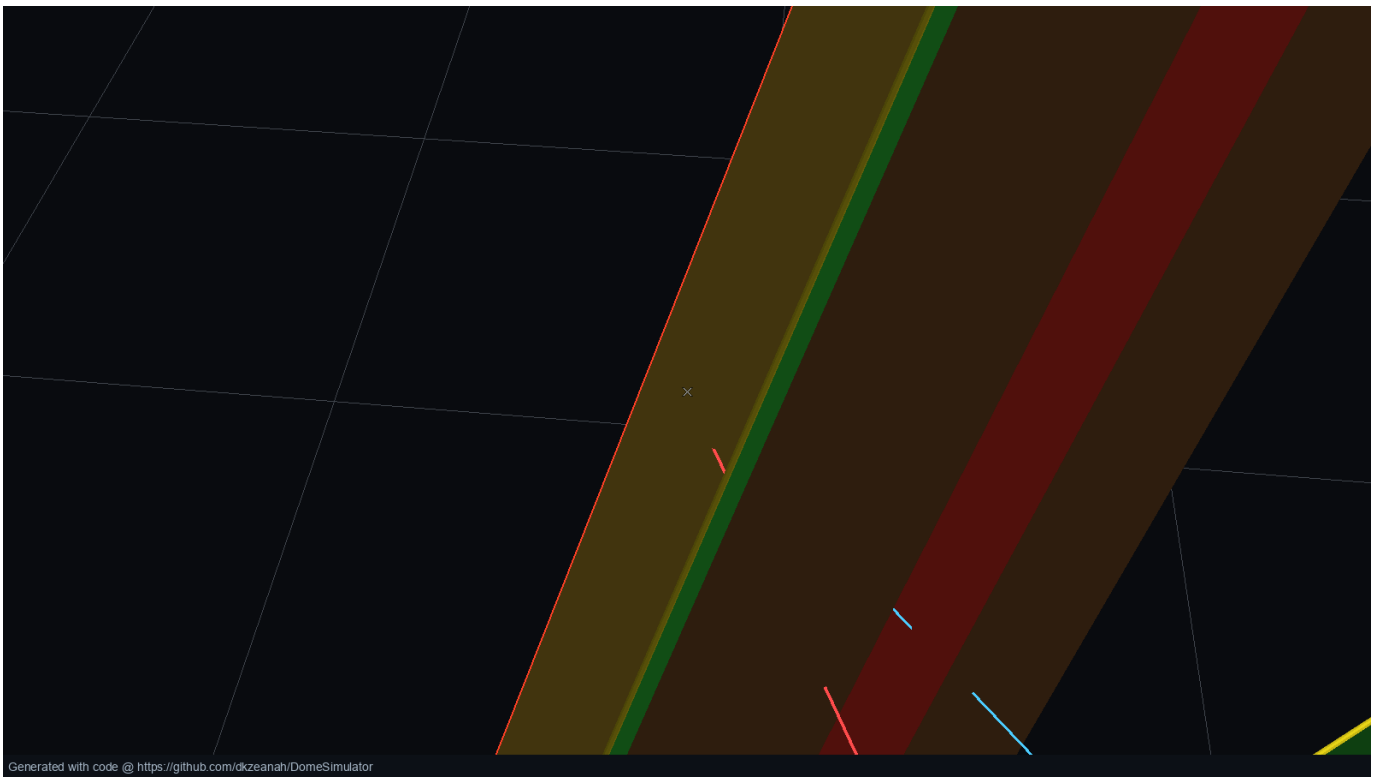

Figure 52. The raw sector, point truncated

The raw sector, point truncated. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Internal brackets

An internal bracket sits inside the joint — between the members, hidden once assembled.

WHAT THEY ARE FOR

Alignment, mostly. A small internal plate or dowel that locates one member against another makes assembly faster and stops a joint sliding out of position while you drive the fastener.

On a 120-joint frame, anything that makes a joint self-locating is worth real money in time.

THE DOWEL VERSION

The simplest internal bracket is a dowel: a hole in the butting face, a matching hole in the receiving face, a hardwood pin between them.

The pin does not carry much load. What it does is hold the joint in exactly one position while you fasten it, which means every joint ends up in the same place rather than wherever it drifted to.

That is a repeatability tool disguised as a fastener.

THE COST

Every internal bracket is invisible once the dome is together, which sounds like an advantage and is also a problem: you cannot inspect it.

If an internal bracket corrodes, splits the wood around it, or works loose, you will find out from the outside — a joint that has moved — long after it started.

So: internal brackets for alignment, external fasteners for load. Then anything carrying the building can be looked at and tightened without taking the building apart.

THAT IS THE RULE

Hidden parts should not be load-bearing parts.

Not because hidden parts fail more often. Because when they do, you cannot tell.

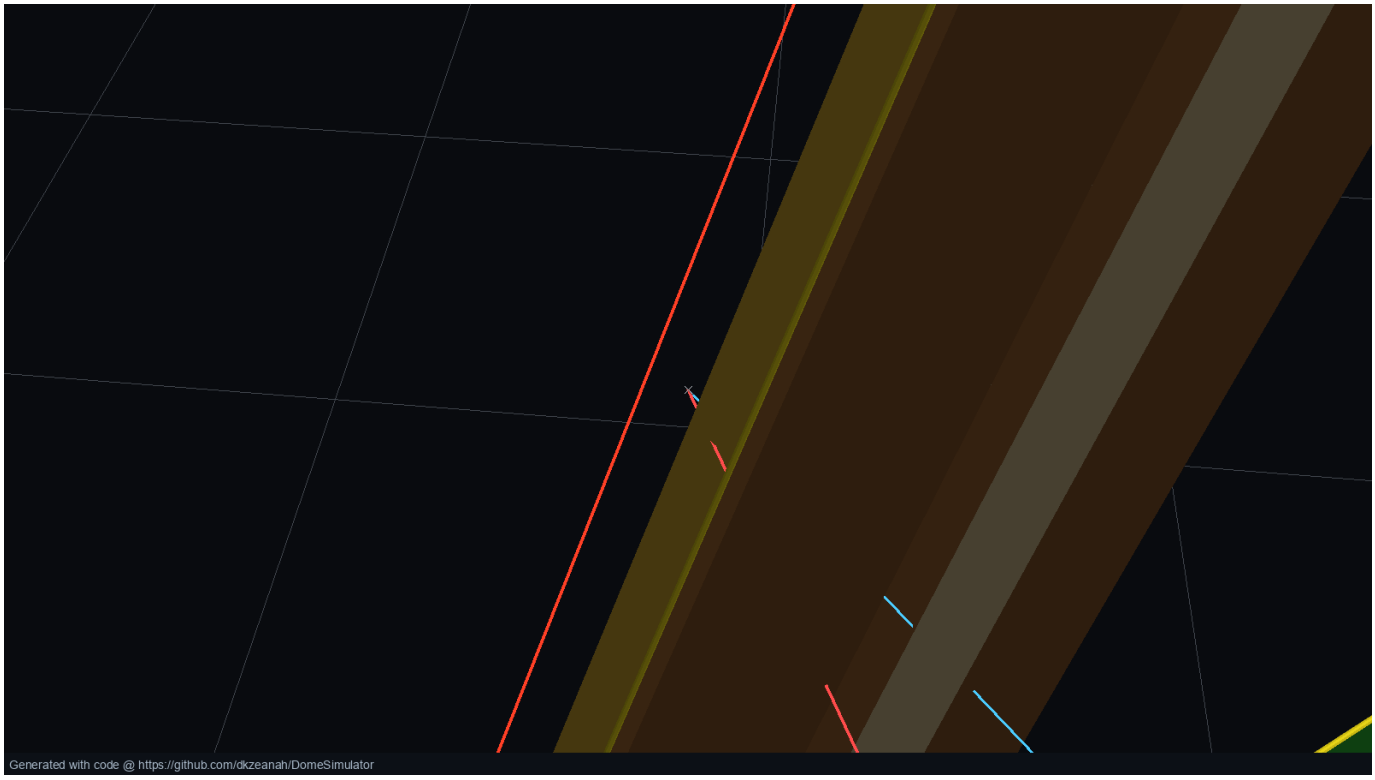

Figure 53. A machined flat key

A machined flat key. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

External brackets

An external bracket is fastened on the outside of the joint, visible and reachable.

THE ARGUMENT FOR THEM

You can see them and you can get at them.

Every fastener that carries load should be one you can inspect, tighten and replace without dismantling anything. An external bracket is the most honest version of that.

On a building you intend to take apart and move — which this one is — that matters more than on a permanent house. A dome that comes down in a weekend needs every structural fastener reachable with a driver.

THE ARGUMENT AGAINST

They are on the outside, which means they are in the weather and they are in the way.

Weather: any bracket outside the envelope needs to be corrosion-rated for the climate, and the fastener holes through the timber are water paths unless they are sealed or shielded.

In the way: the outside of the frame is where the skin goes. A bracket standing proud of the member is a lump the panel has to accommodate, and forty panels each accommodating three lumps is a lot of fiddling.

THE COMPROMISE THE REFERENCE BUILD MAKES

Fasteners on the inside face, where they are reachable from within the room and out of the weather entirely.

The inside of a dome is accessible everywhere — there is no roof space and no wall cavity. So the inside face is both the sheltered position and the convenient one, which is unusual and worth taking advantage of.

The outside stays clean for the skin to land on.

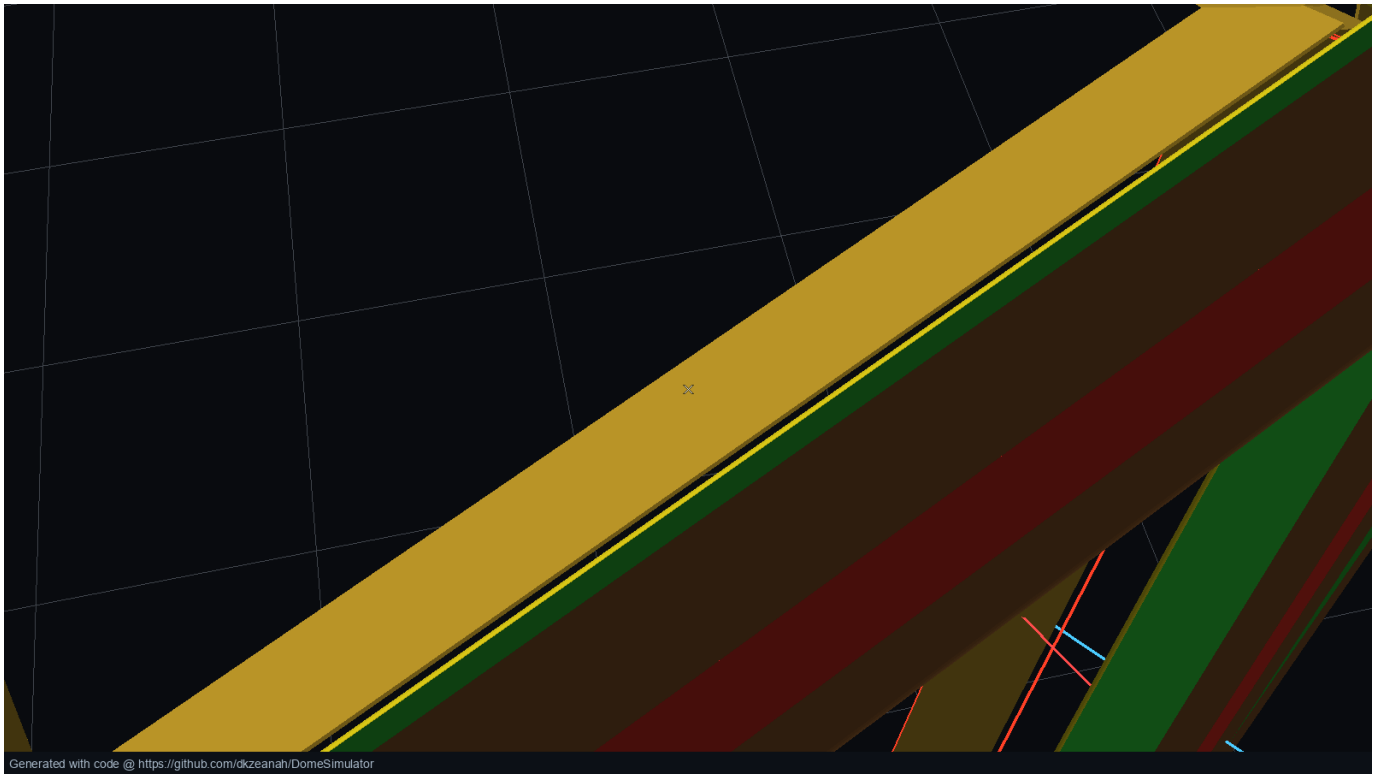

Figure 54. A rigid spline

A rigid spline. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Temporary assembly joints

A dome does not stand up until it is nearly finished.

Partway through assembly you have a shape that is not yet closed, not yet stable, and entirely dependent on whatever is propping it. Temporary joints are how you survive that.

WHAT NEEDS PROPPING

The base ring, until it closes. Ten panels standing on edge in a circle want to fall inward, and until the tenth one goes in there is nothing stopping them.

Each subsequent ring, until it closes. Same reason.

The whole dome, until the top ring goes in. A dome without its apex is an arch with a hole in it, and arches push outward at the base.

THE SIMPLEST RIG

A post in the middle with a temporary platform on it, at the height of each ring as you build it.

You build the base ring around the post, resting on nothing. You build the next ring resting on the platform. You raise the platform. Repeat.

That is how domes have been put up by one person for sixty years and there is no reason to improve on it.

TEMPORARY FASTENERS

Use a screw that comes out easily and a different colour of screw from the permanent ones.

Different colour matters more than it sounds. At the end of the build you need to remove every temporary fastener and leave every permanent one, and there will be several hundred fasteners in the building. If they all look the same you will either leave temporaries in or take permanents out, and both are bad in different ways.

THE ORDER TO REMOVE THEM

From the top down, once the dome is closed.

The dome becomes self-supporting when the last panel goes in, not before. Removing bracing from the bottom while the top is still open takes the props out from under the thing that is still pushing outward.

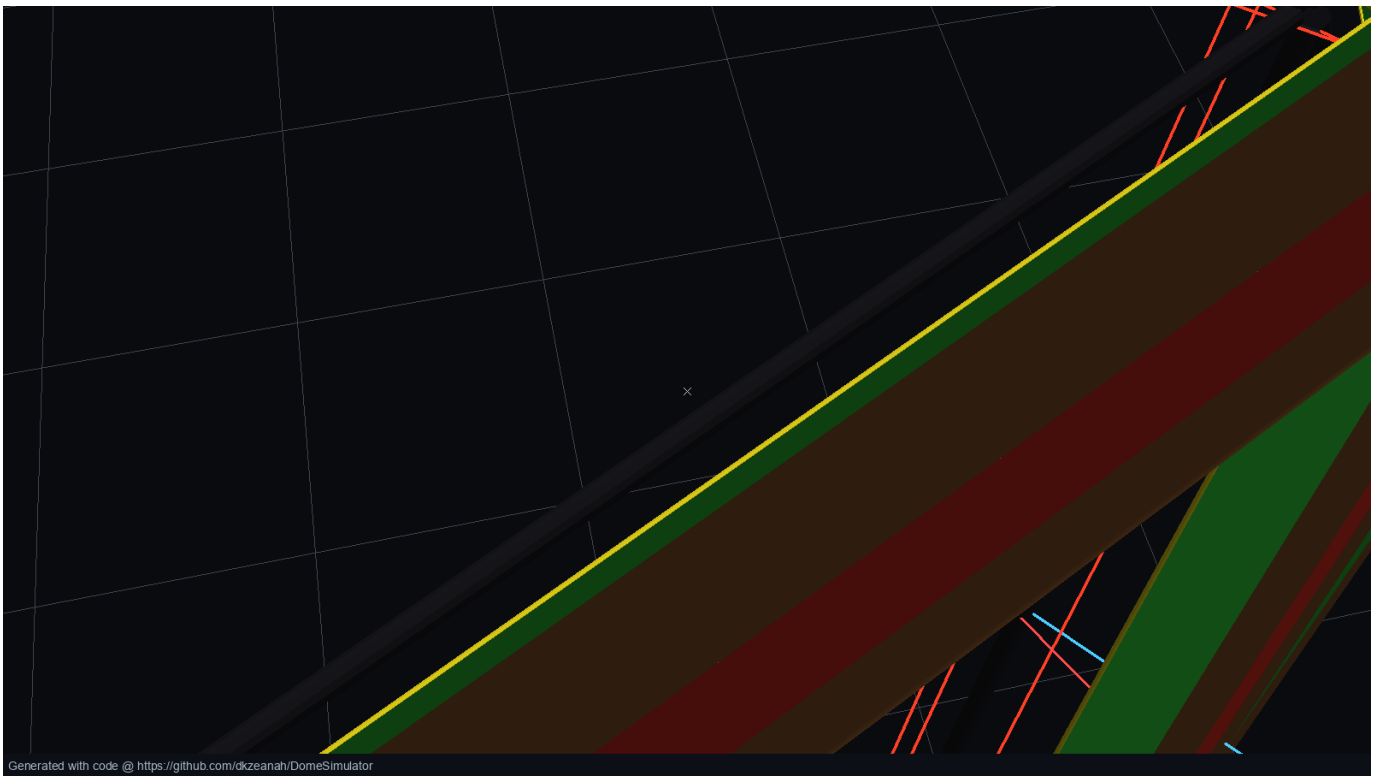

Figure 55. A hose, which is also the air and water route

A hose, which is also the air and water route. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Replaceable connections

Design every joint on the assumption that somebody will undo it.

This is the section that separates a dome you build from a dome you can live with for twenty years.

WHY ANYTHING GETS UNDONE

Wood moves. Green timber especially — it dries, it shrinks across the grain, and joints that were tight in month one are loose in month eighteen. That is not a failure; it is what wood does, and it means every

joint gets tightened at least once.

Things get damaged. A member splits, a panel gets hit, water finds a way in. Replacing one member should not require dismantling a quarter of the building.

Buildings change. A door moves, a window is added, a partition goes in.

WHAT MAKES A CONNECTION REPLACEABLE

Three things, and they are all decisions you make before assembly.

Access. The fastener can be reached with a driver, with the building standing and the skin on. If it cannot, it is not replaceable no matter what it is made of.

A thread that does not live in the wood. A screw straight into timber can be removed and refitted a few times before the hole is chewed. A threaded insert can be done indefinitely, because the thread is in metal. That is the whole argument for inserts and it is worth the extra cost per joint.

Standard parts. If the fastener is something you can buy in ten years, the joint is replaceable. If it is a custom bracket from a supplier who may not exist, it is not.

THE ORDER OF DISASSEMBLY

A dome comes apart in the reverse of the order it went together, and it is worth knowing that before you need it.

Top ring first, then down. If you take out a middle ring first, the rings above it have nothing to stand on.

That is also why the fasteners go on the inside face: at every stage of taking the dome apart, the joints you need to reach are reachable from inside the shrinking building.

THE ONE-LINE TEST

Ask of every joint: if this member split tomorrow, what would it take to replace it?

If the answer is more than an hour and a driver, change the joint now while it is still a drawing.

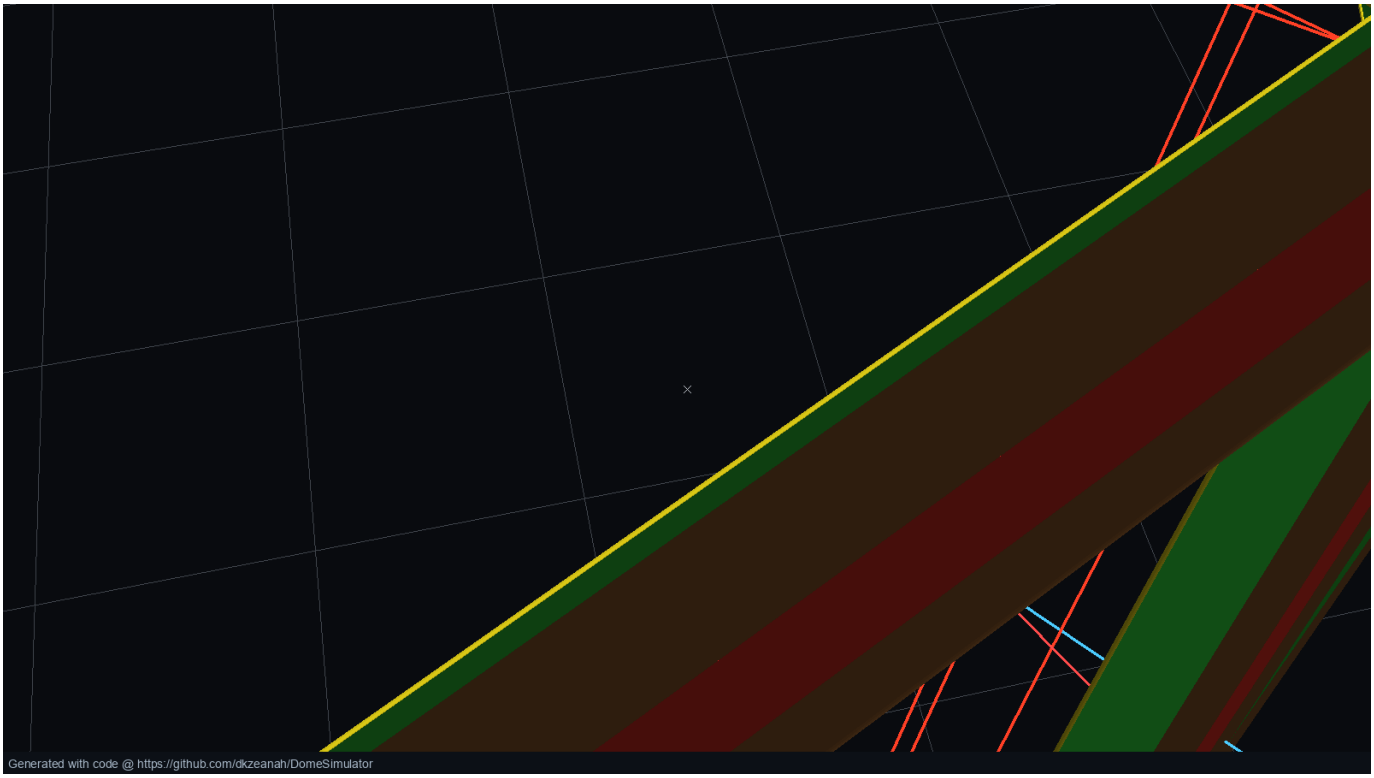

Figure 56. Left open

Left open. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Every other way: what fills the seam

The gap two sawn faces leave is 309 feet long in this dome and it exists whether or not anything is put in it. These are what the solver will put there.

EVERY OTHER WAY OF DOING IT

```
rigid            a rigid spline; the part is a key 2.80 in across its base   <- the reference build
hose             a hose, which is also the air and water route; the part is a line up to 2.88 in outside diameter
none             left open; the part is nothing; the channel stays open
raw_trapezoid    the raw sector, point truncated; the key is 2.80 in wide   <- the reference build
shaved_flat      a machined flat key; the key is 1.00 in wide
```

These change the frame itself. Every other choice in this book can be made after the dome is standing; these cannot be made twice.

WHERE THESE NUMBERS COME FROM

the raw-wedge solver's own configuration space, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

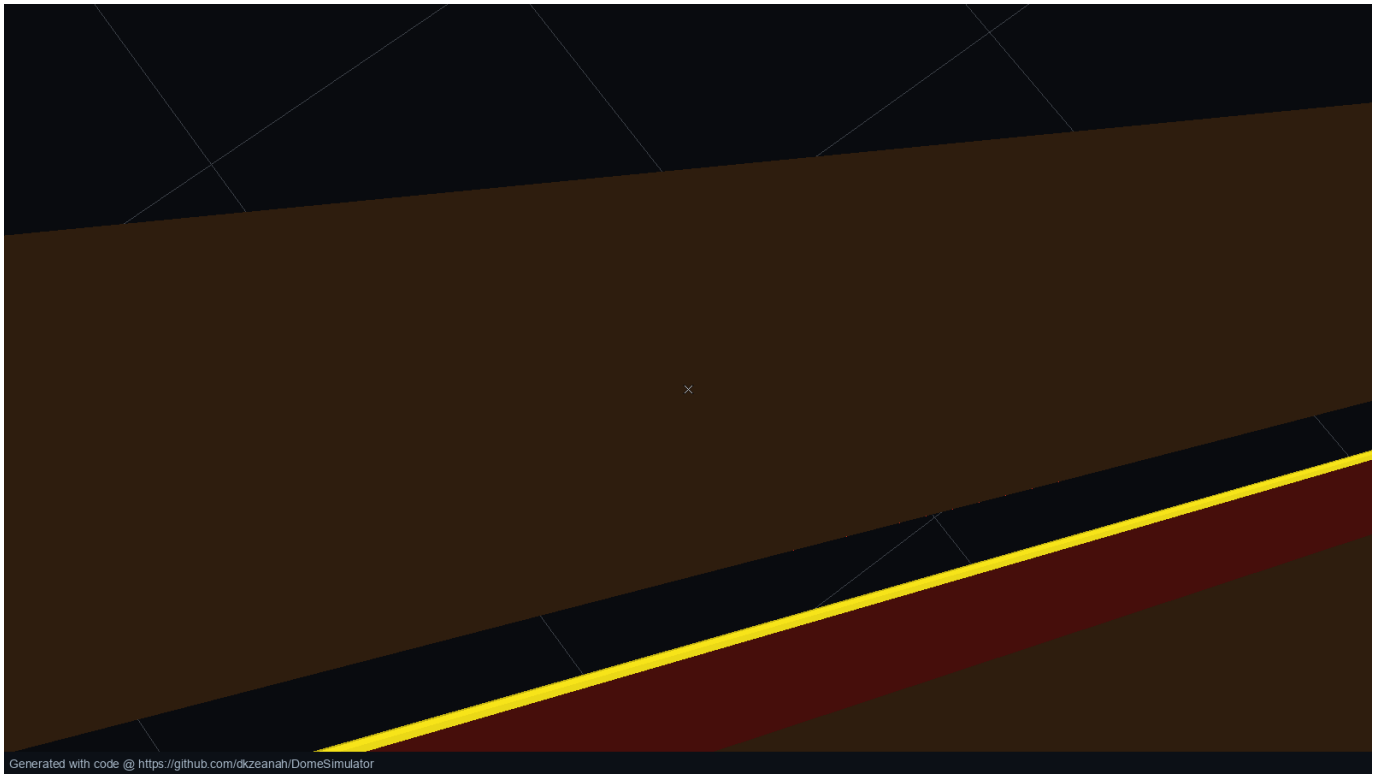

Figure 57. Barely truncated

Barely truncated: a fine point, easy to split. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

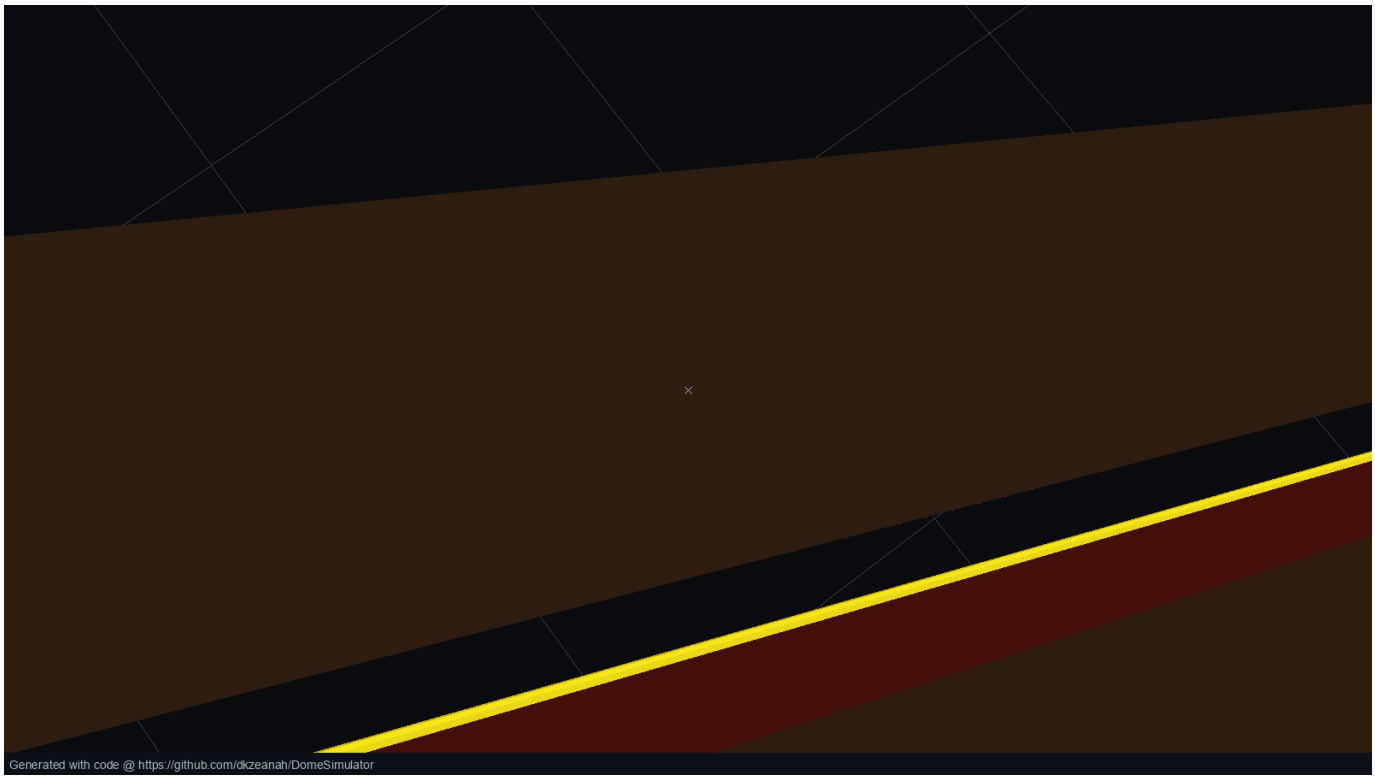

Figure 58. The reference build

The reference build. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

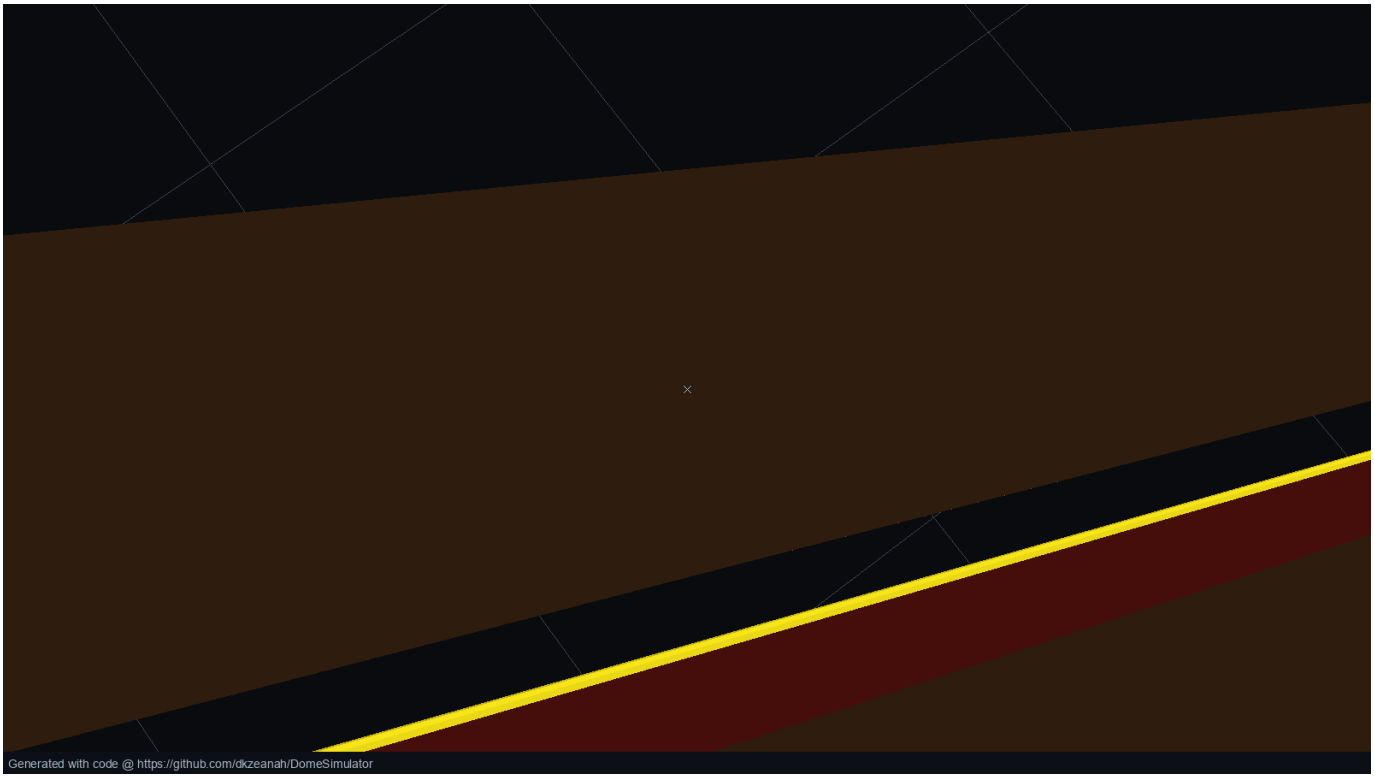

Figure 59. A deep flat

A deep flat: strong, and it stands proud of the ridge. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

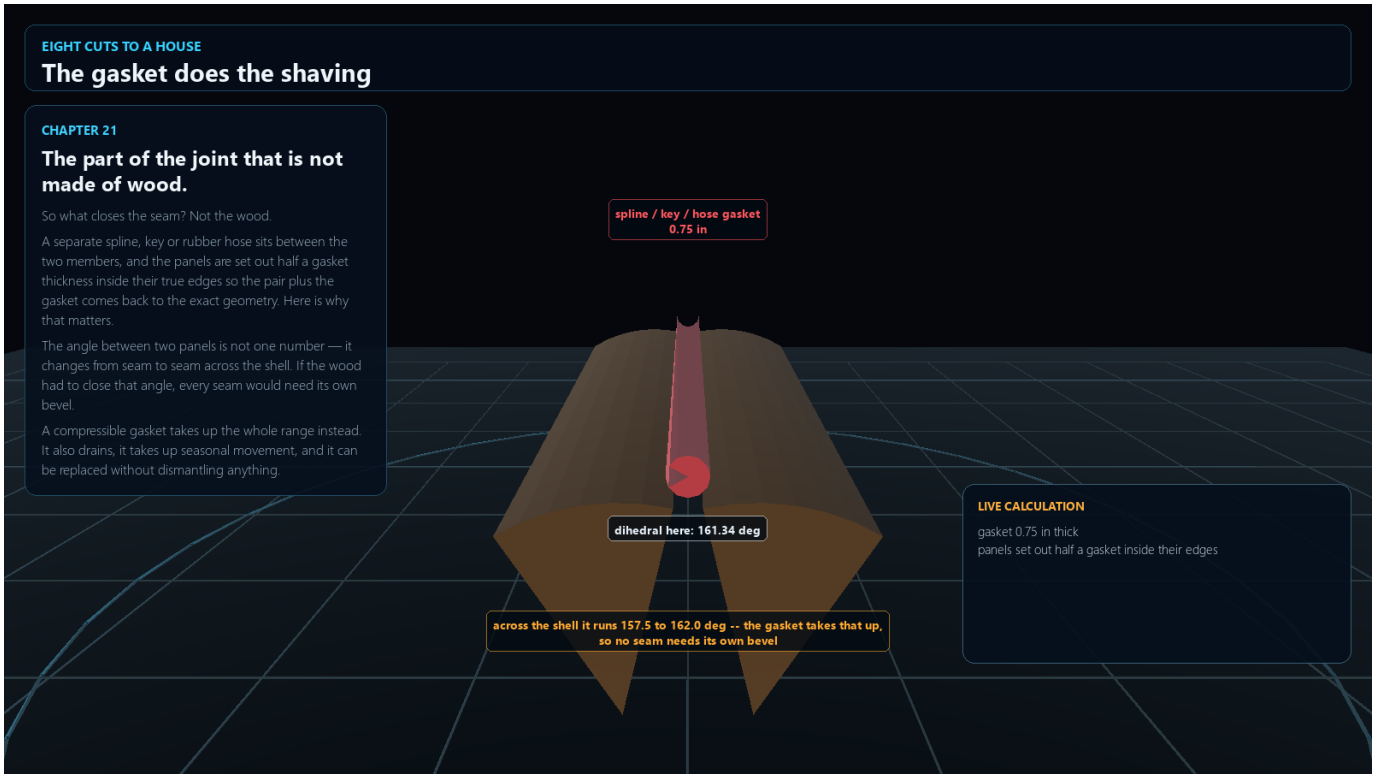

Figure 60. The gasket does the shaving

The part of the joint that is not made of wood. So what closes the seam? Not the wood. From Eight Cuts to a House: A Dome Straight From the Tree, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

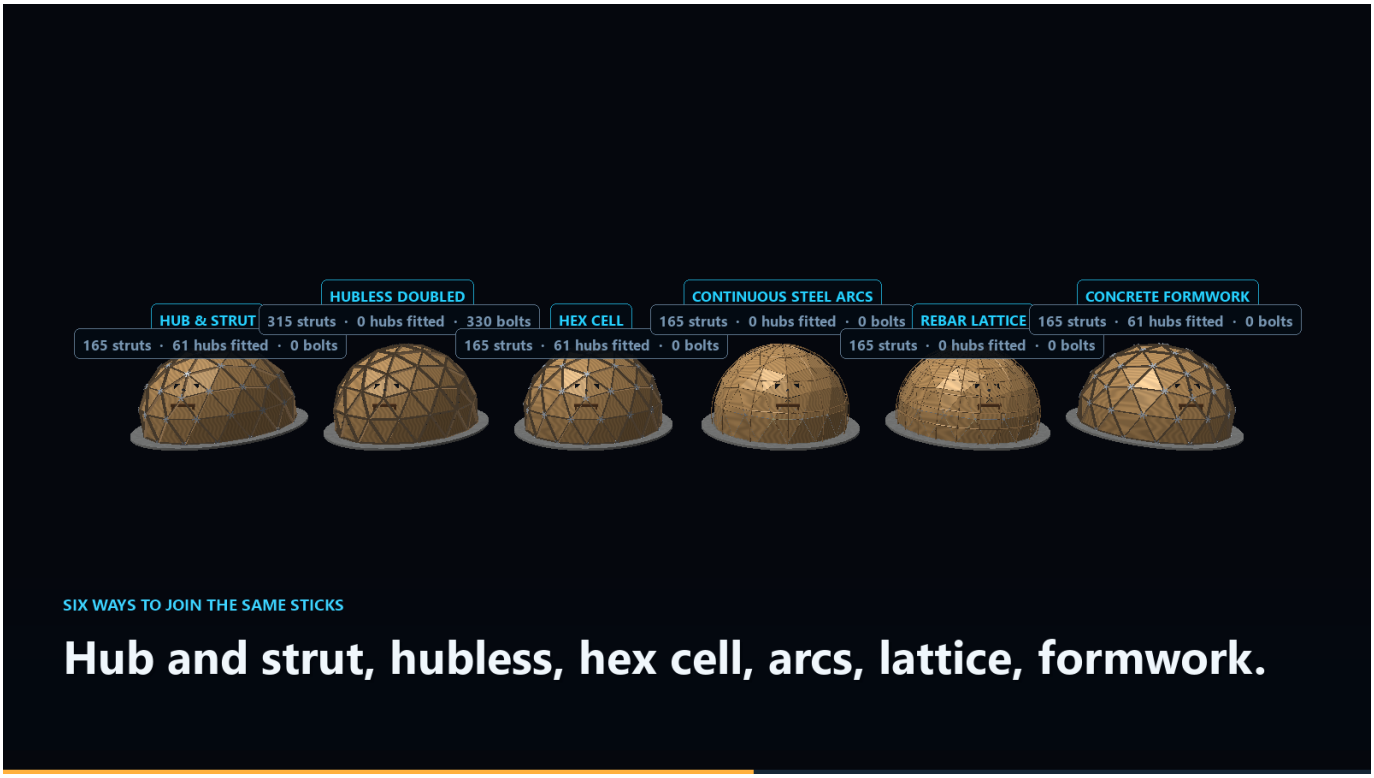

Figure 61. Six ways to join the same sticks

Hub and strut, hubless, hex cell, arcs, lattice, formwork. The frame style menu changes how the members meet, and it changes the. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Part III - Building the Dome

8. Laying Out the Base

A dome is laid out from one point and one distance. Get those two right and the rest of the site work is bookkeeping.

The rule that runs through this chapter: measure everything from a datum, never from the last thing you measured. Chained measurements accumulate error. A datum does not.

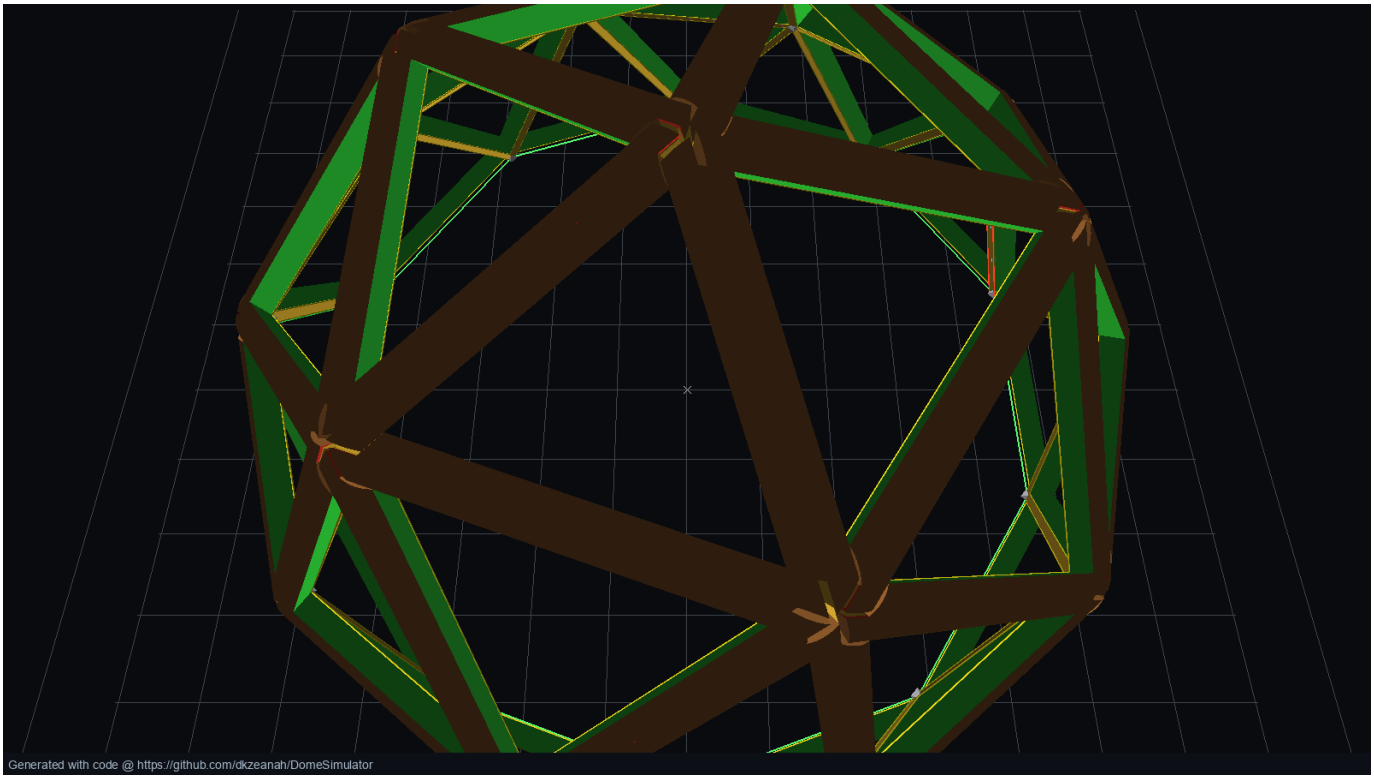

Figure 62. The base ring from above

The ten base edges seen from overhead: a regular decagon 72.000 inches to a side, 60 feet around, which is the whole of the setting-out. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Finding center

Everything on the site is measured from one stake.

Put it in properly, because you cannot move it later without moving the building.

WHAT THE CENTRE IS

It is the point directly below the apex, and it is the point every base vertex is the same distance from.

On the reference build every base vertex is 116.4984 inches from it — the sphere radius — measured horizontally along the ground.

HOW TO SET IT

Drive a stake. Put a nail in the top of it, dead centre, and use the nail as the point rather than the stake. A stake is an inch across; a nail is a point.

Then leave it alone. Everything references it and anything that knocks it moves the whole building.

WHAT TO CHECK FIRST

Before you commit: stand at the centre and turn round.

Is there room? The dome is 19.416 feet across, which means 9.708 feet in every direction from where you are standing, plus room to work outside that, plus room for the eaves of whatever you skin it with.

Check the low side. Ground that looks flat is rarely flat, and a dome needs its base ring level all the way round.

Check what is above. A dome is 9.708 feet tall at the apex and branches do not move.

THE THING PEOPLE SKIP

Check the centre against the services.

The reference build brings power, water and drain up through a single port in the middle of the floor. That port is at the centre stake. If there is a rock, a root or an existing pipe there, you want to know before the frame is up and not after.

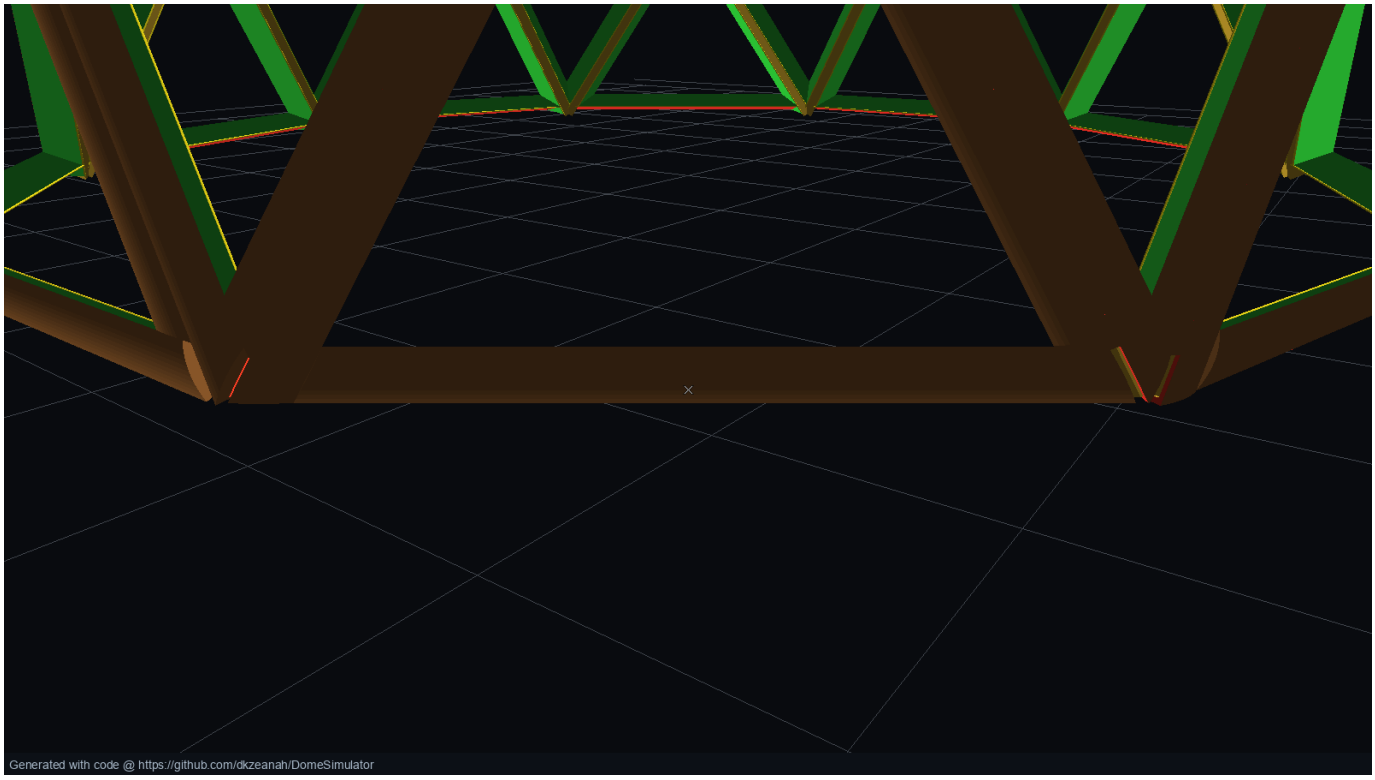

Figure 63. One base edge, on the ground

Where a panel lands on the rim. The base edge is an A chord and the two members standing off it are the panel's own, so the ground plate carries two sticks and not a vertex. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Radius layout

A string and a stake will lay out a dome more accurately than most survey equipment, provided the string does not stretch.

THE METHOD

Tie a non-stretch line to the centre nail. Mark it at the radius — 116.4984 inches on the reference build, which is 9 feet 8.5 inches.

Swing it round and scribe a circle.

Every base vertex lands on that circle.

WHY THE LINE MATTERS

Ordinary string stretches. Over nine and a half feet, under the tension you pull it at, mason's line can add most of an inch — and it will add a different amount each time depending on how hard you pull.

Use a steel tape, a fibreglass tape, or a wire. Something that returns the same number twice.

Better still: use a rigid stick. Cut a piece of timber to exactly the radius, pivot it on the centre nail, and mark from the end of it. A stick does not stretch at all, and for a dome this size a stick is entirely practical.

CHECK IT BY CLOSING

Scribe the whole circle and come back to where you started. The line should meet itself.

If it does not, your pivot has moved, and you should find out why before you put ten vertices on it.

THE SECOND CHECK

Measure across the circle through the centre. It should be twice the radius: 19.416 feet.

Do that in four directions. All four should agree. If one is different, the centre is not where you think it is.

Two minutes of checking here prevents a base ring that will not close, which is a problem you otherwise discover with ten panels built.

Polygon/base geometry

The base of a 2V hemisphere is a ten-sided polygon, not a circle.

Ten vertices on the circle, and straight members between them.

THE NUMBERS

Ten base vertices, evenly spaced — 36 degrees apart around the circle.

Ten base members between them, and on the reference build the base perimeter is 60.0 feet, which makes each base member 6.0 feet. That is the same as the A chord, and it is not a coincidence: the base edges of a 2V hemisphere are A edges.

The polygon encloses 277.0 square feet. The circle through the same ten points encloses 296.1.

Use the decagon figure for flooring. Use the circle only if you are working out how much space the dome needs on site.

MARKING THE TEN POINTS

Do not step a chord round the circle with dividers. Error accumulates and the tenth point will not land on the first.

Two better ways.

Measure the chord: mark one point, measure 6.0 feet along the circle's circumference to the next, and so on — but check at the halfway point that you are diametrically opposite where you started.

Or divide by halving: mark two points at opposite ends of a diameter. Bisect each half to get four. You cannot get to ten by halving, so this only takes you so far — but establishing an accurate diameter first and working from both ends toward the middle beats stepping round in one direction.

THE CLOSING CHECK

The tenth base member should fit between the tenth vertex and the first without being cut, forced, or persuaded.

If it does not fit, the error is spread through all ten and you should re-lay the base rather than absorb it in one joint. A forced base ring pushes its error up into every ring above it.

Foundation interfaces

The frame has to sit on something, and what it sits on has to do three things: hold it level, keep it off the ground, and hold it down.

LEVEL

Every base vertex at the same height. Not approximately — a dome is a closed shape and a base that is out of level puts a twist into the whole frame.

If the ground is not level, the foundation makes up the difference, not the frame. Do not shim individual members to fix a sloping site.

OFF THE GROUND

Timber in contact with soil rots. Split timber with exposed grain rots faster.

The base members should be clear of the ground with air under them, sitting on piers, blocks or a deck rather than bearing on soil.

The reference build stands the frame on a framed platform, which puts the base ring six hundred-odd millimetres clear of the ground with ventilated space underneath.

HELD DOWN

This is the one people forget.

A dome is light and it is a very good aerofoil. The load case that matters on a dome is not weight pushing down; it is wind lifting up.

Every base vertex needs a hold-down to the foundation — a strap, an anchor, a bolt into the deck. Ten of them.

The reference build uses a strap, ratchet and ground anchor at each of the ten base vertices, because the soft shell it uses has no stiffness of its own and the frame carries the whole uplift.

THE INTERFACE ITSELF

Keep it simple and keep it reachable.

A bracket that bolts to the deck and takes the base member with two fasteners is enough, and it can be inspected and tightened with the building standing.

Anything cast in, buried, or inaccessible is a joint you will never check again, and this is the joint that stops the building leaving.

Figure 64. The ground is not ours and it is not yours

The landowner builds the pad. It stays when the dome leaves. Here is the half of tiny-house pricing that everybody adds together and should not. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Leveling

Level the foundation, not the frame.

The frame is a rigid closed shape. If you try to level it by adjusting members you are distorting the geometry, and the distortion goes somewhere — usually into the apex, which will not close.

HOW LEVEL IS LEVEL ENOUGH

Every base vertex within an eighth of an inch of the others.

That is tighter than it sounds and looser than perfectionism. An eighth over nineteen feet is a very flat platform, and it is achievable with ordinary tools.

TOOLS THAT WORK

A water level — two clear tubes and a length of hose — is accurate over this span, costs almost nothing, and does not care about line of sight.

A rotary laser is faster if you have one.

A spirit level on a straight edge works for short spans and accumulates error over long ones. For a nineteen-foot circle, do not chain spirit-level readings round the perimeter.

THE METHOD

Establish one datum point. Every other point is measured to that datum, not to its neighbour.

That is the same rule as the members: measure from the master, not from the last one. Chaining measurements accumulates error; referencing a datum does not.

Go round twice. The second pass catches the points that moved while you were working on the others, and there will be some.

CHECK AFTER LOADING

A platform that is level empty may not be level with a dome on it, especially on soft ground or new fill.

Check again once the base ring is down and before you go up. Settlement that happens at that stage is easy to shim out. Settlement discovered at the apex is not.

Establishing reference points

A reference point is something you can measure from that will still be there tomorrow.

You need several, because the centre stake disappears under the building.

WHAT TO ESTABLISH

The centre, marked on the foundation itself rather than only on a stake.

A north line, or any fixed bearing, scribed on the foundation. Every base vertex is at a known angle from it — 36 degrees apart — and it is how you re-establish the ten points if a mark gets scuffed off.

Two offset points, well outside the building, on that bearing. If the centre gets covered, two offsets let you find it again by measurement.

The datum height, marked somewhere that will not move. A screw in a nearby post, a mark on a wall.

WHY THIS IS WORTH TWENTY MINUTES

Halfway through the build you will need to answer a question like "is this vertex in the right place" and the answer requires measuring from the centre.

The centre, by then, has a floor over it.

If you have offsets you measure from those and get an answer in a minute. If you do not, you either guess or you take something apart.

WRITE THEM DOWN

On paper, in the shop, not only on the ground.

Radius, datum height, bearing, offset distances. Four numbers. They are the only things that connect the building to the site, and remembering them is not a plan.

THE ONE THAT SAVES THE MOST TIME

Mark the ten base vertex positions permanently on the foundation before the frame goes on — a screw, a scribe, a paint mark.

Then when a panel does not sit where you expect, you can see immediately whether the panel is wrong or the layout was.

9. Assembly

Assembly is the short part. If the panels are made and checked, raising a nineteen-foot dome is a day or two with two people.

Everything here assumes you built all forty panels first. The method is arranged so that nearly all the work happens flat, at waist height, where it can be checked — and then the building goes up quickly because there is nothing left to figure out.

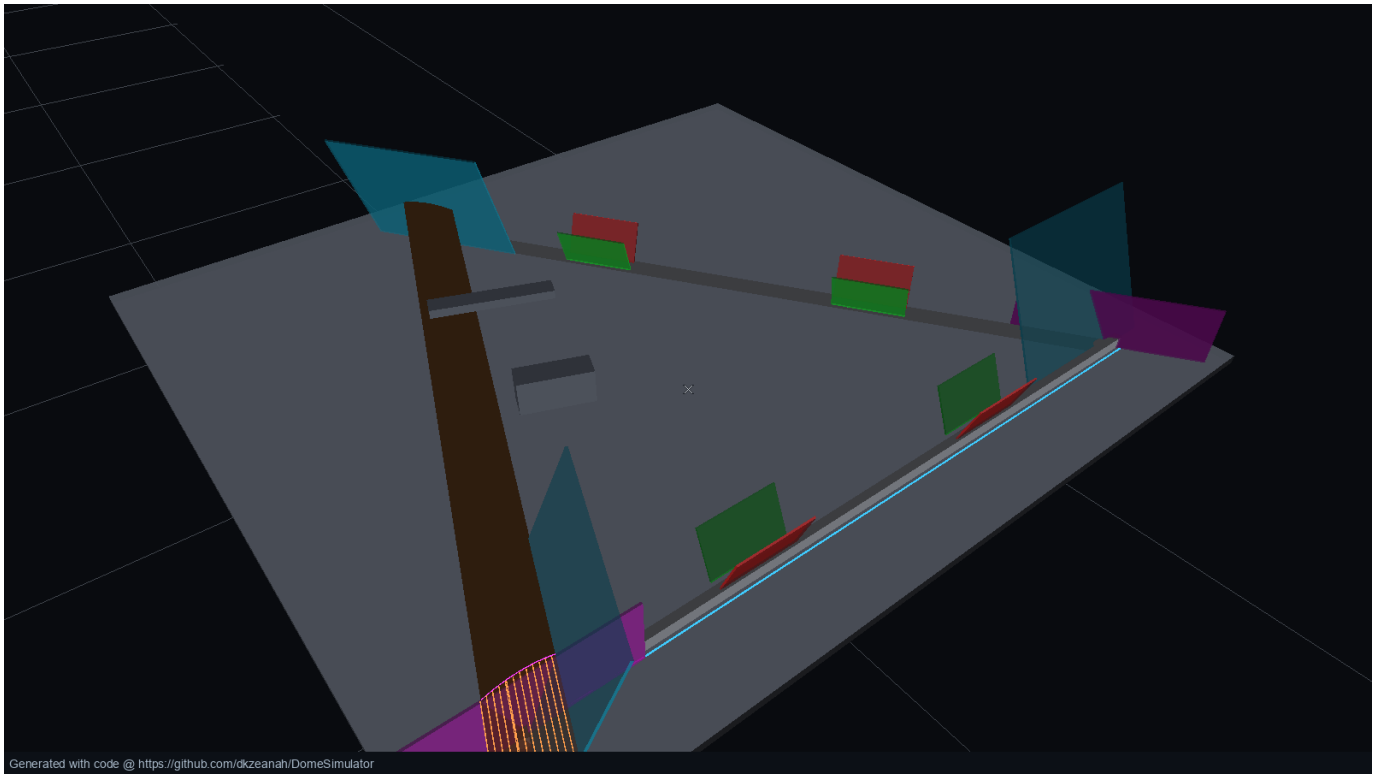

Figure 65. First Stick On The Bench

The same stick, butt now cut, dropped against its rail and its two fences. The head end is still long and hangs off the board. Two ends, two different operations. The butt arrives finished; the head arrives long on purpose and is not touched yet. Generated with code @ https://github.com/dkzeanah/DomeSimulator

First course

The first course is the ring of triangles sitting on the foundation, and it is the one that decides whether the rest goes together.

WHAT IT IS

On a 2V hemisphere the bottom course is ten BAB triangles, alternating point up and point down around the base ring.

Ten panels, and when the tenth one goes in the ring closes on itself.

WHY IT IS THE HARD ONE

Because until it closes, nothing is stable.

Each panel wants to fall inward. There is no completed structure to lean on, and every joint you make is holding two panels that are themselves being held by props.

It is also the course where a layout error announces itself. If the base polygon is out, the tenth panel will not fit, and you will have made nine joints before finding out.

THE ORDER

Set all ten panels in place, loosely propped, before fastening any of them permanently.

Do not build round the ring fastening as you go. That is how the accumulated error ends up entirely in the last joint.

Set them all, check that the ring closes, adjust, and then fasten.

CHECK BEFORE COMMITTING

With all ten standing and loosely held:

every base vertex on its mark the ring closes without forcing the top edge of the course is level all the way round opposite vertices measure the full diameter, 19.416 feet, in several directions

If all four are true, fasten. If any is not, find out why now.

THE HONEST PART

This course takes longer than the four above it put together.

That is normal and it is not a sign you are doing it wrong. The first course is the one with no structure to help you and the most consequences if it is out.

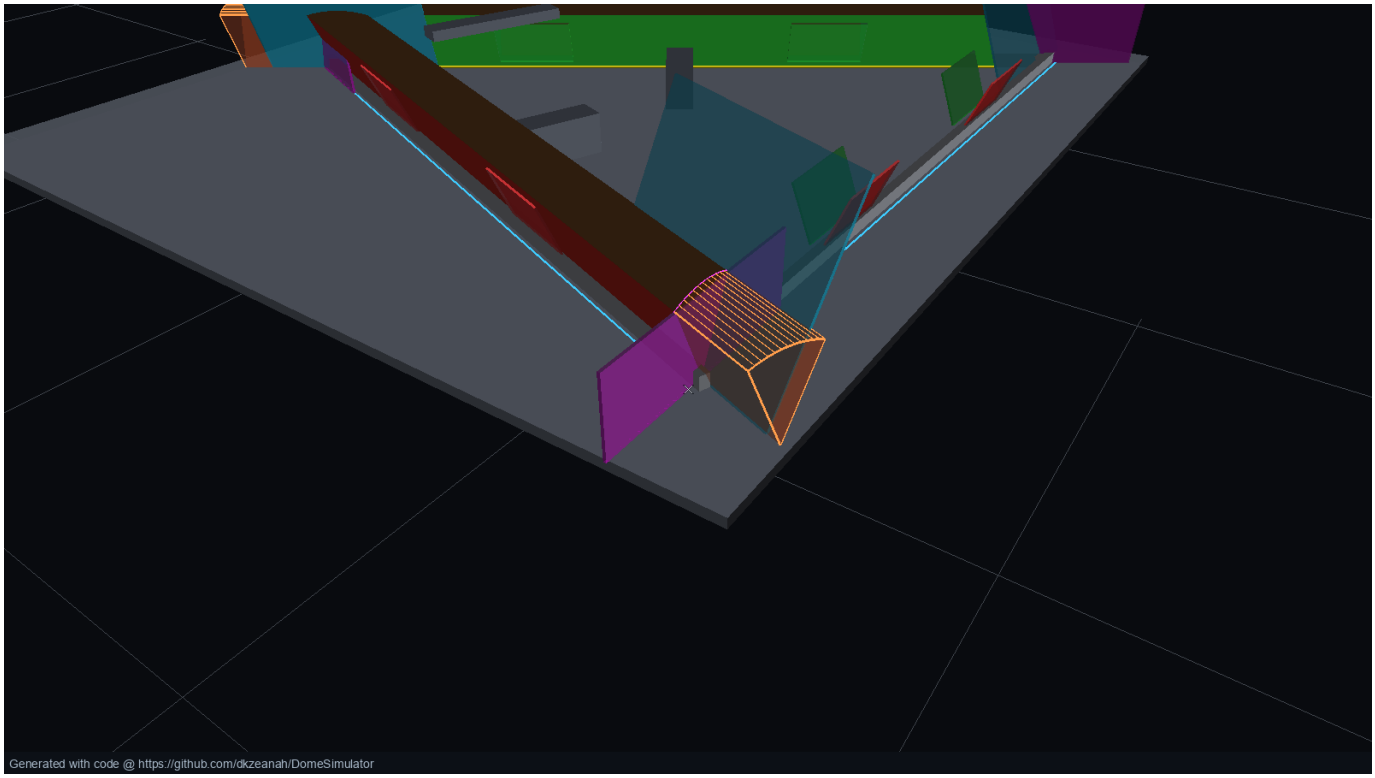

Figure 66. Second Stick, Butt Onto Side

The second stick lands so that the FIRST stick's butt bears on its radial face -- on the side of it, not on its end. End-to-side. Nothing is mitred to meet at a point; the joint is a flat sawn face pressed onto a flat sawn face, which is what a split wedge already has without any further machining. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Triangle assembly

A panel is three members and three pinwheel joints, built flat on a bench.

Build them all before you raise anything.

WHY FLAT AND WHY FIRST

A joint made on a bench, at waist height, with the workpiece held in a jig, is a better joint than the same joint made overhead on a ladder.

Every joint in this building can be made on a bench. That is not true of most construction methods and it is one of the genuine advantages here.

So: forty panels, all made flat, all checked, and then an assembly job that is purely about putting finished things in the right places.

THE SEQUENCE FOR ONE PANEL

Lay the three members in the jig in the pinwheel arrangement. Each one butts into the side of the next, all going the same way round.

Check the handedness. All forty panels must pinwheel the same direction — clockwise seen from outside on the reference build. A panel built the other way is a mirror image and it will not fit anywhere.

Clamp, pilot, fasten. Two or three fasteners per joint.

Take it out of the jig and check it on a flat floor: sits on all three corners, no rock, sides match the chords.

Label it and stack it.

THE STACK

Stack panels flat, not on edge, and separate the AAAs from the BABs.

Ten of one and thirty of the other. Put them where you can get the one you want without moving the other twenty-nine, because you will be doing that forty times.

TIME

About an hour each for the first few. Twenty minutes each once you have the rhythm.

Forty panels is somewhere between fifteen and twenty-five hours of bench work, and it is the most pleasant part of the whole build: repetitive, at a comfortable height, indoors if you want, and entirely checkable.

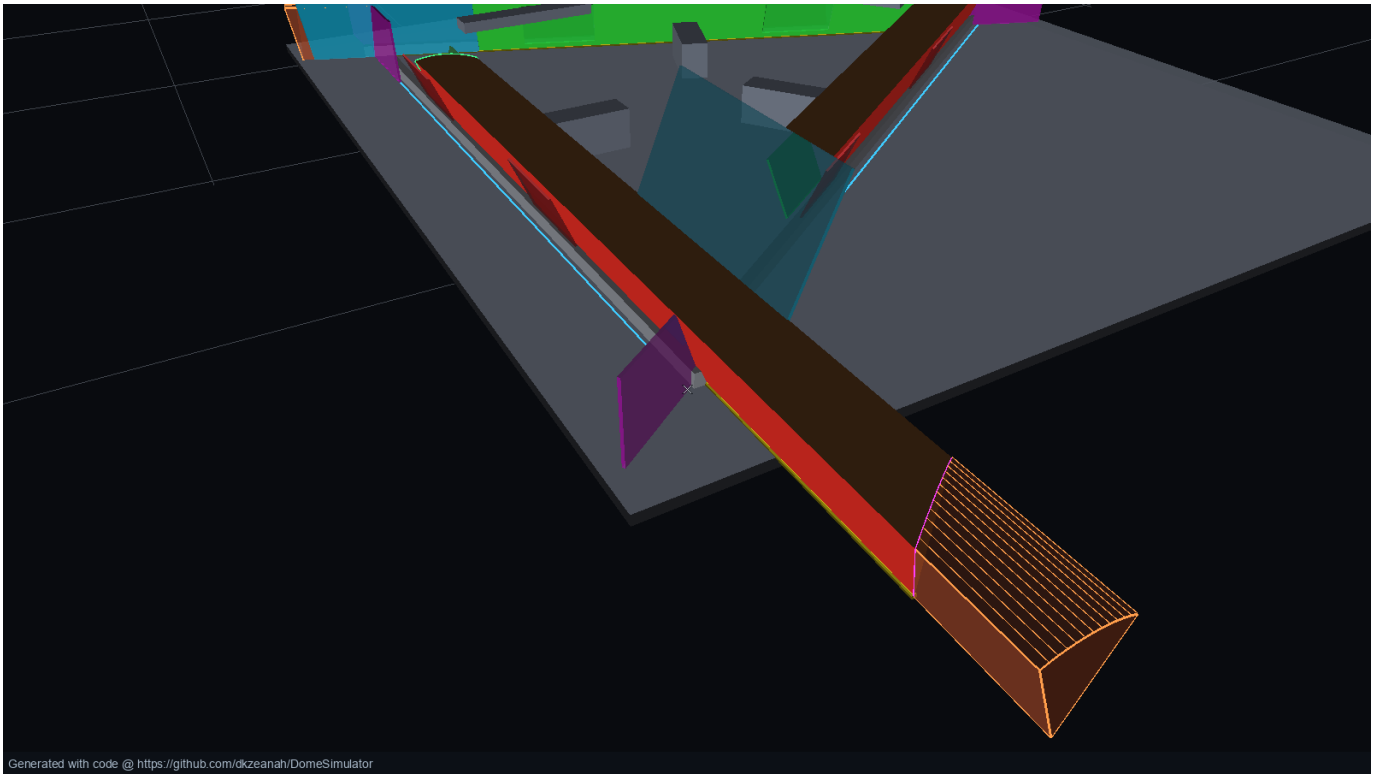

Figure 67. The Last Stick Slides In

The third stick is captured at BOTH ends once the other two are down, so it cannot drop in from above. It is shown backed off along its own axis, ready to slide home into the pocket. The chain is cyclic. This is the one awkward move in the whole method, and it is why the head end is left long: the extra length is what gives the last stick somewhere to come from. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Raising sections

Raising is getting finished panels from the stack into the air and holding them there until they are fastened.

WHAT IT TAKES

For a nineteen-foot dome: two people and a central prop.

A BAB panel out of six-inch split timber is not light but it is not a two-person lift either — one person can carry one. Two people are needed because one holds while the other fastens.

THE CENTRAL PROP

A post at the centre with an adjustable platform on it.

Each course rests on that platform while it is being fastened. When the course closes, it is self-supporting and the platform goes up for the next one.

Simple, cheap, and it turns "hold this panel in exactly the right place while I drive four screws" into "put this panel down on the platform".

ORDER WITHIN A COURSE

Go round the ring in one direction, setting panels loosely.

Do not fasten permanently until the course closes. Same rule as the first course, for the same reason: an accumulated error should be distributed round the ring rather than dumped into the last joint.

WHAT NOT TO DO

Do not raise a partly-built course and leave it overnight.

Wind gets into an open dome and an unclosed course is the least stable thing on the site. Either close the course or brace it properly before you stop.

THE DAY IT TAKES

The raising, for a dome this size, is a day or two with two people — provided every panel is already made.

That ratio is the point of the method. Weeks of comfortable bench work, then a couple of days of assembly. Most of the building happens at waist height.

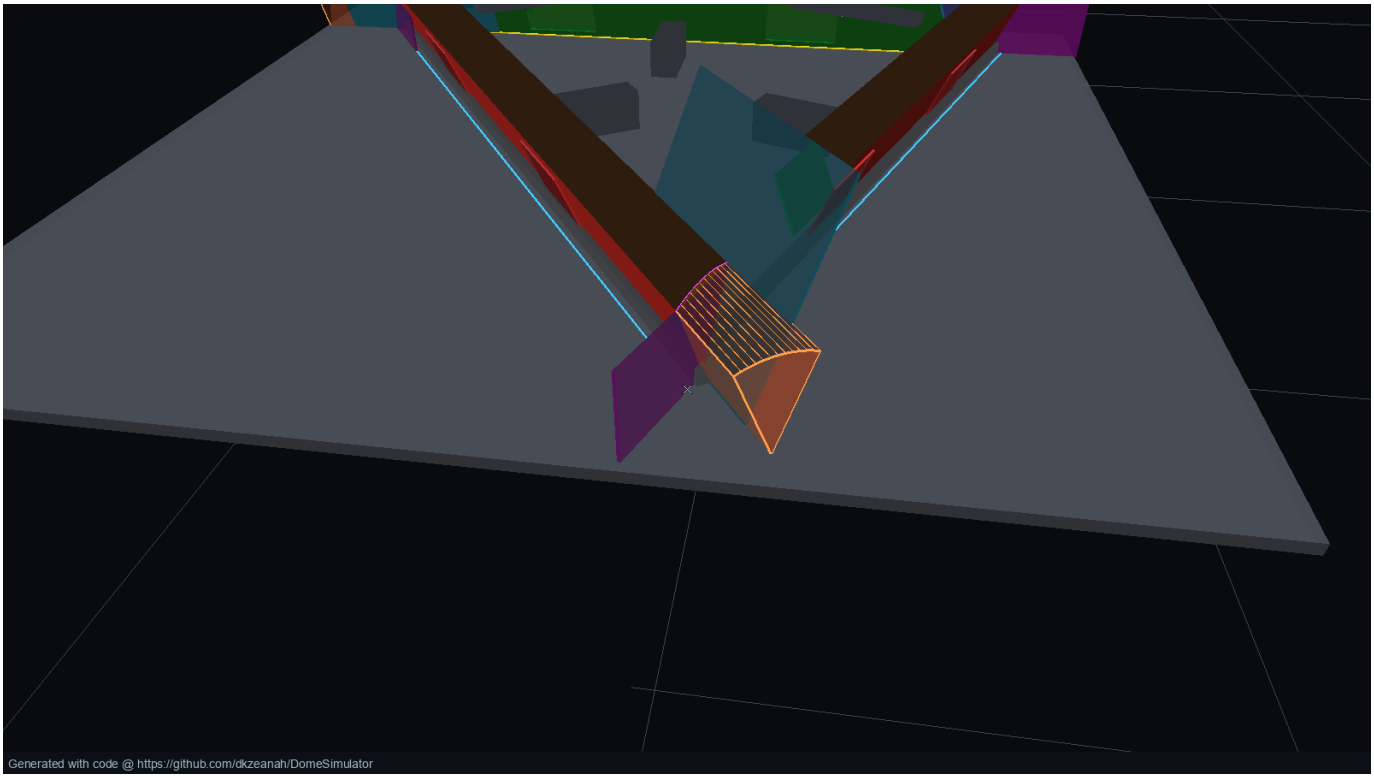

Figure 68. Pulled Up Tight

Home. Three butts bearing on three sides, the green outlines are the actual contact faces, and all three heads still hang off long. Self-locating. With the loop closed the frame holds its own shape and the jig is only keeping it flat while the heads are trimmed. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Temporary bracing

Temporary bracing holds the building in the shape it will hold itself in once it is finished.

Everything here comes out at the end, and every piece of it should be planned to come out.

WHAT NEEDS BRACING

Each course until it closes.

The whole frame until the apex goes in. A dome without its top ring is an incomplete arch and it pushes outward at the base.

Any opening. If you are leaving a door or window out of the frame, the panels either side of the gap have lost a neighbour and need holding.

THE CENTRAL POST

Does most of the work. It carries the courses while they are open and it gives you something to measure from.

Keep it plumb and keep it in the centre. It is also your reference for whether the dome is going up straight.

RADIAL BRACES

Timber from the central post out to individual vertices, where a vertex needs holding at a specific radius.

Use these sparingly. A brace that forces a vertex into position is hiding an error rather than fixing it — if a vertex does not want to be where the geometry says, find out why.

MARK THEM

Use a different colour screw for every temporary fastener.

At the end of the build you have to remove every temporary and leave every permanent one, and there will be several hundred fasteners in the frame. If they all look the same you will either leave temporaries in or take permanents out.

A box of bright screws costs a few pounds and removes an entire category of end-of-build confusion.

WHEN IT COMES OUT

Top down, after the apex closes and not before.

The dome becomes self-supporting when the last panel goes in. Taking bracing out from the bottom while the top is open removes the props from under the part that is still pushing.

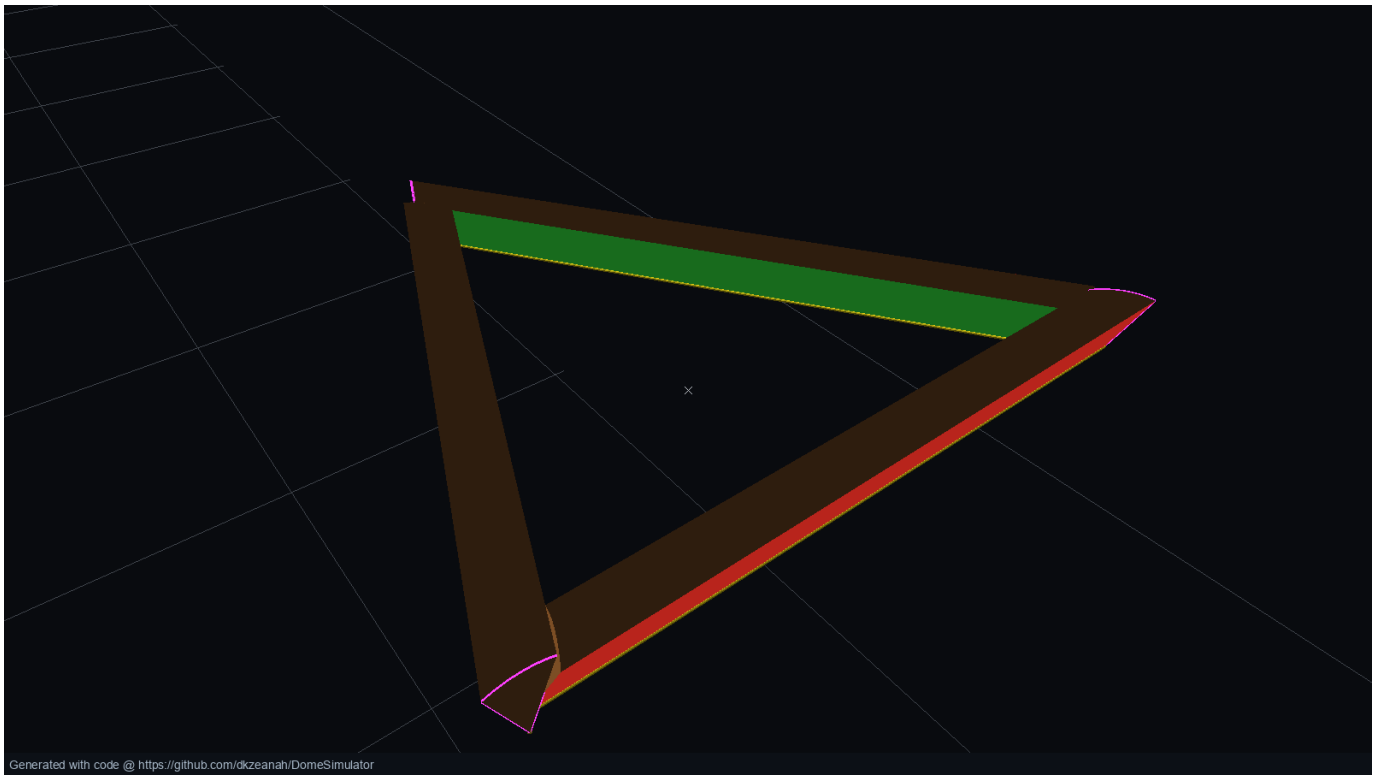

Figure 69. Panel Off The Jig

A finished triangular frame, its three corners clean inside the mathematical triangle. The jig is empty and unchanged. Forty of these come off the same fixture. Nothing on the jig was adjusted between panels, so panel forty is panel one. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Course-by-course construction

A 2V hemisphere goes up in three courses and a cap.

THE COURSES

Bottom: ten BAB triangles on the base ring, alternating up and down.

Middle: more BABs, closing the band between the base and the top ring.

Top: the ten AAA triangles, five of which form the ring immediately around the apex.

Thirty BABs and ten AAAs, in that order, bottom to top.

WHY THIS ORDER

Each course rests on the one below, and each closed course is a rigid ring that supports the next.

An open course is unstable. A closed one is a compression ring and it will hold a lot.

So the rhythm is: set a whole course loosely, check it closes, fasten it, raise the platform, next course.

CHECKING BETWEEN COURSES

Every time a course closes, check three things.

Is it level? The top edge of the completed course should be level all the way round.

Is it round? Measure across, through the centre, in four directions. All four should be the same.

Is it the right height? Compare against the geometry. Each course has a known height and a known radius at its top edge, and the tool will print them.

Those three checks take ten minutes and they stop an error propagating upward.

WHAT TO DO IF A COURSE WILL NOT CLOSE

Stop. Do not force it.

Measure the panels in that course against each other. If one is different, you have found it.

If they all match, the error is below — in the course underneath or in the base layout — and forcing this course will push it higher.

A dome that is forced closed at course two has a fight waiting at the apex, and the apex is the worst place in the building to be arguing with geometry.

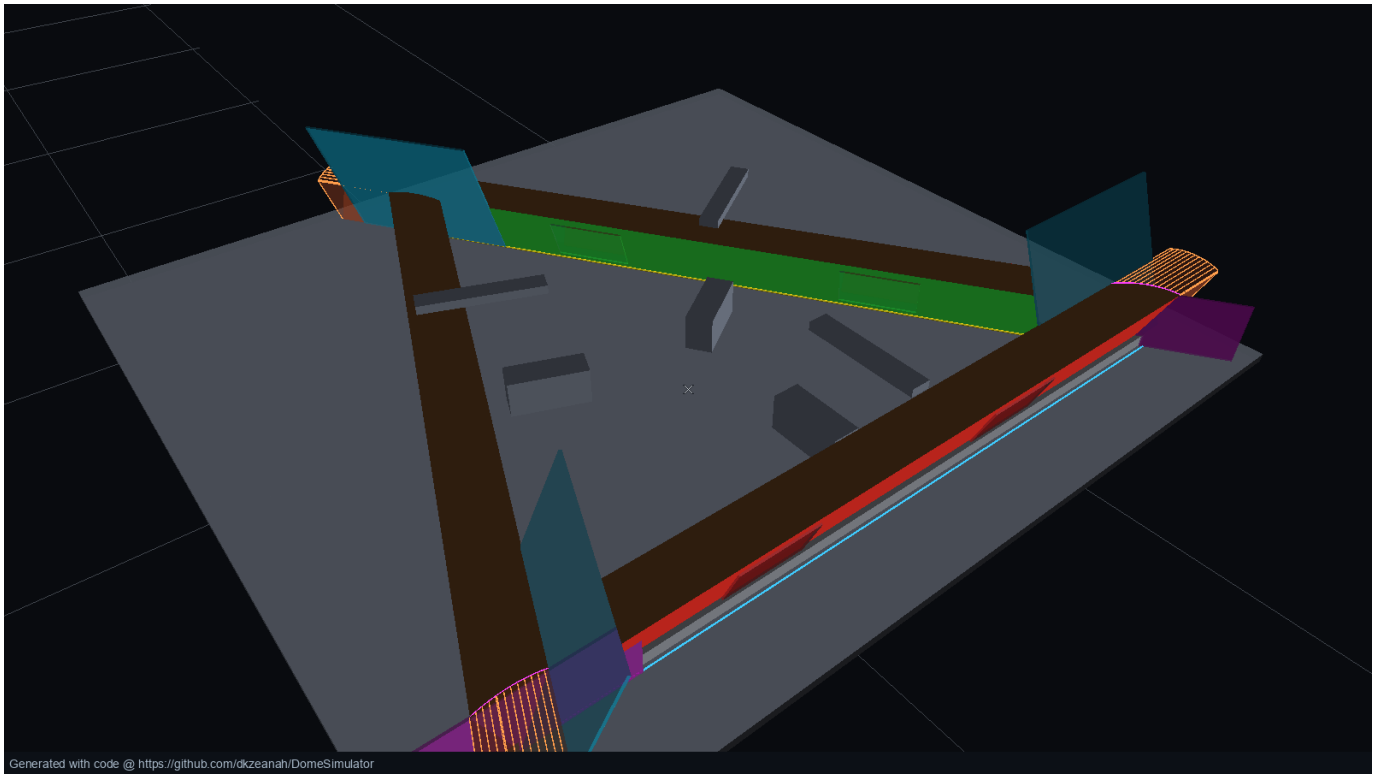

Figure 70. One flat jig, forty identical panels

The fixture with a panel closed on it: three butts bearing on three sides and three heads still hanging off long. Nothing on the jig is adjusted between panels, so panel forty is panel one. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Apex closure

The last panel is the test of everything.

WHAT HAPPENS AT THE TOP

Five AAA triangles meet at the apex vertex. When the fifth one goes in, the dome is closed and self-supporting.

Until then it is an arch. After it, it is a dome.

WHY THE LAST PANEL IS DIFFICULT

Because it has no slack.

Every other panel went into a space with one open side. The last one has to go into a hole that is already bounded on all three sides, and it only fits if everything below it is right.

That is what makes it such a good test and such a bad surprise.

IF IT DOES NOT FIT

Do not modify the panel.

A panel trimmed to fit the hole is a panel that is now the wrong size, and the error is still in the building — it has just moved into the shape of the frame rather than the gap.

Measure the hole. Compare it to what the geometry says it should be. The difference tells you how much error has accumulated and, usually, which direction it came from.

Then loosen the top course, redistribute, and try again. An accumulated error spread over five joints is invisible. Concentrated in one, it is a gap you can put your hand through.

WHEN IT DOES FIT

Fasten it, then take the central post out, and stand underneath.

The dome is now holding itself up with nothing in the middle. That is the moment the method either convinces you or does not, and on a frame that has gone together properly it is genuinely surprising how solid it feels.

THE PENETRATION

On the reference build the apex is not solid. It carries a sleeve — the one hole in the weather surface, with a gasketed cap over it, through which services leave the building.

If you are building that version, the sleeve goes in with the last panel and not after. Fitting it later means working at the top of a finished dome, which is the thing the whole design is arranged to avoid.

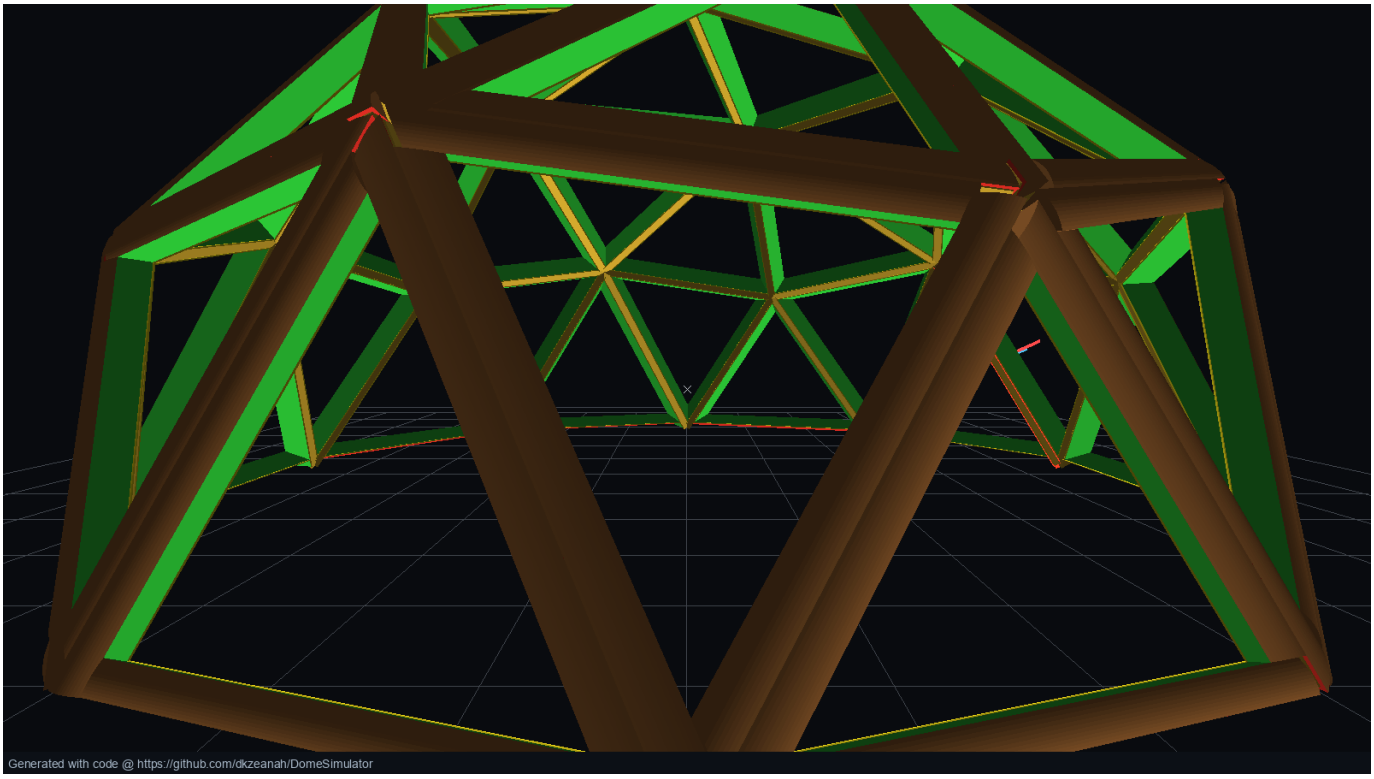

Figure 71. The first course

The ten panels that stand on the ring, framed on their own. They lean in from the start, which is why the first course needs bracing and the second one does not. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Alignment correction

Something will be out. This is what to do about it.

FIND OUT WHETHER IT IS ONE THING OR EVERYTHING

Measure three panels from different parts of the dome.

If one is different and two agree, you have a bad panel. Replace it.

If all three differ from the geometry by the same amount in the same direction, you have a systematic error — a jig setting, a stop block, a fence — and replacing panels will not help because the next one will be wrong too.

That distinction is the whole of diagnosis and it takes five minutes.

CORRECTING A SINGLE PANEL

Take it out and rebuild it. Do not shim it, pack it, or trim it in place.

A dome is a closed geometry and a panel that is the wrong size is the wrong size in three directions at once. You cannot fix it with a shim; you can only move where it is wrong to.

CORRECTING ACCUMULATED ERROR

Spread it.

If a course is half an inch out over ten joints, loosen all ten and let each one take a twentieth of an inch. Nobody will ever find it.

If you concentrate it in one joint, everybody will find it, starting with you every time you walk past.

WHEN TO ACCEPT IT

A dome built to a sixteenth is built better than most houses.

If a vertex is a sixteenth off its mark and everything closes, leave it. Wood moves more than that with the seasons.

If the apex closed, the base ring is level and round, and no joint is held open, the frame is good. Stop measuring and put the skin on.

THE THING WORTH REMEMBERING

Every error you find in the frame is an error you found before it was covered up.

That is worth something. The dome is never going to be easier to correct than it is right now, with nothing on it and every joint reachable from both sides.

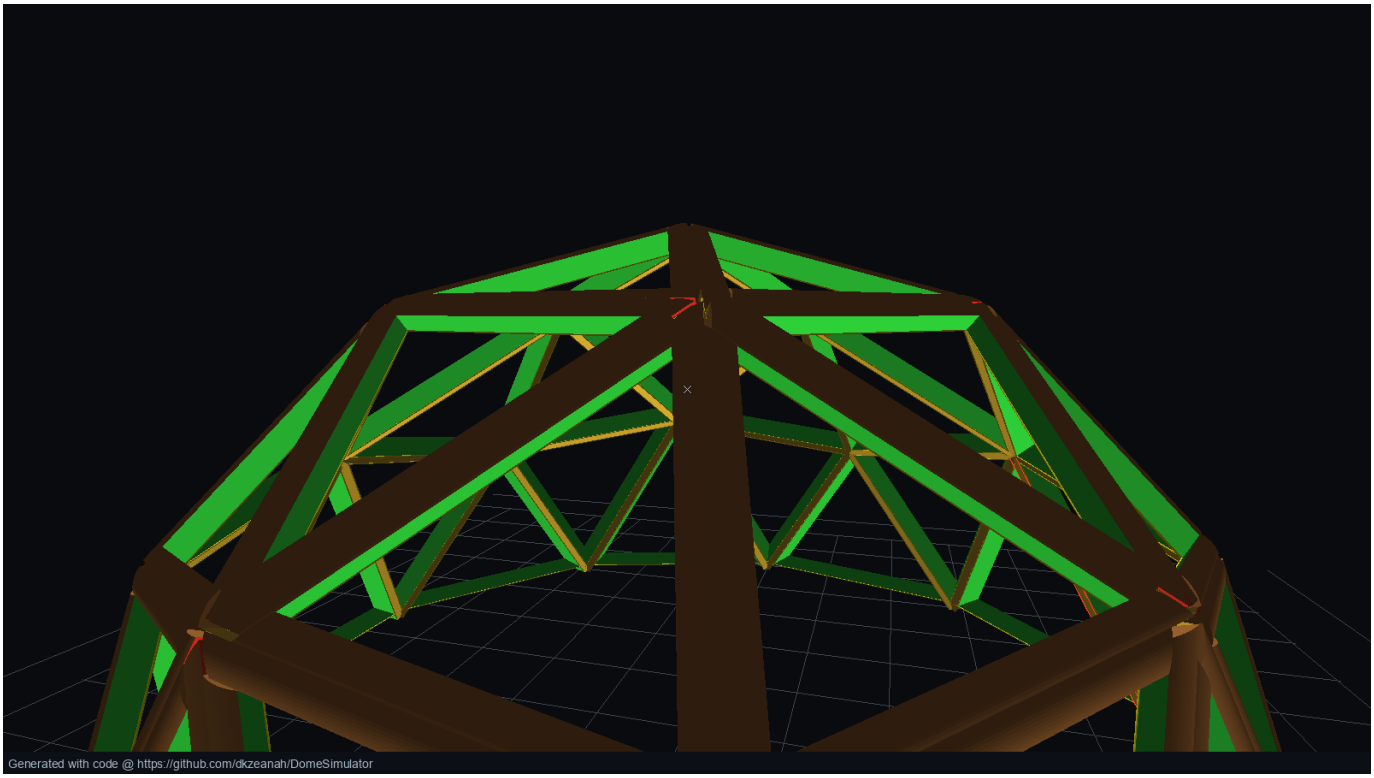

Figure 72. The second course

The band above the first: thirty BAB panels leaning further in. This course is where the dome starts holding its own shape, and it is the point at which the bracing can come out. Generated with code @ https://github.com/dkzeanah/DomeSimulator

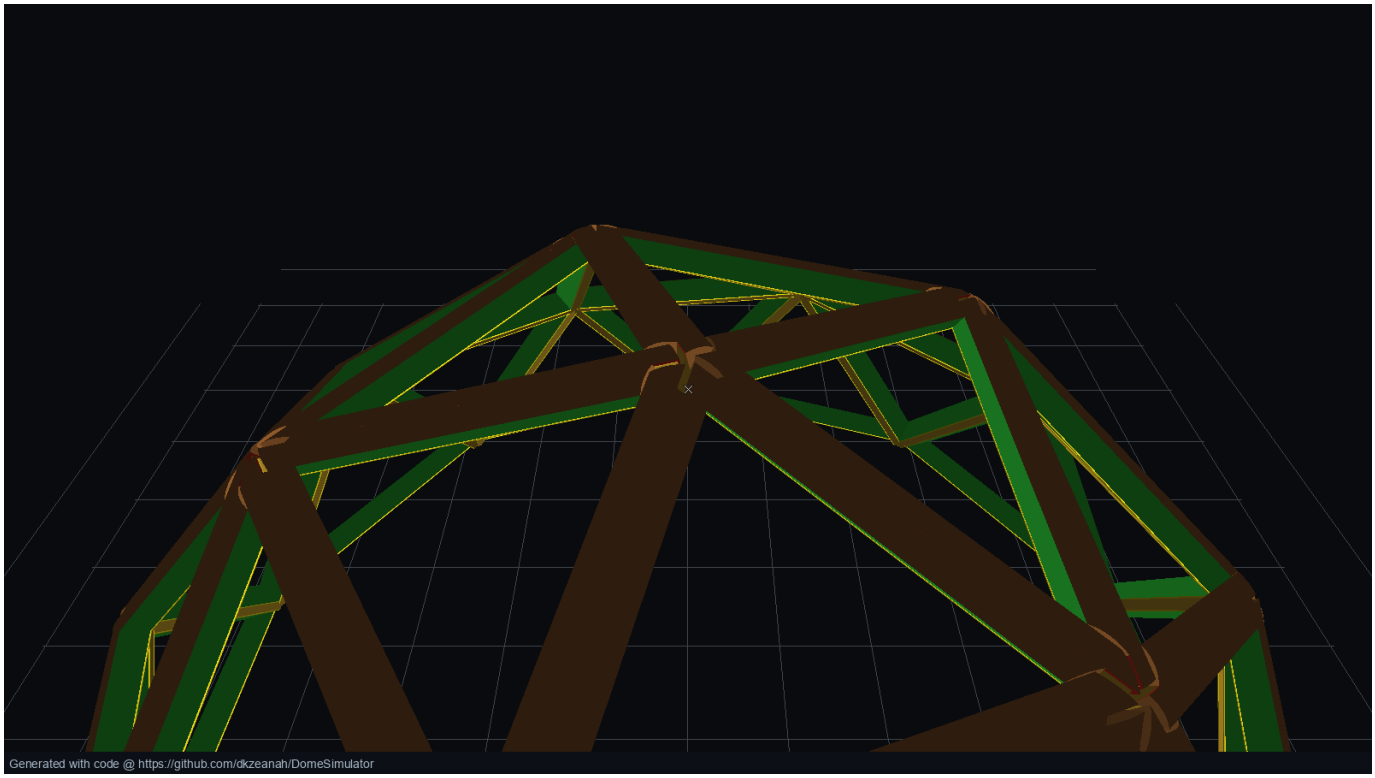

Figure 73. Closing the apex

The top of the dome from below and outside: five panels around one vertex, which is the last thing to go in and the only part of the assembly that has to be held. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Figure 74. One dome, step by step

The tool's own construction order, played end to end. The mesh is not assembled in a random order. The builder emits it the. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Figure 75. Inside one station

A stockpile, a dome, and two people who never leave. Every station holds the same three things: the material waiting in a. From Assembly Line Energy Masterclass, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

10. Doors, Windows, and Openings

Every triangle in a dome is carrying something, so every opening is a decision about where its load goes instead.

The structure is usually the easiest part. An opening is also a hole in the envelope, a hole in the insulation, a path for water and an edge that has to be finished, and those four take longer to solve than the framing does.

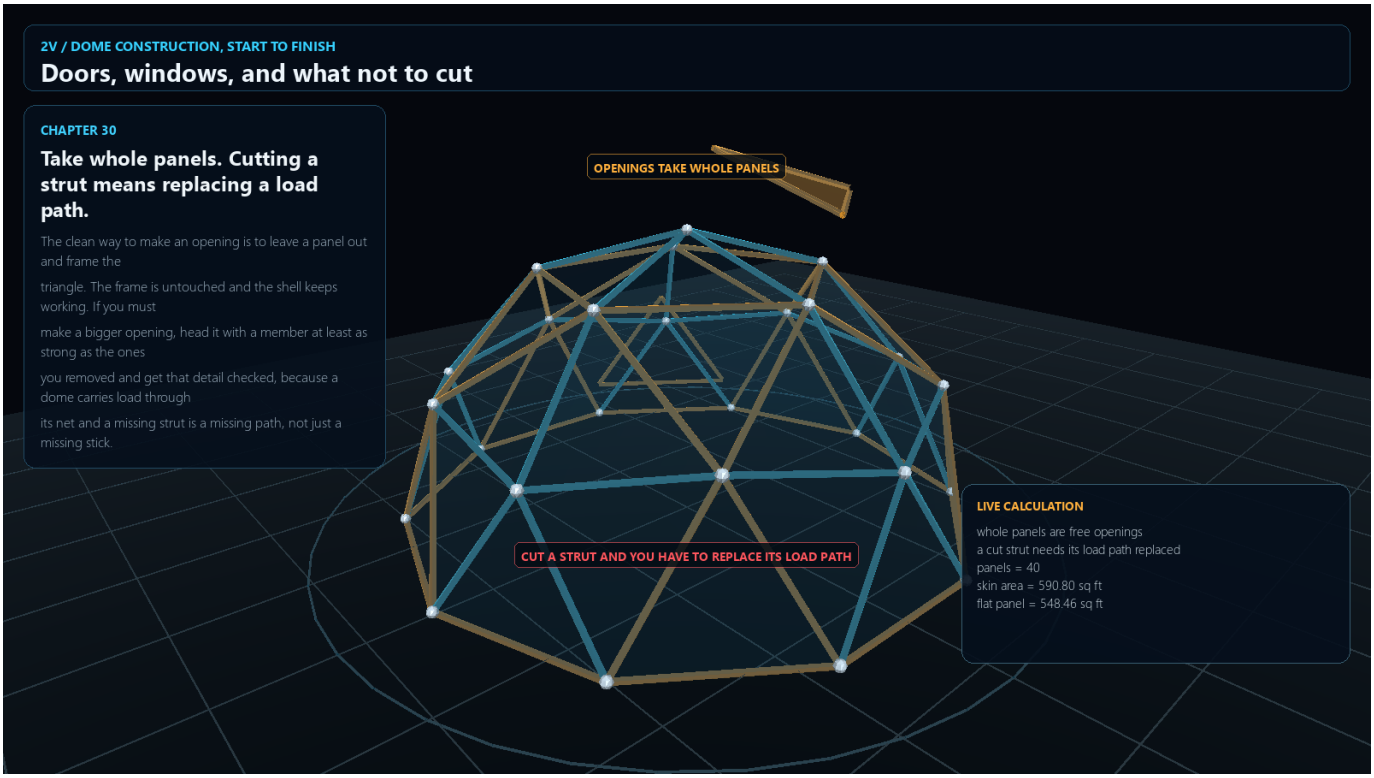

Figure 76. Doors, windows, and what not to cut

Take whole panels. Cutting a strut means replacing a load path. The clean way to make an opening is to leave a panel out and frame the. From 2V Dome Construction Masterclass, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Removing structural triangles

Every triangle in a dome is carrying something. Take one out and its load goes to the ones around it.

That is the whole problem with openings, and it is worth stating before any of the solutions.

WHAT A TRIANGLE ACTUALLY DOES

A geodesic shell works because it is triangulated. A triangle cannot change shape without changing the length of a side, and the sides are timber, so it does not change shape.

Remove one and you have left a quadrilateral or worse in the shell — a shape that can change without stretching anything. It can rack.

The dome does not fall down. The rest of the triangulation is still there and it is redundant. But the shell around the hole is now doing a different job than it was designed for, and the members immediately either side of the opening are the ones doing it.

HOW MANY YOU CAN TAKE

On a 2V hemisphere with forty triangles, taking one is unremarkable. Taking two adjacent ones is fine with a frame round the hole. Taking three in a row at the base ring — which is what turns a dome into a garage — wants deliberate reinforcement and is the largest opening this book will describe without sending you to an engineer.

WHERE TO TAKE THEM

Low, and at the base ring, in preference to anywhere else.

A base triangle has one edge on the ground, which is already an edge with nothing on the far side of it. Removing one interrupts less than removing a triangle from the middle of the shell, where five other panels are relying on it in five directions.

The apex is the worst place. Five triangles meet there and they are all holding each other.

THE PART THAT IS EASY TO FORGET

An opening is not only a structural hole. It is a hole in the envelope, a hole in the insulation, a place water can get in, and an edge that has to be finished.

The structure is usually the easiest of those four to solve.

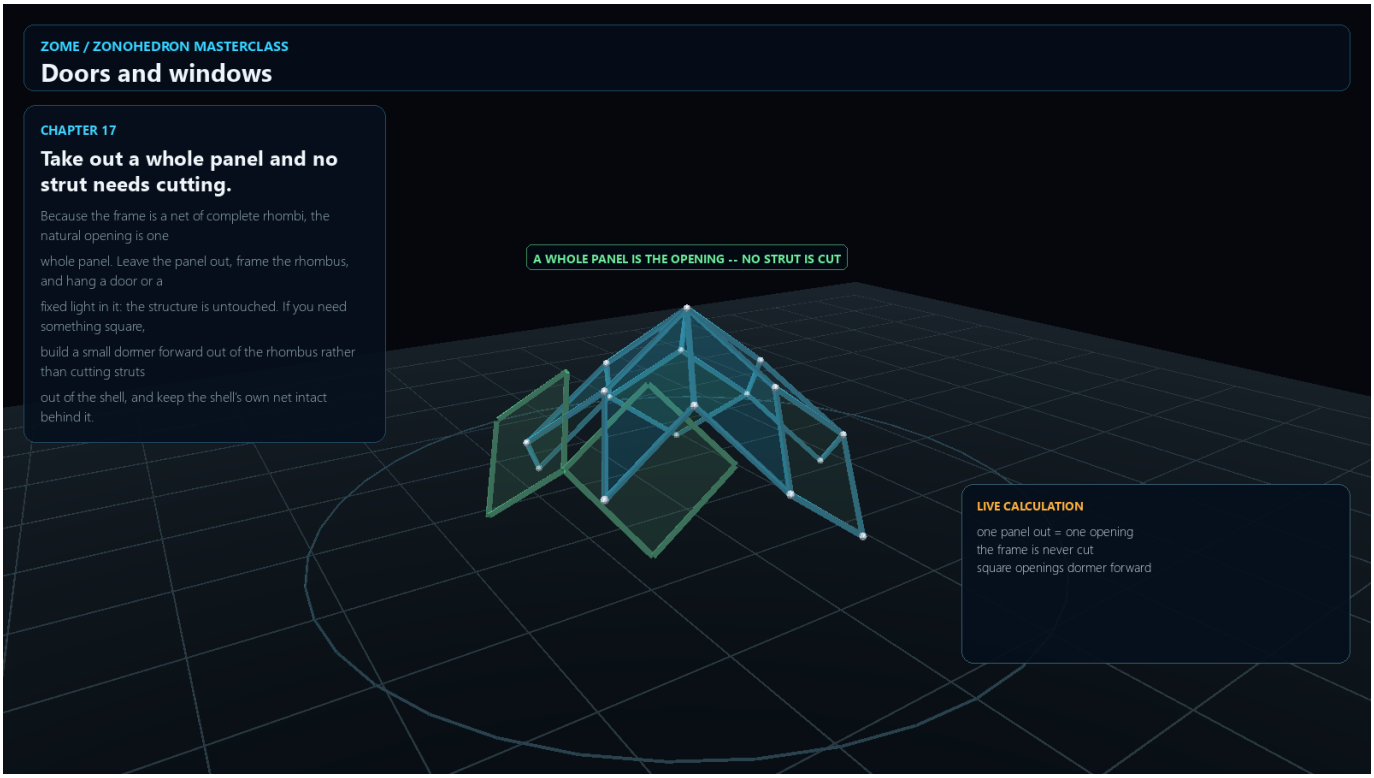

Figure 77. Doors and windows

Take out a whole panel and no strut needs cutting. Because the frame is a net of complete rhombi, the natural opening is one. From Zome Construction Masterclass, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Reinforcing openings

The job of reinforcement is to give the load a path around the hole.

WHAT THAT MEANS PHYSICALLY

The members that used to run through the space now stop at its edge. Whatever they were carrying has to get to the members that are still continuous, and it gets there through whatever you put around the opening.

So the reinforcement is a ring. Not a lintel over the top — a ring all the way round.

THE SIMPLE VERSION

Double the members that border the opening.

A second stick beside each bordering member, fastened to it along its length, roughly doubles the section and is the cheapest reinforcement there is. It costs three or six extra members and an afternoon.

For a single-triangle opening in a 2V hemisphere, that is usually enough.

THE HEADER

For a multi-triangle opening, add a member straight across the top of the hole, from one continuous vertex to another.

That is a header, it is doing what a header does in a stud wall, and it is the piece that stops the shell above the opening from settling into it.

Size it generously. A header is one stick and it is carrying what several used to.

WHAT NOT TO DO

Do not cut a member and leave the cut ends unsupported. A member that stops in mid-air is not carrying anything and its neighbours are carrying it.

Do not rely on the door or window frame as structure unless you designed it as structure. A window frame is a weather part.

Do not open the shell in more than one place on the same ring without thinking about it. Two openings opposite each other on the base ring halve that ring's ability to act as a compression ring, and the compression ring is what stops the dome spreading.

THE HONEST LIMIT

Everything above is reasoning, not engineering.

A single triangle out of a forty-triangle shell, with its border doubled, is within what a careful builder can decide. Three in a row with a header over them is at the edge of it. More than that, or anything in a place where people sleep, wants somebody who can put a stamp on it.

Door frames

A door is the opening you will use ten times a day for twenty years, and it is worth more care than any other hole in the building.

WHERE IT GOES

At the base ring, taking one or two triangles.

The base triangles are the only ones where the wall is close to upright, and a door in a leaning wall is a door you duck through. The dome is 9.708 feet at the apex and the wall is already tilting in noticeably by the time you are head height at the perimeter.

That is a real constraint and it is the first thing people are disappointed by. A nineteen-foot 2V hemisphere does not have a full-height vertical wall anywhere.

THE EXTENSION ANSWER

The usual solution is not to fight it. Build the door in a small extension — a porch, a vestibule, a box on the side — with vertical walls of its own, joined to the dome at the opening.

That gives you a proper door in a proper frame, a place to take boots off, and an airlock, which matters more than it sounds in a building with this little thermal mass.

It also moves the weather detail away from the curved surface, which is the hardest place to make a door seal.

THE FRAME ITSELF

Build it as a box, independent of the shell, and fix the box to the reinforced opening.

That way the frame is square even though the hole is triangular, the door is a normal door, and the joint between the two is a piece of flashing and a gasket rather than a piece of carpentry nobody has done before.

Do not try to hang a door directly off the dome members. They are at odd angles, they are split timber with irregular faces, and the door will not stay square.

THE THRESHOLD

The floor is a platform above ground and the door sill is at platform level.

There will be a step. Plan where it goes and make it a proper step rather than discovering it at the end, and if the building needs to be accessible, this is the detail that decides whether it is.

Window frames

Windows are easier than doors, because nothing walks through them and they can follow the shell's geometry instead of fighting it.

THE TRIANGLE WINDOW

The simplest window in a wedge dome is a glazed panel in a bay.

The bay is already a frame — three members with a lip cut into them by the shape of the wedge. Put a triangular pane in instead of a solid panel and you have a window with no new framing at all.

That is the option the reference build uses, and it is why the fit-out catalogue can turn a dome into a studio by changing which bays are glazed.

WHAT IT COSTS YOU

Triangular glass is a special order and it is more expensive per square foot than rectangular.

It does not open. A triangular opening light is possible and it is not a standard product.

And a triangle high in the shell is a skylight whether you meant it or not — it faces upward, it gets sun all day, and it leaks heat all night.

THE OPENING WINDOW

If you want a window that opens, put a square one in a square sub-frame inside a triangular opening, the same way the door goes in a box.

You lose some glass area to the frame. You gain a normal window with normal hardware that a normal person can replace in fifteen years.

WHERE TO PUT THEM

Low for view, high for light.

A dome's upper panels face the sky and will light the room brilliantly and overheat it in summer. The lower panels face out and give you something to look at.

Most people want more of the second than they expect and fewer of the first.

THE SEAL

Every window is a hole in the envelope with a perimeter, and on a curved shell that perimeter is not in one plane.

That is the detail to get right, and the way to make it easy is the same as everywhere else in this book: make the opening flat and local — a sub-frame — and seal to the sub-frame rather than to the curve.

Entry extensions

An entry extension is a small structure joined to the dome at an opening, and it solves several problems at once.

WHAT IT SOLVES

A vertical door, in a building with no vertical walls.

An airlock, in a building with very little thermal mass, where opening the door in January dumps the room's heat in ten seconds.

Somewhere to leave boots, coats and firewood, in a round room with no corners and no hallway.

And a place to put the weather detail, off the curved surface.

WHAT IT IS

A box. Stud-framed, or built from the same wedge members if you want the consistency, with a flat roof or a small pitch, joined to the dome around a reinforced opening.

It does not need to be big. Four feet deep is enough for a door, a step and a pair of boots.

THE JOINT

This is the only difficult part and it is the same difficulty as any roof penetration: a flat thing meeting a curved thing.

Two approaches.

Flash it. Treat the joint like a chimney flashing — an apron under the shell covering above, sealed at the top, open at the bottom so anything that gets behind it drains out.

Or gasket it. Build the extension with a collar cut to the dome's curve, and compress a gasket between the two. This is tidier, more expensive, and depends on the collar being accurate.

The reference build's shell is removable, which makes the second option awkward — anything gasketed to the shell has to come off with it. So: flash to the frame, not to the shell, and let the shell overlap.

THE THING WORTH SAYING

An extension makes the dome look less like a dome.

Some people mind that a great deal. If you are one of them, the answer is a door straight into the shell and a life of ducking, and that is a legitimate choice as long as it is a choice rather than a surprise.

Avoiding weak discontinuities

A discontinuity is any place the shell stops being a shell — an opening, a join, a change of material, a place where the triangulation is interrupted.

They are not avoidable. They are manageable, and the management is mostly about not putting them next to each other.

THE RULES THAT MATTER

Do not stack them. Two openings in the same vertical line, one above the other, leave a strip of shell with nothing bracing it sideways.

Do not mirror them. Two large openings directly opposite each other across the dome halve the base ring's ability to work as a compression ring — and it is the compression ring that stops the dome spreading at the bottom.

Do not cluster them. Three openings within a few panels of each other make one large weak region rather than three small ones.

Spread them round. If you need four windows, put them at roughly 90 degrees to each other rather than in a row.

THE APEX

Leave it alone.

Five triangles meet at the top and they are the most interdependent group in the building. The reference build puts one penetration there — a gasketed sleeve for services — and that is a hole through a panel, not a missing panel.

THE BASE RING

The base ring is a compression ring and it is the single most important continuous element in the dome. It is what turns outward thrust into hoop stress and stops the dome spreading.

Interrupting it is the one thing to avoid if you can. A door that takes a base triangle interrupts the panels but should not interrupt the ring member itself — keep the member at the bottom continuous and take the opening out above it, with a threshold over the member.

That is a small detail and it is worth more than most of the reinforcement in this chapter.

THE TEST

Walk round the finished frame and look for any place where you could push the shell sideways and it would move.

If there is one, that is a discontinuity that needs bracing. If there is not, the openings you have cut are fine.

Part IV - Turning the Frame Into a Building

11. Panels and Shells

The frame holds the building up. The skin keeps the weather out. They are separate jobs and this method keeps them separate, which is why the skin can be changed without touching the structure.

One number governs the whole chapter: no panel of this dome fits on a four-foot sheet in any rotation. The narrowest is 52.520 inches and a sheet is 48. Everything about how the skin gets built follows from that.

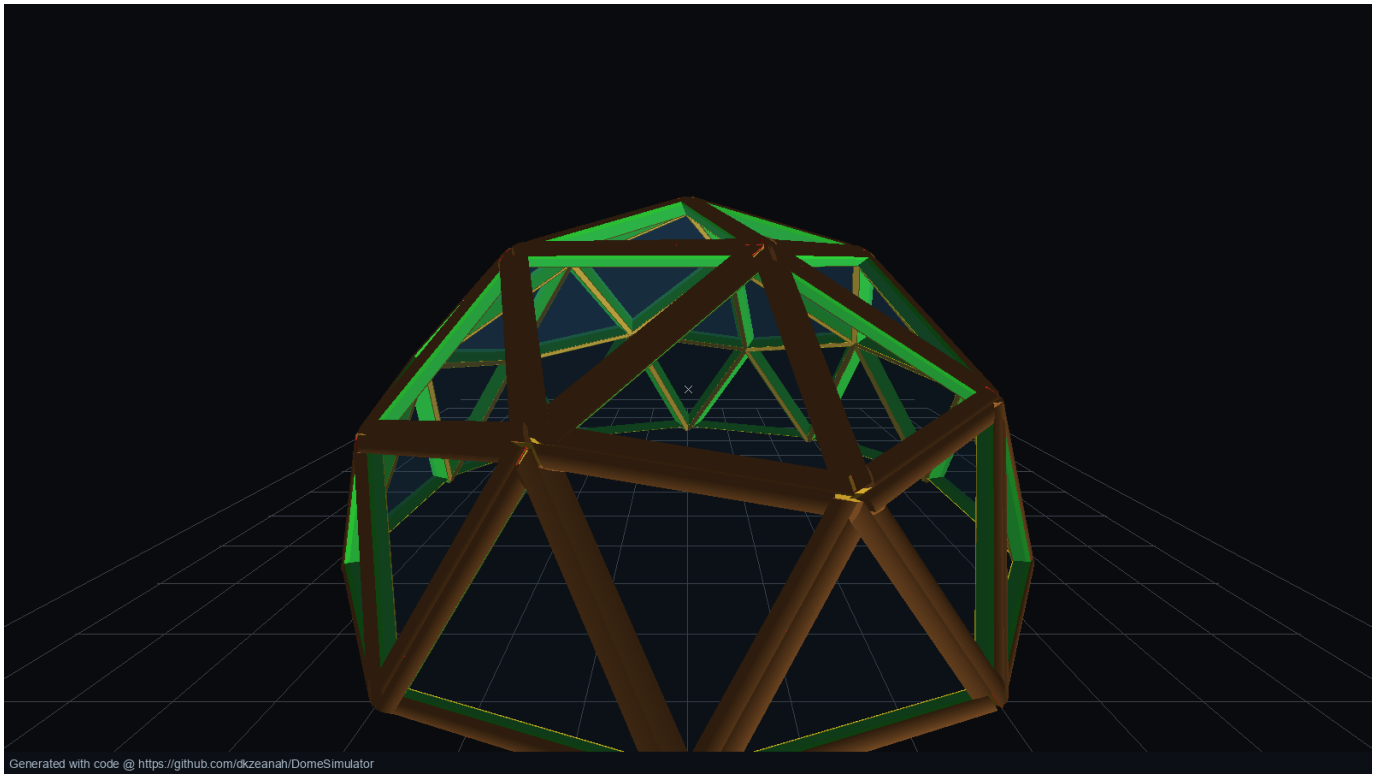

Figure 78. The frame with its skin on

The same frame with the shell offset 4.10 inches off the members, which is where a panel actually lands: clear of the outer faces and clear of the key standing proud of the ridge. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Plywood

Plywood is the default and it has one problem on this building, which is worth getting out of the way first.

NO PANEL FITS A SHEET

The narrowest panel in a 2V hemisphere on a six-foot member is 52.516 inches across its shortest altitude. The widest AAA panel is 62.354.

A sheet of plywood is 48 inches wide.

There is no rotation, no nesting and no clever layout that gets a panel of this dome out of a standard sheet. The narrowest one is four and a half inches too wide.

That is not a small inconvenience. It decides how the entire skin gets built, and it is the sort of thing that is obvious in a car park and invisible on a drawing.

WHAT TO DO ABOUT IT

Three options.

Buy oversized sheet. Some suppliers carry 5x10 or 4x10; a 60-inch sheet takes a BAB panel and a 5-foot sheet takes both. It is a special order and it costs more per square foot.

Sheet across the frame. Run ordinary sheets over several panels at once, following the curve in facets, and cut at the frame lines. You get joints in the middle of panels instead of at their edges, which is structurally irrelevant and visually obvious.

Build the panel up from boards. Narrower stock, laid edge to edge inside the bay. More labour, no special order, and it looks like a timber building rather than a sheet-goods building.

THE REFERENCE BUILD

Sheets across the frame, taped at the joints, because it is the option that does not require anybody to order anything unusual.

GRADE AND GLUE

If it is inside the envelope and dry, interior-grade is fine.

If it is outside, or if it is part of the removable shell that lives in the weather, it wants exterior glue at minimum and marine ply if you can afford it. The failure mode of the wrong glue is delamination, and delamination on a curved surface is not repairable.

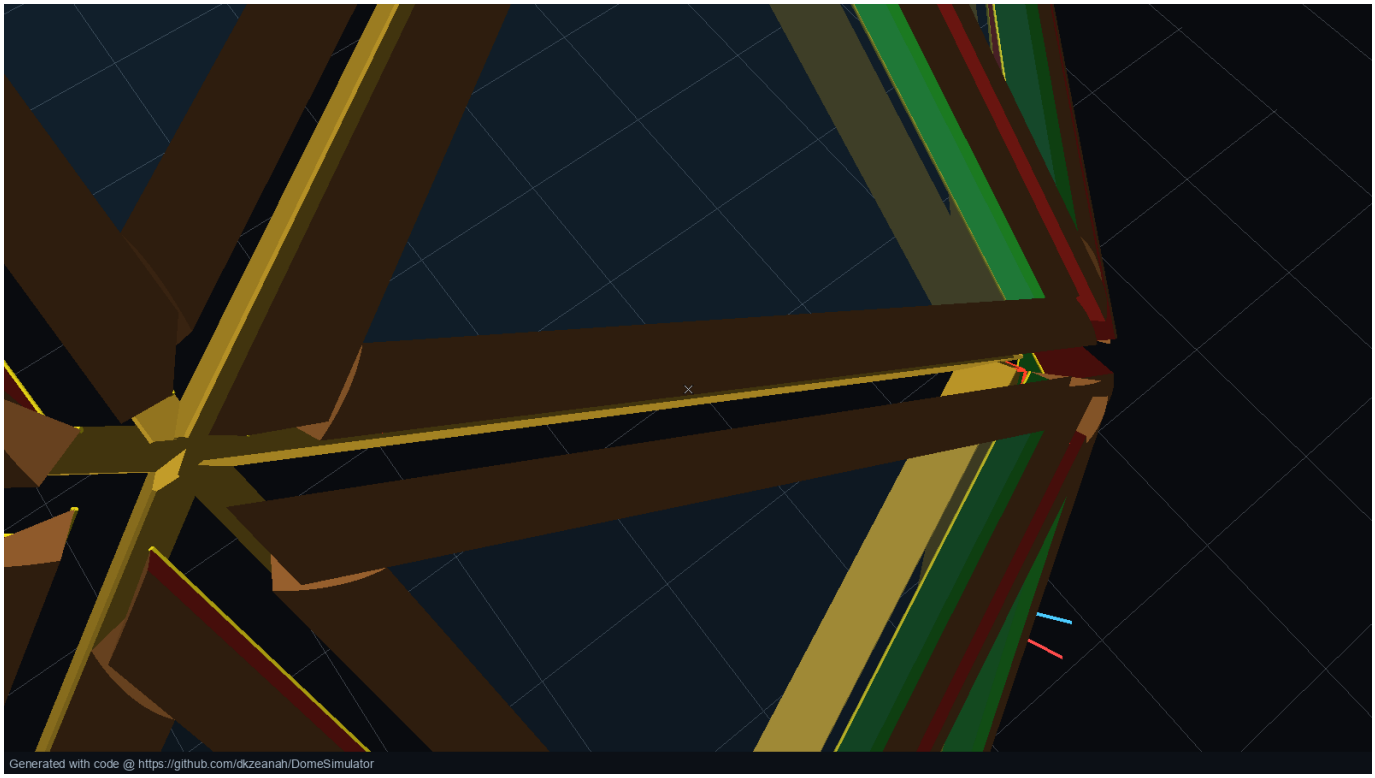

Figure 79. Where the panel meets the seam

The shell's edge over an exploded seam, which is the detail every weather problem in this building happens at. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Composite panels

A composite panel is two skins with a core between them, and the core is doing most of the work.

WHY THEY SUIT A DOME

A dome panel is a triangle spanning about five feet, carrying wind and snow and its own weight, and it is not carrying the building — the frame does that.

That is exactly the load case a composite is good at: stiff across a span, light, and not required to be strong in the way a structural member is.

THE COMMON CORES

Foam. Light, insulating, cheap, and it will not take a fastener. Anything screwed to it needs a hard insert.

Honeycomb. Stiffer for its weight than foam, more expensive, and it does not insulate.

Timber. A ply skin either side of a timber frame is the oldest composite there is and the one you can make in a shed.

THE JOIN PROBLEM

Composites are excellent panels and awkward edges.

A foam-cored panel has an exposed core all the way round, which has to be capped, sealed or rebated. On a triangle that is three edges per panel and 120 edges for the dome.

That edge detail is where composite panels get expensive, and it is the reason people who cost a composite skin off the panel price alone are always surprised.

WHERE THEY EARN IT

Where you want insulation and structure in one operation and you are not planning to take the panel off again.

The reference build does not use them, because the reference build's whole argument is a shell that comes off and layers that stack underneath. A composite panel decides its own insulation value at the factory and cannot change its mind afterwards.

That is the trade: a composite is better on the day it goes on, and it is the last decision you get to make about that bay.

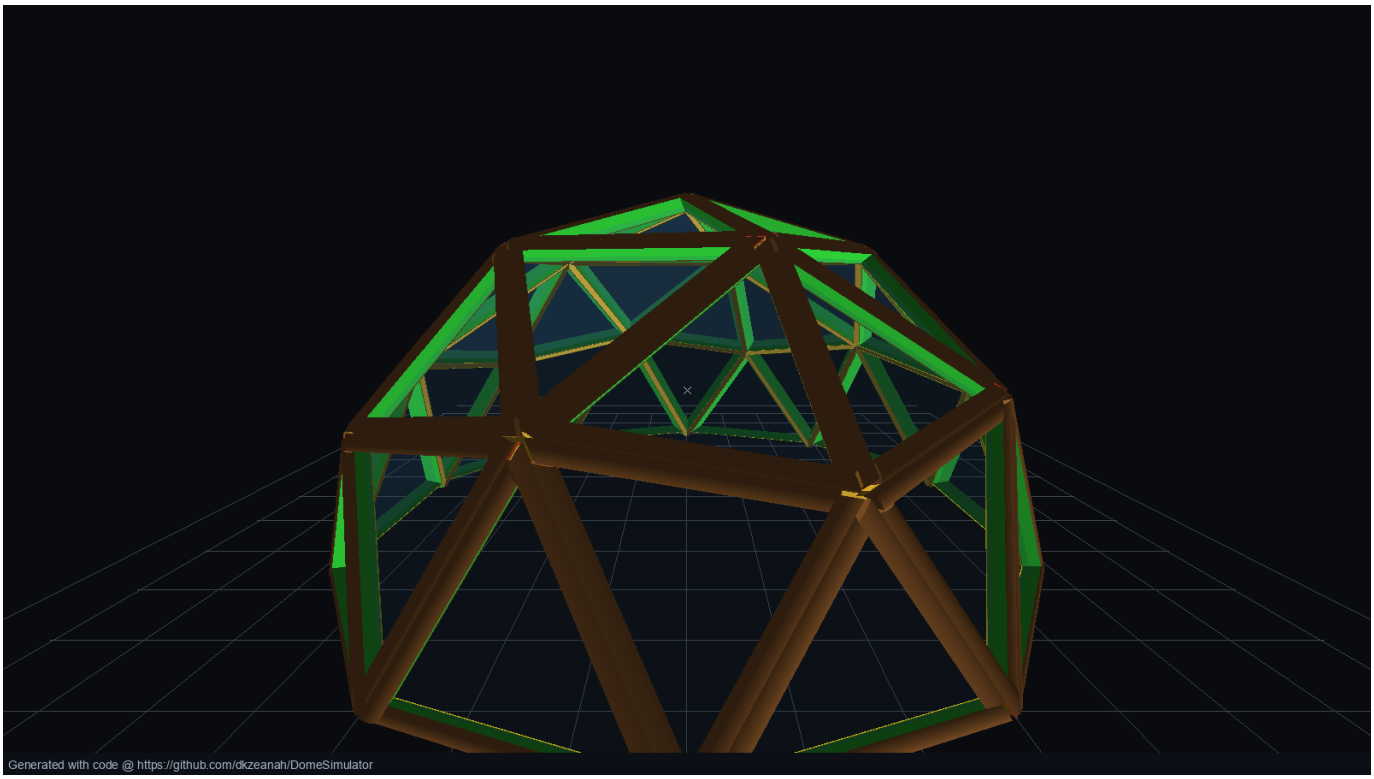

Figure 80. Tight to the frame

Tight to the frame. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Fiberglass

Glass fibre in resin is the material the reference build's hard shell is made of, and it is worth understanding because it sets the cost ceiling for the whole building.

WHAT IT IS

Cloth — woven or stitched glass — wetted out with a resin that cures hard. Layers build thickness. A core between layers makes it stiff without making it heavy.

HOW IT IS PRICED

By weight, not by area. Ounces of glass per square foot, times a resin-to-glass ratio.

That is the industry's own method and it is the thing most amateur costings get wrong. A rule of thumb in resin-per-square-foot will be right for one schedule and wrong for every other.

WATCH THE UNITS. Woven and stitched fabrics are sold by ounces per square yard. Chopped strand mat is sold by ounces per square foot. Getting that backwards understates a laminate by a third.

WHAT IT COSTS ON THIS DOME

Over 1,325 square feet of laminate, four named systems:

- sheathed ply in epoxy: $5,395
- boatyard polyester: $6,045
- premium marine polyester: $6,668
- vinyl ester: $9,144

That is the most expensive line in the building by a wide margin — more than the frame, the core, the services and the fit-out together.

WHY IT IS STILL WORTH DESCRIBING

Because it is a sixty-year object.

A gelcoated hull outlives every other option in this chapter several times over, it is genuinely waterproof rather than nearly waterproof, and it is a trade with real suppliers, real published prices and sixty years of people knowing how to do it.

THE ARGUMENT AGAINST, WHICH THE NEXT CHAPTER MAKES PROPERLY

It is made once, at one size, and it cannot grow.

The cavity behind it holds four quilted layers and then it is full. After that, insulating further means losing floor area or building a second shell outside the first, and neither happens.

So the hull quietly caps how good the building is ever allowed to get. That is the problem the shower cap exists to solve.

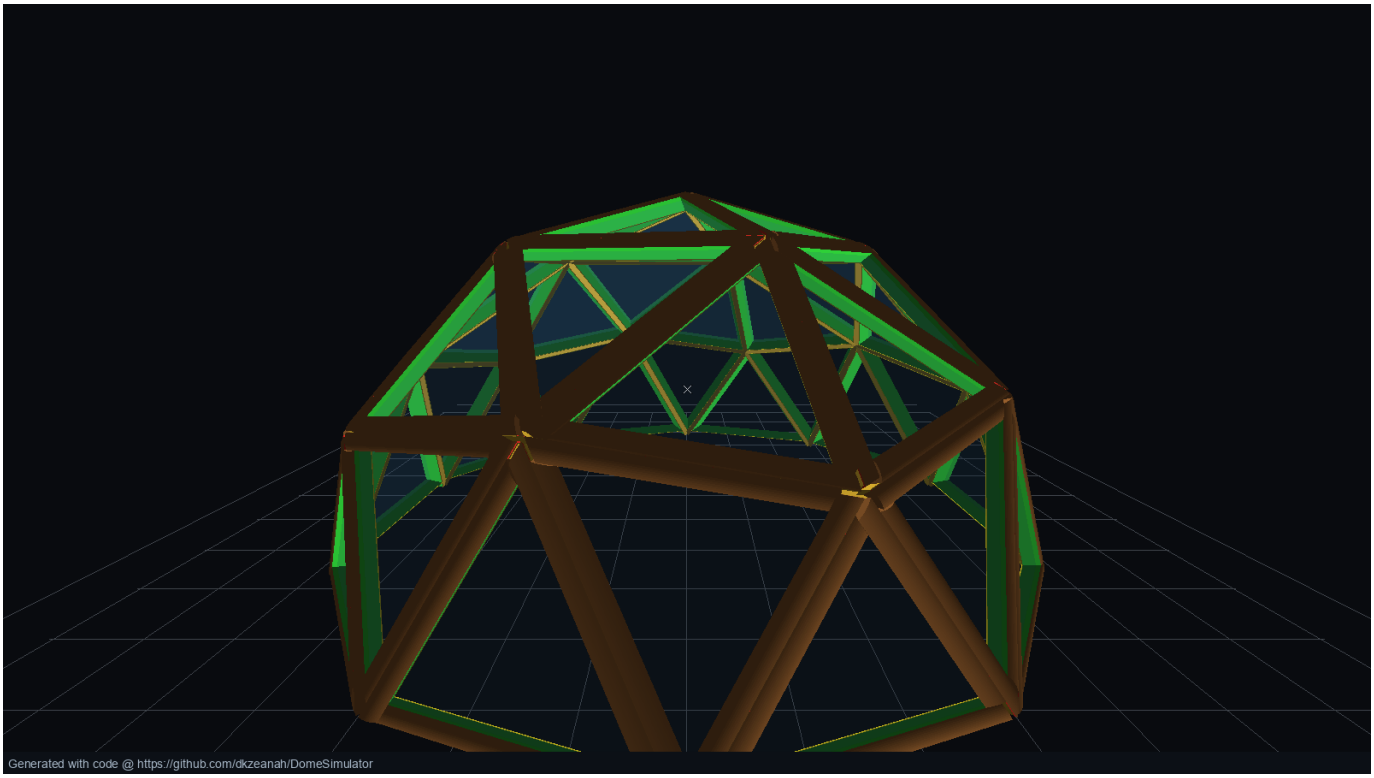

Figure 81. The reference build

The reference build. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Metal

Sheet metal over a dome is common, cheap and worth knowing the failure modes of.

WHAT IT IS GOOD AT

Water. A metal roof sheds it and keeps shedding it for decades.

Fire. In a place where that matters, nothing else in this chapter compares.

Cost per square foot, if you buy it flat and cut it yourself.

WHAT IT IS BAD AT

Compound curves. A dome is not a cone and it is not a cylinder: you cannot wrap flat sheet round it without either cutting it into small facets or stretching it.

Small facets means a great many pieces and a great many laps.

Noise. Rain on metal over an uninsulated shell is memorable, and a dome is a very good acoustic reflector — it focuses sound at the centre of the room.

Heat. Uninsulated metal in sun radiates into the space below it.

Condensation. Metal is cold, inside air is damp, and the underside of a metal sheet is where that meets. Without a vented cavity or a vapour control layer you get water dripping off the inside of the roof and everybody blames the roof for leaking.

THE PANEL-BY-PANEL APPROACH

The one that suits this dome: a flat triangle of sheet per bay, fixed to the frame with the edges lapped or capped at the seams.

Forty pieces, all one of two shapes, all flat. That is a manageable job and it fits the method's logic of doing everything forty times.

THE DETAIL THAT DECIDES IT

The seams. Forty triangles is 120 edges, of which 55 are interior seams where two panels meet at a dihedral angle.

Every one of those is a lap that has to shed water downhill and accommodate a fold of 18 or 22 degrees. Get the seam detail right and metal is excellent. Get it wrong and you have 55 places for water to get in.

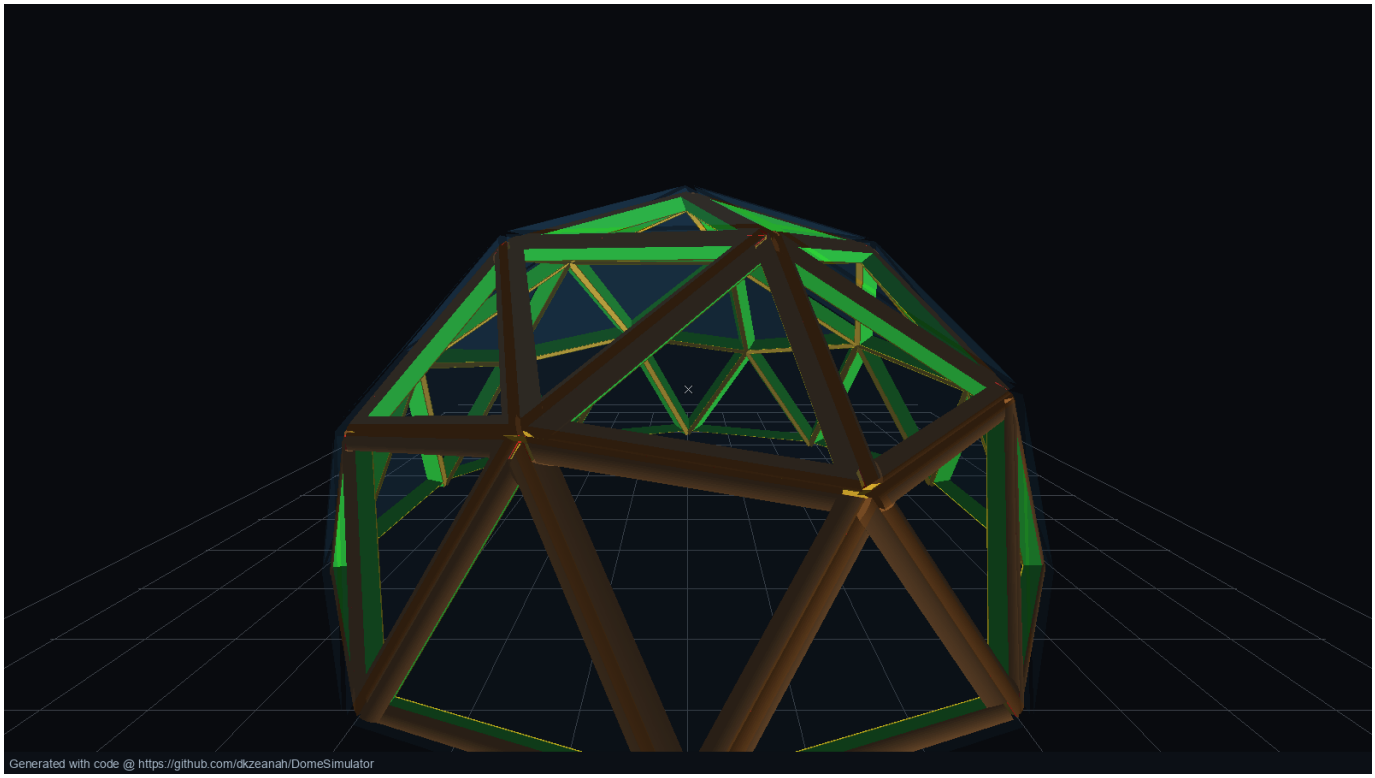

Figure 82. Standing well off, which is where a vented cavity goes

Standing well off, which is where a vented cavity goes. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Transparent panels

Glazing a dome is easy to overdo and easy to get wrong in one specific way.

THE OPTIONS

Glass. Best optically, heaviest, most expensive, and a special order in triangles.

Polycarbonate, twin-wall or solid. Light, tough, insulating in the twin-wall form, and it yellows and goes brittle under UV unless it has a coated face — and the coated face has to go outward, which people get

wrong.

Acrylic. Clearer than polycarbonate, more brittle, scratches easily.

Polyester or PVC corrugated sheet. Cheap, agricultural, and short-lived.

WHERE TO PUT THEM

Low, mostly.

A dome's upper panels face the sky. A glazed panel up there is a skylight: it will light the room beautifully, cook it in summer and lose heat all night. The reference build's own performance model finds the shape is already worth 41 percent less envelope than a box, and putting glass in the top gives a chunk of that back.

Lower panels face outward and give you a view, which is usually what people actually wanted.

THE THERMAL PROBLEM

Any glazing is a hole in the insulation.

A single-glazed panel is roughly R-1. A quilted layer is R-1.6, and the reference build's seven-layer stack is R-12.7. So one glazed bay undoes rather a lot of stacked quilt.

That is not a reason to avoid glass. It is a reason to count the bays.

THE ONE THAT CATCHES PEOPLE

A dome is a reflector. A curved glazed surface facing a room concentrates light and heat toward the centre, and in a small dome the centre is where you are standing.

Two or three glazed bays are pleasant. Eight is a greenhouse, and it will behave like one.

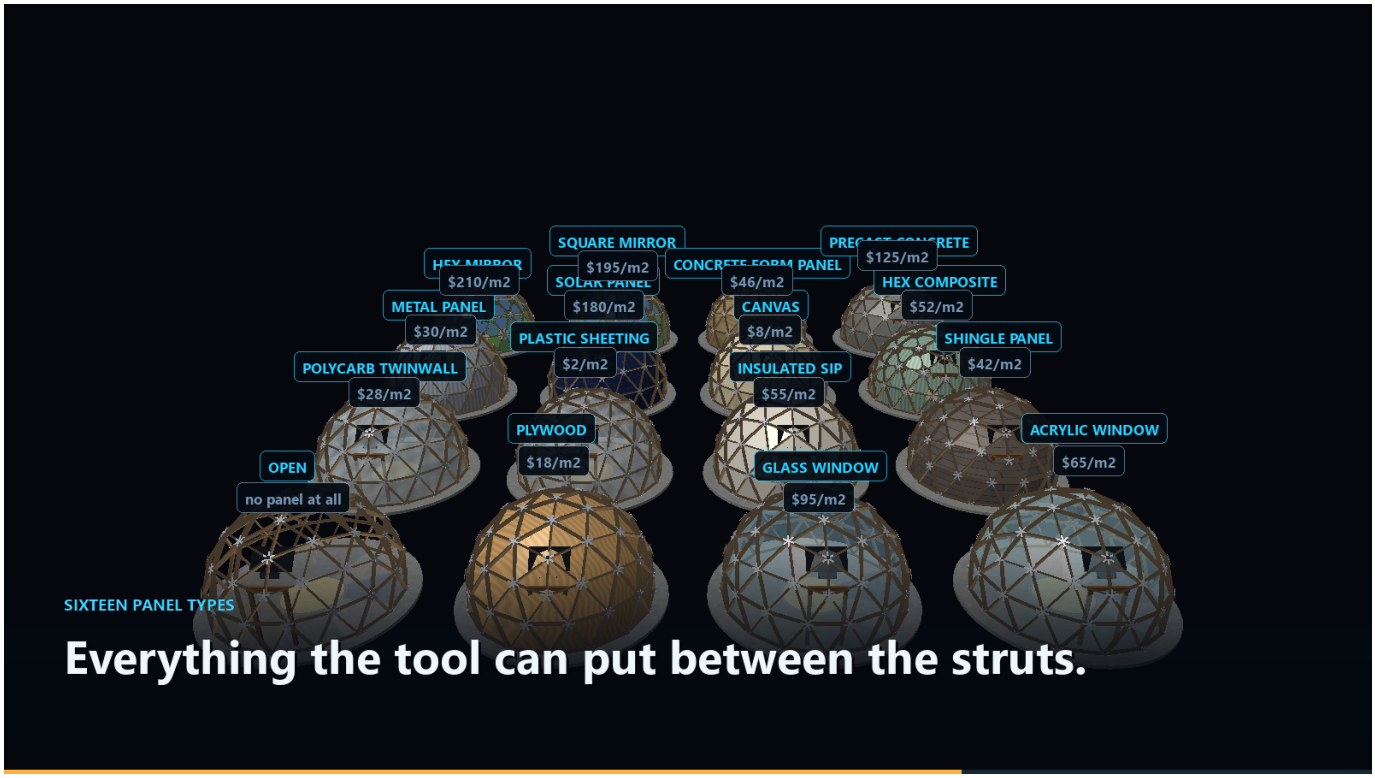

Figure 83. Sixteen panel types

Everything the tool can put between the struts. This is the menu that decides both what a dome looks like and what it. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Insulated panels

An insulated panel puts the insulation in the bay rather than on top of the shell, and that is a real architectural choice rather than a detail.

THE BAY AS A CAVITY

The wedge gives you the cavity for free.

Because the member's inward face stands proud of where the panel sits, every bay has a lip already cut into it by the shape of a split log — three-quarters of an inch on the reference build. An inner panel drops onto that lip, and behind it is a cavity as deep as the member: six inches.

Forty bays of six-inch cavity is a substantial volume of insulation if you want it there.

WHY THE REFERENCE BUILD SHIPS IT EMPTY

Two reasons, and the first is the honest one: money. A dome that ships insulated is a dome that costs more on day one, and the campaign's entire argument is a low entry price.

The second is better. An empty cavity is a duct, and the seam channels connect all forty of them. Fill the bays and you lose the airflow experiment completely.

WHAT GOES IN, IF ANYTHING DOES

Quilted recycled fabric, a layer at a time. Roughly R-1.6 a layer.

Blown cellulose, if you accept that it settles and that you will never get it out again.

Rigid foam, cut to triangles. Best performance per inch, most waste, and forty triangles of offcut.

Nothing. Which is a legitimate answer for a workshop, a store or a summer building.

THE DECISION UNDERNEATH

An insulated panel decides its R-value at the factory. A cavity you can open does not.

This book's bias is toward the second, and it is the same bias that runs through the whole method: prefer the thing you can change your mind about.

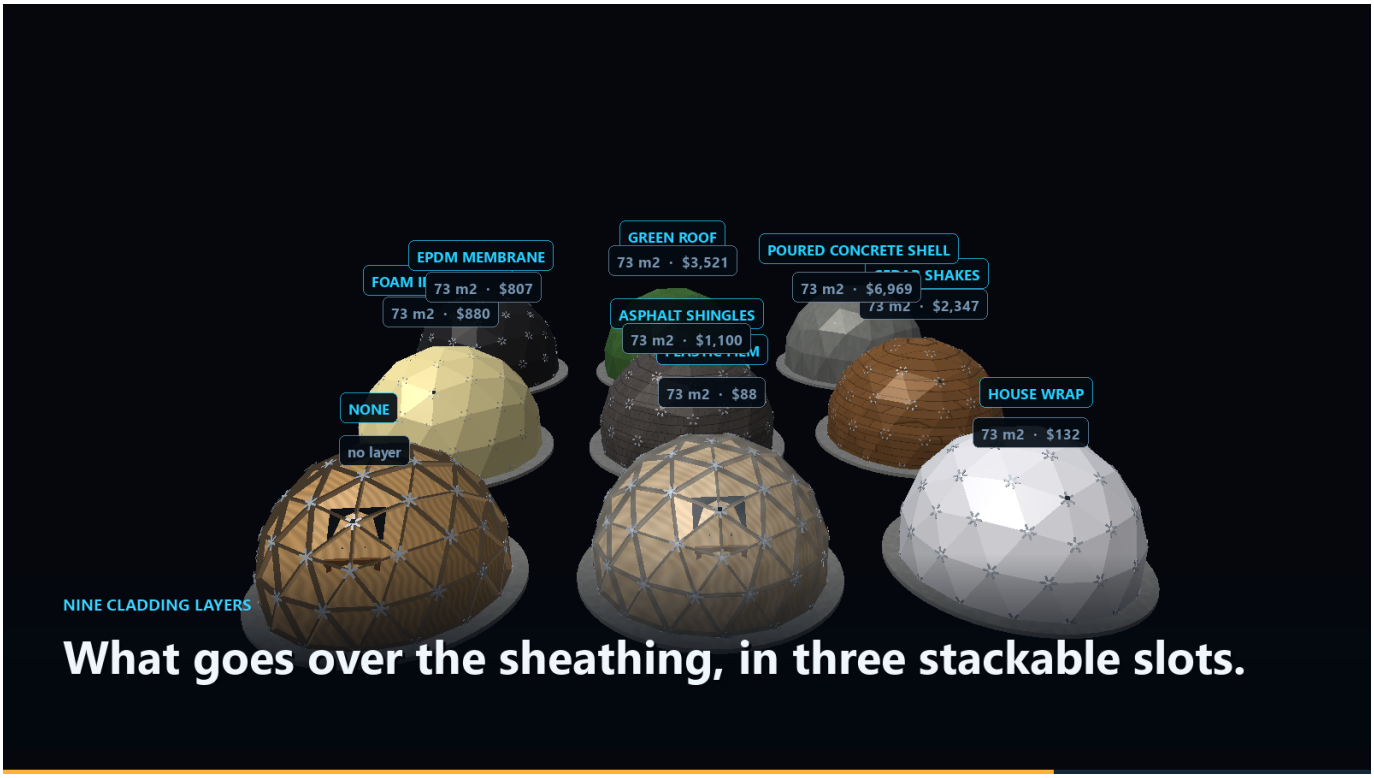

Figure 84. Nine cladding layers

What goes over the sheathing, in three stackable slots. Over the panels the tool will stack up to three cladding layers:. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Removable panels

A panel you can take off is a different object from a panel you screwed on, and the difference shows up in year three rather than on the day.

WHY REMOVABLE

Because the cavity behind it is where the insulation goes, and the insulation is bought a layer at a time over years.

Because a panel gets damaged and replacing one should not mean dismantling six.

Because a bay's function changes — a blank becomes a window, a window becomes an exhaust, a solid panel becomes a solar bay — and the whole fit-out argument depends on that being an afternoon's work.

HOW THE REFERENCE BUILD DOES IT

The panel sits on the outside face of the frame and is pulled into it by screws landing in threaded inserts set in the wedges.

Four inserts per bay, 160 for the dome.

That arrangement matters more than it sounds. The older version had the panel compression-fitted into the bay from outside, held by friction — and most of a dome's bays are overhead, so a friction fit overhead spends the life of the building arguing with gravity.

Screwed in, gravity works with the fixing. And the panel comes off with a driver rather than a pry bar.

THE SECOND JOB THE INSERTS DO

They are a grid of structural anchor points on a known pitch.

Three inserts together hold about 540 pounds, which means a shelf, a bunk, a hanging rail or a kitchen run can span two or three triangles and bolt to them.

So a bay does not have to be used to be useful. It is a barrier with fixings in it, and that is worth more than the panel.

THE COST

Inserts and flange screws come to a couple of hundred dollars across the dome, and about four hours of build time to set them.

That is the price of every future change to the building being a driver job.

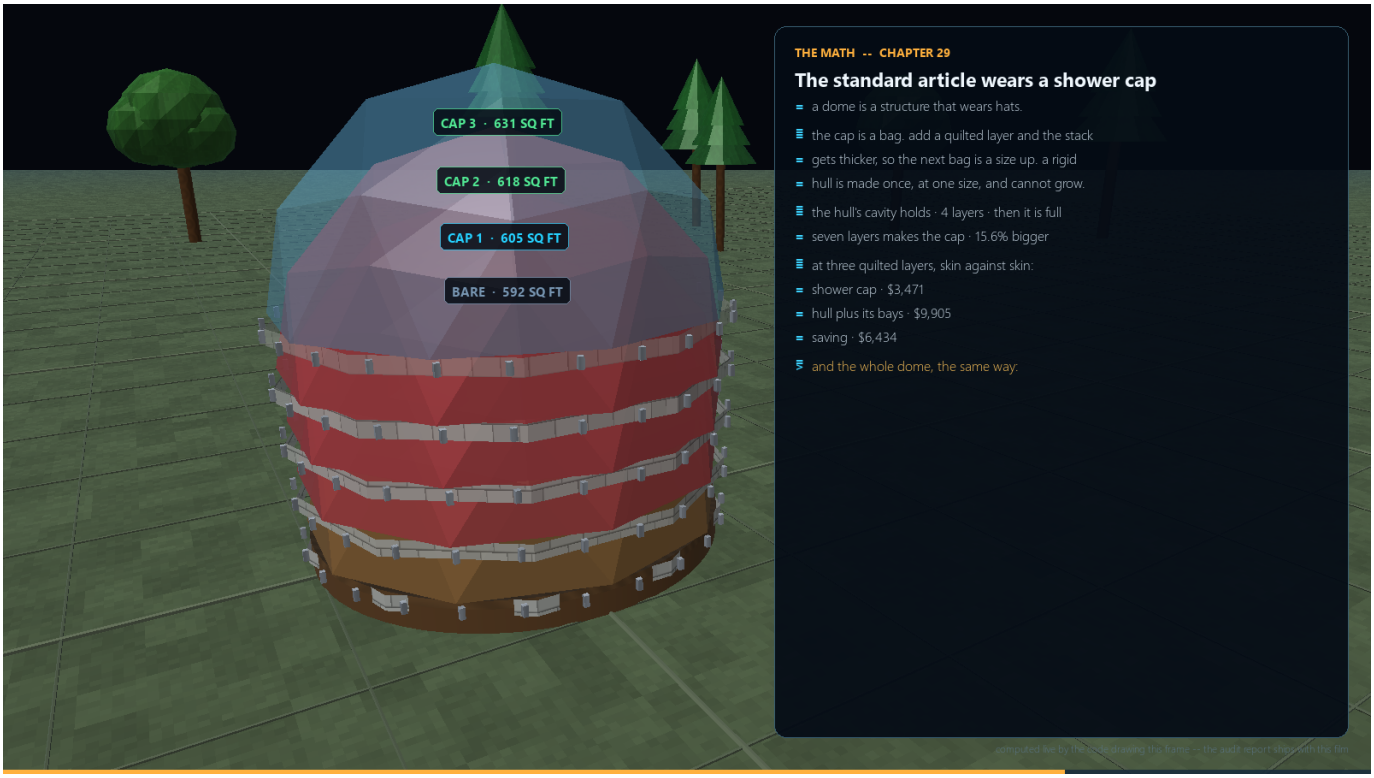

Figure 85. The standard article wears a shower cap

The cap replaces the hull and the bays, and it stacks hats. We changed what goes over the frame. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Snap-in concepts

A snap-in panel goes in without tools. It is the direction this method is heading and it is not where the reference build is yet, so this section is honest about which is which.

WHAT WOULD MAKE IT WORK

A gasket that compresses when the panel is pushed home, and a catch that holds it against the gasket's spring.

That is a door seal, and door seals are a solved problem — but they are solved on rectangular doors in rigid frames, and this is a triangle in a frame made of split logs.

THE PROBLEM THAT HAS TO BE SOLVED FIRST

Split timber is not dimensionally identical bay to bay.

A snap fitting needs the opening to be the same size every time, to within a millimetre or two, or the catch either does not reach or does not release.

Forty bays made from real trees are not that consistent, and making them that consistent means machining the inside faces of every member — which is exactly the work the wedge method exists to avoid.

THE HONEST POSITION

Snap-in is the right answer for a manufactured panel in a manufactured frame.

For a frame split out of a buyer's own trees, screws into inserts are the right answer, and the four screws per bay are not the thing slowing anybody down.

WHAT IS WORTH BORROWING FROM THE IDEA ANYWAY

Captive fasteners. A screw that stays in the panel when it is undone means no bag of screws to lose and no dropped fastener at the top of a dome.

A single-handed catch at one corner, with screws at the others, so a panel can be held with one hand while the first fixing goes in.

Both of those give you most of the convenience without needing the frame to be machine-consistent.

The reference build should probably do the first. It does not yet.

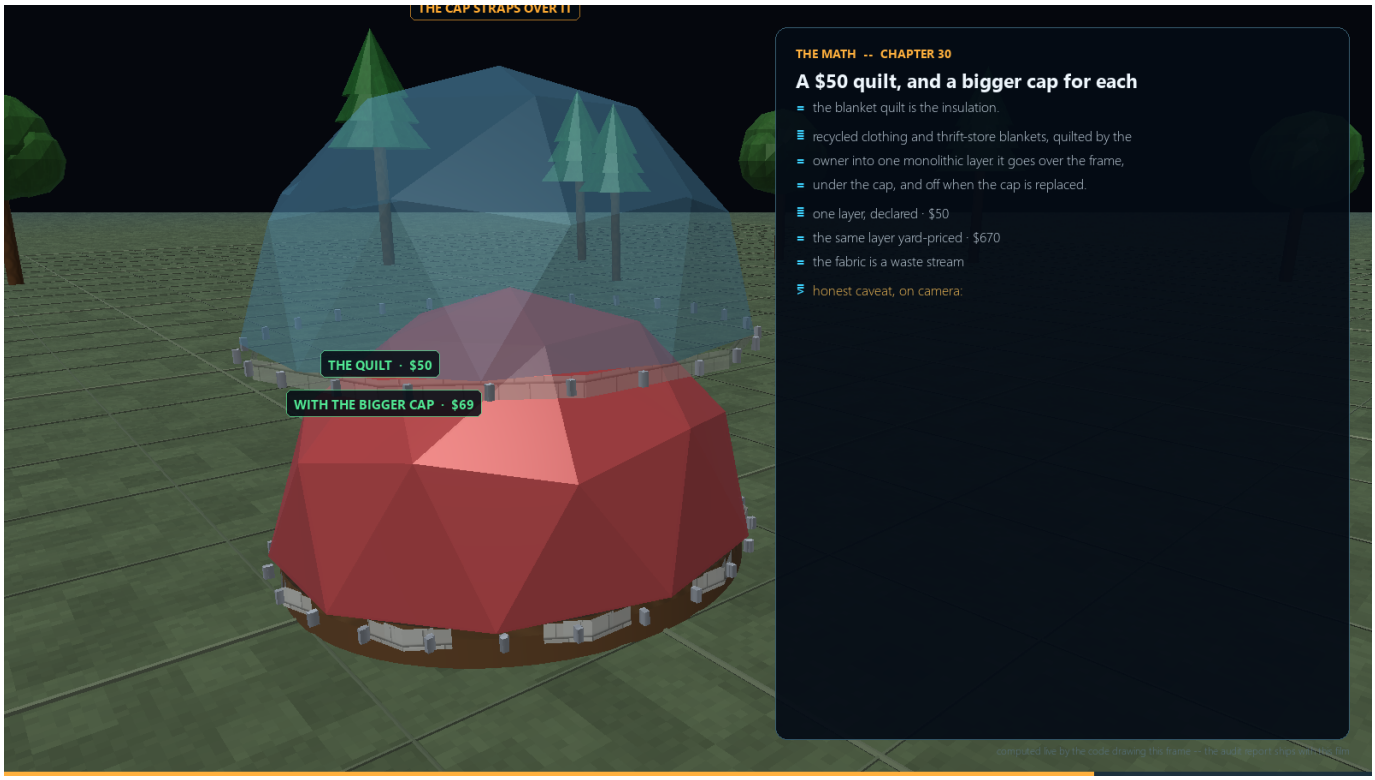

Figure 86. A $50 quilt, and a bigger cap for each

Recycled clothing, quilted by the owner, into one layer under the cap. The insulation is not a thing we sell you. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Weather seals

The dome has 55 interior seams, 10 base edges, one apex penetration and however many openings you cut. Every one of them is a place water can get in.

THE GENERAL RULE

Shed, do not seal.

A detail that relies on a bead of sealant is a detail with a maintenance schedule. A detail where water runs downhill, over a lap, and away needs nothing from anybody for twenty years.

Use sealant as a backup to a shed detail, never as the detail.

THE SEAM

This is the interesting one, because the reference build does something unusual with it.

Two sawn faces meeting at a dihedral angle do not close flush. There is a wedge-shaped gap along every one of the 55 interior seams — 309 feet of it, reaching every vertex.

Every other way of building a dome machines that out. The reference build caps it and uses it: as a conduit for services, and as an air duct, blown to keep the frame dry or drawn to take what lands on the shell down into a tank.

So the seam detail is not only a weather detail. It is a channel with a cap on it, and the cap is doing the shedding.

THE SHELL OVERLAP

The removable shell lands on the frame, not on the panels, and it overlaps every seam.

That is what makes the seams a secondary line of defence rather than the primary one. The shell sheds; the seams catch what gets past.

Two lines is the whole strategy and it is the same one used in the shower: an inner barrier that takes the spray and an outer one that catches what gets through.

THE APEX

One penetration in the whole weather surface, closed by a gasketed cap with over-centre catches.

Meant to stay shut for years and come off in ten minutes. Every service that gets added later leaves under that cap and runs down the outside, so nothing ever cuts a new hole in a weathertight surface.

That is the single best weather decision in the building and it is worth protecting. Do not let anybody drill the shell.

Figure 87. Four ways to skin it, and what each one costs

Priced by weight of glass and resin, the way a yard prices it. So what does a hull skin cost?. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Every other way: what fills the triangles

Sixteen panel types over 550 square feet of shell. The reference build is plywood, and the range from canvas to solar is a factor of a hundred and twenty.

EVERY OTHER WAY OF DOING IT

	cost	weight	
Open	$ 0	0 lb	no panel at all
Plywood	$ 919	743 lb	<- the reference build
Glass Window	$ 4,852	1,689 lb	glazing
Acrylic Window	$ 3,320	676 lb	glazing
Polycarb Twinwall	$ 1,430	191 lb	glazing
Plastic Sheeting	$ 77	17 lb	glazing
Insulated SIP	$ 2,809	1,351 lb	
Shingle Panel	$ 2,145	1,802 lb	
Metal Panel	$ 1,532	563 lb	
Solar Panel	$ 9,193	1,351 lb	190 W/m2
Canvas	$ 409	56 lb	
Hex Composite	$ 2,656	957 lb	
Hex Mirror	$ 10,725	1,464 lb	
Square Mirror	$ 9,959	1,464 lb	
Concrete Form Panel	$ 2,349	2,027 lb	
Precast Concrete	$ 6,384	10,359 lb	

The dearest of these is 140.0 times the cheapest. That is the size of the decision, and it is why the book states which one it assumes rather than leaving it to a reader to infer.

This book assumes plywood. Nothing else in it depends on that: change the row and the rest of the method is unchanged.

WHERE THESE NUMBERS COME FROM

materials.PANEL_TYPES, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

Figure 88. A thing we found out the hard way

No panel of this dome fits a four-foot sheet. Not in any rotation. One finding, because it changed how the thing gets made. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Every other way: what goes over the panels

Nine claddings over the same shell. A green roof is 24 times the weight of house wrap and nine times the cost, and the frame has to carry whichever one is chosen.

EVERY OTHER WAY OF DOING IT

	cost	weight	
None	$ 0	0 lb	bare panels
Plastic Film	$ 61	17 lb	10 mm
House Wrap	$ 92	23 lb	10 mm
Foam Insulation	$ 613	135 lb	50 mm
Asphalt Shingles	$ 766	1,351 lb	30 mm
Cedar Shakes	$ 1,634	1,013 lb	40 mm

```
EPDM Membrane          $      562      203 lb   15 mm
Green Roof             $    2,452    6,193 lb   90 mm
Poured Concrete Shell  $    4,852   13,512 lb   80 mm
```

The dearest of these is 79.2 times the cheapest. That is the size of the decision, and it is why the book states which one it assumes rather than leaving it to a reader to infer.

WHERE THESE NUMBERS COME FROM

materials.LAYER_TYPES, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

Every other way: how it is made watertight

One watertight layer or a laminated hull, and what the quilted layers under the cap add. The reference build is the cap, because two impermeable skins with insulation between them is a trap.

EVERY OTHER WAY OF DOING IT

```
                                        cost     weight
Shower cap, bare                  $    3,265              wood panels, a breather and one rain-slick cap   <- the reference
Shower cap, 1 quilted layer       $    3,333              R-2
Shower cap, 3 quilted layers      $    3,471              R-5
Shower cap, 7 quilted layers      $    3,748              R-11
Laminated hull, sheathed ply      $    5,395      222 lb   2 plies, $2.14/sq ft of materials
Laminated hull, boatyard polyester $   6,045      631 lb   3 plies, $2.63/sq ft of materials
Laminated hull, premium marine polyester $ 6,668 631 lb   3 plies, $3.10/sq ft of materials
Laminated hull, vinyl ester hull  $    9,144      747 lb   4 plies, $4.97/sq ft of materials
```

The dearest of these is 2.8 times the cheapest. That is the size of the decision, and it is why the book states which one it assumes rather than leaving it to a reader to infer.

This book assumes shower cap, bare. Nothing else in it depends on that: change the row and the rest of the method is unchanged.

WHERE THESE NUMBERS COME FROM

soft_shell and hull_laminate, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

12. Floor and Foundation Systems

A dome puts its whole weight on ten points of a circle and nothing in between, which makes most of what is known about foundations slightly wrong for it.

Every figure in this chapter is for the same 22.4-foot pad, priced off the same takeoff the campaign film reads from, so the five options can be compared against each other rather than against somebody's memory of a quote.

Figure 89. Meeting the ground

Piers, a ring beam, or a slab. Whichever you choose has to be level. A dome puts a concentrated outward-and-down load at each base hub, so the. From 2V Dome Construction Masterclass, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Slab

A concrete slab is the most expensive foundation in this chapter and the only one that is also the finished floor.

WHAT IT COSTS

$2,468 under a 22.4-foot dome.

That is the whole platform: excavation, gravel, forms, reinforcement, concrete and finishing. It is not the cheapest and it is not far off the framed alternatives, because a slab has no timber in it and timber is what the others spend their money on.

WHAT YOU GET FOR IT

A floor. Nothing else on this list is a floor — the others are platforms that a floor goes on top of, and that second floor is a cost the comparison tends to forget.

Thermal mass. A slab holds heat and gives it back, which in a building with as little mass as a timber dome is worth having in some climates and a liability in others.

Permanence. It does not rot, it does not move, and it does not need looking at again.

WHAT IT COSTS YOU

The dome can never move. That is the whole trade, and in a method built around a building you can take apart, it is a large thing to give up.

Services are cast in. What you did not put in the pour, you cut for later.

It is a contractor job, or a very organised weekend with help. Concrete does not wait for you to think.

Damp, if there is no membrane. Put one in.

WHEN TO CHOOSE IT

When the building is never moving, when the climate rewards mass, and when you want the floor and the foundation to be one purchase rather than two.

Otherwise, read on.

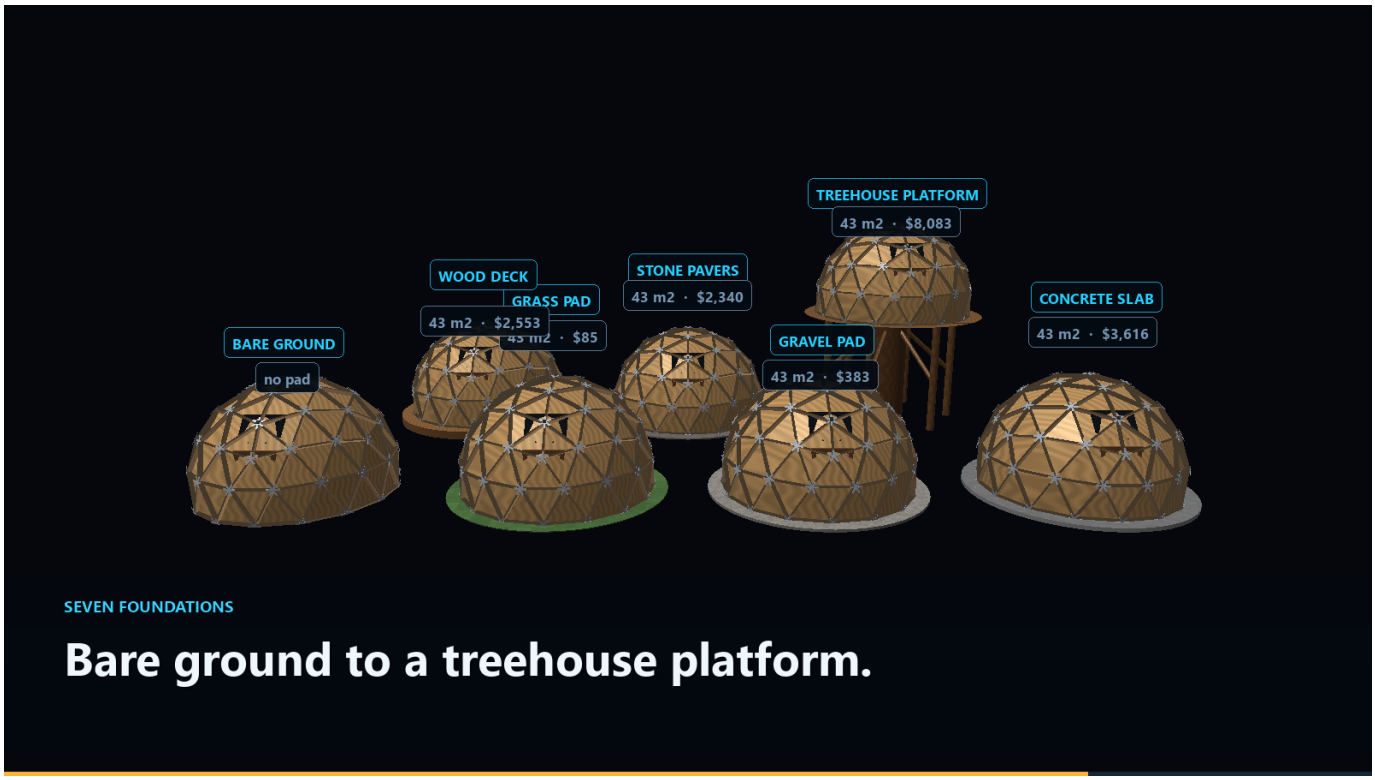

Figure 90. Seven foundations

Bare ground to a treehouse platform. Underneath, seven choices, and the tool builds each one properly:. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Ring foundation

A ring foundation is concrete where the load lands and lumber where it does not, and on a dome that turns out to be most of the floor.

THE REASON IT SUITS THIS BUILDING

A 2V dome delivers its whole weight to ten points on a circle.

Not to a perimeter wall, not to the middle — ten points, evenly spaced, where the base vertices sit. Everything inside that circle is carrying furniture and people, which is a fraction of the load and a job timber does perfectly well.

So pouring the middle is pouring concrete that nothing stands on.

WHAT IT COSTS

$3,039 for the ring and the framed middle, on 82 boards of lumber.

That is more than the slab's $2,468 — and that is worth saying plainly rather than hiding, because the intuition is that half a pour must be cheaper than a whole one. It is not. The saving on concrete is smaller than the cost of the timber deck that replaces it, and the ring's formwork is fiddlier per cubic yard than a slab's.

WHAT YOU ACTUALLY BUY

A floor you can get under.

That is the thing. Services run in the void, they stay reachable, and a mistake is an afternoon rather than a demolition saw. On a slab every pipe is a decision you made once.

It is also the honest way to build on ground that is not flat, because the ring can step and the deck inside it stays level.

WHO ELSE BUILDS THIS WAY

Grain bins and yurt platforms, both of which are round buildings that put their load on a perimeter. It is not an invention; it is the standard answer to this shape, arrived at by people who were not thinking about domes.

WHEN TO CHOOSE IT

When you want permanence at the load points and access everywhere else, and when the ground is uneven enough that a slab would need a lot of fill.

Not for the money. It does not save any.

Pier foundation

Piers are the cheapest permanent thing you can put a building on, and for a dome they are close to ideal.

WHY

Ten load points. Ten piers. There is nothing in between for a foundation to do.

A dome does not need a continuous footing because it does not have a continuous wall. Every other building on a pier foundation is compromising; this one is matching.

WHAT IT COSTS

$2,916 for piers, beams, joists and a sealed 2x6 deck — 116 boards.

That is the reference build's staple and the bare minimum that hosts a dome. Two people and a weekend.

Add $4,263 if you want a floor you can mop: the same platform with plywood and epoxy over it. The ply is not optional. Epoxy over gapped deck boards cracks at every joint, every time.

WHAT A PIER IS

A precast block on compacted gravel, a poured footing, a screw pile, or a treated post in the ground, in rising order of cost and permanence.

Blocks on gravel are what the reference build prices, because they are what somebody can actually buy on a Saturday.

THE TWO THINGS THAT GO WRONG

Frost. A pier that does not reach below the frost line will lift, and it will lift unevenly, and a dome on an uneven ring racks. Know your depth.

Settlement. Ten piers on undisturbed, compacted ground settle together. Ten piers where one sits on backfill settle nine ways. Compact properly or find different ground.

WHEN TO CHOOSE IT

Almost always, for a first building. It is cheap, it is quick, the dome can leave, and everything under the floor stays reachable.

Raised deck

A raised deck is a pier foundation with the piers made taller, and it changes what the building is.

WHAT IT BUYS

Level, on ground that is not.

Airflow underneath, which on a timber building in a wet climate is worth more than most of the things people spend money on instead.

Height, which gets the floor above snow, above runoff, and above the wildlife that would otherwise investigate the underside.

Storage and services in the void, at a height you can stand in rather than crawl through.

WHAT IT COSTS

Above the $2,916 for the standard platform, the extra is posts, bracing and a stair. Height is cheap in material and expensive in the bracing it demands — a deck two feet up needs almost none, and one six feet up needs diagonal bracing in both directions or it will sway.

THE THING THAT CATCHES PEOPLE

A tall deck is a structure in its own right and it wants engineering the dome above it does not.

The dome is a triangulated shell that resolves its own loads. A post-and-beam platform under it is a frame that racks, and the taller it gets the more of a problem that is. The dome will not save it.

Brace it properly, in both directions, and do not let the bracing stop at the fun part.

THE DETAIL WORTH GETTING RIGHT

A step is not a stair. Anything over about eighteen inches needs real treads, real risers and a handrail, and putting that in at the start is a fraction of the cost of retrofitting it after somebody falls.

Figure 91. The platform, built one course at a time

Piers, beams, joists, boards. A hundred and sixteen of them. Before any of that lands anywhere, somebody builds the thing it lands on -- and this is the part people imagine is expensive. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Portable platform

A platform that comes apart is the foundation that matches a building that comes apart, and it is the least-built option on this list for reasons that are mostly habit.

WHAT IT IS

A deck built in panels, bolted together, sitting on adjustable feet or blocks rather than anything poured.

Four to ten panels for a 22.4-foot pad. Two people carry a panel. The whole thing goes together in a morning and comes apart in an afternoon.

WHAT IT COSTS

Broadly the $2,916 of the framed platform, plus the hardware that makes the joints demountable and minus the piers you no longer pour.

It is not cheaper. It is reversible, which is a different kind of value and the one this method keeps reaching for.

WHERE IT EARNS ITS KEEP

Rented land. A platform that leaves no trace is a platform a landowner will say yes to.

Permitting. In many places a structure on a non-permanent foundation is a different animal from one on a poured footing, and the difference is sometimes the difference between allowed and not. Ask locally; do not assume.

Trial siting. Put the dome where you think you want it, live with it a season, and move it fifty feet when you find out you were wrong about the morning sun.

THE COMPROMISES

It moves. Adjustable feet need adjusting, seasonally, and somebody has to remember.

It does not like slopes. Every inch of fall is another inch of adjustment range you need at one edge.

And the services have to be demountable too, which is easy if the design allowed for it and infuriating if it did not. The reference build's utility core exists partly for this reason.

Prepared building pad

The pad is what goes under every other option here, and it is the one people skip.

WHAT IT IS

Strip the topsoil. Cut and fill to level. Compact. Lay graded stone. Compact again. Drain away from the building.

WHAT IT COSTS

$443 for a 22.4-foot pad.

That is the cheapest line in this whole chapter by a factor of five, and it is the one that decides whether any of the others work.

WHY IT MATTERS MORE THAN THE THING ON TOP

Every foundation on this list transmits its load into the ground. If the ground moves, the foundation moves, and it does not matter what the foundation is made of.

Uncompacted fill settles. It settles for years, it settles unevenly, and a dome on an uneven base ring racks — which shows up as doors that stop closing and panels that will not sit in their bays.

Concrete does not fix bad ground. It spreads the load over more of it, which helps, and then cracks where the ground moved anyway.

DRAINAGE IS THE OTHER HALF

Water under a building is what turns settled ground into moving ground.

Fall away from the pad in every direction. A swale uphill if the site has one. Gutters, if there is anything to catch.

None of it is expensive and all of it is boring, which is exactly why it gets left out.

THE HONEST SUMMARY OF THE CHAPTER

Cheapest to dearest, on the same 22.4-foot pad:

- gravel base only: $443 — not a floor, and nobody lives on it
- full concrete slab: $2,468 — and it is the floor
- framed deck on piers, sealed: $2,916 — the staple
- concrete ring, timber middle: $3,039
- framed deck, ply and epoxy: $4,263 — a floor you can mop

Two things in that list are worth stopping on. The slab is not the expensive option people assume, because it is the only one that does not then need a floor built on it. And the ring, which sounds like it ought to be the clever economy, costs more than the slab it half-replaces.

Neither of those is the answer the argument wanted. They are what the takeoff says.

Every other way: what it stands on

Five platforms taken off board by board, and seven priced at a rate per square metre. The two lists disagree, which is what an estimate and a take-off do; the book uses the take-off.

EVERY OTHER WAY OF DOING IT

		cost	weight	
Compacted gravel base	$	443		taken off, not rated: 1 lines; not a floor
Full concrete slab	$	2,468		taken off, not rated: 3 lines
Framed deck on piers, sealed	$	2,916		taken off, not rated: 7 lines <- the reference build
Concrete ring, wood middle	$	3,039		taken off, not rated: 9 lines
Framed deck on piers, ply and epoxy	$	4,263		taken off, not rated: 8 lines
Grass Pad (rate)	$	51	227 lb	$2/m2, a rate rather than a take-off
Gravel Pad (rate)	$	232	9,077 lb	$9/m2, a rate rather than a take-off
Concrete Slab (rate)	$	2,187	17,020 lb	$85/m2, a rate rather than a take-off
Wood Deck (rate)	$	1,544	1,589 lb	$60/m2, a rate rather than a take-off
Stone Pavers (rate)	$	1,415	7,375 lb	$55/m2, a rate rather than a take-off
Treehouse Platform (rate)	$	4,889	3,971 lb	$190/m2, a rate rather than a take-off

The dearest of these is 95.0 times the cheapest. That is the size of the decision, and it is why the book states which one it assumes rather than leaving it to a reader to infer.

This book assumes framed deck on piers, sealed. Nothing else in it depends on that: change the row and the rest of the method is unchanged.

WHERE THESE NUMBERS COME FROM

pad_deck.compare() and materials.FOUNDATION_TYPES, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

13. Utilities

A conventional house runs its services in wall cavities. A dome does not have a wall cavity, and the attempts to pretend it does are where most of the trouble comes from.

This method puts everything in one column standing in the middle of the building — $1,405 of it, in through one port in the floor and out through one sleeve at the apex. Three designed penetrations in the whole envelope, all of them gasketed, none of them cut later.

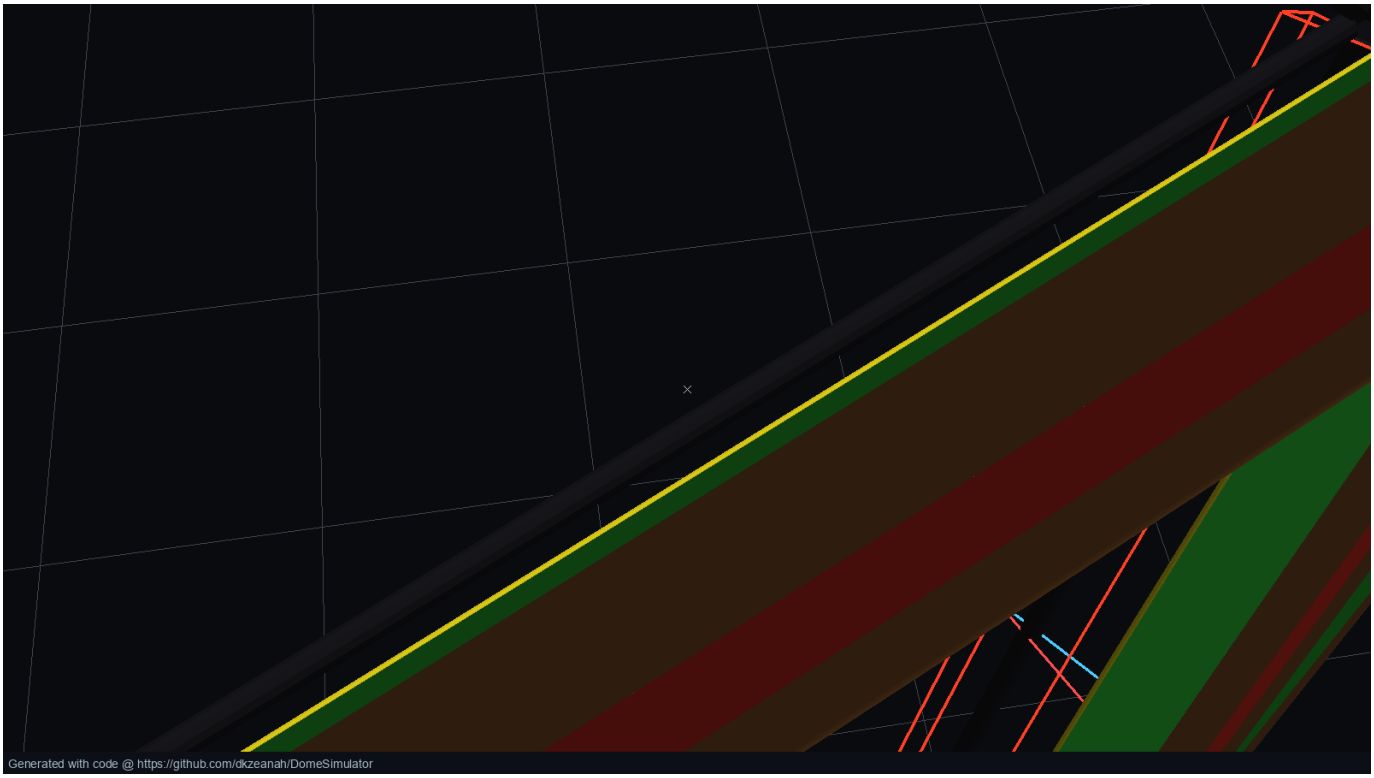

Figure 92. The seam as a service route

A hose in the seam channel instead of a rigid key. The same space that has to be filled anyway takes a 2.88-inch line, so air and water run in the structure rather than beside it. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Electrical entry

Electricity comes into this building at one point, low, and goes up.

WHERE

The floor port, in the middle of the pad, under the dome.

Not through a wall. A dome does not have a wall in the sense a service entry wants — every surface is curved, sloping and part of the weather envelope, and a penetration through it is a penetration through the roof.

The middle of the floor is flat, sheltered, and already where everything else arrives.

WHAT IT IS

A sub-panel with breakers and an outlet ring, priced at $310 in the reference build. That is the equipment, not the supply.

The supply is somebody else's number: a spur from a meter, a generator, a battery bank, or the shared hub the campaign film prices at a quarter of a four-way panel. Which of those you have changes the cost by thousands and changes nothing about the dome.

THE RULE THAT MATTERS

Everything electrical goes up the column, and the column is metal and grounded.

That gives you one chase, one route, and one place to look. The alternative — cable run round the inside of a curved shell — is how you end up drilling into something you should not have.

SEPARATION

Water rises in the same column. They are in the same object and they are not in the same chase: the electrical side is its own conduit with its own clearance, and the wet side drains downward past it.

Three feet of clearance between the electrical and anything wet is the reference build's rule, and it is the one number in this chapter nobody should shave.

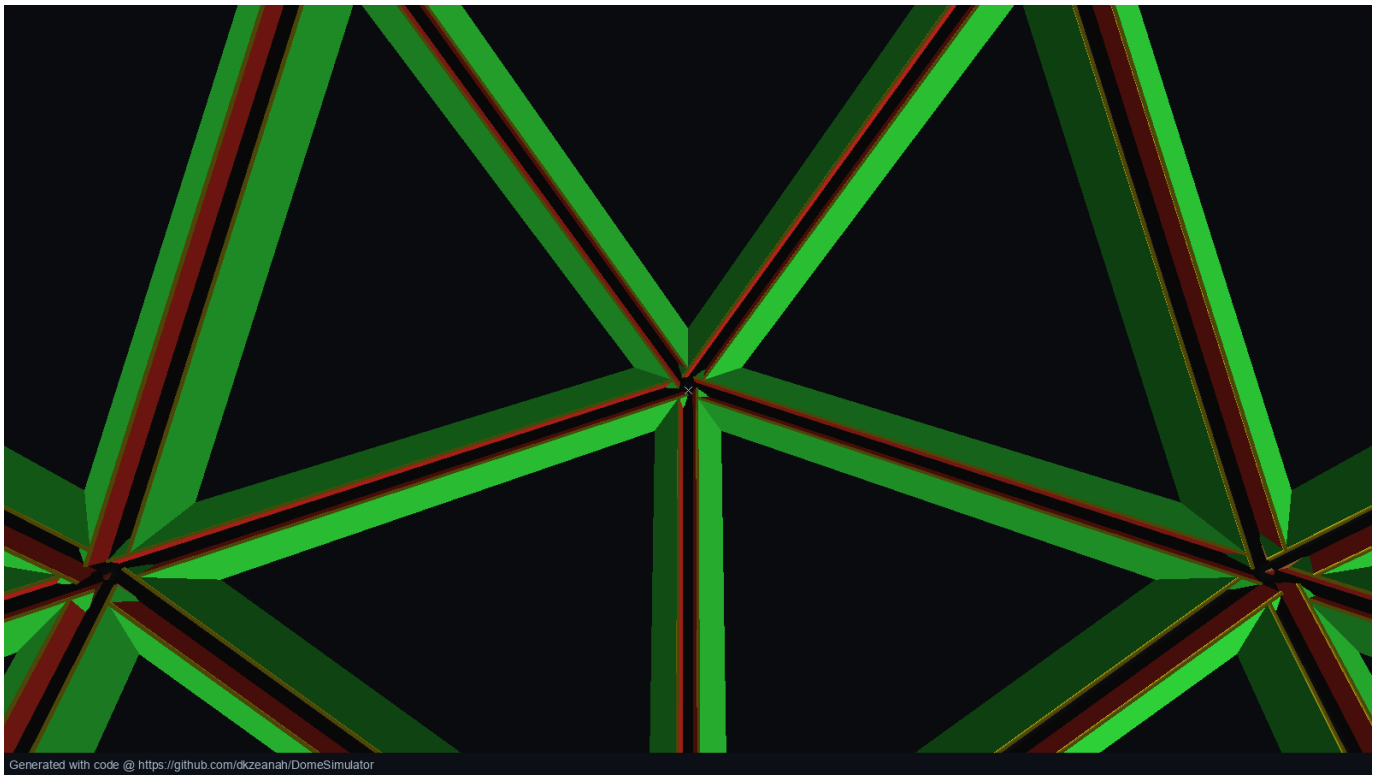

Figure 93. The apex from inside

Where the utility column arrives and the seal cap closes. Five seams converge here, which is why this is the one penetration the design allows. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Water entry

Water arrives where the electricity does: at the pad port, in the middle of the floor, and rises from there.

THE MANIFOLD

A water manifold and rise from the pad, $240.

One inlet, one shut-off, and branches off it at working height. That is the whole thing, and the reason it is one object rather than a run of pipe is so that disconnecting the dome from the pad is one action.

WHERE IT COMES FROM

Four honest options, in rising order of independence:

A spur from a mains supply. Cheapest to run, most dependent.

A tank, filled by truck or hose. The reference build prices 120 gallons under the floor at $1.35 a gallon of capacity, which is a few hundred dollars and buys a week or two depending entirely on how you live.

Rainwater off the shell. A dome is a very good catchment — it is a smooth curved surface with a single low edge — and the seam network can be run as the collector.

A well. Different chapter, different trade, and the only one on this list that needs a rig.

FREEZING

A pipe in an unheated void freezes. The under-floor space is an unheated void.

Either the tank and the rise are inside the insulated envelope, or they are heat-traced, or the system drains down when the building is empty. Pick one deliberately. Discovering in February which one you picked by accident is a bad afternoon.

Figure 94. The top of the dome is a socket

Power and water come up the middle and out through the apex. Now the part the whole system hangs off. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Plumbing

Waste is the part of a small building people plan last and regret first.

THE DRAIN STACK

A drain stack and floor-port tie-in, $165.

Everything wet in the dome drains to the middle of the floor and out through the pad. That is the opposite of a conventional building, where waste runs to a wall and out sideways — and it is the right way round here, because the middle of the floor is the only place the geometry guarantees is reachable.

FALL

Waste moves because it is running downhill, and in a building on a platform there is not much downhill available.

A quarter-inch of fall per foot is the number, and the shower tray in the reference build's wet area is built to exactly that over a three-foot run. Under a quarter-inch the water moves and the solids do not, which is the failure people describe as "the drain keeps blocking" and is actually "the drain was never going to work".

WHERE IT GOES

Off the pad and into whatever the site has: a sewer connection, a septic field, a holding tank, or a greywater system with a composting toilet handling the rest.

That last combination is the one that suits this building best, because it halves the waste problem and removes the one part of it that needs treating.

THE WET AREA

Water and electricity in the same small round room is the thing to design rather than hope about.

The reference build's answer: a sloped tray that redirects to the drain, two barriers round the shower rather than one — a double glass wall or a double curtain — and three feet of clearance between anything wet and anything electrical.

Two barriers is not belt and braces. The inner one takes the spray and the outer one catches what gets past, which is the same two-line strategy the shell uses against rain.

Figure 95. Build the chase on the bench

3 steps, 95 minutes. The chase first, and on a bench, upside down. From Build one utility core, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Central utility column

The utility column is the single best idea in this building and the one hardest to explain on paper.

WHAT IT IS

A metal column standing in the middle of the dome, floor to apex, carrying everything: water up, waste down, electrical up, and out through the top.

Priced at $1,405 complete, over 6.5 hours to assemble on a bench.

WHY THE MIDDLE

Because a dome has no good walls and one excellent middle.

Every service in a conventional house runs in a wall cavity. A dome's equivalent of a wall cavity is a triangular bay in a curved shell, and running services there means they follow the curve, cross the seams, and end up somewhere different from where you left them.

The middle is vertical, straight, and the same at every height.

WHY IT IS ONE OBJECT

Because then it can leave.

The column unbolts from the pad port at the bottom and the apex sleeve at the top, and goes into the next dome. The campaign film prices that move at $260 and rates the core for 25 years, which is longer than most of the buildings it will stand in.

That is the modular argument, and it only works because the services are one thing rather than forty things buried in a shell.

WHAT IS ON IT

The housing and apex sleeve, $380. The water manifold. The drain stack. The sub-panel. A service chase running 10.5 feet from floor to apex at $11 a foot. And the seal cap over the apex penetration at $195.

Six items. That is the entire service infrastructure of the building, and you can walk round it.

Figure 96. Close it, prove the cap, stand it up

2 steps, 50 minutes. Close it, prove the cap, stand it up. From Build one utility core, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Utility routing through the floor

Everything arrives through one hole in the floor, and that hole is the most important detail in the building's services.

THE PORT

A service port up through the deck, $340.

One opening, lined, sealed to the deck, with the water, waste and electrical passing through it and the column landing on top of it.

WHY ONE HOLE

Because a hole in a floor is a place for air, water and animals, and one of them is manageable.

Also because the dome has to be able to leave. Lift it off and what is left is one capped port in a platform, not a scattering of stubs.

THE VOID UNDERNEATH

On a framed platform there is a crawl space, and everything upstream of the port lives in it: the tank, the incoming spur, the shut-off, the drain run out to wherever it goes.

That space is unheated, so see the freezing note two sections back, and it is where the reference build puts $420 of storage because the space exists whether you use it or not.

On a slab, none of this is true. Everything is cast in, it is in its final position, and the crawl space does not exist. That is the slab's real cost and it does not appear in its price.

THE ONE THING TO GET RIGHT

Seal the port to the floor, not to the column.

The column comes out. If the seal belongs to the column, taking the column out opens the floor. If it belongs to the floor, the floor stays closed and the column lifts through it.

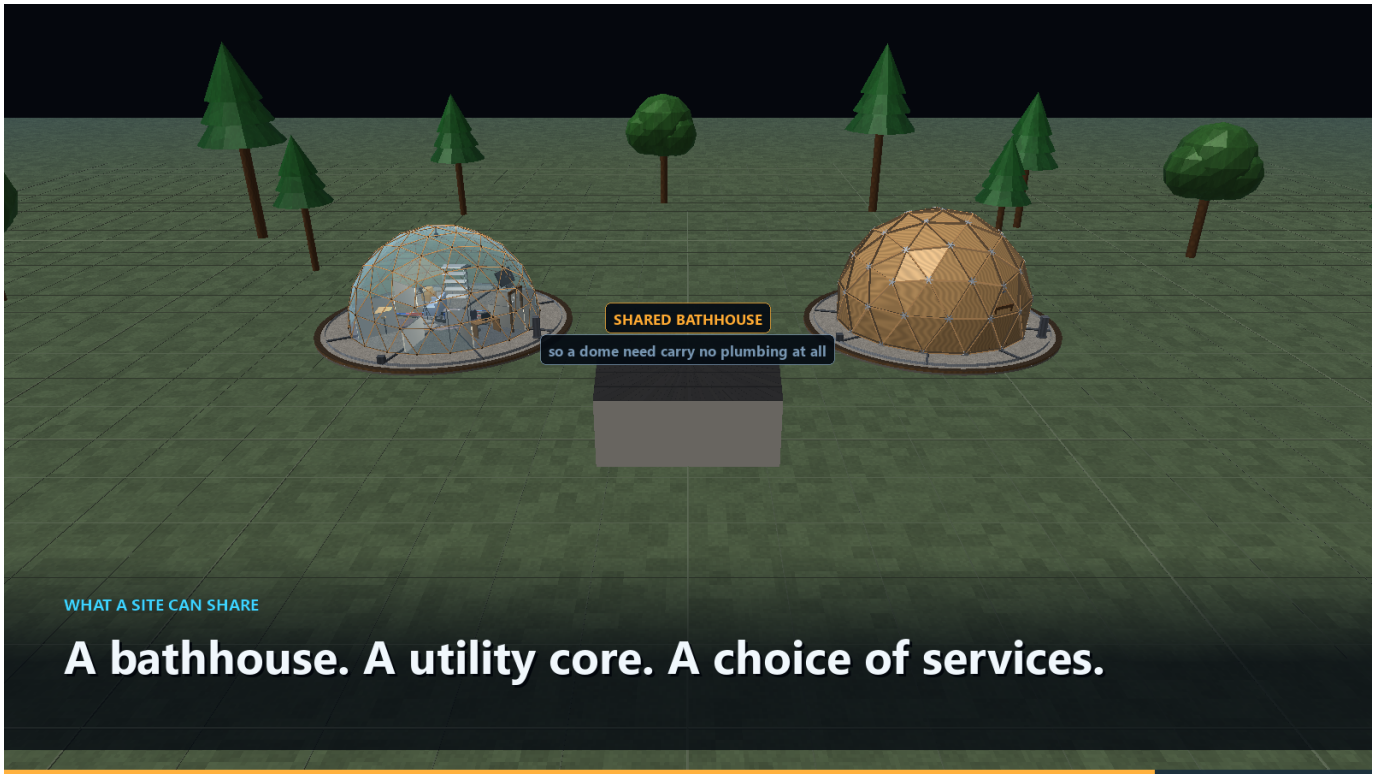

Figure 97. What a site can share

A bathhouse. A utility core. A choice of services. Some hosts can build around communal services. A bathhouse means a small dome may not need its own bathroom. An on-pad utility column is another option: sink, shower, toilet, outlets, and drain at the known connection point. From Bring Your Own Dome, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Apex routing

The top of the dome is a socket, and everything the building grows later leaves through it.

WHAT IS UP THERE

A sleeve through the apex panel, the riser passing through it, and a gasketed seal cap over the lot, $195.

The riser does not stop at the apex. It stands proud of it, and the part standing proud is the interface: a place a line can be picked up from outside without opening the building.

WHY THIS IS WORTH THE TROUBLE

Because it means nothing ever cuts a new hole in a weathertight surface.

Solar goes on later, and it comes down from the apex. A satellite dish, an aerial, a weather station, a vent, a flue for a stove that was not there originally: all of them leave under the cap and run down the outside.

The alternative — one penetration per addition — is how a roof becomes a leak.

THE CAP

Gasketed, with over-centre catches, meant to stay shut for years and come off in ten minutes.

It is the single best weather decision in the building, and the one to protect. Anyone who drills the shell to save walking a cable up has undone it.

THE PART PEOPLE DOUBT

A hole in the highest point of a roof sounds like the worst possible place for one.

It is the best, for the same reason a boat's deck fittings are on the high side: water does not collect there. A penetration halfway down the curve sits in the path of everything running off the surface above it. The apex has nothing above it.

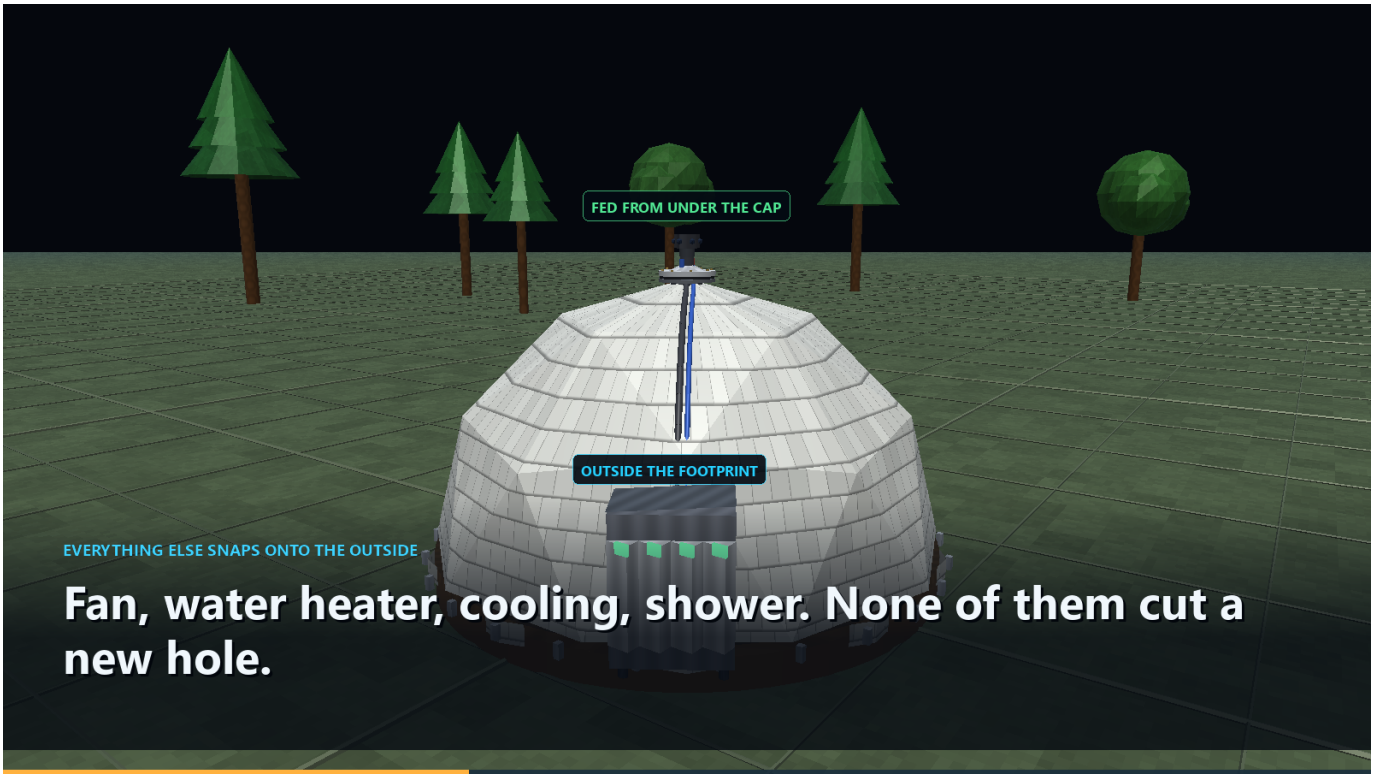

Figure 98. Everything else snaps onto the outside

Fan, water heater, cooling, shower. None of them cut a new hole. Under that cap, a line runs down the outside of the shell to a panel hanging off the rim. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Exterior routing

Once a line is outside the shell, it runs down the outside, and that is a deliberate choice rather than an admission of defeat.

WHAT IT COSTS

$6.40 a foot for conduit, clips and weathering.

WHY NOT INSIDE

Because inside means through the insulation, through the vapour control, and through a bay somebody will want to open later.

And because the thing a service usually connects to — a solar panel, a tank, a second building — is outside anyway. A line that goes out at the apex and down the outside crosses the envelope once. A line routed internally crosses it twice and does the second crossing somewhere less convenient.

HOW IT LOOKS

Like conduit on the outside of a building, which is to say, industrial.

Some people mind. The honest answer is that the alternative is worse and the conduit can be painted the colour of the shell, which is what the reference build does.

THE PANELS IT FEEDS

Utility panels — the reference build calls them polyps — bolt to the outside of the dome and take their service from this routing. A wall port at $88 and the service connection at $95.

Each one is a bathroom, a kitchen, a plant room or a battery bank hung on the outside of the shell, fed from the top, occupying none of the floor.

That is what the exterior routing is for. It is not tidying-up; it is the delivery mechanism for everything the building becomes later.

Gasketed interface penetrations

A penetration is a hole with a promise attached, and the promise is what costs money.

THE THREE THIS BUILDING HAS

The floor port, where everything arrives.

The apex sleeve, where everything leaves.

The wall ports, where utility panels connect.

That is it. Three kinds of hole in the entire envelope, all of them designed, none of them cut later.

WHY GASKETED AND NOT SEALED

A sealed penetration is permanent. A gasketed one comes apart.

Every one of these three is somewhere the building expects to be opened: the floor port when the dome moves, the apex cap when a service is added, the wall port when a panel is swapped. Sealant at those points has to be cut out and remade every time, which means it will not be, and the thing will be open.

A gasket compresses when the fitting is done up and springs back when it is undone. It can be done a hundred times and the hundredth is as good as the first.

WHAT MAKES A GASKET WORK

Compression, evenly. That means a fitting that clamps rather than one that is screwed down at one edge.

A face to compress against that is flat and clean. On a curved shell that means a flange, and the flange is the expensive part.

And a groove for the gasket to sit in, so it cannot roll out under compression. A gasket squeezed between two flat faces will find its way out.

THE FAILURE TO KNOW ABOUT

Gaskets take a set. A gasket left compressed for years does not spring back as far, and a joint that has been open and shut a dozen times wants a new one.

They cost a few dollars. Keep spares, and change them when the building is open anyway.

Replaceable service connections

Every service in this building is meant to be disconnected, and designing for that is what makes the rest of the method possible.

THE PRINCIPLE

Nothing is plumbed or wired in permanently. Everything lands on a connector.

Water on a union or a quick-connect. Waste on a slip joint. Electrical on a plug or a terminal block rather than a spliced run. Each of those costs a few dollars more than the permanent version and takes the same time to fit.

WHAT IT BUYS

The dome moves. The column comes out and goes into the next one — $260, rated for 25 years of service.

A fault is isolated rather than hunted. A connector is a place you can test either side of.

A part is replaced rather than repaired. The fitting that failed comes off and a new one goes on, by somebody who is not a tradesman.

And an upgrade is an afternoon. Better pump, bigger tank, more circuits: all of them land on the same connectors.

THE ARGUMENT AGAINST, WHICH IS REAL

A connector is a joint, and every joint is a place that can leak or work loose. A permanently plumbed system has fewer failure points than a demountable one, and that is simply true.

The trade is between fewer failures and easier fixes, and this building chooses easier fixes — because it is a building that expects to be altered, and one that cannot be altered without a tradesman will not be.

THE DISCIPLINE THAT MAKES IT WORK

Label everything. Standardise the fittings so there are three sizes in the building and not eleven. Keep the spares in the under-floor storage that is already there.

None of that is clever. All of it is the difference between a demountable building and a building somebody once took apart and could not put back.

14. Environmental Control

A hemisphere has about 41 percent less envelope than a box of the same floor area, which is the biggest thermal fact about this shape and the reason everything in this chapter is smaller than the equivalent for a conventional building.

The shape gives that back in one place: the upper third of a dome's wall faces the sky, so it behaves like roof in summer. Most of what follows is about spending the saving well and not losing it through the top.

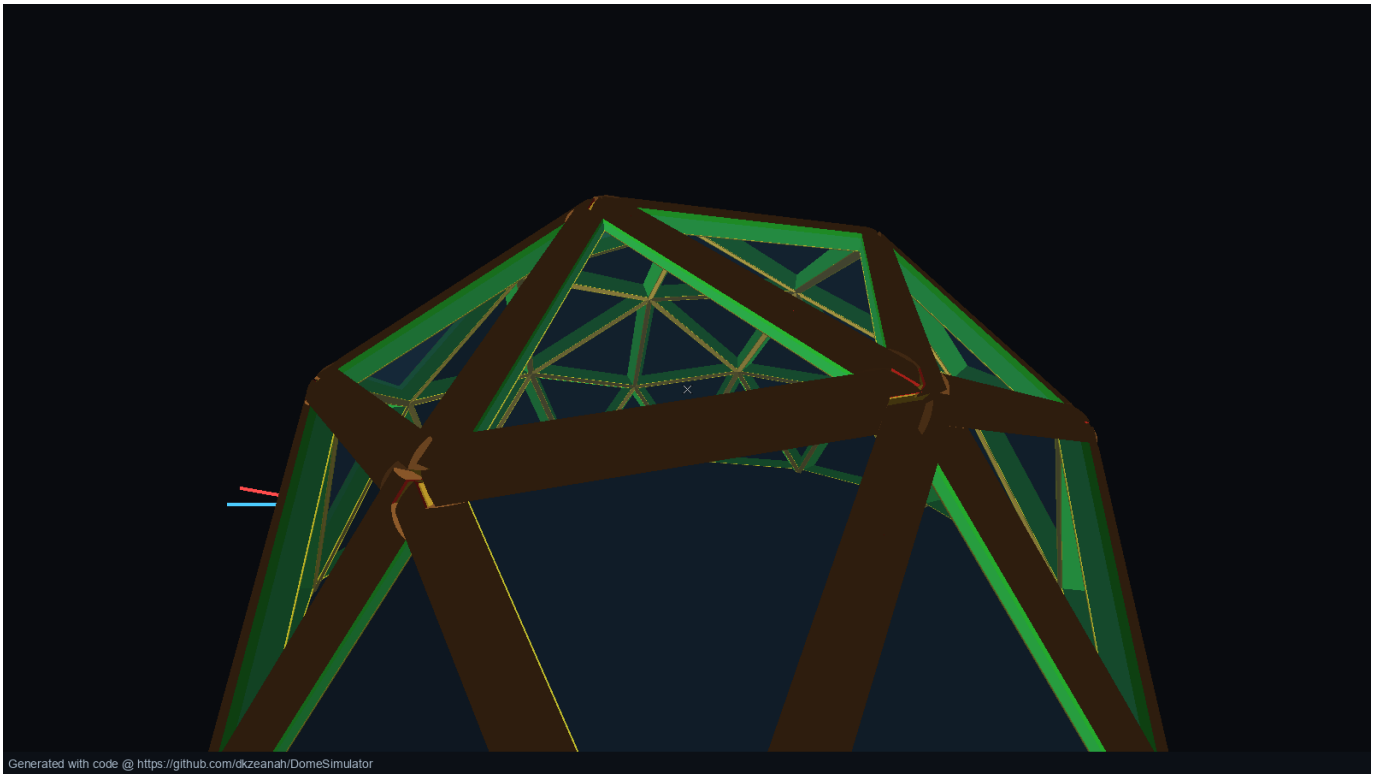

Figure 99. Inside, with the shell on

The interior with the skin in place: the cavity between the members is the insulation's, and the seam channel behind the key is the only path through the envelope. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Insulation

A dome's advantage is that it has less outside than a box does. Its disadvantage is that the outside it has is curved, and insulation comes flat.

THE SHAPE ARGUMENT, FIRST

For the same floor area, a hemisphere has substantially less envelope than a rectangular building — the reference build's own model puts it at 41 percent less.

That is real and it is the single biggest thermal thing about this shape. Less surface means less heat leaving, before anybody has installed anything.

It also means the insulation you do buy goes further, because there is less of it to buy.

THE STACK

Bare, the reference build's skin is about R-1.5. That is panels, a membrane and the air films, and it is nearly nothing — it is a tent with opinions.

Each quilted layer adds about R-1.6. Seven of them reach R-12.7, which is a real wall.

TWO WAYS TO BUY IT

In the bays, behind the panels. The frame gives six inches of cavity for free, because the wedge's inward face stands proud of where the panel sits. Fill it and it is done.

Or over the frame, under the cap. This is the reference build's route: quilted recycled clothing, sewn by the owner, at a declared $50 a layer.

The difference matters. A bay holds 4 layers and then it is full. A cap is a bag — every layer makes the next cap a size up, and there is no limit. Seven layers makes the cap 15.6 percent bigger and nothing else changes.

That is why this method insulates outside. Not because it performs better — at the same R it does not — but because it can keep going.

THE DEFERMENT ARGUMENT

Insulation bought a layer at a time is insulation bought when there is money.

The building works at R-1.5. Badly, in January. But it works, and every winter after that it is warmer, and nothing has to come apart for that to happen.

Figure 100. It gets warmer every winter

Lift the shell. Add a layer. Put the shell back. The cavity ships empty, and that is on purpose. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Condensation

This is the failure mode most likely to ruin a building like this one, and the reference build has an unproven answer to it. Both halves of that sentence matter.

THE PHYSICS, BRIEFLY

Warm air holds more water than cold air. Inside air in an occupied building is warm and wet — people, cooking, washing, breathing.

That air moves outward through the envelope. Somewhere in the thickness of the wall it reaches a temperature where it cannot hold its water any more, and the water comes out as liquid.

That place is the dew point, and it moves depending on the weather.

WHY THIS BUILDING IS EXPOSED

The reference build's shower cap is a plastic membrane, then quilted fabric, then a rain-slick cap.

Two impermeable layers with fabric between them is a moisture trap. Water arrives in the fabric, cannot get out through either side, and stays. Wet insulation does not insulate and wet fabric against timber rots it.

That is said out loud in the campaign film, on camera, because it is a real objection and pretending otherwise would be dishonest.

THE ANSWER, AND ITS STATUS

The seam network.

Every interior seam of a wedge dome is a wedge-shaped channel — 309 feet of it, reaching every vertex, connecting all forty bays. Blow air through it and the frame stays dry. Draw air through it and what condenses gets taken down into a tank.

It is a duct that the geometry gives you free, and it is exactly the right shape for this problem.

It is also unproven. Nobody has built it and measured it. The reference build prices the fans and ports and says plainly that the performance is an expectation rather than a measurement.

WHAT TO DO IN THE MEANTIME

Ventilate generously. Vent moisture at source — a cooker hood, a bathroom fan — because air that never gets wet cannot deposit water in a wall.

And if you build this before anybody has measured the duct, open a bay in spring and look.

Ventilation

A small, tight, well-insulated building needs deliberate ventilation, and a round one needs it more than most.

WHY MORE

A dome has no corners and no natural cross-draught. Air in a sphere wants to circulate in a loop rather than pass through, and a room with one opening does not ventilate at all — it just breathes slowly at that opening.

Two openings, at different heights and on opposite sides, is the minimum that produces actual airflow.

THE STACK EFFECT IS YOUR FRIEND HERE

Warm air rises, and a dome's highest point is dead centre and nine feet up.

An opening at the apex and an opening low at the perimeter gives you a chimney: warm air leaves at the top, cool air is drawn in at the bottom, and it happens without a fan for as long as there is a temperature difference.

That is free ventilation and it comes from the shape.

THE APEX IS ALREADY A HOLE

The seal cap is the obvious place for a controllable vent, and it costs almost nothing to make it one.

THE SEAM DUCT, AGAIN

Reversible airflow through the seam network is the reference build's ambition: blow to keep the frame dry, draw to collect what lands on the shell.

Unproven, as the last section said. But the ports and the fans are priced, the channel exists whether it is used or not, and a building that has the ducting in it can experiment later.

THE RULE

Ventilation is not the same as draughts, and insulating a building without ventilating it is how you get mould.

Tight envelope, deliberate openings, controllable. In that order.

Heating

A small dome is easy to heat and easy to overheat, and the second is more common.

HOW LITTLE IT TAKES

At R-12.7 with 277 square feet of floor, the heat load is small enough that most domestic heaters are oversized for it.

That is not a boast; it is a warning. An oversized heater cycles — full blast, off, cold, full blast — and a building with almost no thermal mass follows it exactly. The room is alternately too hot and too cold and never comfortable.

THE WOOD STOVE ARGUMENT

The reference build prices a small wood stove, and it suits this building for reasons beyond the obvious.

It is mass, in a building that has none. A stove and its shield hold heat and give it back slowly, which is the smoothing the envelope cannot do.

It is independent of the grid, which matters on the kind of land this building goes on.

And its flue leaves through the apex, which is already a designed penetration, so it does not cut a new hole.

The cost is floor area and a clearance zone, in a room that has 277 square feet of it.

THE ALTERNATIVES

Electric resistance: cheapest to install, dearest to run, trivial to control.

Heat pump: dearest to install, cheapest to run, and the same unit does the cooling.

Radiant, in the floor: only really available if the foundation is a slab, which is one more thing the slab quietly buys.

THE DOME-SHAPED PROBLEM

A curved ceiling focuses. Heat rising from a stove in the middle collects at the apex, and the apex is nine feet up where nobody is.

A slow ceiling fan pushing down solves it, and the reference build prices a central light and fan as one fitting for that reason.

Cooling

The dome's shape helps in winter and hurts in summer, and it is worth being straight about which.

WHAT THE MODEL SAYS

The reference build sizes a window unit at a computed 3,365 BTU/h and buys a 6,000 BTU/h one, because that is the smallest thing actually sold.

Design temperature difference 20 degrees, solar gain 2.6 BTU/h per square foot of glazing, SEER 14.

Three thousand BTU is very small. A single room in a conventional house is often twice that. The 41 percent envelope saving is doing most of the work.

THE SUMMER PROBLEM THE SHAPE CREATES

Upper panels face the sky.

In a box, the roof is the only surface facing up and it is the one everybody insulates hardest. In a dome, the upper third of the wall faces up too, and it is getting sun for the whole day rather than for the part of it when the sun is on that side.

That is the real summer penalty of this shape and it does not get talked about much.

WHAT ACTUALLY WORKS

Shade the top, which the next two sections are about.

Ventilate at night, when the outside air is cooler than the building, and shut it up during the day. In a building with this little mass that works less well than it does in a stone house, but it is free.

Do not glaze the top. A glazed upper bay is a skylight, it will light the room beautifully, and it will cook it from May to September.

THE THING THAT SURPRISES PEOPLE

A dome is a reflector. A curved glazed surface facing a room concentrates light and heat toward the centre, and in a nineteen-foot dome the centre is where you are standing.

Two or three glazed bays are pleasant. Eight is a greenhouse, and it will behave like one.

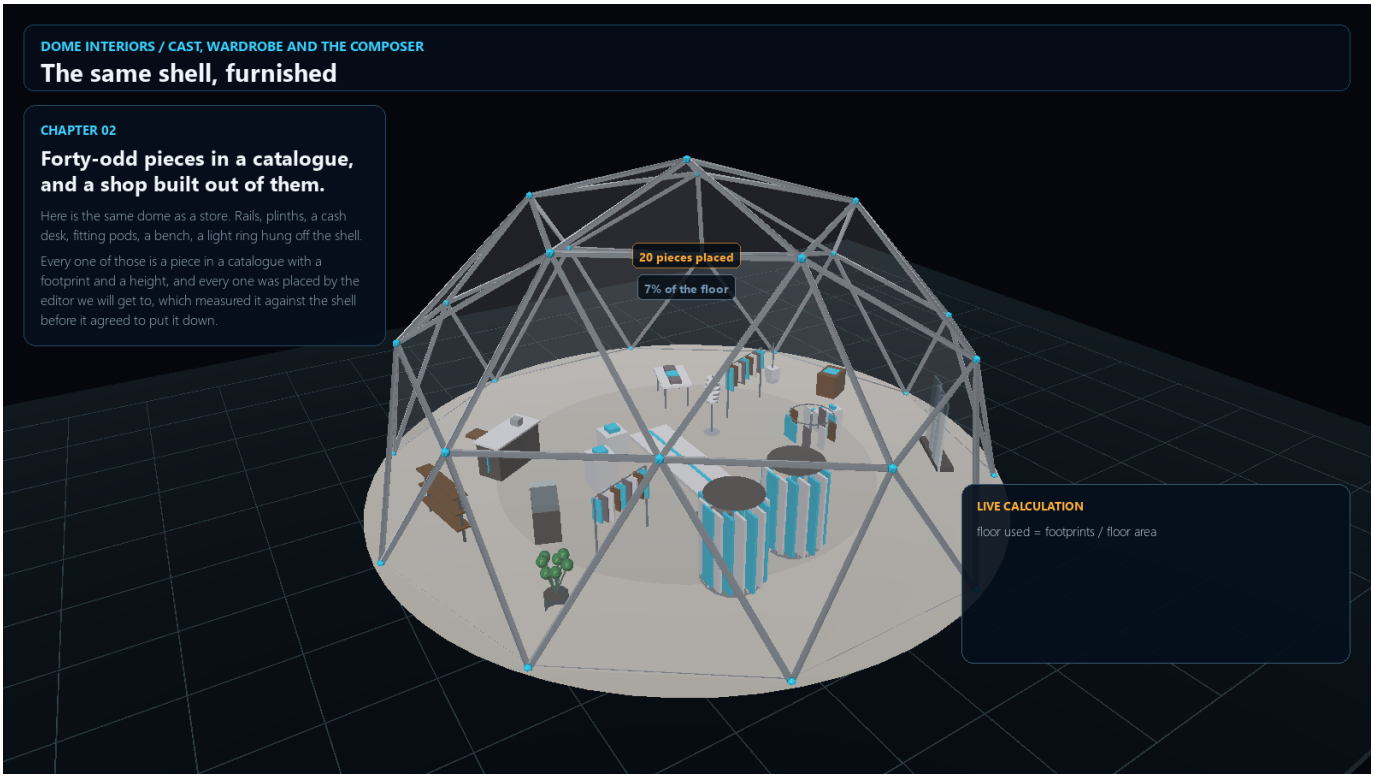

Figure 101. The same shell, furnished

Forty-odd pieces in a catalogue, and a shop built out of them. Here is the same dome as a store. Rails, plinths, a cash desk, fitting pods, a bench, a light ring hung off the shell. From The Dome House Lookbook, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Solar gain

Solar gain is heat arriving through glass, and in this shape it needs counting rather than guessing.

THE NUMBER

2.6 BTU/h per square foot of glazing is what the reference build's cooling model allows.

Multiply by the glazed area and you have the summer penalty in the same units as the heater and the cooler. Three glazed bays at roughly 15 square feet each is 45 square feet, which is about 117 BTU/h — small. Ten bays is 390, which is a meaningful fraction of the whole cooling load.

That is the calculation, and it takes a minute.

WHICH BAYS

Low bays face outward and give a view. Their gain is seasonal and modest, because in summer the sun is high and strikes them at a glancing angle.

Upper bays face the sky. Their gain is worst exactly when you least want it, because in summer the sun is high and strikes them square on.

That is the whole geometry of it, and it is why the answer to "where should the windows go" is low.

THE WINTER SIDE OF THE LEDGER

Solar gain is free heat in January.

A dome with glazing on its equator-facing side collects usefully in winter, when the sun is low and strikes those panels square. The same panels that are glancing in July are perpendicular in December.

So: glazing low on the sun side is good twice. Glazing high is bad once and good never.

THE HONEST LIMIT

None of this is a substitute for a real energy model. It is enough to stop somebody putting eight glass triangles in the roof because it looked good in a render.

Shading

Shading is the cheapest cooling there is and the one that suits this shape best, because the surface you need to shade is the one you were going to put something over anyway.

THE CAP IS ALREADY THERE

The reference build's outer layer is a rain-slick cap strapped over the building.

Its colour is a free decision and it is worth making deliberately. A light cap reflects; a dark cap absorbs. On a surface this size that is a substantial difference in summer and a small one in winter, because winter sun is low and weak.

Light, unless there is a specific reason.

EXTERNAL BEATS INTERNAL, ALWAYS

A blind inside a window stops light after it has already entered the building. The heat is in the room; the blind only decides where it lands.

Anything outside stops the heat before it arrives. An external shade is several times more effective than an internal one and it is the straightforward thing people skip because internal blinds are easier to buy.

WHAT SUITS A DOME

A brise-soleil over the upper bays: a panel standing off the shell, shading the glass and letting air move behind it.

Deciduous trees. Shade in summer, bare branches in winter, and the site probably has some. This is the best shading device available and it costs nothing if you site the building thoughtfully.

A removable panel over a glazed bay. This one is specific to the method: the bays are demountable, so a glazed bay can become a solid bay for the summer and back again in October. That is twenty minutes with a driver.

THE SHADING THE SHAPE GIVES YOU FREE

A dome shades itself. The curve means one side is always turned away from the sun, and the transition is gradual rather than the hard shadow line a box casts on itself.

Moisture management

The whole of environmental control in this building comes down to where the water is, so this section collects what the others have been circling.

THE FOUR PLACES WATER ARRIVES

Rain, on the outside. Handled by the cap and by the seams — shed, do not seal, two lines of defence rather than one.

Ground, from below. Handled by the pad, the drainage and a membrane under the floor. 40 inches a year is what the reference build assumes falls, and all of it has to go somewhere that is not under the building.

Occupants, from inside. Cooking, washing, breathing. Handled by ventilation at source.

Condensation, in the middle of the wall. Handled by — as the section on it says — an unproven duct, and until somebody measures it, by generous ventilation and by looking.

THE ORDER OF PRIORITY

Keep it out. Cheapest, most effective, and where nearly all the effort belongs.

Let it escape. A wall that can dry in both directions forgives a lot. A wall sealed on both faces forgives nothing, which is the shower cap's structural weakness as a design.

Manage what is left. Fans, dehumidifiers, and opening a window.

THE CATCHMENT, WHICH TURNS THE PROBLEM AROUND

A dome is an excellent rainwater catchment: a smooth curved surface, one low edge all the way round, and a seam network that already reaches every part of it.

The reference build assumes 72 percent of what falls can be caught. On this footprint, with 40 inches a year, that is thousands of gallons.

Which is the useful way to think about the whole chapter. The water is going to arrive regardless. The only question is whether the building is arranged to get rid of it or arranged to keep it.

Part V - Variations and Future Systems

15. Variations of the Wedge Method

The wedge is a sector of a log, and the reason the method works is that the 45-degree sector you split out of a tree is exactly the seam budget the dome needs — both pairs of seam angles sum to 45 degrees.

Everything in this chapter is a way of making that member some other way, and each one trades a piece of that argument for something else. It is worth knowing which piece.

Figure 102. Four sectors

Four sectors: a quarter of the log, and a very deep wall. The A seam then wants a key 7.05 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Solid wedges

A solid wedge is one piece of timber, sawn to a triangular section, and it is what this whole method assumes unless something says otherwise.

WHAT IT IS

A sector of a log. Split a round log into 8 radial pieces and each one is a 45-degree wedge — flat on two faces, curved on the third, and pointed along one edge.

The reference build's member is 4.59 inches across the wide face and 6.0 inches deep.

WHY THE SHAPE IS THE POINT

Two sawn faces meeting at an angle is a joint you can make with a saw.

The 45-degree sector is not arbitrary: it is the whole seam budget of the dome. The A seams fold 18.029 degrees and open 26.971; the B seams fold 22.459 and open 22.541. Both pairs sum to 45, which is the raw sector you split out of the tree.

That is the closest thing this method has to a theorem, and it is why a split log is the right stock for this building rather than a convenient one.

THE ADVANTAGES

Strength. One piece of continuous grain, no glue line, no fastener in the section.

Cost. It is what comes out of a tree. There is no processing between the splitting and the using.

Simplicity. Nothing to delaminate, nothing to come apart, nothing that needs a shop.

THE LIMITS

Size. The member cannot be bigger than the log. Two trees give 160 members where the dome needs 120, which is comfortable at this size and would not be at twice it.

Movement. Solid timber moves as it dries, and a wedge sawn green will not be the same wedge in a year. Build green and accept the movement, or dry and accept the wait.

Defects. A knot in a solid member is a weakness in that member. There is no lamination to spread the load round it.

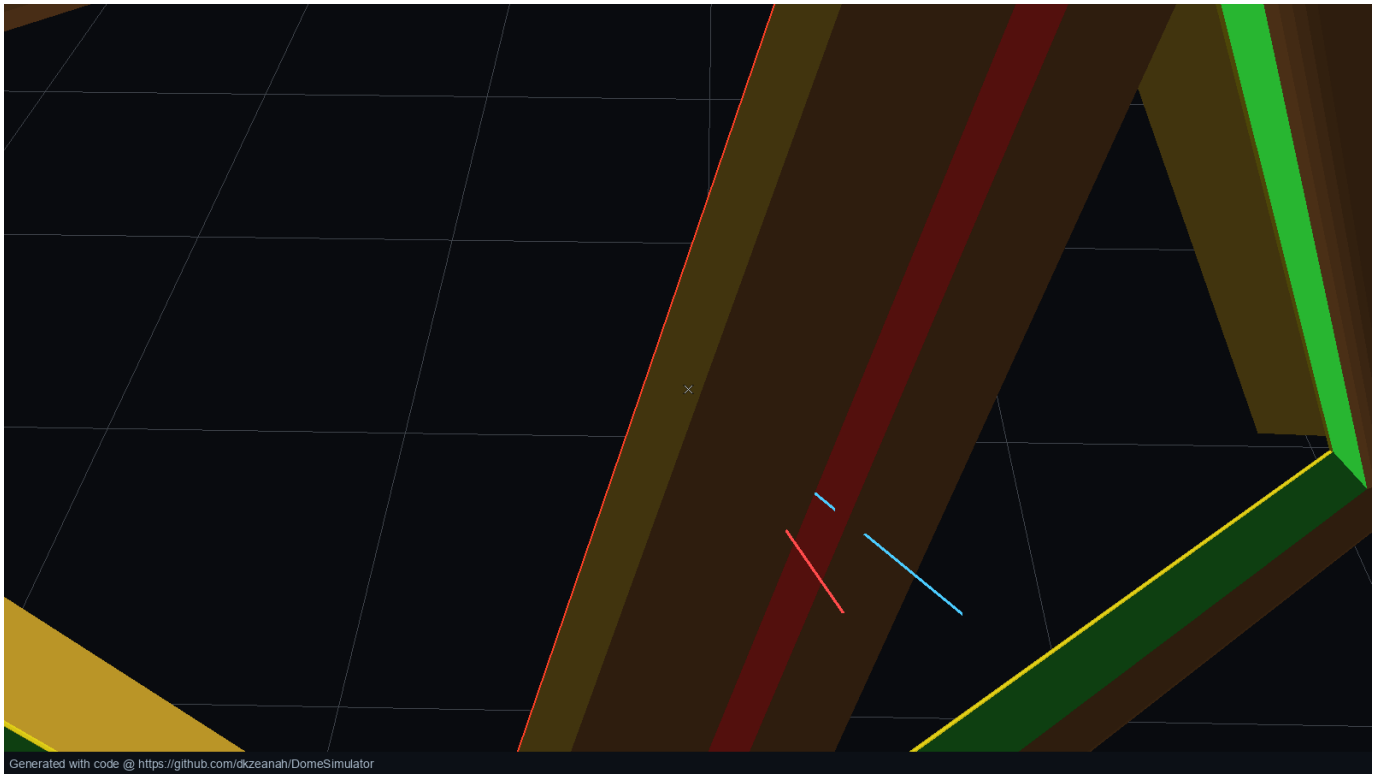

Figure 103. Six sectors

Six sectors: a fat wedge, a deep wall. The A seam then wants a key 4.30 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Hollow members

A hollow member is a box: four thin walls enclosing air, in the same outside shape as the solid one.

THE ARGUMENT FOR

Stiffness lives near the surface. In bending, the material at the middle of a section is doing very little, and taking it out costs less strength than it saves weight.

At 15.3 square inches of solid section, a hollow equivalent might use half the material for most of the stiffness.

The frame weighs 2,478 pounds in green pine. Halving that changes what two people can lift, which changes how the dome gets built.

THE ARGUMENT AGAINST, WHICH IS DECISIVE HERE

You cannot make one with a saw.

A hollow triangular member is four pieces, ripped accurately, glued or fastened along four long joints, in a jig, by somebody with a shop. That is the exact work the wedge method exists to avoid.

And the whole harvest advantage disappears. The method's economy comes from using 88 percent of a trunk instead of the 45 percent a mill gets. A hollow member is made from milled boards, so it starts from the 45.

WHERE IT WOULD MAKE SENSE

A manufactured dome, in a factory, where the jig is a fixed cost spread over hundreds of frames and the weight saving compounds through shipping.

Not for somebody with a chainsaw and two trees.

THE MIDDLE OPTION NOBODY TRIES

Solid wedges with the middle bored out along the length. It keeps the sawn faces, keeps the continuous grain at the surface, and removes some of the weight.

It is fiddly, it needs a long auger, and nobody has measured what it does. Listed here as an idea rather than a recommendation.

Laminated wedges

A laminated wedge is built up from thinner pieces glued face to face, then cut to the wedge section.

WHAT IT BUYS

Size beyond the tree. A laminated member can be any length and any section, which removes the constraint that the log sets.

Straightness. Laminating averages out the movement in each lamina, and a glued-up member stays where you put it in a way a solid one does not.

Defects spread. A knot in one lamina is surrounded by six that are clear, so its effect on the whole is small. That is why glulam is graded higher than the timber it is made from.

Grain direction, chosen. You can put the strong direction where the load is.

WHAT IT COSTS

Boards, which means milling, which means the 45 percent recovery rather than the 88.

Glue, clamps, a flat surface long enough, and the time for it to cure.

And a level of shop practice that most of this book is written to avoid needing.

WHERE IT EARNS ITS PLACE

Big domes. Above a certain span the member a tree can give you is not enough, and lamination is the only way to keep going with timber.

Long-span headers over openings, where one member is carrying what several used to.

And repairs: a failed member replaced with a laminated one, because the right tree is not standing there today.

THE HONEST POSITION

Lamination is better engineering and worse economics, and this method is an economic argument. It belongs in the book because at some sizes there is no alternative — not because it is an upgrade to reach for at this one.

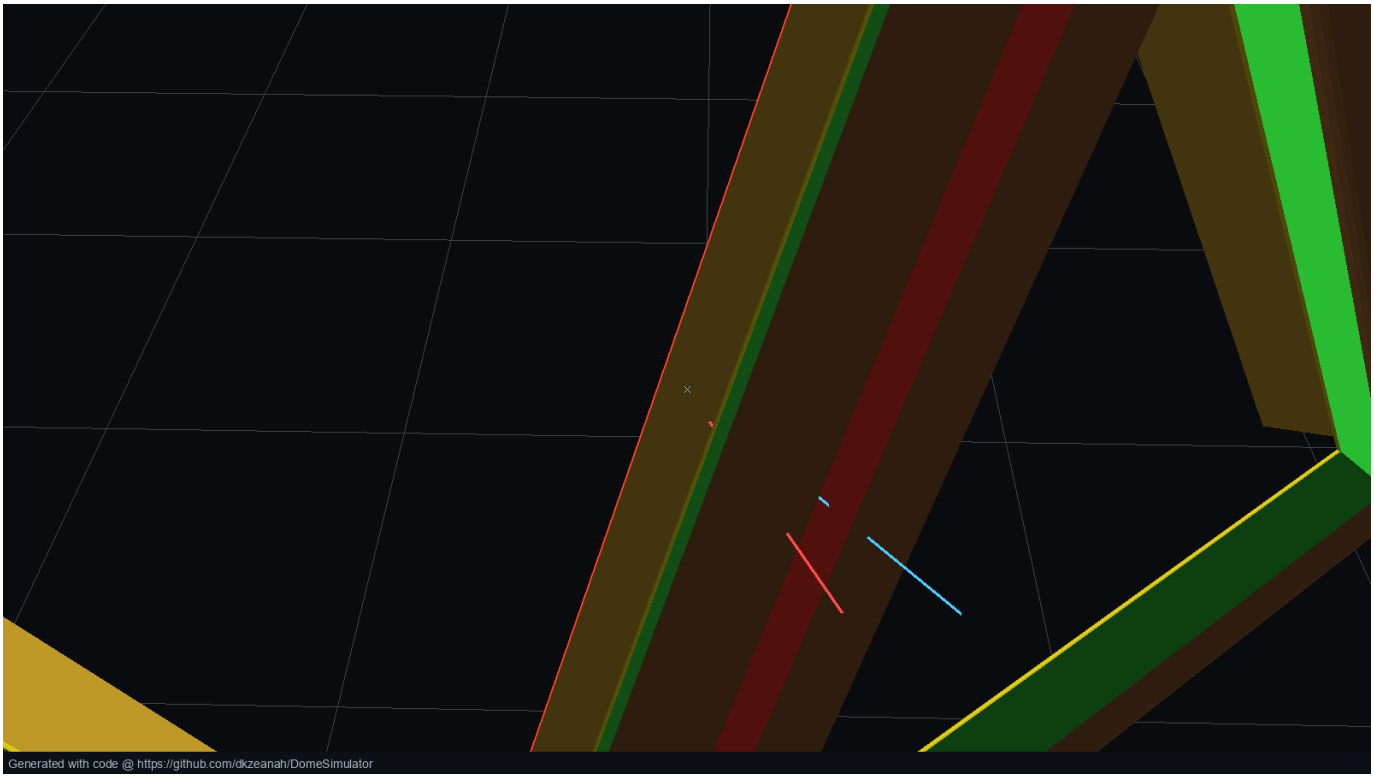

Figure 104. Eight sectors

Eight sectors: the reference build. The A seam then wants a key 2.80 in across its base. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Split wedges

Splitting is different from sawing, and the difference is the oldest argument in woodworking.

SPLIT VERSUS SAWN

A saw cuts through the fibres wherever the blade goes. A split follows them.

Split stock is therefore stronger along its length, because no fibre is cut short — which is why axe handles and riven oak gates are split, and why a sawn one of the same dimensions breaks first.

It is also faster and needs no power.

WHAT IT COSTS

Accuracy. A split follows the grain, and the grain in a real tree is not perfectly straight. A split wedge is approximately the shape you wanted, and the amount of approximately depends on the tree.

Kerf, though: none. The reference build's model loses 51.2 board feet per log to a 0.25-inch kerf — out of 442 solid — and splitting loses none of it.

THE METHOD THAT ACTUALLY WORKS

Split radially, into 8 pieces, then clean the two faces that matter with a saw or a plane.

That gets the strength of split stock and the accuracy of sawn faces on the two surfaces that have to mate. The curved back is left as it comes, because nothing bears on it.

That is the reference build's assumption and it is the reason the numbers work: it is a split log with two tidied faces, not a machined beam.

WHERE IT FAILS

Twisted grain. Some trees spiral, and a spiral log splits into spiral wedges. Test one section before committing sixty feet.

Big knots. A split runs into a knot and goes round it, or stops.

Species. Straight-grained softwoods split beautifully. Interlocked tropical hardwoods do not split at all.

Log-derived wedges

This is the reference build, and it is worth setting out as a variation rather than as the default, because the numbers only work for a particular kind of tree.

THE LOG THE MODEL ASSUMES

15 inches at the butt, 5.5 at the top, 60 feet usable, cut into 12-foot sections.

That is an ordinary plantation softwood. Not a veteran, not a specimen, not anything anybody would object to felling.

WHAT ONE GIVES YOU

442 board feet solid. Take off 51.2 for kerf and 391 board feet end up in wedges — 40 of them, making 80 struts at 4.89 board feet each.

Two trees make 160 members. The dome needs 120. So a 2V hemisphere of this size is two trees, with a third of the stock spare for mistakes.

THE COMPARISON THAT MATTERS

Milled, the same log yields about 45 percent of itself as usable lumber. The rest is slab, edging and sawdust.

Split into wedges, it yields about 88 percent. Nothing is edged because nothing needs a rectangular section.

That is a 1.95x harvest advantage. Against it, the pinwheel frame uses 1.80x more stock than a mitred one, because every joint overlaps rather than meeting. Net: a wedge dome uses about 92 percent of the trees a mitred dome of the same size would.

Eight percent, which is small — and the point was never the eight percent. It is that the eight percent comes with no mitre saw, no jig, no shop, and a log instead of a lumber order.

THE CONSTRAINTS TO CHECK BEFORE COUNTING ON IT

Straight grain, adequate diameter, and enough of them. A stand of crooked scrub will not do this, however many stems it has.

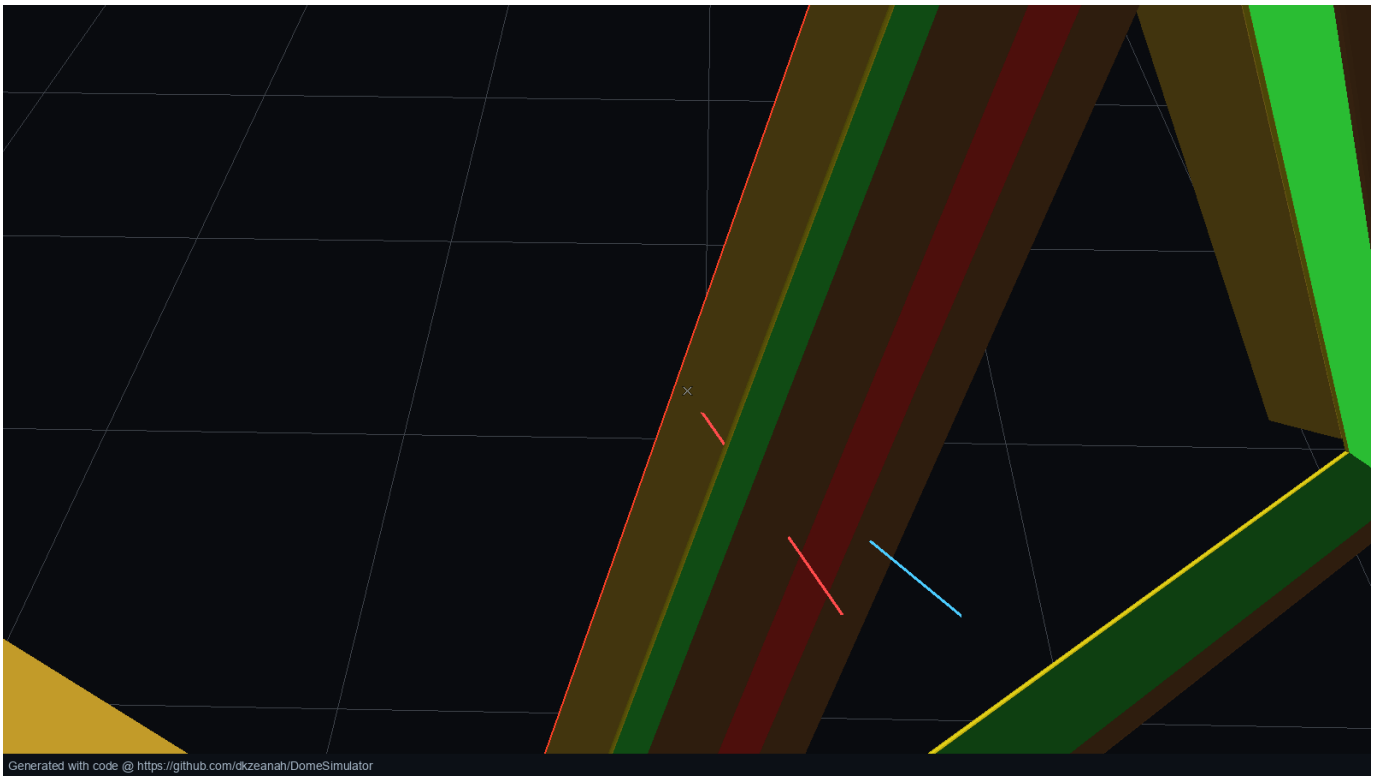

Figure 105. Twelve sectors

Twelve sectors: thin sticks, more of them. The A seam then wants a key 1.25 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Manufactured wedges

A manufactured wedge is one made to a specification, in quantity, by somebody else — and it is the version of this method that could actually be sold.

WHAT CHANGES

Consistency. Every member identical to a thousandth, which makes everything downstream easier: panels that fit, snap fittings that work, assembly instructions that are true.

Grading. Stress-graded stock with a number on it, which is what an engineer needs and what a building inspector will ask for.

Treatment. Pressure treatment, fire retardant, or preservative, applied properly rather than brushed on.

Species choice. The right timber rather than the one growing on the site.

WHAT IS LOST

The tree. The method's central economic claim — your frame is already standing on your land — does not survive being bought in a bundle.

The recovery advantage. A factory making rectangular-ish stock into wedges starts from milled lumber and the 45 percent.

The independence. A manufactured wedge is a supply chain.

THE HONEST ASSESSMENT

Manufactured wedges make a better product and a worse argument.

For a kit sold to somebody who wants a dome and does not want a chainsaw, they are obviously right. For the person this book is mostly addressed to — somebody with land, trees and time, and not much money — they remove the only reason the method was interesting.

Both markets are real. They are not the same market, and a design that tries to serve both usually serves neither.

Hybrid metal/wood systems

Steel where the strength is and timber everywhere else is a sound principle and the reference build already uses it in one place.

WHERE IT ALREADY APPEARS

The mast: a steel core running the height of the utility column, clad in timber.

The floor: a steel hub clamping that mast, steel spokes radiating to the base ring, timber decking over them.

In both cases the metal is doing a job timber is bad at — carrying a concentrated load through a small section — and the timber is doing the job metal is expensive at, which is covering area.

WHERE ELSE IT WOULD WORK

A steel base ring. The ring is a compression ring, it works in pure hoop stress, and that is exactly what a steel band is for. It would also stop the ring being the thing that limits how many openings you can cut.

Steel connectors at high-load vertices. The apex, and the base vertices where the dome delivers its load to the pad.

Steel headers over openings.

WHERE IT WOULD NOT

Everywhere. A steel-framed dome is a different building with different economics, and it loses the harvest argument entirely.

THE JOINT IS THE PROBLEM

Timber moves and steel does not.

A timber member bolted rigidly to a steel plate will, over a few seasons of shrinking and swelling, either crush around the bolt or split from it. The fix is a connection that allows movement in the direction timber moves — slotted holes, or a single central fastener rather than a rigid pattern of them.

That is well understood and routinely got wrong.

CORROSION

Wet timber and steel together is a battery. Use stainless or hot-dip galvanised at every point where the two meet, and particularly at the base, where water collects.

The reference build specifies stainless throughout its bolted connections, which costs more and removes a whole category of failure from the building's future.

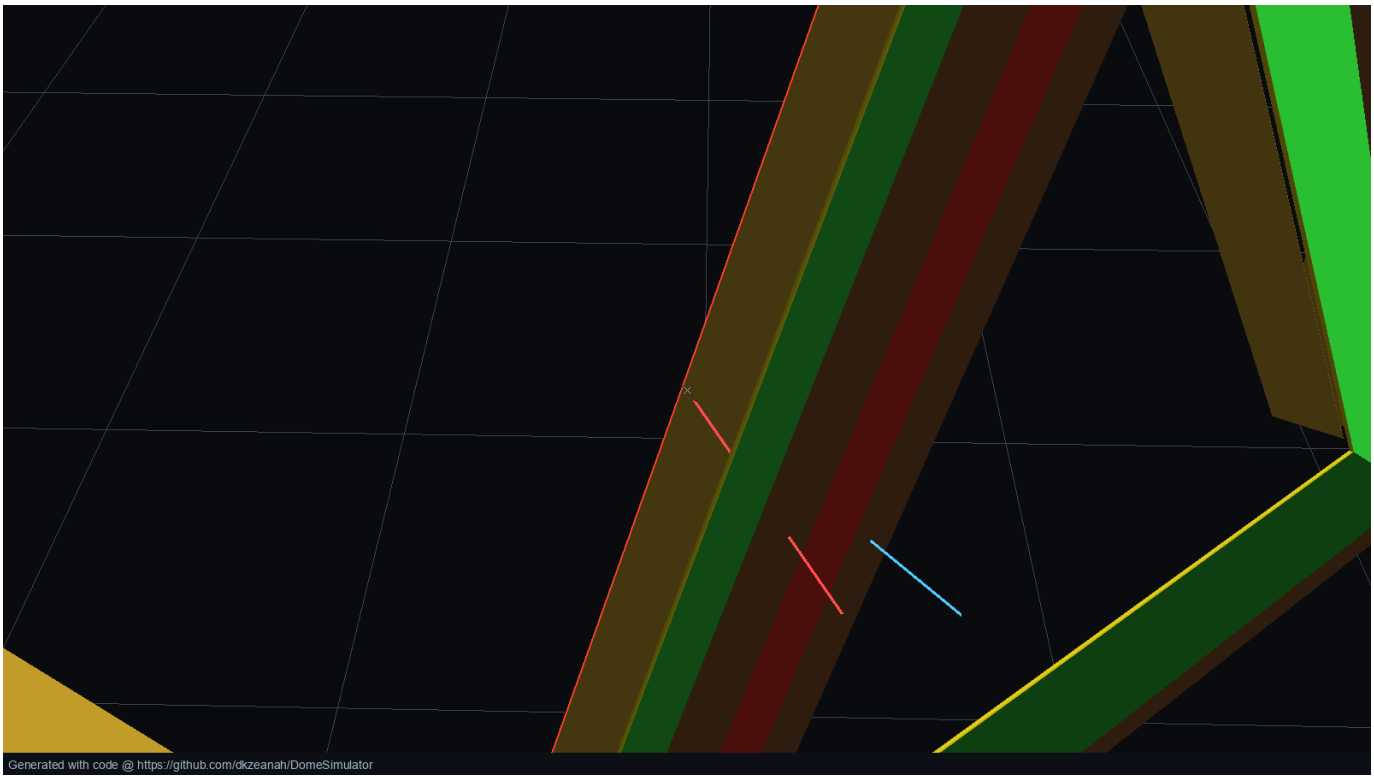

Figure 106. Sixteen sectors

Sixteen sectors: staves, and a seam channel under half an inch. The A seam then wants a key 0.47 in across its base. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Every other way: the four decisions that change the frame itself

Which way the wedge points, how many sectors the log splits into, how big the tree was, and which way the triangle pinwheels. Every one of them is irreversible once the first panel is cut, and every one has a figure in this book.

EVERY OTHER WAY OF DOING IT

```
wedge orientation = point_dome_in          both points inward, toward the centre; seam key 2.80 in   <- the reference build
wedge orientation = point_panel_in         points apart, each into its own triangle; seam key 11.67 in
wedge orientation = point_dome_out         both points outward, at the sky; seam key 6.27 in
wedge orientation = point_panel_out        points toward each other, across the seam; seam key 11.67 in
radial splits = 4                          four sectors: a quarter of the log, and a very deep wall; seam key 7.05 in
radial splits = 6                          six sectors: a fat wedge, a deep wall; seam key 4.30 in
radial splits = 8                          eight sectors: the reference build; seam key 2.80 in   <- the reference build
radial splits = 12                         twelve sectors: thin sticks, more of them; seam key 1.25 in
```

```
radial splits = 16                          sixteen sectors: staves, and a seam channel under half an inch; seam key 0.47 in
trunk diameter in = 8.0                     an eight-inch trunk: a slender frame; seam key 1.87 in
trunk diameter in = 12.0                    a twelve-inch trunk: the reference build; seam key 2.80 in   <- the reference build
trunk diameter in = 16.0                    a sixteen-inch trunk: heavy section, fewer trees; seam key 3.73 in
panel joint handedness = clockwise          clockwise, seen from outside; seam key 2.80 in   <- the reference build
panel joint handedness = counterclockwise   counterclockwise, seen from outside; seam key 2.80 in
```

These change the frame itself. Every other choice in this book can be made after the dome is standing; these cannot be made twice.

WHERE THESE NUMBERS COME FROM

the raw-wedge solver's own configuration space, applied to the reference build's own quantities. Nothing on this page is a rate quoted from outside the project, and nothing on it was typed: regenerate the whole table with `py -3.12 -m wedge_book.alternatives --apply`.

There is no straw-bale row and no earthbag row, because this project cannot cost one. A table with a plausible number in it that came from nowhere is worse than a table with a gap.

16. Different Dome Configurations

Frequency decides how many pieces you cut. Where you cut the sphere decides what it is like to live in.

Most of the disappointment people report with domes comes from the second of those, and nearly all the writing about them is about the first.

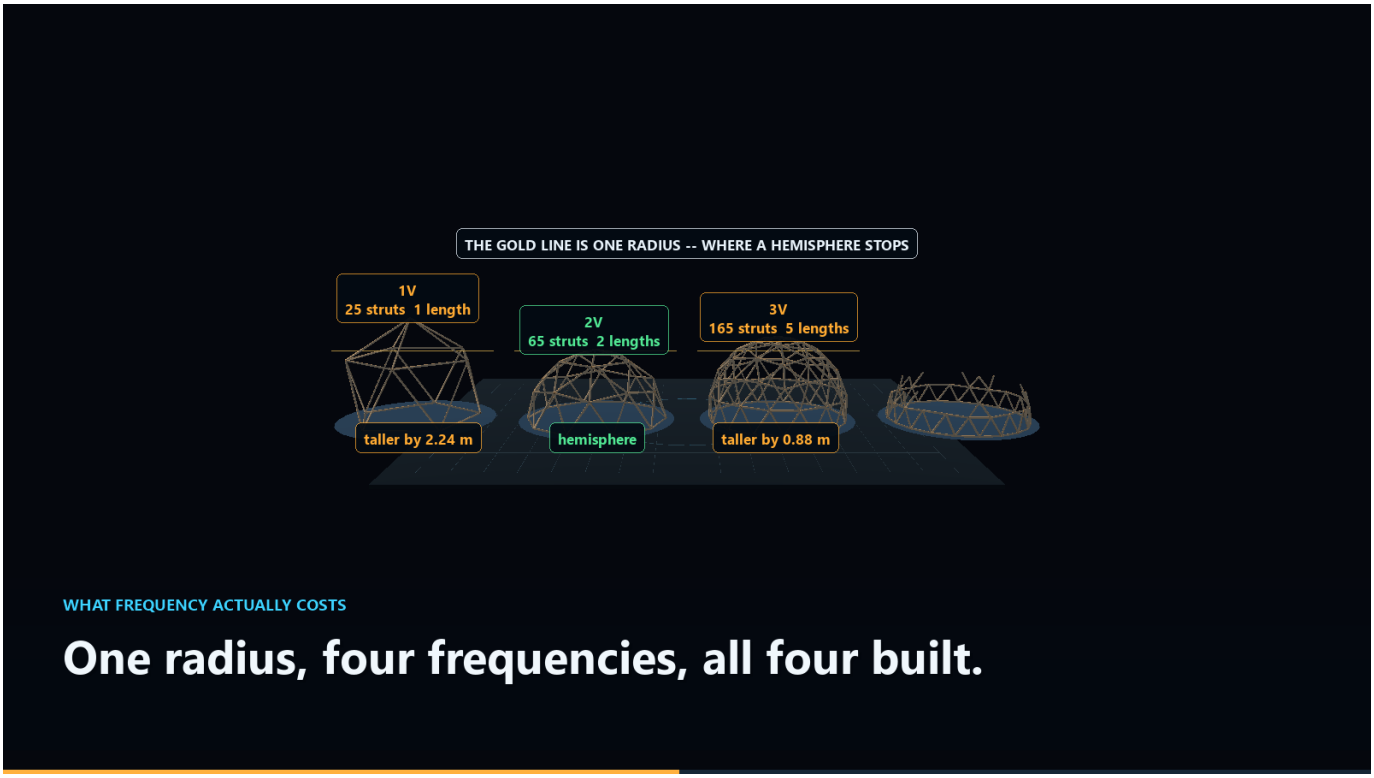

Figure 107. What frequency actually costs

One radius, four frequencies, all four built. You have now seen one, two, three and four frequency domes. From Every Dome In The World, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

1V

A 1V dome is an icosahedron cut in half: twenty triangles on the whole sphere, and every one of them identical.

WHAT IT IS GOOD FOR

Learning. One strut length, one triangle, one angle. You can build a model of one in an evening out of drinking straws and understand more about geodesic geometry than a week of reading will give you.

Cheapness. Fewer members, fewer joints, less of everything.

Speed. There is nothing to sort, because every piece is the same piece.

WHAT IT IS BAD AT

Being round. A 1V dome is visibly a faceted polyhedron, not a sphere. The flat triangles are large relative to the building and the shape reads as angular.

That matters structurally as well as visually: a large flat panel in a shell carries bending that a smaller, more curved arrangement would resolve as membrane action.

Headroom. The facets cut in aggressively, so the usable floor is a smaller fraction of the footprint than in a higher frequency.

WHERE IT ACTUALLY BELONGS

Small structures. A greenhouse, a play structure, a chicken house, a shed. At small spans the faceting does not matter and the simplicity is everything.

And as the first thing anybody builds. Build a 1V, in a weekend, before committing to something that takes a month. Everything you learn transfers.

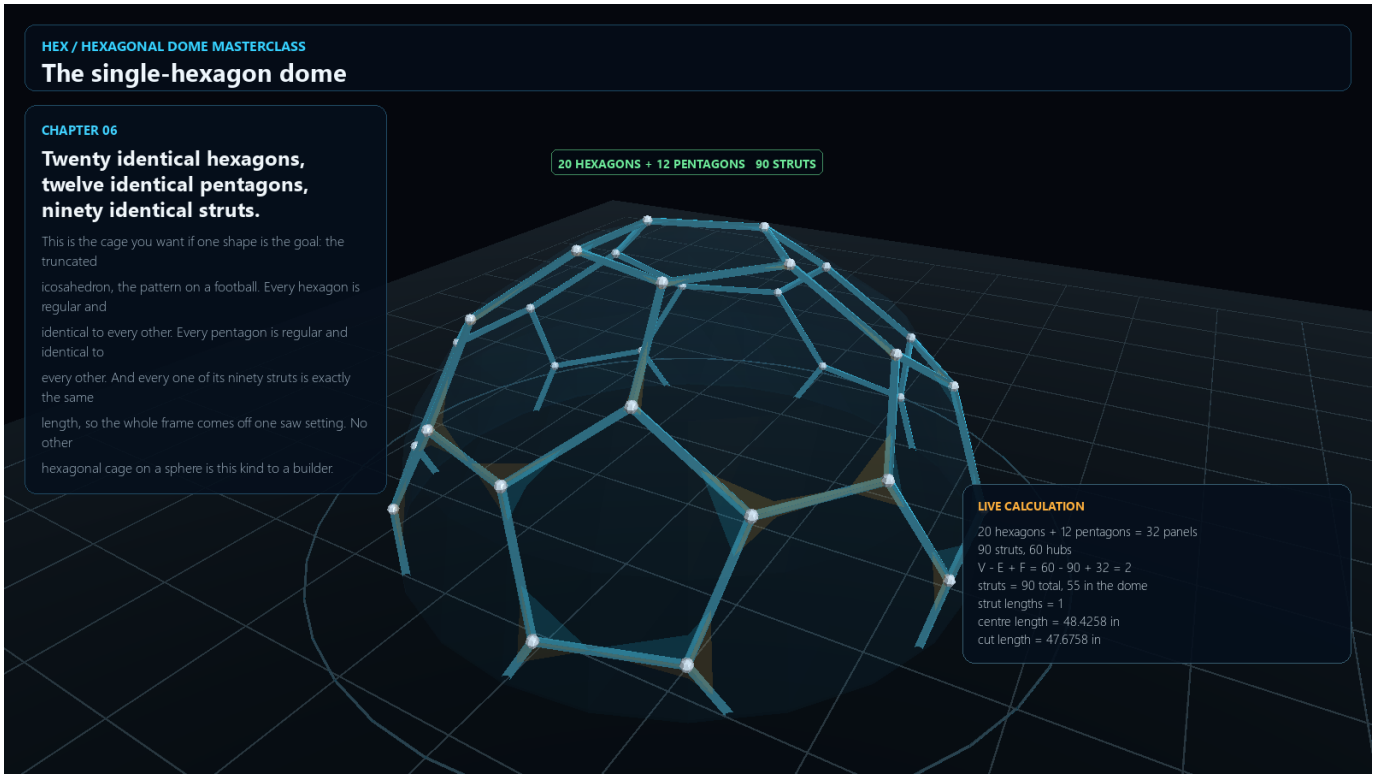

Figure 108. The single-hexagon dome

Twenty identical hexagons, twelve identical pentagons, ninety identical struts. This is the cage you want if one shape is the goal: the truncated. From Hexagonal Dome Masterclass, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

2V

The 2V is the reference build and the shape the rest of this book assumes.

WHAT IT IS

Each face of the icosahedron divided into four, giving 80 triangles on the sphere and 40 faces on the hemisphere — 10 of one kind and 30 of the other.

Two strut lengths, A and B, rather than one.

WHY IT IS THE SWEET SPOT

Two lengths is manageable. You can sort two piles in your head; five you cannot, and above that people start colour-coding and still get it wrong.

It is round enough. The faceting is visible but not dominant.

It is big enough to live in. The reference build is a 19.42-foot sphere cut at the equator — 296 square feet inside the circle, or 277 inside the ten-sided figure the base ring actually makes.

And the parts count is buildable by hand: 26 vertices, 120 members, 55 interior seams and 10 base edges.

THE NUMBERS THAT DEFINE IT

Radius 116.4984 inches. The A chord is 72.000 inches and the B chord 63.6703.

Note what that does not say. The chord is the distance between two vertices and the stock you cut is neither of those. An A member is 70.182 inches of timber against a 72-inch chord, and a B member 62.681 against 63.670.

Neither end of a member lands on a vertex. One end butts into the side of its neighbour, short of the corner; the other runs on past the mathematical vertex so the previous member can butt into its side. The corners of a triangle are reference points, not wood endpoints, and the two effects nearly cancel — leaving the A member about an inch and three-quarters shorter than its chord.

Confusing the chord with the cut is the most expensive mistake available in this method, and it is available on the very first cut.

Dihedral angles fall between 157.5 and 162.0 degrees — a spread of under five degrees across the whole building, which is why one gasket thickness works everywhere. That is topology, so it holds at any diameter.

309 feet of interior seam, reaching every vertex.

THE THING IT DOES NOT SOLVE

No panel of a 2V at this size fits a four-foot sheet.

The AAA faces are 62.354 inches across their narrowest altitude and the BAB faces 52.516. A sheet is 48. The narrowest panel in the building is four and a half inches too wide, in every rotation.

That is entirely a consequence of picking this frequency at this diameter. A different diameter moves it, and this one was picked for a six-foot member rather than for a sheet of plywood.

3V

A 3V divides each icosahedral face into nine, giving 180 triangles on the sphere.

WHAT IT BUYS

Roundness. A 3V reads as a sphere rather than as a polyhedron, and the difference at any size above a shed is obvious.

Smaller panels, which means sheet goods start to fit, which quietly solves the 2V's sheet problem.

Better structural behaviour. More, smaller triangles distribute load more evenly and rely less on any single member.

And a proper base. A 3V can be cut at several heights, including one that gives a near-vertical lower wall — which is the thing people most often want from a dome and cannot get from a 2V.

WHAT IT COSTS

Three strut lengths, and the member count roughly triples from the reference build's 120. That is three piles to keep straight and three times as many joints to make.

More seams, more gaskets, more sealing, more of every per-joint cost.

And at this scale the wedge method starts to strain: a hundred and twenty members is a weekend of sorting and four hundred is a different kind of job.

WHERE THE LINE IS

Roughly: up to about 20 feet, 2V. From about 25 feet upward, 3V. Between them it is a judgement about whether you would rather cut more pieces or live in a more faceted room.

The reference build sits at 19.4 feet, just under the line, which is not a coincidence.

THE WEDGE-SPECIFIC CATCH

A higher frequency means shorter members, and shorter members mean the pinwheel overlap is a larger fraction of each one.

At the reference build's size an A member is 70.2 inches of stock for a 72-inch chord. Halve the chord and the overlap does not halve with it, so the ratio gets worse.

The pinwheel already carries a 1.80x stock penalty against a mitred frame, covered by a 1.95x harvest advantage. As the members get shorter that penalty grows and at some frequency the harvest stops covering it.

Nobody has worked out where. It is one of the open questions this book would rather name than paper over.

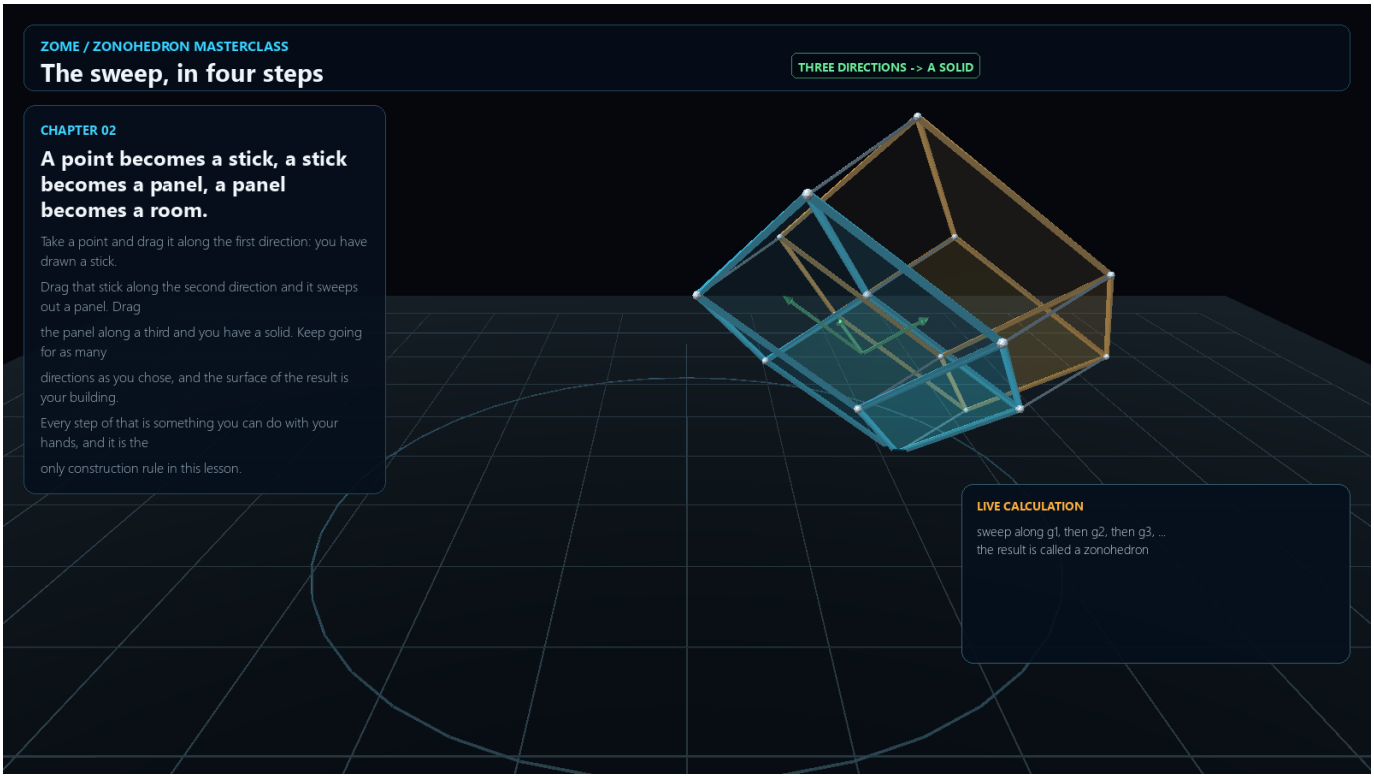

Figure 109. The sweep, in four steps

A point becomes a stick, a stick becomes a panel, a panel becomes a room. Take a point and drag it along the first direction: you have drawn a stick. From Zome Construction Masterclass, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Hemispheres

A hemisphere is a sphere cut at its equator, and it is the default for good reasons and one bad one.

THE GOOD REASONS

It sits flat. The cut is a plane, the base is a circle, and the building lands on level ground without anything below it.

It is the most enclosed volume per unit of surface of anything you can build. That is where the 41 percent envelope saving over a box comes from.

The base ring is a compression ring. Cut at the equator, the ring is in pure hoop stress, which is the most efficient thing a ring can be doing. Cut above or below and there is a vertical component the ring has to resolve.

The maths is clean. Strut lengths and angles for hemispheres are published; a cut anywhere else has to be solved.

THE BAD REASON

Habit. Most domes are hemispheres because most domes are hemispheres.

WHAT A HEMISPHERE IS ACTUALLY LIKE TO BE IN

The reference build is 19.42 feet across and 9.708 feet to the apex. The floor inside the base ring is 277 square feet.

That sounds generous and in the middle it is. At the perimeter the wall is leaning in hard: a hemisphere's surface is 45 degrees from vertical at half its height, so by the time you are at head height near the edge, the usable floor is well inside the footprint.

So the floor area a hemisphere gives you and the floor area you can stand upright on are different numbers, and the second is the one that matters.

It is the first thing people are disappointed by, and it is the reason the next three sections exist.

Partial spheres

A partial sphere is cut somewhere other than the equator, and which way you cut changes the building completely.

CUT BELOW THE EQUATOR

More than half a sphere. The walls come back in below the widest point, so the floor is smaller than the widest part of the building.

Nobody does this deliberately for a house — you lose floor and gain an overhang that has to be supported. It appears in water tanks and in structures where the enclosed volume is the point rather than the floor.

CUT ABOVE THE EQUATOR

Less than half a sphere: a shallow cap. More floor for the height, a lower profile, less material.

This is a good shape for something that does not need headroom at the perimeter — a workshop with benches round the edge, a store, a cover over something. It sheds snow less well and it is less stable in wind, because there is less depth to resist overturning.

THE TRADE, PLAINLY

Cut low and you get a tall building with a small floor. Cut high and you get a wide building with a low ceiling. The equator is the balance and that is most of why it is the default.

THE STRUCTURAL CATCH

A cut above the equator has a base ring that is not in pure hoop stress. The shell above it leans outward, so the ring is being pushed out as well as squeezed, and it needs either more section or a tie across the floor.

A ring beam and a tension ring in the floor deck handle it. It is not difficult. It is just not free, and the hemisphere gets it free.

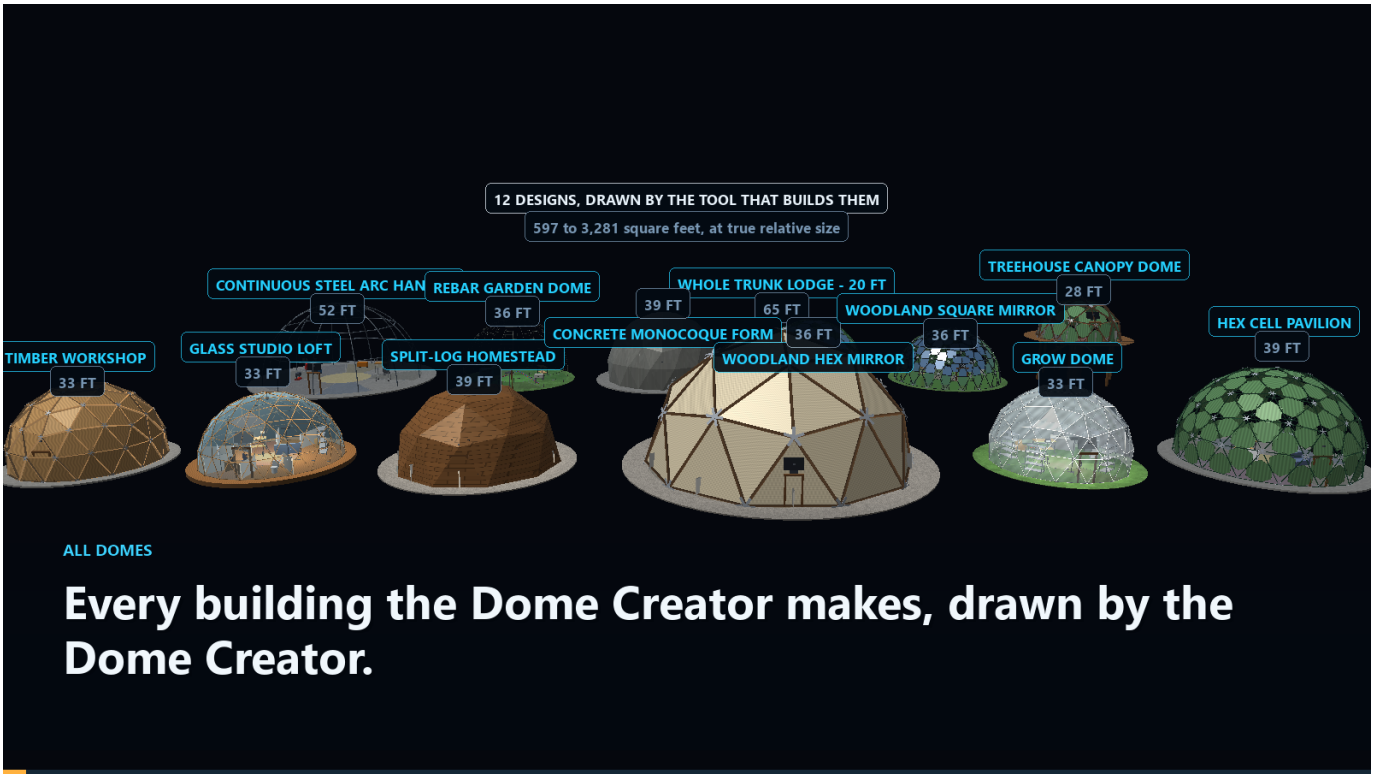

Figure 110. All domes

Every building the Dome Creator makes, drawn by the Dome Creator. The Dome Creator is a walkable customizer with twelve finished. From All Domes, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Knee-wall domes

A knee wall is a short vertical wall between the foundation and the dome, and it is the single most effective change you can make to how a dome feels inside.

WHAT IT FIXES

The lean. Every complaint about living in a dome is a complaint about the perimeter — furniture that will not go against a wall, a door you duck through, a room whose usable area is smaller than its floor.

Raise the shell three feet on a vertical wall and all of those improve at once. The wall is vertical where furniture is, the door is a normal door, and the usable floor grows by more than the wall's own area.

WHAT IT COSTS

The wall itself, which is conventional construction and cheap.

Some of the shape's efficiency: more envelope for the same floor, so the 41 percent saving shrinks.

And the base ring stops being the dome's ring. The dome now lands on a wall, and the wall has to resolve the outward thrust the ring used to take in hoop stress. That means a top plate acting as a ring beam, properly joined at every splice, and a tie across the floor or a foundation stiff enough to resist spreading.

That last point is the one that gets missed. A dome on a knee wall is pushing that wall outward all the time, and the wall is not a dome and cannot resolve it by itself.

HOW HIGH

Two to four feet is the useful range.

Under two, it does not change the feel enough to be worth the complication. Over four, the building stops reading as a dome and starts reading as a round house with a domed roof — which is a perfectly good building and a different one.

THE HONEST RECOMMENDATION

If somebody is going to live in it full time, and they have not lived in a dome before, a three-foot knee wall is probably the right answer and this book's reference build does not have one.

Raised domes

A raised dome is one lifted clear of the ground on posts, piers or — in this method's more speculative corner — cables.

WHY

Flood. Getting the floor above water is the whole reason in some places.

Slope. A raised platform gets level out of ground that is not, without moving earth.

Airflow underneath, which keeps a timber building dry.

The view, and the space beneath, which becomes storage, parking or shade.

WHAT IT DEMANDS

The platform becomes the structure. The dome above it is a shell that resolves its own loads; the posts and beams below are a frame that racks and must be braced in both directions.

Wind matters more. A raised building has more exposed area, and the overturning moment is the wind load times the height — so going up compounds.

Access is a stair, and the stair is not an afterthought.

THE VERSION THIS METHOD REACHES FOR

Hung, not stood.

The reference build prices a mast through the utility column, a floor that clamps to that mast, an apex lifting ring and three cables to saddles on two trees. The rig is $1,490, the mast and the dome's own floor are $622 and $1,899, and the whole floating upgrade is $4,011 — against a frame weighing 2,478 pounds.

Saddles, not holes: nothing is drilled into the trees.

THE SENTENCE THAT HAS TO GO WITH IT

That is a design possibility, not an engineered structure.

The loads on the trees, the cables and the mast need an engineer before anybody stands under it. The model prices the parts; it does not rate them, and the difference between those two things is the whole of somebody's safety.

17. Modular Dome Construction

The frame is the same in every building this method makes. What changes is the infill, and that is the whole manufacturing argument.

What makes it hold is a very short list of interfaces — about six — that are fixed while everything else is free to vary. The proof it works is that the most expensive component in the design was swapped for an entirely different technology and nothing else in the building had to change.

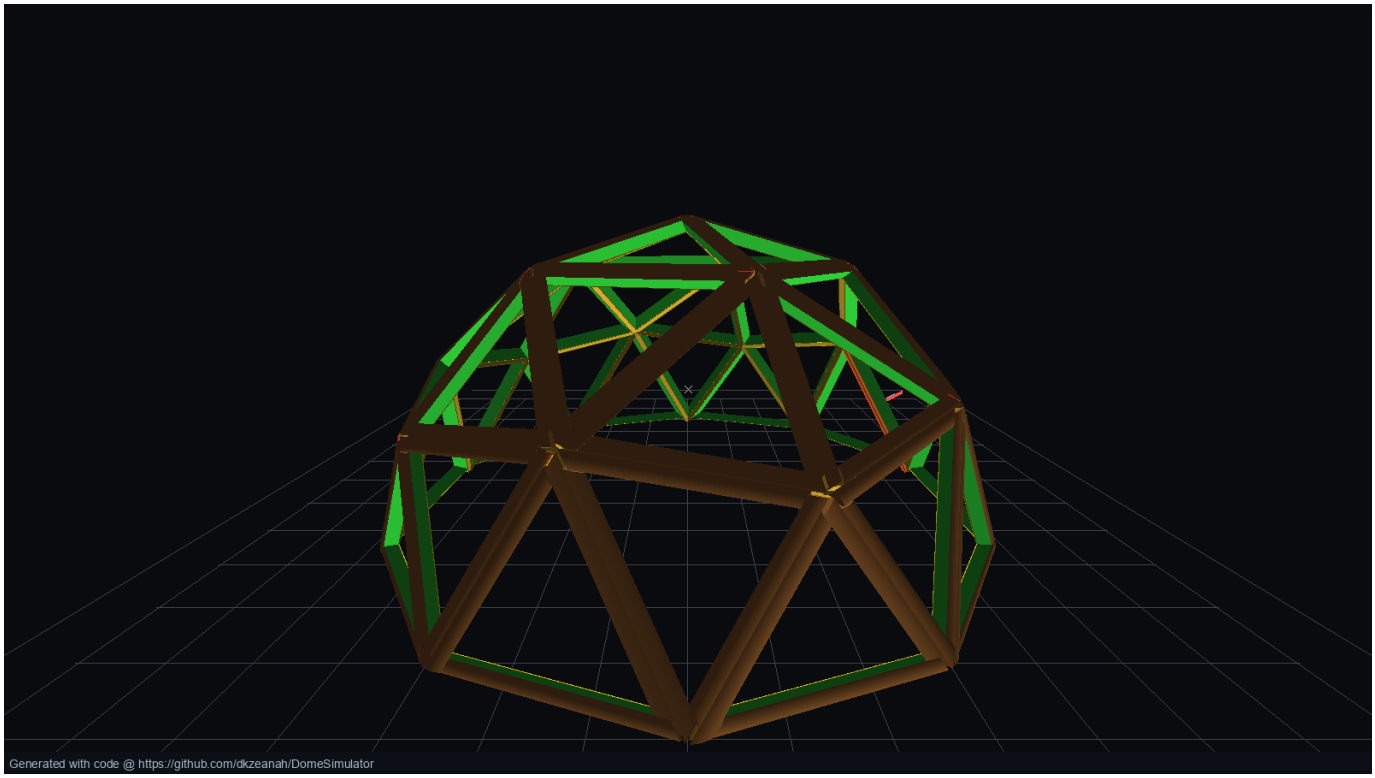

Figure 111. Assembled

Assembled: the real building. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Prefabricated triangles

A prefabricated triangle is one bay assembled on a bench and carried to the dome as a unit, rather than built in place stick by stick.

WHY IT IS TEMPTING

Work at waist height instead of overhead. That alone changes the job from uncomfortable to pleasant, and most of a dome is overhead.

Jigs. A bench with stops on it makes every triangle the same, and every triangle being the same is worth more than any single one being accurate.

Parallel work. Two people making triangles produce more than two people climbing a frame.

Weather. Triangles can be made in a shed in February and the dome raised in April.

WHY THE PINWHEEL RESISTS IT

The wedge method does not build triangles. It builds a pinwheel: each triangle has its own three members, and they run past each other rather than meeting at the corners.

That is why an A member is 70.182 inches of stock against a 72-inch chord — one end stops short of the corner and the other runs past it, and neither lands on a vertex.

But the pinwheel is inside one triangle, not between two. Neighbouring panels do not share a stick at all: each carries its own member along the shared edge and the two meet through a gasket.

So a triangle can be prefabricated. What it cannot do is be adjusted afterwards — its three members are locked into each other, and the panel's overall size is fixed the moment the third one goes in.

WHAT WORKS INSTEAD

Sub-assemblies of two or three triangles, where the overlaps are internal to the group and only the group's edges stick out.

Five triangles at the apex as one cap, lifted whole. That is the worst part of the build done at ground level, and it is the sub-assembly worth making.

And the base ring as a unit, because it is a single closed loop and it wants to be accurate more than anything else in the building.

THE ORDER THAT FOLLOWS FROM THIS

Base ring first, accurately. Then upward in rings. Apex cap last, or first and lifted.

Not triangle by triangle from one side, which is how a pinwheel frame ends up with the last member not fitting.

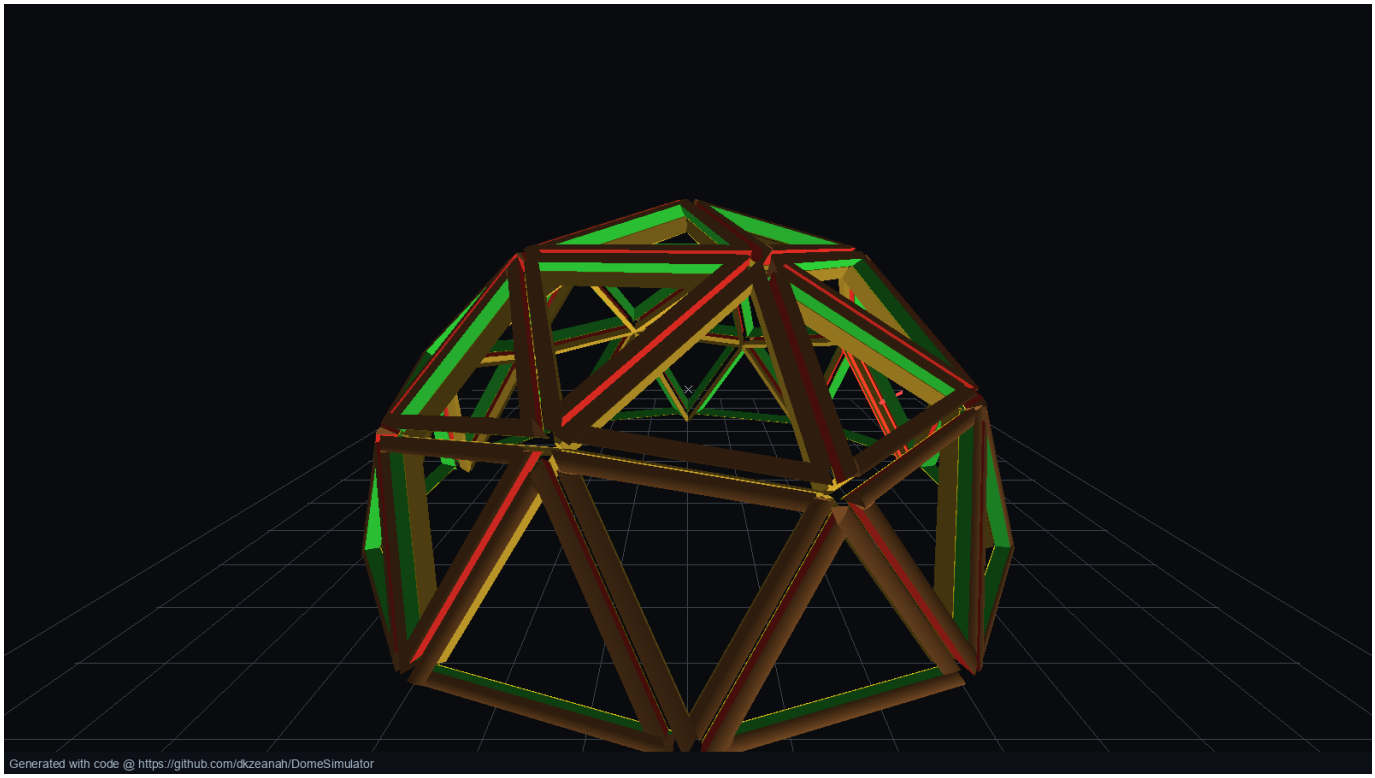

Figure 112. Opened enough to see the seam

Opened enough to see the seam. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Panel modules

A panel module is a bay's infill treated as a product: made once, fitted in minutes, and swapped later for a different one.

THE REFERENCE BUILD'S VERSION

The panel sits on the outside face of the frame and is pulled into it by screws landing in threaded inserts set in the wedges. 4 per bay, 160 for the dome.

That arrangement is not a detail. An earlier version compression-fitted the panel into the bay from outside, held by friction — and most of a dome's bays are overhead, so a friction fit overhead spends the life of the building arguing with gravity.

Screwed in, gravity works with the fixing.

THE TWO SIZES

Ten AAA bays at 15.59 square feet and thirty BAB at 13.13. Two panel types, forty panels, and every one of them interchangeable with its own kind.

That is the manufacturing argument in one sentence: a gym and a garage differ only in which panels are in the bays.

WHAT A PANEL CAN BE

Solid. Glazed. Louvred. Insulated. A door. A hatch. A solar bay. A blank with a shelf bolted to its inserts.

Each of those is the same outline, the same four fixings and the same twenty minutes.

THE SECOND JOB THE INSERTS DO

They are a grid of structural anchors on a known pitch. Three together hold about 540 pounds, which is a shelf, a bunk, a hanging rail or a kitchen run spanning two or three triangles.

So a bay does not have to be used to be useful. It is a barrier with fixings in it, and that is worth more than the panel.

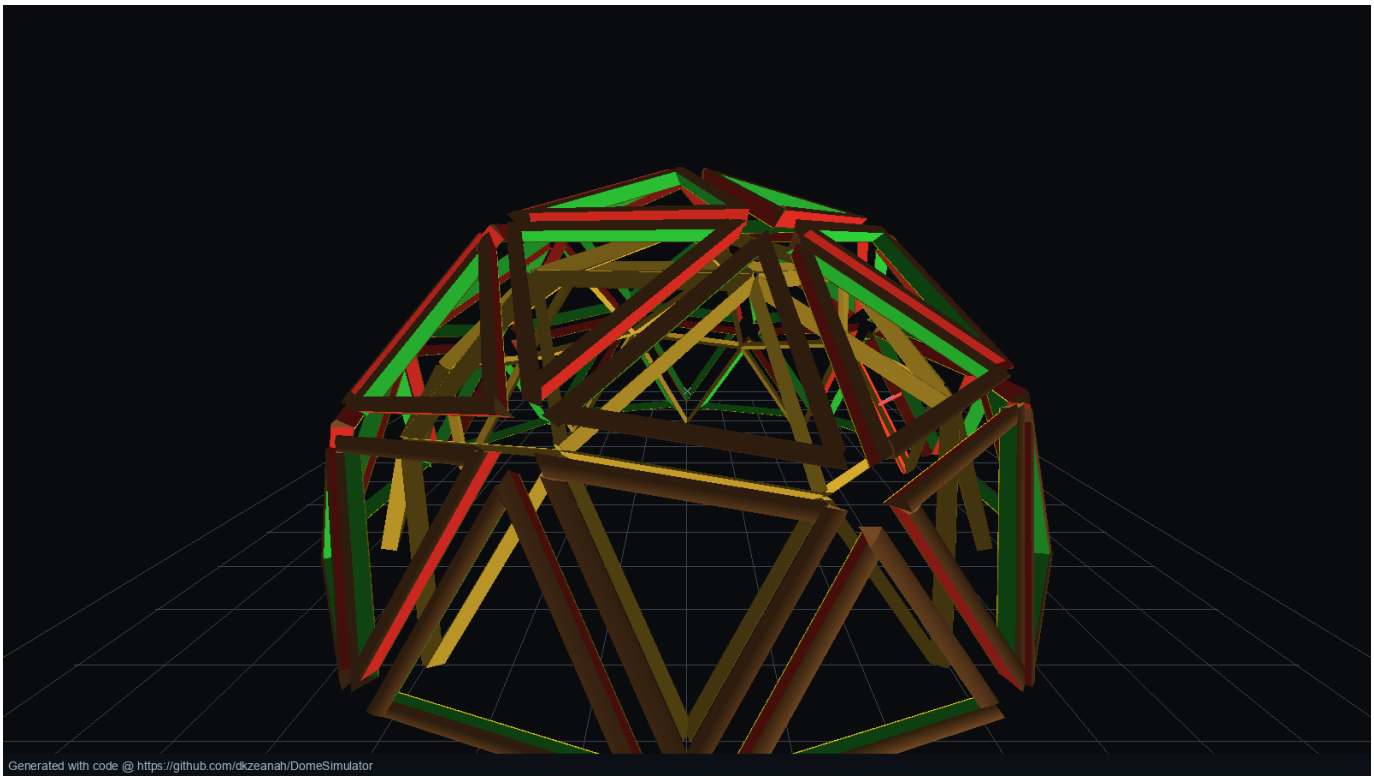

Figure 113. Forty separate panels and the keys that join them

Forty separate panels and the keys that join them. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Replaceable exterior shells

The shell is a consumable and the frame is not, and separating them is the single most consequential decision in this method.

THE TWO SHELLS THIS BOOK PRICES

A laminated hull: glass fibre in resin over the whole envelope. Sixty years, genuinely waterproof, and $4,826 more than the alternative.

A shower cap: wood panels, one membrane, quilted layers added over the years, and a rain-slick cap strapped over the lot. $3,265 bare.

WHY THE CAP WON

Because a hull is moulded once, at one size, and cannot grow.

Its cavity holds 4 quilted layers and then it is full. After that, insulating further means losing floor area or building a second shell outside the first, and neither happens.

A cap is a bag. Every layer under it buys the next cap a size up — seven layers makes the cap 15.6 percent bigger and nothing else about the building changes.

THE FRAME DOES NOT CARE

That is the point. The frame is the same frame under either, and under whatever comes third.

A shell that fails, or dates, or turns out to have been the wrong choice, is a replacement rather than a demolition. The building underneath is untouched.

WHAT MAKES IT POSSIBLE

The shell lands on the frame, not on the panels, and it overlaps every seam. It latches to the pad at the rim. It has one penetration, at the apex, closed by a gasketed cap.

Take off the latches and the cap and it lifts. Nothing else in the building is attached to it.

THE HULL IS NOT GONE

It became the upgrade — the fifty-year option, sold to the buyer who wants it, at $21,513 against $12,036.

That is the shape of every decision in this method: ship the cheap thing that works, and let the expensive thing remain available.

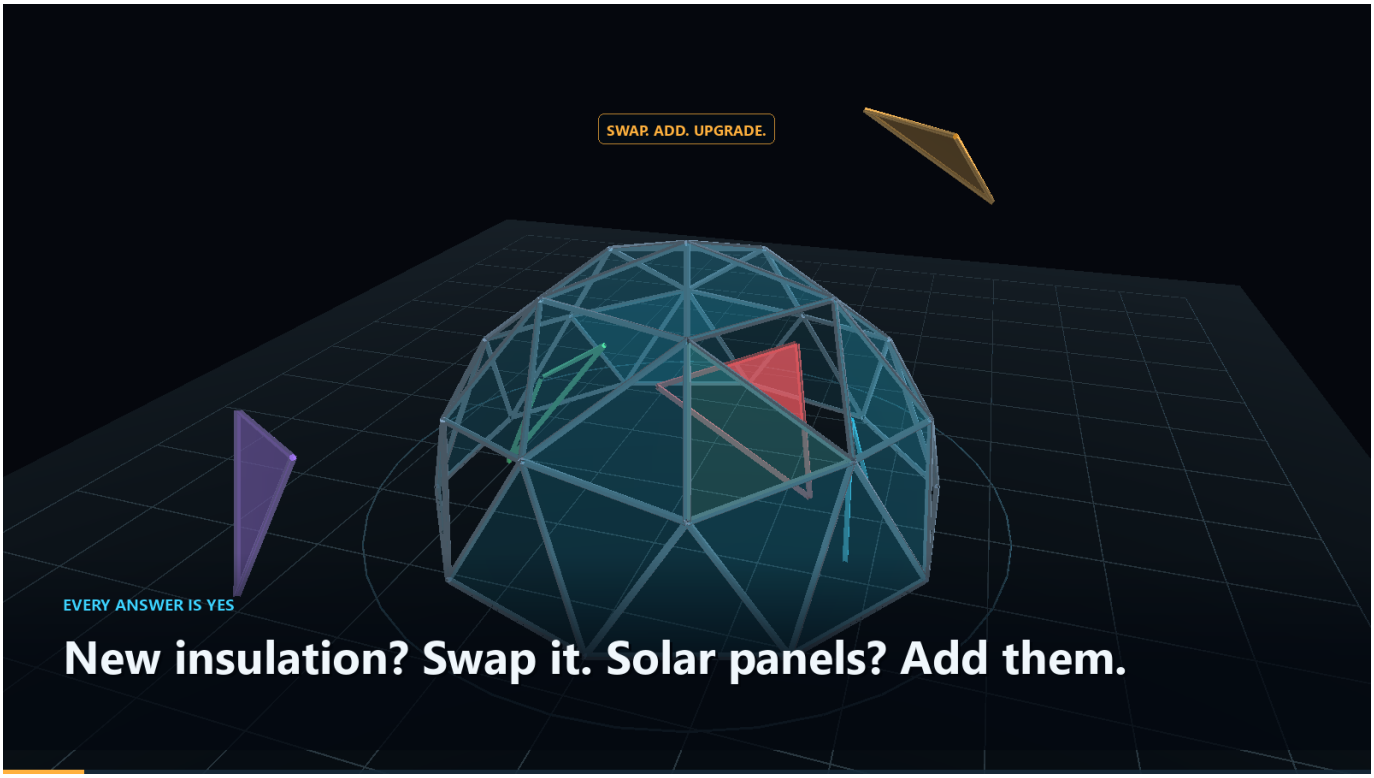

Figure 114. Every answer is yes

New insulation? Swap it. Solar panels? Add them. New insulation? Swap it. Solar panels? Add them. Greenhouse wall? Sure. From Frankendome, Version Six, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Transportable components

Everything in this building is sized by what two people can carry, and that constraint did more design work than any aesthetic decision.

THE PIECES, AND WHAT THEY WEIGH

A member: a six-foot wedge, comfortably one person.

A panel: 15.6 or 13.1 square feet of sheet. Two people, awkwardly, overhead.

A shell slice: the hull splits into slices along S-lip seams. This is the one that caught the film out — an early cut said two people could carry a slice and the model said 427 pounds. They cannot. It is a machine lift or more slices.

The utility column: one object, on a trolley, out of one dome and into the next.

WHY IT MATTERS MORE THAN IT SOUNDS

A component that needs a crane needs a crane on the day it goes in and on every day it might come out. A building assembled by machine is a building that cannot be altered by its owner.

The whole deferment argument — buy it cheap, improve it over years — depends on every future step being something the owner can do.

THE SHIPPING VIEW

120 members at six feet, forty panels at about five feet, a shell in slices and a column.

None of it is longer than a pickup bed. That is deliberate and it means the building can reach a site that a truck cannot.

THE ONE THAT IS NOT TRANSPORTABLE

The pad. $2,916 of platform that stays where it is when the dome leaves, because it is the host's.

That is the right division. The expensive, immovable, site-specific thing belongs to the land, and the light, demountable, valuable thing belongs to the person.

Figure 115. The core moves to the next dome

You buy the plumbing once. The shell is the thing that grows. And because the services are a column rather than a system built into the walls, they come out. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Kit construction

A kit is what this method becomes when somebody else does the cutting, and it is a different product with a different customer.

WHAT IS IN ONE

120 members, cut and sorted. 40 panels in two shapes. The utility column, assembled and tested. Fixings, gaskets, a shell, and a set of drawings.

WHAT IT COSTS TO INCLUDE THE FRAME

The frame is the part a buyer with trees does not need.

That is the whole harvest argument: a milled log yields about 45 percent of itself as usable lumber, and split into wedges it yields about 88. A kit buyer is paying for somebody else's 45 percent.

So a kit is either expensive or it is not a wedge kit — and a kit that ships milled rectangular stock has abandoned the method's central claim while keeping its shape.

THE HONEST SPLIT

Two products, not one.

For somebody with land and trees: everything except the frame. The column, the panels, the shell, the fixings and the drawings, and a weekend with a chainsaw for the rest.

For somebody with money and no trees: the whole thing, at a price that reflects the timber, and no pretence that it is the cheap version.

Trying to sell one product to both customers is how a good idea gets a reputation for over-promising.

WHAT A KIT IS GENUINELY GOOD AT

The column. It is fiddly, it wants 6.5 hours on a bench, and it is the same object every time. That is exactly what should be made by somebody who has made forty.

Panels, gaskets and fixings, for the same reason.

The frame is the part that should stay local. The rest is the part that should not.

Standardized interfaces

An interface is a promise about where two things meet, and this building has about six of them. That number is the design.

THE LIST

The pad port: where services enter, and where the column lands.

The apex sleeve: where services leave, under a gasketed cap.

The bay: two triangles, forty openings, four inserts each.

The rim: where the shell latches to the pad.

The wall port: where a utility panel connects to the exterior routing.

The mast: where the floor's hub clamps, and where a hoist lifts.

WHY SO FEW IS THE POINT

Every interface is a thing that has to be documented, manufactured consistently, and kept compatible forever.

Six is a number a person can hold in their head and a drawing can show on one page. Sixty is a system that needs a manual and a version number, and it will drift.

WHAT STANDARDISING THEM BUYS

Parts from different years fit. A panel made now fits a frame made in five years, because the bay did not change.

A supplier can make one part without making the whole building.

And an owner can make a part. A plywood blank with four holes on the right pitch is a panel, and anybody can cut one.

THE DISCIPLINE

Fix the interface, vary everything else.

The bay outline does not change, so panels can be anything. The pad port does not change, so the column can be improved. The rim does not change, so a shell can be replaced with a different kind of shell entirely — which is exactly what happened when the hull became the cap and nothing else in the building needed touching.

That last one is the proof the discipline works. The most expensive component in the design was swapped for a completely different technology, and the frame, the panels, the column and the pad did not notice.

18. The Dome as a Platform

A dome is not a house. It is a volume with services in it, and what it becomes is decided by which panels are in the bays and what is bolted to the inserts.

This chapter is the catalogue of what people actually do with them, including the two uses where the shape's disadvantages stop mattering and the one where they get worse.

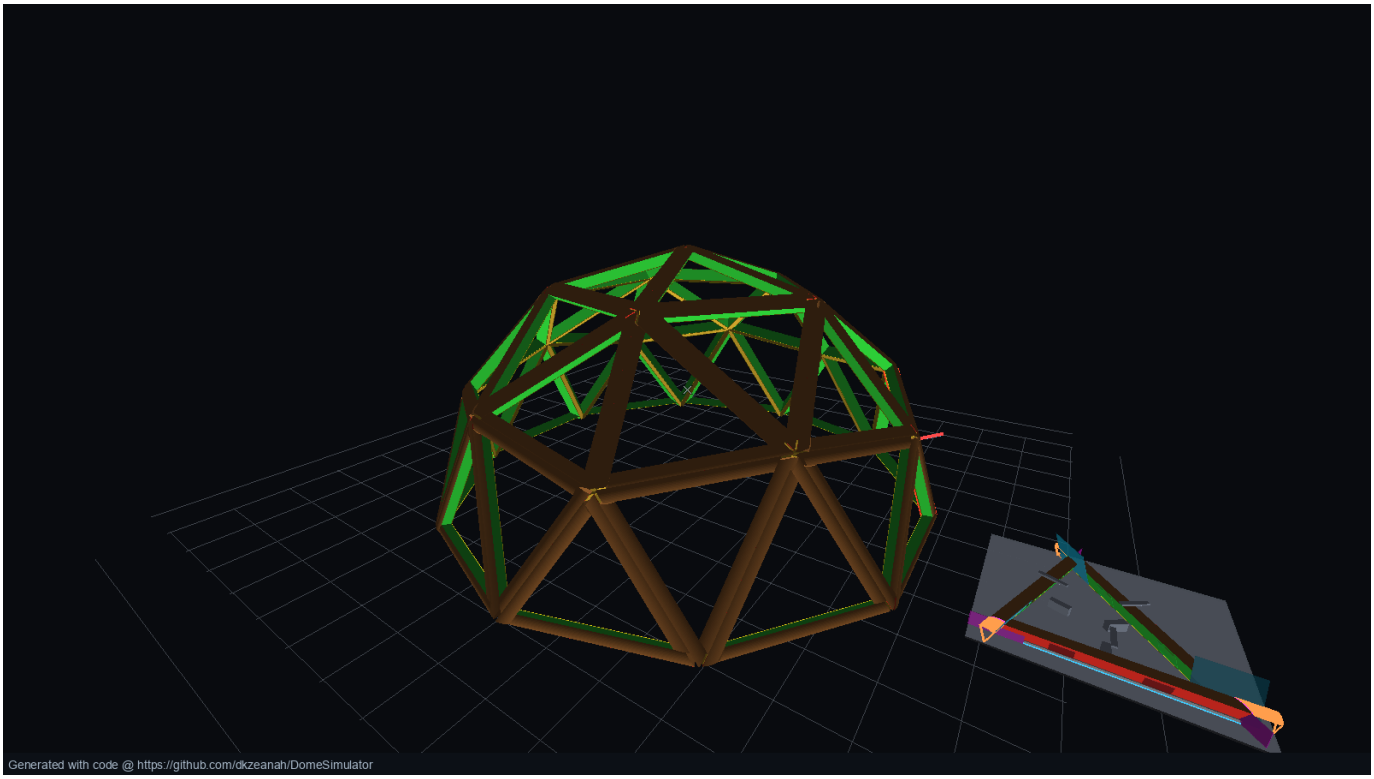

Figure 116. The building and the machine that makes it

The dome and its jig in one frame, at the same scale. The fixture is a sheet of plywood and five cleats, and it is the entire manufacturing plant. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Workshops

A workshop is the best first dome anybody builds, for reasons that have nothing to do with domes.

WHY

Nobody lives in it, so the things this shape is bad at do not matter. The leaning perimeter is where the benches go. The lack of a vertical wall is irrelevant when the storage is shelving bolted to the frame.

And the things it is good at are exactly what a workshop wants: one open volume, no columns, 277 square feet of clear floor, and daylight from every direction.

THE FIT-OUT

Benches round the perimeter, under the lean, where the headroom is lowest and you are sitting or standing at a bench anyway.

Machines in the middle, where the ceiling is 9.7 feet.

Storage on the inserts. Three of them hold about 540 pounds, and there are 160 of them round the building.

WHAT TO SPECIFY DIFFERENTLY

Power, generously. A workshop wants circuits a house does not, and adding them later means opening the column.

Dust extraction, which needs a route out. The apex is already a penetration and it is the obvious one.

Light. Glazed bays low, for working light without glare, and a lot of artificial light because a dome's curved ceiling scatters it beautifully.

Heat that comes on quickly, because a workshop is heated when occupied rather than continuously, and this building has almost no thermal mass to fight.

THE REASON TO BUILD THIS ONE FIRST

Because everything you get wrong is recoverable, and you will have learned the method on a building whose standards are forgiving before you build one whose standards are not.

Figure 117. A mast through the column, and a floor that comes later

Steel where the strength is, wood everywhere else. The floor clamps on. One more upgrade, and it changes what the dome can do. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Cabins

A cabin is the hardest thing on this list and the one most people want, so it is worth being straight about what it is like.

WHAT YOU GET

277 square feet, one round room, 9.7 feet at the middle.

The reference build's standard article lists at $12,036.

WHAT IS HARD ABOUT IT

The perimeter. Furniture is rectangular and the wall is not, so a bed against the wall has a triangle of dead space behind it and a wardrobe will not stand upright at the edge.

Subdivision. Putting a wall inside a round room wastes more of it than the same wall in a rectangular one, and a bathroom in a corner of a dome takes a disproportionate bite.

Sound. A curved ceiling focuses sound at the centre of the room, which is noticeable and occasionally unpleasant.

Doors. There is no vertical wall for one, which is why the entry extension exists.

WHAT SOLVES MOST OF IT

A knee wall. Three feet of vertical between the pad and the dome fixes the furniture, fixes the door and grows the usable floor by more than its own area. The reference build does not have one, and for somebody living in it full time it is probably the right first change.

Built-in furniture, following the curve. This is genuinely better than free-standing and it is what people who have lived in domes end up doing.

The utility panels. A bathroom hung on the outside of the shell takes none of the floor, which is the reference build's answer to the subdivision problem.

THE HONEST SUMMARY

A dome cabin is a lovely single volume and an awkward set of rooms. Build it if you want the first. Do not expect it to become the second without a fight.

Greenhouses

A greenhouse is the use case where a dome's disadvantages become advantages, and it is underrated.

WHY THE SHAPE SUITS IT

Glazing the top is the right answer here, not the wrong one. Everything the cooling chapter warned about — upper panels facing the sky, gain all day, the curve concentrating light toward the centre — is what a greenhouse wants.

Light from every direction. A dome has no north wall in the way a pitched greenhouse does, so plants get even light and do not lean.

Strength in wind and snow, which matters for a structure that is mostly glass and usually out in the open.

And the volume-to-surface efficiency means it holds heat better overnight than the same floor area under glass in a box.

WHAT CHANGES IN THE SPECIFICATION

Glazing most bays, which is the opposite of every other use here.

Ventilation, urgently. A glazed dome overheats faster than anything else in this book. The apex is the natural exhaust and it wants to be a proper automatic vent, not a cap somebody remembers to open.

Frame protection. Timber in a permanently humid glasshouse needs treating or it will not last.

Shading, seasonal. Removable panels over selected bays, or a shade cloth over the whole thing in July.

THE VERSION WORTH BUILDING FIRST

A small one. A 1V or a small 2V greenhouse is a weekend, it costs very little, and it teaches the method on a building where a leak is a nuisance rather than a disaster.

Figure 118. Hang it between two trees

Three cables, three saddles, one winch. And an engineer first. The floor is what makes this next part possible. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Temporary housing

Temporary housing is the use this method was arguably designed for without setting out to be.

WHAT MAKES IT SUIT

Speed. A frame goes up in a day with help. The whole building is a few weekends.

Cost. $12,036 for the standard article, and a cheaper version is possible by leaving out everything the deferment ladder defers.

It comes apart. That is not a nice feature here, it is the requirement: the building has to leave when the situation that needed it ends.

It travels. Nothing longer than a pickup bed.

And it does not need a foundation. A prepared pad at $443, or a portable platform, and no concrete.

WHAT IT NEEDS THAT THE REFERENCE BUILD DOES NOT

Speed of erection over cost of parts, which flips several decisions: bolted connections over anything glued, pre-cut members over site-cut, and a shell that goes on in one piece.

Storage between deployments. A folded cap, stacked panels and bundled members want a container, and the design should suit the container.

A documented assembly sequence somebody can follow who has not seen one before.

THE HONEST CAVEAT

A dome is unfamiliar, and unfamiliar is expensive in exactly the situations where temporary housing is needed — the people putting it up are usually tired, untrained and working against a clock.

That argues for a kit with very few decisions in it, not for the version of this method that starts with a chainsaw.

Semi-mobile structures

Semi-mobile means it moves, but not often and not quickly — which is a genuinely different category from either a building or a caravan.

WHAT IT LOOKS LIKE

The dome comes apart and travels as parts. It goes up on a new pad in a weekend. It does that perhaps once every few years.

The utility column moves with it — the campaign film prices that move at $260 and rates the core for 25 years, which is longer than several of the buildings it will stand in.

The pad stays. That is the host's.

WHY THIS IS THE METHOD'S NATURAL SHAPE

Every decision in this book has been pushing toward it: demountable panels, a removable shell, a column that unbolts at both ends, connectors instead of splices, gaskets instead of sealant.

None of those were chosen for mobility. They were chosen so the owner could change their mind. Mobility is what you get for free when everything can be undone.

THE PRACTICAL LIMITS

Gaskets take a set and want replacing after a few cycles.

Timber that has dried in place does not go back exactly as it came apart. Label everything, and expect the second assembly to be a little fussier than the first.

And the services at the new site have to match. A standard pad port is what makes that true, and a site without one is a site that needs one built.

THE ARRANGEMENT THIS ENABLES

Pads as infrastructure and domes as tenants — which is the next section but one, and the most commercially interesting idea in this book.

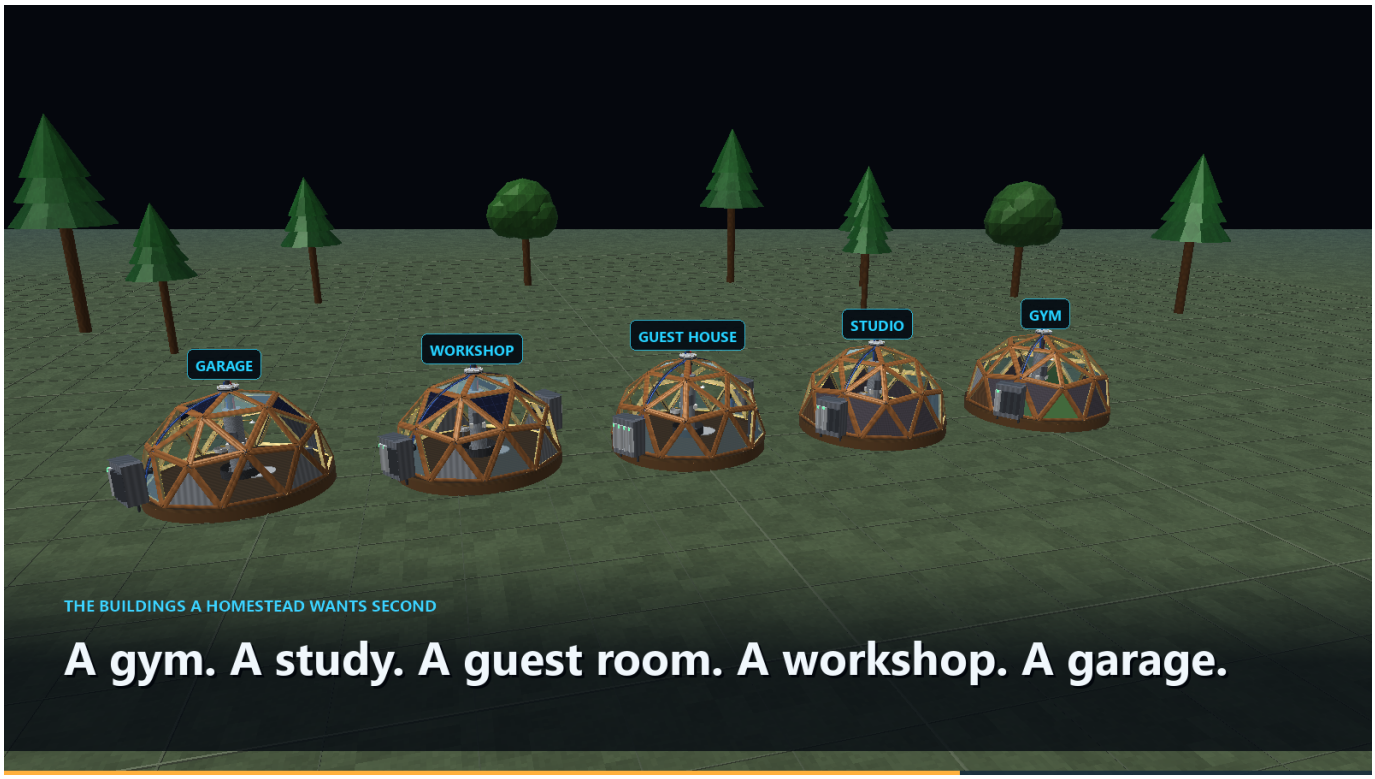

Figure 119. The buildings a homestead wants second

A gym. A study. A guest room. A workshop. A garage. Last thing, and it is the one most people actually want first. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Prepared dome pads

A prepared pad is a serviced platform waiting for a building, and it turns a piece of land into something that earns.

WHAT IS IN ONE

A compacted base. A platform — $2,916 for the framed version, $2,468 for a slab.

A service port in the middle with water, waste and power brought to it.

Latching points at the rim for a shell, and a footprint sized for a known dome.

WHY A HOST WOULD BUILD ONE

Because the platform is the expensive, immovable, permission-requiring part, and the dome is not.

A landowner who builds pads is renting the part of the building that has to stay. A tenant who brings a dome owns the part that can leave. Neither is renting from the other in the way a normal tenancy works, and that changes the relationship.

The upkeep is the other half of it. A pad host maintains a platform and a service connection — not a roof, not a boiler, not somebody's kitchen. The tenant's building is the tenant's problem.

THE NUMBER THAT DECIDES IT

Foundations run from 8 to 63 percent of the cost of the dome that stands on them, depending which foundation and which dome.

At the low end the pad is a rounding error and the model is obviously attractive to the host. At the high end the host has spent most of a building and owns none of one.

That spread is the whole business case, and it is worth working out for the specific pad and the specific dome before anybody builds ten of them.

THE HONEST DIFFICULTY

It is an unusual arrangement and the law in most places does not have a word for it. Whether a dome on a pad is a building, a vehicle, a tenancy or a chattel varies by jurisdiction, and the answer determines the tax, the permit and the insurance.

Find that out first, locally, before the second pad.

Multi-dome compounds

One dome is a room. Several domes are a house, and the connection between them is where the design problem lives.

WHY MULTIPLE RATHER THAN BIGGER

Because the frame line does not change.

Six second buildings priced off the same model differ only in which panels are in the bays. A gym, a garage, a studio and a guest room are the same frame, the same 120 members, the same column, and different infill.

That is the manufacturing argument and it only works at one size. A bigger dome is a different frame, a different member length and a different everything.

Building the same dome four times is cheaper per square foot than building one dome four times the size, which is the opposite of how conventional construction works.

THE LAYOUT

Domes want to be far enough apart that the ground between them drains and close enough that walking between them in the rain is acceptable.

A cluster round a shared yard reads well and shelters the yard. A row reads badly, because a row of round things looks accidental.

WHAT IS SHARED

The services. One connection, one tank, one panel, spurred to each pad — the reference build prices a share of one hub panel at four ways.

The expensive machinery. One workshop with the good tools. One kitchen, if the domestic arrangement allows it.

WHAT IS NOT

The heating. Four small buildings heated separately cost more than one large one heated throughout, and this is the compound's real penalty. Four envelopes is more envelope than one.

THE ARGUMENT THIS LOSES

The 41 percent envelope saving is per building, and four buildings have four envelopes. A compound gives up a good part of the shape's thermal advantage in exchange for modularity.

Worth saying, because the compound is usually presented as all upside.

Connected structures

Joining two domes is the detail that decides whether a compound is a compound or a house, and it is harder than it looks.

THE THREE WAYS

A covered walk. Two separate buildings, a roof between them, and you go outside-but-sheltered to move from one to the other.

This is the easy one. No structural connection, no thermal bridge, no difficult junction, and each dome stays a dome.

A short link. A small rectangular structure joining the two, with a door at each end.

Harder. Two junctions of a flat thing to a curved thing, which is the same problem as the entry extension, twice. Flash to the frame, not to the shell, and let the shell overlap.

Direct intersection. Two spheres overlapping, with the shared region open.

This is the one everybody draws and almost nobody builds, because the intersection of two spheres is a circle in three dimensions and every member meeting it is cut at a different angle. It is beautiful and it is a geometry problem rather than a carpentry one.

WHAT THE METHOD PREFERS

The link, with a proper structure of its own, joined at reinforced openings in both domes.

It keeps each dome intact and independent — which means each can still be taken apart, which is the thing this whole book is protecting.

A direct intersection welds two buildings into one and gives up every demountable property they had.

THE THERMAL DETAIL

A link is a corridor, corridors are unheated in practice, and an unheated corridor between two heated buildings is four cold walls where there were two.

Either heat it and accept the cost, or insulate the two doors properly and treat the link as outside. Deciding by accident gives you the worst of both.

19. Reference Designs

Five domes, from one you build to get it wrong to one nobody has built at all.

Every figure is solved from the same factors at the diameter it claims, so a reader who checks one against the simulator gets the same answer. Where a size has not been built, it says so.

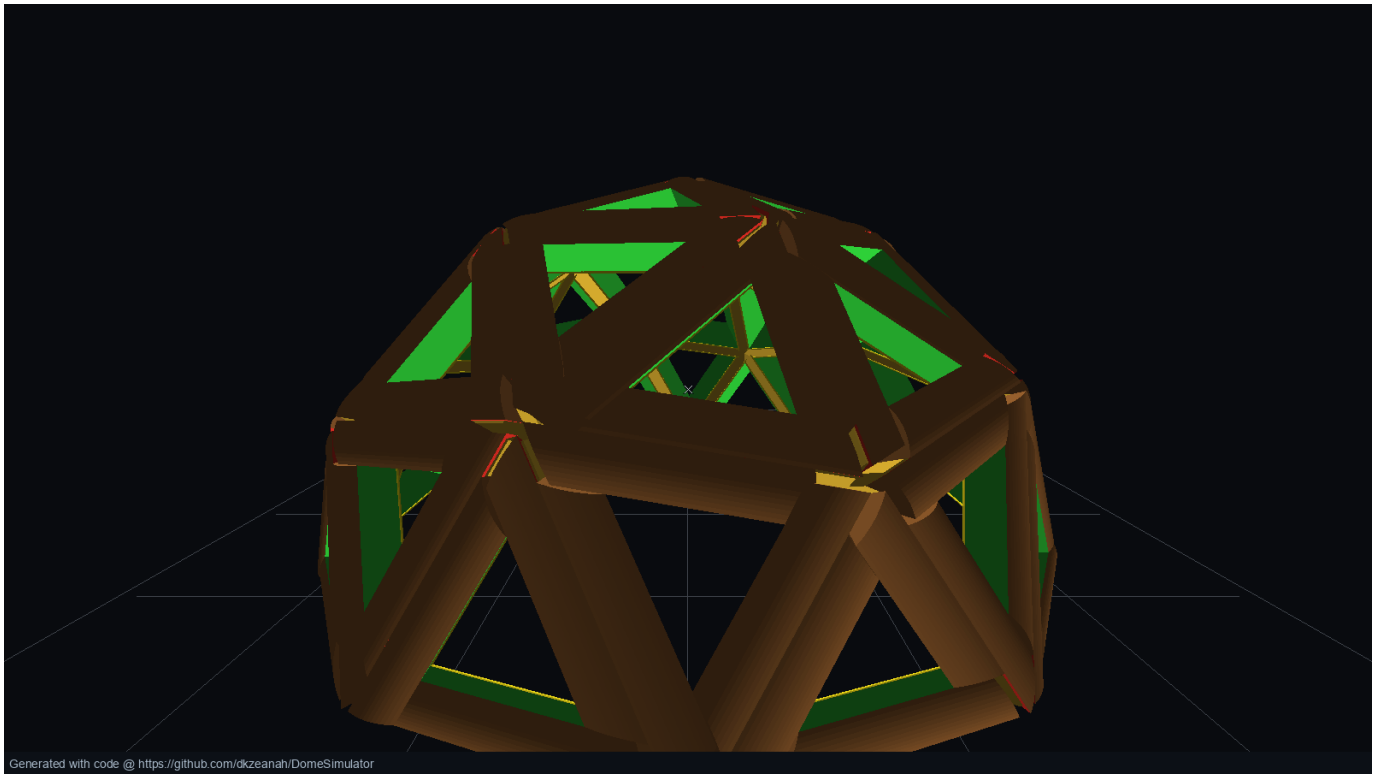

Figure 120. A 4-foot radius

A 4-foot radius: the demonstration dome. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Small demonstration dome

Build this one first. It exists to be got wrong.

THE SIZE

diameter	8.00 ft
apex height	4.00 ft
A strut, cut length	27.848 in
B strut, cut length	25.244 in
floor, inside the ring	47.0 sq ft
shell surface	93.3 sq ft
timber in the frame	265 ft

At eight feet you can reach the apex from a step. The whole frame is 120 members under three feet each, which is one afternoon of cutting and one of assembly.

WHY IT IS WORTH THE TIME

Because everything in this method that is hard is hard at every size, and everything that is expensive is only expensive at full size.

The pinwheel is the same pinwheel. The compound butt cut is the same 22.5-degree bevel and the same 3 mitres. The assembly order is the same. The mistake you are going to make with the base ring is the same mistake.

Make it here, on 120 short sticks, rather than on a hundred and twenty six-foot ones.

WHAT TO USE IT FOR AFTERWARDS

A cold frame. A dog house. A play structure. Firewood storage. Or take it apart and keep the members as a demonstration of the joint.

WHAT NOT TO LEARN FROM IT

How it feels to be inside one. At eight feet it does not feel like a building, and any conclusion you draw about living in a dome from standing in this one will be wrong.

THE ONE RULE

Build it to the same tolerance you intend to build the real one. A demonstration built sloppily teaches you that sloppy works, right up until the fortieth panel on the real dome.

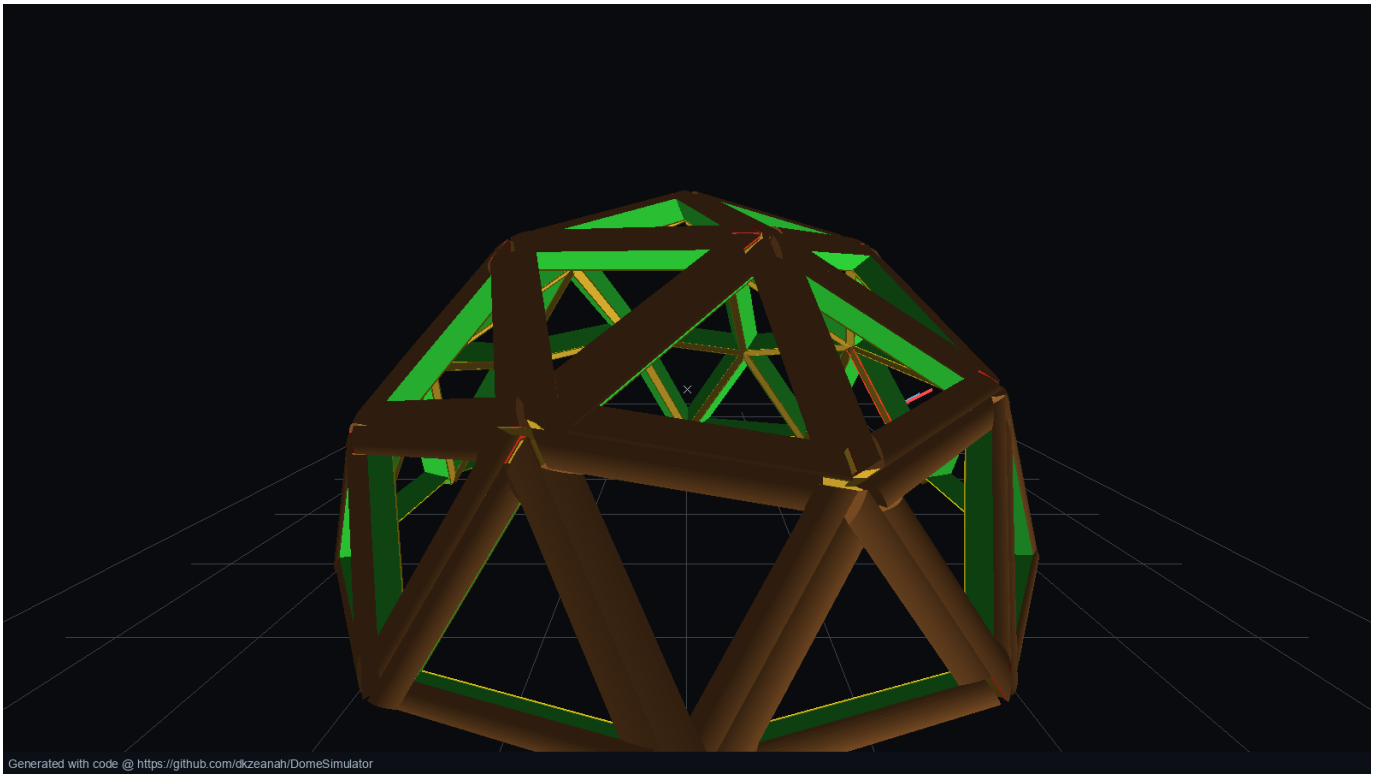

Figure 121. A 6-foot radius

A 6-foot radius: the twelve-foot dome. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

12-foot dome

Twelve feet is the smallest dome that is genuinely useful, and the largest one person can build alone without much trouble.

THE NUMBERS

diameter	12.00 ft
apex height	6.00 ft
A strut, cut length	42.681 in
B strut, cut length	38.361 in
floor, inside the ring	105.8 sq ft
shell surface	210.0 sq ft
timber in the frame	405 ft

WHAT IT IS FOR

A greenhouse. A workshop for one person. A studio. A shed that is nicer than a shed.

Not somewhere to live. At six feet to the apex the standing room is a circle in the middle and the rest is bench height.

THE ADVANTAGES OF THIS SIZE

Every member is under four feet, so the stock comes out of small trees or short offcuts, and the harvest argument works on timber a bigger dome could not use.

Nothing needs lifting equipment or a second pair of hands.

And panels fit sheets. At this diameter the narrowest face is under four feet across, so ordinary plywood works with no special order — which is the reference build's one real inconvenience, absent.

THE DISADVANTAGES

The perimeter is almost unusable. The wall leans in so sharply that only the middle third of the floor has headroom.

A knee wall helps disproportionately here. Two feet of vertical on a twelve-foot dome is a bigger change than three feet on a twenty.

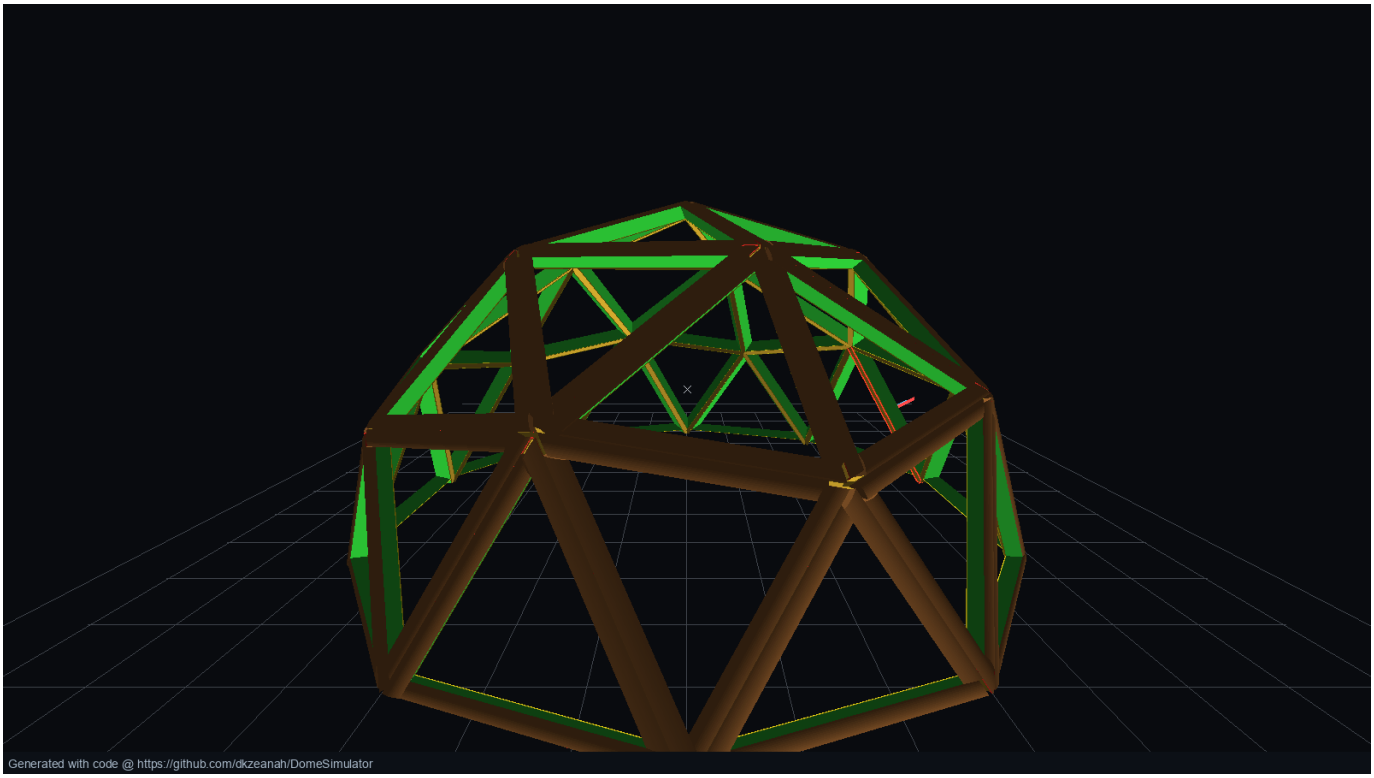

Figure 122. The reference build's six-foot member

The reference build's six-foot member. the reference build uses this. Generated with code @ https://github.com/dkzeanah/DomeSimulator

16-foot dome

Sixteen feet is the size most people should build and fewer do, because twenty sounds better.

THE NUMBERS

```
diameter                  16.00 ft
apex height                8.00 ft
A strut, cut length       57.513 in
B strut, cut length       51.478 in
floor, inside the ring     188.1 sq ft
shell surface              373.3 sq ft
timber in the frame          545 ft
```

WHAT IT IS FOR

A cabin for one or two people. A serious workshop. A guest building. A studio with room for the work and the person.

WHY IT IS THE PRACTICAL SWEET SPOT

Members under five feet. Still one-person handling, still short enough to come out of modest trees, still transportable in anything.

Eight feet to the apex, which is a normal ceiling. You can stand anywhere in the middle half of the floor.

188 square feet, which is a real room.

And the whole build is within a few weekends rather than a season.

THE HONEST COMPARISON WITH TWENTY

Going from sixteen to twenty adds about 89 square feet of floor and about 117 feet of timber, and it moves every member past five feet.

That is worth it if the extra floor is needed. It is not worth it for the sound of the number, and a sixteen-foot dome finished well is a better building than a twenty-foot one abandoned at the shell.

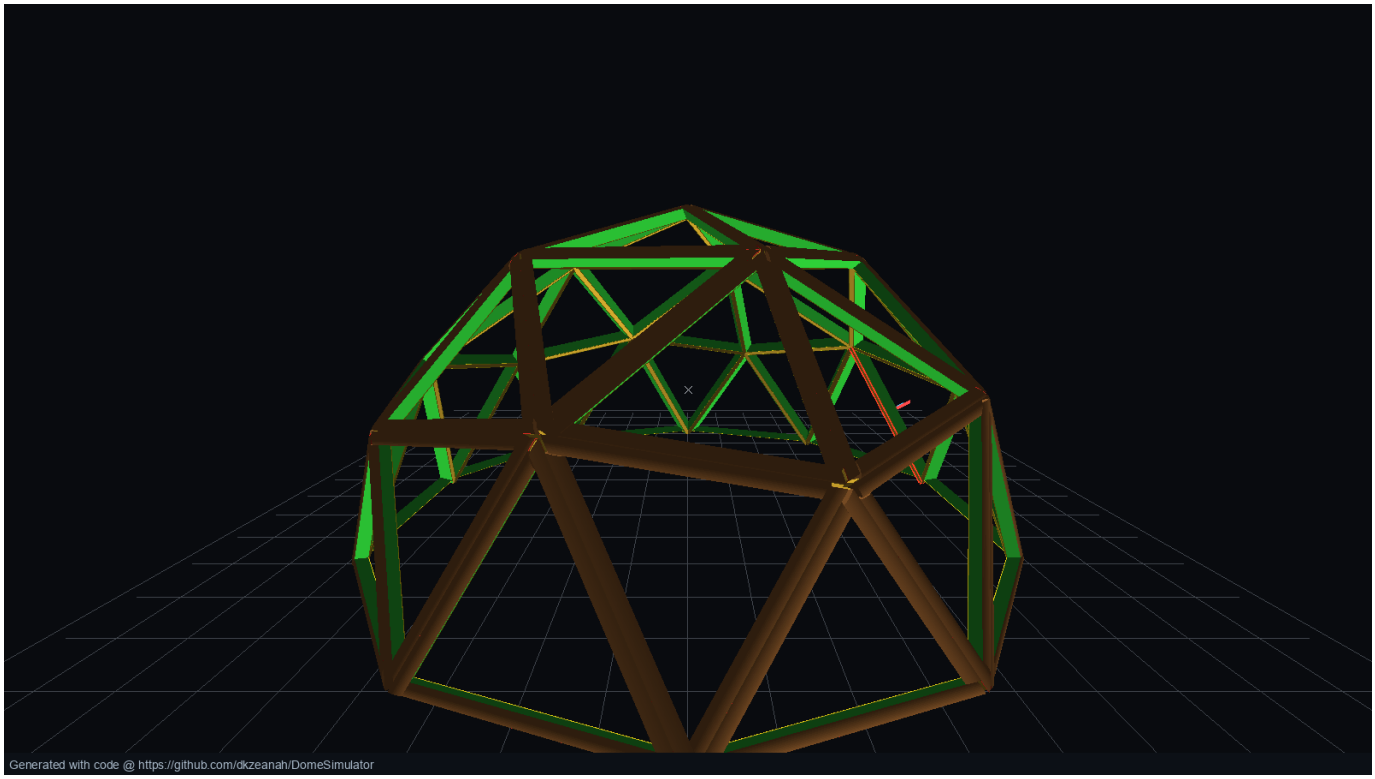

Figure 123. A 12-foot radius

A 12-foot radius: where the 2V starts to strain. an alternative to the reference build. Generated with code @ https://github.com/dkzeanah/DomeSimulator

20-foot reference dome

This is the dome the rest of this book prices, the one the simulator solves, and the one every figure in the campaign film comes from.

THE NUMBERS

```
diameter                  19.42 ft
apex height                9.71 ft
A strut, cut length       70.182 in
B strut, cut length       62.681 in
floor, inside the ring     277.0 sq ft
shell surface              549.8 sq ft
timber in the frame          664 ft
```

Precisely: a radius of 116.4984 inches, so 19.416 feet across and 9.708 to the apex.

THE PARTS

26 vertices. 40 faces — 10 AAA at 15.59 square feet and 30 BAB at 13.13.

120 members: 60 A at 70.182 inches and 60 B at 62.681, against chords of 72.000 and 63.6703.

55 interior seams, 10 base edges, 309 feet of seam channel.

WHY THIS SIZE

Because the member is six feet.

Everything else follows from that. A six-foot stick is what comes comfortably out of a twelve-foot log section, it is what one person carries, and it fits in a pickup. The diameter is the answer to the member, not the other way round.

WHAT IT COSTS

12,036 dollars for the standard article, shower cap and all.

THE INCONVENIENCE

No panel fits a four-foot sheet. The narrowest face is 52.516 inches across and a sheet is 48.

That is a direct consequence of sizing the dome from the member rather than from the sheet, and it is the price of the decision. Anybody who would rather have the sheets than the six-foot member should build a different diameter, and the tables in the next chapter will give them one.

Figure 124. What one costs, line by line

Materials, labour, overhead, margin. Nothing hidden in a lump. So here is the whole thing. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Larger examples

Above about twenty feet the 2V starts to strain, and the strain shows up in three places.

WHAT A 24-FOOT 2V LOOKS LIKE

```
diameter                  24.00 ft
apex height               12.00 ft
A strut, cut length      87.179 in
B strut, cut length      77.711 in
floor, inside the ring    423.2 sq ft
shell surface             839.9 sq ft
timber in the frame         824 ft
```

THE THREE PROBLEMS

Member length. Over seven feet, which is past comfortable one-person handling and past a pickup bed.

Faceting. The panels are over 20 square feet and they are flat. At this size a 2V reads as a faceted polyhedron rather than a dome, and the flat panels start carrying bending they should not.

Load. The same 2V geometry at a bigger span means bigger forces in the same number of members, and at some point the section has to grow — which means a bigger log, which the harvest argument may not be able to supply.

WHAT TO DO INSTEAD

Go to 3V. More, smaller triangles, three strut lengths, panels that fit sheets, and a shape that reads as round.

The member count roughly triples from 120, which is the cost.

OR BUILD TWO

Two twenty-foot domes have more floor than one twenty-eight-foot one and cost less per square foot, because the frame line does not change between them.

That is the manufacturing argument again, and it is the reason this method would rather make the same building four times than make a bigger one.

THE LIMIT NOBODY HAS TESTED

At what span does a wedge frame need engineering rather than reasoning?

This book does not know. Nothing here has been load-tested, and everything above about twenty feet is extrapolation from a building that has been solved but not built. That is worth knowing before anybody scales it.

Figure 125. Three things this says against itself

Ships with an empty cavity. Wastes wood. And the duct is unproven. Three things we are going to say against our own pitch, because a proposal that only shows its good numbers is not a proposal. From A house you can take apart, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

20. Builder's Reference

Everything in this chapter is generated from the solved geometry rather than copied from a table, which means it can be checked and it means it extends.

The two ratios at the top of the strut table are the whole of it: the A chord is 0.618034 times the radius and the B chord 0.546533. Every other number here is those two, a member width, and arithmetic.

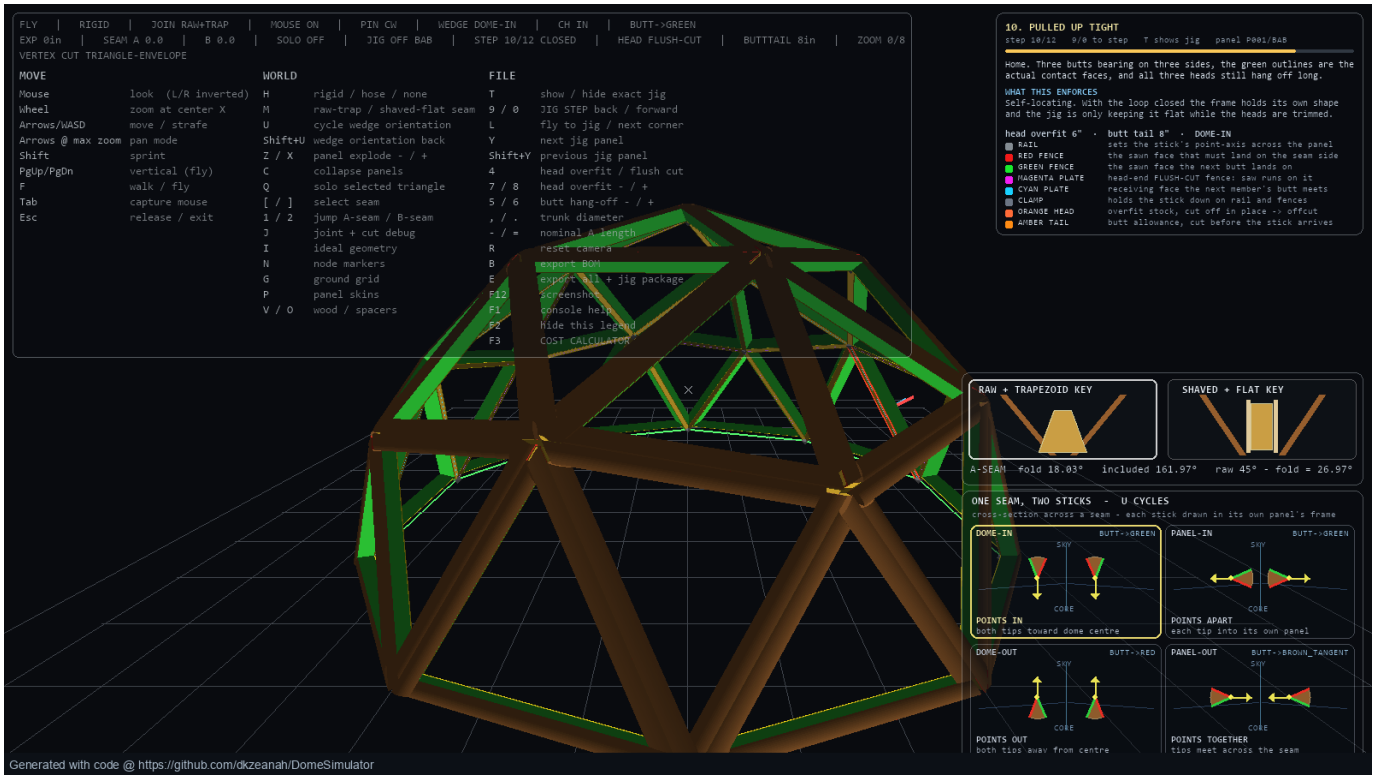

Figure 126. Reading the tool

The solver as it opens, with its own heads-up display on. Every figure in this book was taken in this program, and this is the one figure where the display is the subject rather than the clutter. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Strut tables

Two strut lengths per dome, and the cut is not the chord.

diameter	radius	A chord	B chord	A cut	B cut	
8.00'	4.00'	29.666"	26.234"	27.848"	25.244"	
10.00'	5.00'	37.082"	32.792"	35.264"	31.802"	
12.00'	6.00'	44.498"	39.350"	42.681"	38.361"	
14.00'	7.00'	51.915"	45.909"	50.097"	44.919"	
16.00'	8.00'	59.331"	52.467"	57.513"	51.478"	
18.00'	9.00'	66.748"	59.026"	64.930"	58.036"	
19.42'	9.71'	72.000"	63.670"	70.182"	62.681"	<- reference
20.00'	10.00'	74.164"	65.584"	72.346"	64.594"	
22.00'	11.00'	81.580"	72.142"	79.763"	71.153"	
24.00'	12.00'	88.997"	78.701"	87.179"	77.711"	
26.00'	13.00'	96.413"	85.259"	94.596"	84.270"	
30.00'	15.00'	111.246"	98.376"	109.428"	97.386"	

HOW TO READ IT

The CHORD is the straight-line distance between two vertices of the sphere. It is the number every geodesic reference publishes and it is not the number you cut.

The A chord is 0.618034 times the radius and the B chord 0.546533. Those two ratios are properties of a 2V icosahedron, they are exact at every size, and everything else on this page is derived from them.

The CUT is what a pinwheel member is, once the joint has taken its bite. One end butts into the side of a neighbour, short of the corner; the other runs past the mathematical vertex so the previous member can butt into its side. Neither end lands on a vertex.

THE THING THAT CAUGHT THIS TABLE OUT

The bite does not scale with the dome.

It is set by how wide the member is, and a member does not get wider because the dome does. Solve the same pinwheel at a 4-foot radius and at a 15-foot one with the same stick and the bite comes out identical to four decimal places.

So the cut is CHORD MINUS A CONSTANT, not chord times a ratio. An earlier version of this table used a ratio — correct at the reference build and wrong at every other row, by more than eight inches at the small end.

If you take one thing from this chapter, take that. Scale the chord; subtract the bite.

THE BITE USED HERE

```
A members    1.818 in
B members    0.990 in
```

Both at the reference build's 4.592-inch member. A wider member takes a bigger bite, roughly in proportion, so a frame cut from heavier stock wants its own solve rather than this table.

TWO MODELS, AND THEY DISAGREE

This repository solves the pinwheel twice.

The raw-wedge solver treats a member as a real 45-degree log sector, and that is where the bites above come from. It is the live model, it is what the simulator draws, and it is what the reference build is cut to.

A second, simpler model treats a member as a rectangular band lying with its bark face on the edge line. It gives a bite of 9.253 inches for the same stick — five times larger.

Neither is a mistake. They are answers about differently shaped sticks, and the difference is the shape of a split log versus a sawn board.

The numbers in this book are the first. If you are cutting rectangular stock rather than log sectors, this table does not describe your frame, and the difference is nine inches a stick.

THE ROW TO CHECK YOURSELF AGAINST

19.42 feet: A chord 72.000, A cut 70.182. If your own arithmetic gives that, the rest of the table will behave.

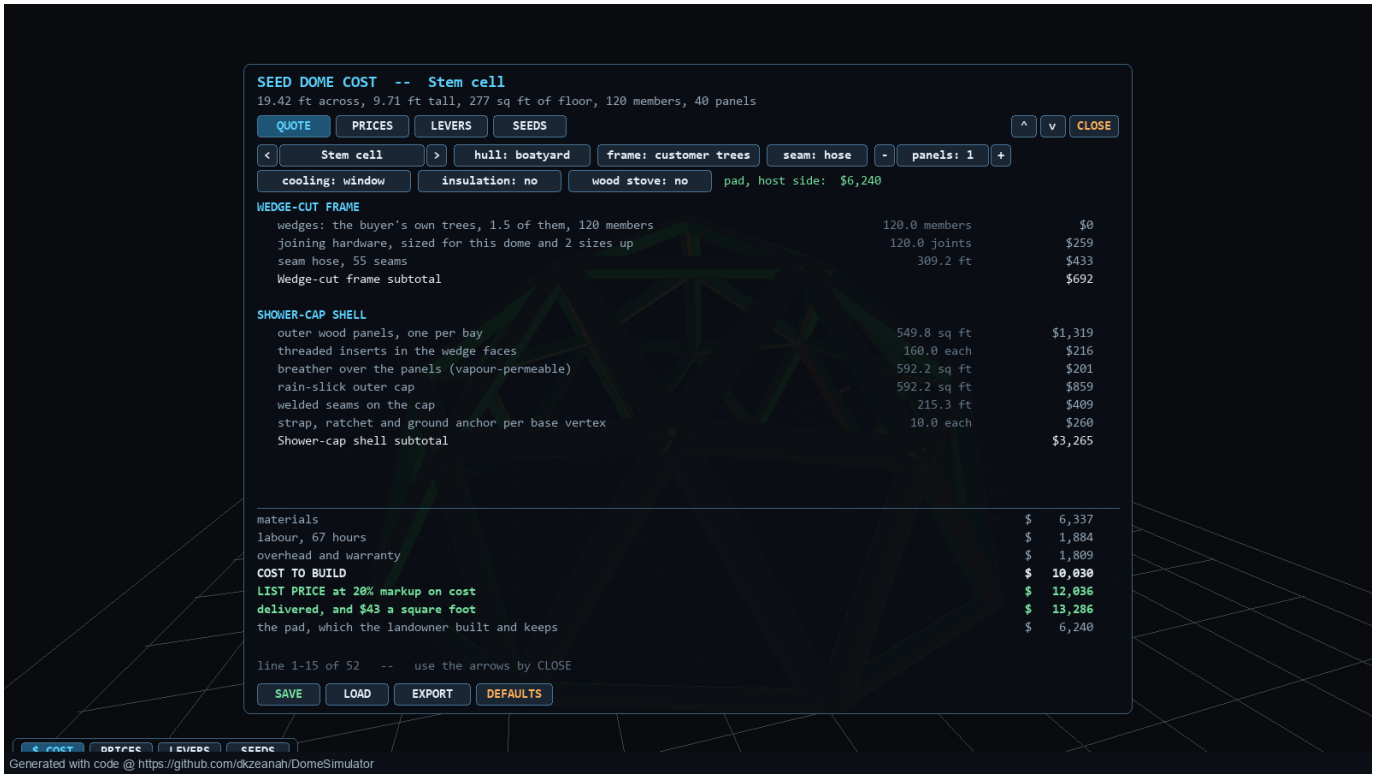

Figure 127. The same tool, pricing the dome

The cost console open over the solved dome. The geometry and the price are one program, which is why no number in this book has to be carried between two of them by hand. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Triangle tables

Two triangles, forty faces, and one of them will not fit a sheet of plywood.

diameter	AAA face	BAB face	shell	floor ring	floor circle
8.00'	2.65	2.23	93.3	47.0	50.3
12.00'	5.95	5.01	210.0	105.8	113.1
16.00'	10.59	8.92	373.3	188.1	201.1
20.00'	16.54	13.93	583.3	293.9	314.2
19.42'	15.59	13.13	549.8	277.0	296.1
24.00'	23.82	20.06	839.9	423.2	452.4
30.00'	37.21	31.34	1312.4	661.3	706.9

THE FACES

AAA: 10 of them, equilateral, made of three A struts. Perimeter 216 inches at the reference diameter, and 62.354 inches across the narrowest altitude.

BAB: 30 of them, isosceles, two B struts and one A. Perimeter 199.34 inches, narrowest altitude 52.516.

THE SHEET PROBLEM, IN ONE LINE

A sheet of plywood is 48 inches wide, and the narrowest altitude of a BAB face scales directly with the dome.

At the reference build it is 52.516 inches, so the crossover is at 17.75 feet of diameter.

Under about 17 feet, ordinary sheets work. Over it, no panel of the dome fits a sheet in any rotation, and the skin has to be sheeted across the frame, built up from boards, or ordered oversize.

That single fact decides how the skin of any dome in this book gets built, and it is worth checking before the diameter is chosen rather than after. The reference build is over the line — because it was sized from a six-foot member rather than from a sheet of plywood, and that was a deliberate trade.

WHAT THE TWO TYPES MEAN FOR ORDERING

Ten of one and thirty of the other. If panels are being cut from sheet, the BAB is the one to work out the nesting for, because there are three times as many of them.

Surface area

The shell, the cap and the paint are all priced per square foot, so this is the table the money comes out of.

diameter	AAA face	BAB face	shell	floor ring	floor circle
8.00'	2.65	2.23	93.3	47.0	50.3
12.00'	5.95	5.01	210.0	105.8	113.1
16.00'	10.59	8.92	373.3	188.1	201.1
20.00'	16.54	13.93	583.3	293.9	314.2
19.42'	15.59	13.13	549.8	277.0	296.1
24.00'	23.82	20.06	839.9	423.2	452.4
30.00'	37.21	31.34	1312.4	661.3	706.9

WHICH NUMBER TO USE

For the FRAME's shell — panels, membrane, paint — use the shell surface. It is the sum of the forty faces and it is the area of material that has to exist.

For the CAP, use more than that. A cap stands off the frame, so its area is larger: the reference build's bare cap is 592.2 square feet against 549.8 of face area, and every quilted layer under it makes the next one bigger again.

For LAMINATE, add waste. Glass fibre over compound curvature wastes a great deal, and the reference build's model carries an explicit waste fraction for exactly this reason.

THE COMPARISON WORTH MAKING

At the reference diameter the shell is about 550 square feet over 277 square feet of floor — a ratio of 1.98.

A rectangular building of the same floor area with an eight-foot wall and a flat roof has substantially more, which is where the 41 percent envelope saving comes from. Less surface is less material, less insulation and less heat leaving, all at once.

It is the strongest argument this shape has and it is entirely contained in this table.

Floor area

Two floor numbers, and the difference between them catches people out.

diameter	AAA face	BAB face	shell	floor ring	floor circle

8.00'	2.65	2.23	93.3	47.0	50.3
12.00'	5.95	5.01	210.0	105.8	113.1
16.00'	10.59	8.92	373.3	188.1	201.1
20.00'	16.54	13.93	583.3	293.9	314.2
19.42'	15.59	13.13	549.8	277.0	296.1
24.00'	23.82	20.06	839.9	423.2	452.4
30.00'	37.21	31.34	1312.4	661.3	706.9

THE CIRCLE AND THE RING

The FLOOR CIRCLE is the full circle at the dome's widest point. It is what you get if you multiply pi by the radius squared, and it is the number most dome literature quotes.

The FLOOR RING is the ten-sided figure the base ring actually makes. It is what the building encloses, and at every diameter it is about 6 percent smaller.

Quote the second one. The first is a circle nobody built.

THE THIRD NUMBER NOBODY PUBLISHES

Usable floor — the part with headroom.

A hemisphere's surface is 45 degrees from vertical at half its height, so standing room shrinks fast toward the perimeter. At the reference build's 19.42 feet, the circle where the shell is six feet up is considerably smaller than the base.

There is no table for this because it depends on what you call headroom. Work it out for your own height before you decide a diameter, and work it out again if you are considering a knee wall — which moves the whole curve up and is the single biggest change available to this number.

Figure 128. How much of the tree survives

The same two trees, converted both ways, counted in board feet. This is one pine measured at the butt and at the top, bucked into twelve foot sections, and then converted twice on paper. Split radially it keeps almost everything, losing only the kerf of the passes down each section. From Why Wedges: A Dome That Does Not Need a Sawmill, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Member count

The counts do not change with diameter. They are properties of the frequency.

```
vertices      26
faces, AAA      10
faces, BAB      30
faces, total      40
geodesic edges      65
members, A      60
members, B      60
members, total     120
interior seams      55
base edges      10
```

THE TWO NUMBERS THAT LOOK WRONG TOGETHER

There are 65 edges and 120 members.

In a shared-strut dome those would be the same number, because one stick serves both triangles along an edge. In a pinwheel they are not: every triangle carries its own three members, so the count is three times the face count rather than the edge count.

That difference is the 1.80x stock penalty, and it is the price of never cutting a stick to suit its neighbour.

THE SEAMS

55 interior seams and 10 base edges is 65 joints, which is fewer than the edge count because the base ring's own edges are counted once.

Every interior seam is also a length of duct, 309 feet of it at the reference diameter, reaching every vertex. The count is the same at any size; only the length scales.

Material quantities

Timber, by diameter, for a 2V hemisphere with a 4.592-inch member.

diameter	A stock ft	B stock ft	frame ft	seam ft
8.00'	145	129	274	127
12.00'	217	194	411	191
16.00'	289	258	547	255
20.00'	361	323	684	318
19.42'	351	313	664	309
24.00'	434	387	821	382
30.00'	542	484	1026	478

WHAT IS AND IS NOT IN IT

This is the stock in the frame. It does not include waste, and the reference build's model carries an explicit waste fraction on top.

It does not include the base ring's blocking, the panels, the shell or anything bought rather than cut.

HOW MANY TREES

The model's assumed log — 15 inches at the butt, 60 feet usable — yields about 391 board feet of wedges, enough for 80 struts.

At the reference diameter the dome needs 120, so two trees, with a third of the stock spare for mistakes. Read that as two trees for anything up to about twenty feet and three above it.

THE HARVEST COMPARISON

Milled, a log yields about 45 percent of itself as usable lumber. Split into wedges it yields about 88, because nothing is edged.

That is a 1.95x advantage. Against the pinwheel's 1.80x stock penalty, the net is about 92 percent of the trees a mitred dome would take.

Eight percent, which is small — and the point was never the eight percent. It is that the eight percent arrives with no mitre saw, no jig, no shop, and a log instead of a lumber order.

Conversion tables

The handful of conversions this method actually needs.

LENGTH

```
1 inch  25.4 mm exactly
1 foot  304.8 mm exactly
1 metre  39.3701 in
```

The reference build's radius is 116.4984 inches, which is 2.9591 metres. The simulator works in metres and the tables in inches, and that is the conversion between them.

ANGLE

```
the log sector  45 degrees
the butt bevel  22.5 degrees (half the sector)
A seam fold / gap  18.029 / 26.971 degrees
B seam fold / gap  22.459 / 22.541 degrees
```

Both seam pairs sum to 45. That is not a coincidence and it is the closest thing this method has to a theorem: the sector you split out of the tree is exactly the seam budget the dome needs.

TIMBER

```
1 board foot  144 cubic inches
a 12-ft 2x6  12 board feet nominal
```

AREA AND VOLUME

```
1 square metre  10.7639 sq ft
1 cubic foot  7.4805 US gallons
```

That last one is for tanks. The reference build's under-floor water tank is 120 gallons, which is 16.0 cubic feet of space to find.

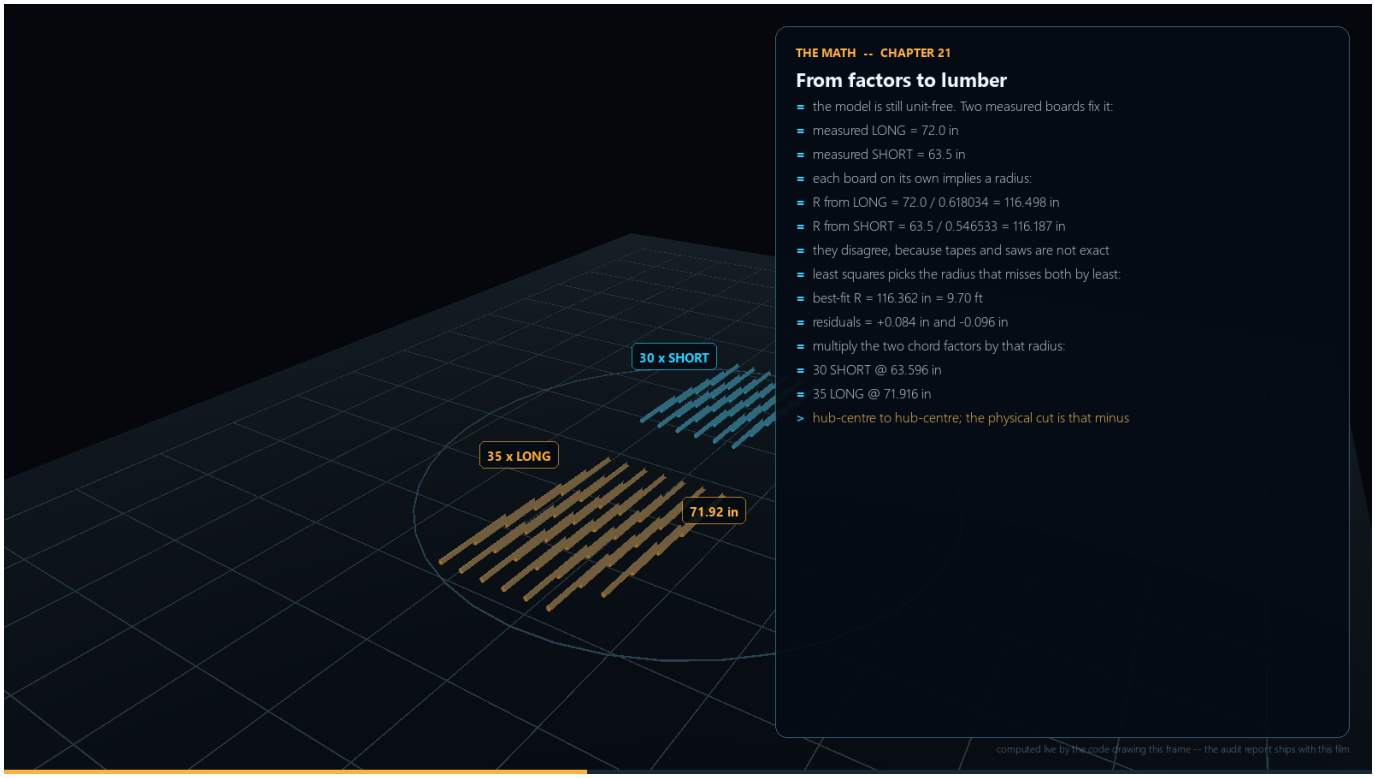

Figure 129. From factors to lumber

Two boards, one best-fit radius, sixty-five cuts. The fit, in full. Each measured board on its own implies a radius, and the two disagree by about a third of an inch. From From Scratch: Every Calculation Behind a Dome on Screen, a film in this project. Generated with code @ https://github.com/dkzeanah/DomeSimulator

Cut worksheets

One page, filled in before any wood is touched.

THE SETTINGS

```
bevel, every member  22.5 deg
mitre, setting 1  21.138 deg
mitre, setting 2  30.000 deg
mitre, setting 3  34.431 deg
```

3 settings for 120 members. The bevel is constant because it comes from how the log was split; set it on the first stick and do not touch it again.

THE CUT LIST, REFERENCE BUILD

```
A members    60  at   70.182 in
B members    60  at   62.681 in
total   120       664 ft of stock
```

Add the waste fraction before buying or felling anything.

THE BLANK LENGTH

Longer than the cut, on the butt end, because the compound cut runs 2.23 to 2.91 inches along the stick — the sector's point reaches further than its bark corner.

Add three inches to every blank and trim to the line. It is cheaper than finding out on the fortieth stick.

THE ORDER

Cut all the A members. Check three of them against each other before cutting the rest. Then all the B.

Do not alternate. Every change of setting is a chance to get it wrong, and 3 settings done once is a different job from 3 settings done a hundred and twenty times.

THE FIRST STICK

Cut one. Cut a second. Put them together as the joint will hold them, and look at the gap.

If there is one, the setting is wrong and you have wasted two sticks. If there is not, cut the other hundred and eighteen.

Blank project sheets

Copy these. Fill them in with a pencil before anything is cut.

THE DESIGN SHEET

```
diameter  _________ ft
apex height  _________ ft   (half the diameter)
A chord  _________ in   (radius x 0.618034)
B chord  _________ in   (radius x 0.546533)
A cut length  _________ in   (A chord MINUS 1.818)
B cut length  _________ in   (B chord MINUS 0.990)
member width  _________ in   (changes the bite, not the chord)
narrowest face  _________ in   (under 48 to use sheets?)
floor, ring  _________ sq ft
frame stock  _________ ft   (+ waste)
trees needed  _________
```

THE FOUNDATION SHEET

```
pad diameter  _________ ft
type  gravel / piers / ring / slab / portable
frost depth  _________ in
fall away from pad  _________ in per ft
service port position  _________
```

THE SERVICES SHEET

```
water from  mains / tank / rain / well
tank capacity  _________ gal
waste to  sewer / septic / holding / greywater
fall to drain  _________ in per ft   (min 0.25)
power from  mains / generator / solar / hub
wet-to-electrical clearance  _________ ft   (min 3)
```

THE PANEL SCHEDULE

Forty rows. Bay number, type (AAA or BAB), infill, and a column for what it became later — because it will become something later, and the sheet is the only record of what the building used to be.

```
bay  type  infill  changed to  date
___  ___  _______  _________  _____
```

SUBTRACT, DO NOT SCALE

The bite is a fixed length set by the member's width. It is the same at every diameter, so the cut is the chord minus a constant and never the chord times a fraction. The bites above are for the reference build's 4.592-inch member; heavier stock wants its own solve.

THE ONE THAT MATTERS MOST

The last column. Everything else in this book can be worked out again from the geometry. What you did to a particular bay in a particular spring cannot.